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ABSTRACTS

# Abstracts of the 54th Annual Meeting of the Behavior Genetics Association, London, UK

## Program chair: Prof. Chandra Reynolds

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### Associations Between Genome-Wide Autozygosity and Complex Traits in Two Ancestrally Diverse US Cohorts

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**Abstract**
The cumulative effect of rare, recessive variants acting on a trait can be estimated by genome-wide autozygosity (the proportion of the genome that is homozygous due to inheritance of alleles identical by descent). Prior studies have demonstrated associations of greater autozygosity with generally worse outcomes on presumed fitness-related complex traits (e.g., lower height and reduced fertility), suggesting that these phenotypes have been under directional selection pressures over evolutionary time. Studies have also identified associations between autozygosity and risk-taking behaviors (e.g., likelihood of tobacco smoking and alcohol drinking), although the mechanisms driving these associations are less clear. In this talk, I will present new data linking genome-wide autozygosity to complex traits across two large cohorts of diverse ancestries: the All of Us Research Program (AoU, max N = 206,819) and the Million Veteran Program (MVP, max N = 622,497). In a meta-analysis across five broad population groups (African, Admixed American, Central/South Asian, East Asian, and European ancestries), we found significant associations between greater autozygosity and fewer years of education, shorter height, less frequent alcohol drinking, and lower likelihood of smoking tobacco, consistent with prior studies. Forthcoming analyses will include a meta-analysis with published results from the ROHgen consortium and an investigation into ancestry-specific differences in some trait associations. Overall, genome-wide autozygosity analyses are a useful tool to understand evolutionary forces and population dynamics influencing the genetic architecture of complex behavioral phenotypes. Future research leveraging the whole genome sequence data available in these cohorts will provide a more comprehensive understanding of these associations.

**Keywords** Substance use, Autozygosity, Genome-wide, Runs of homozygosity

### Do Children Who Develop Faster Go on to Age Faster in Midlife?

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**Abstract**
**Aim**
Quantify the association between child development and midlife biological aging.
**Design**
The Dunedin Study is a prospective, longitudinal, population-representative birth cohort (N = 1037) in Dunedin, New Zealand. Participants were recruited at birth and followed up at ages 3, 5, 7, 9, 11, 13, 15, 18, 21, 26, 32, 38 and 45 years.
**Child Development Measures**
Height and weight were measured at birth-age 26 to estimate birth weight, weight change from birth-age 3, timing of toddler adiposity

rebound, and timing/tempo of adolescent growth spurt. Early childhood milestones (e.g., age at first instance of walking, talking, toilet training) were reported by parents of Study members. Age at first menstrual period was self-reported by Study members.

**Midlife Aging Measure**

Midlife biological aging was measured using a composite score of multi-organ system decline at ages 26–45 years. This measurement was distilled into a DNAm biomarker (DunedinPACE).

**Results**

Children who reached the nadir of a BMI growth curve (i.e., adiposity rebound) earlier had more advanced biological age at 45 (ß, − 0.17; 95% CI, − 0.24 to − 0.11; $P < 0$-0.001). Children who grew faster during adolescent growth spurt had more advanced biological age at 45 (ß, 0.09; 95% CI, 0.01 to 0.17; $P < 0.05$). Girls who experienced their first menstrual period earlier had more advanced biological age at 45 (ß, − 0.13; 95% CI, − 0.22 to − 0.04; $P < 0.01$).

**Conclusions**

Children who developed earlier had faster aging in midlife. Findings support the conceptualization of development and aging as continuous rather than distinct biological processes during one's lifespan.

**Keywords**

Biological aging, Puberty, BMI, Developmental psychology, Lifespan

**Grants**

**Grants**

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## How Early Do Socioeconomic Gradients in Biological Aging Emerge? Investigating the Role of Early Life Stressors Using DNAm in Children

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**Abstract**

Epigenetic processes, such as DNA methylation, play a vital role in human growth and development (Schmitz and Duque 2022). Numerous so-called epigenetic clocks, individual-level measures that summarize DNA methylation information at sites dispersed widely across the genome, have been developed to index biological aging (Wang et al. 2022). In addition, recent work has used such epigenetic clocks to highlight inequality in rates of biological aging across the socioeconomic spectrum. However, the extent to which existing epigenetic clocks—which have been developed using adult samples—index biological aging in children remains unclear. Moreover, exactly how early in the life course detectable socioeconomic inequalities in biological aging begin to emerge is largely unknown. This study uses unique data from the Future Families and Child Wellbeing Study (FFCWS). The FFCWS follows a cohort of 4898 children born between 1998 and 2000 and includes repeated measures of children's DNA methylation age profiles at ages 9 and 15. To my knowledge, it is the only dataset with repeated biological data on a representative child sample, making it vital to answer our main questions: **Do adult-trained biological clocks appear to meaningfully index aging in samples of children, and to what extent does maternal socioeconomic status explain differences in biological aging among children?** We find that first and second-generation clocks trained on adult samples predict chronological age in children at ages 9 and 15. Results also show maternal income at birth, at ages 9 and 15, are associated with changes in DNAmGrimAge and Dunedin clocks across childhood and adolescence.

**Keywords** DNA methylation, Epigenetics, Childhood, Adolescence, Adulthood, Maternal income, Aging

## Early Developmental and Psychiatric Manifestations of Rare Copy Number Variants in Children and Adolescents: A Population-Based Investigation

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**Abstract**

There is growing clinical interest in identifying neurodevelopmental copy number variant (CNV) carriers, due to the impact of these CNVs on short and long-term health. Neurodevelopmental CNVs have been consistently associated with intellectual disability, autism spectrum disorder (ASD), and attention deficit hyperactivity disorder (ADHD), yet their broader manifestations in childhood remain understudied, particularly in the general population.

Carriers of 54 known neurodevelopmental CNVs were identified in the Avon Longitudinal Study of Parents and Children (ALSPAC) and the Millennium Cohort Study (MCS, total carriers = 295, total controls = 12,776). Logistic regressions were used to measure the association between CNV carrier status and: birth complications, early developmental delay, low cognitive ability, neurodevelopmental conditions, mental health, and social/educational outcomes. Primary analyses were conducted in ALSPAC, with replication in MCS and meta-analysis, where corresponding phenotypes were available.

In ALSPAC, neurodevelopmental CNV carriers had a greater likelihood of preterm birth, early motor delay, low general, performance, and verbal cognitive ability, reading difficulties, language difficulties, coordination difficulties, ADHD, ASD and special educational needs (SEND). Meta-analysis identified associations between CNV carrier status and low birthweight, low performance cognitive ability, ADHD, ASD, reading difficulties, SEND and peer problems, but did not support associations with preterm birth or low verbal cognitive ability.

Our study provides an extensive population-wide assessment of the neurodevelopmental and psychiatric presentation of CNVs in childhood and adolescence, and highlights developmental conditions and phenotypes that may be useful for clinicians to review when suspecting a CNV or considering genetic testing.

**Keywords** Copy number variants, Neurodevelopmental conditions, Childhood development, ALSPAC, MCS

**Grants**

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## Heritability of Humor Production Ability: A Twin Study

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**Abstract**
Sense of humor is a universal human attribute, enjoyed daily by people all over the world. Researchers agree that humor has an evolutionary basis, but the nature of the adaptive function of humor is disputed. Relatively little is known about the sources of individual differences in humor and what might contribute to developing a great sense of humor. The current study is the first attempt to examine the heritability of humor production ability by evaluating the relative contributions of genetic and environmental factors to the development of humor ability, based on a twin study design. Four hundred and eighty-one pairs of MZ twins and 219 pairs of DZ twins ($\bar{x}$ age = 60 years, mostly female) from the Twins UK registry completed a humor production task requiring them to compose a funny caption for a captionless cartoon. Analyses of zygosity-stratified twin correlations suggest little presence of additive genetic effects, with substantial shared environmental effects. Formal twin modeling analyses are ongoing, including bivariate analyses with cognitive data. While previous studies have found a substantial genetic component for humor appreciation and humor styles, humor production ability is more complex and harder to assess. Other limitations are discussed, as well as suggestions for improved tools to assess humor ability.

**Keywords** Humor, Bivariate, Twin model

**Grants**
We acknowledge Julie Aitken Schermer from the University of Western Ontario, Canada, whose departmental funding contributed to data collection.

## Genome-Wide Meta-regression Analysis of Social Behaviour from Toddlerhood to Early Adulthood

**Lucía de Hoyos[1], Fenja Schlag[1], Daniëlle Admiraal[1], Cognitive Working Group for the EAGLE Social Behaviour Project[1], Beate St Pourcain[1,2,3]**

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**Abstract**
Social behaviour represents a wide spectrum of inter-individual interactions, partially influenced by underlying genetic factors that are not yet well understood. Our social behaviour changes across development and social contexts, and failure to meet developmental expectations can lead to poor mental health. Social difficulties can, for example, reflect an individual's low prosocial behaviour (LPB) but also peer/social problems (PSP).

Within the EArly Genetics and Lifecourse Epidemiology (EAGLE) Consortium, we carried out a genome-wide meta-regression analysis of LPB and PSP spanning infancy to early adulthood (2–29 years, 73,321 European-descent individuals, 25 cohorts). Repeated measures ($N_{observations}$ = 491,246) of LPB and PSP were ascertained using parent-, teacher- and self-reports assessed with the Strengths and Difficulties questionnaire, the Child Behaviour Checklist, and other closely related instruments. Random-effects meta-regression analysis was carried out accounting for repeated observations, estimating differences in single-variant effects as explained by social phenotype, reporter, age and age squared (fixed covariate effects) and predicting genome-wide single-variant effects. Genetic correlations of social behaviour at different ages, varying by reporter and social phenotype, were largely positive and modest to strong, except for an inverse relationship between early-life parent-reported PSP and both later-life parent- and teacher-reported LPB ($r_g < -0.44$). Meta-analysed LPB and PSP across reporters and ages had a low single-nucleotide polymorphism heritability [LPB-$h^2_{SNP}$ = 0.04 (SE = 0.01), PSP-$h^2_{SNP}$ = 0.03 (SE = 0.01)] and measures were strongly genetically correlated [$r_g$ = 0.58 (SE = 0.10)].

These results highlight that the genetic architecture of social behaviour is multi-factorial and that social relationships across different developmental stages, reporters and phenotypes can be accurately modelled using genome-wide meta-regression approaches.

**Keywords** Social behaviour, Meta regression, EAGLE, Low prosociality, Peer problems

**Consortia**
Elizabeth Corfield, Andrea G. Allegrini, Ellen Verhoef, Sofia Weidle Scatolin, Simone van den Bedem, Ebba Du Rietz, Ashley Thompson, Natalia Llonga, Maria Soler Artigas, Eveline de Zeeuw, Priyanka Choudhary, Anni Heiskala, Sergi Marí Alemany, Sílvia Alemany, Natalia Vilor, Alexander Neumann, Mannan Luo, Ilja M. Nolte, Teemu Palviainen, Katherine Lange, Melissa Wake, Elisabeth Thiering, Carol Wang, Anke R. Hammerschlag, Ehsan Motazedi, Zhijie Liao, Jean Shin, Sheena Reilly, Else Eising, Casper-Emil Pedersen, Maria Hernandez Lorca, Nora Fernandez-Jimenez, Polina Girchenko, Jari Lahti, Jose Ramon Bilbao, Tarunveer Singh Ahluwalia, Klaus Bønnelykke, Zdenka Pausova, Tomas Paus, Tanja Vrijkotte, Christel Middeldorp, Andrew Whitehouse, Marie Standl, Angela Morgan, Jaakko Kaprio, Eero Vuoksimaa, Catharina Hartman, Charlotte Cecil, Henning Tiemeier, Mariona Bustamante, Jordi Sunyer, Alina Rodriguez, Marjo-Riitta Jarvelin, Sylvain Sebert, Dorret I. Boomsma, Marta Ribases Haro, Henrik Larsson, Paul Lichtenstein, Simon E. Fisher, Robert Plomin, Kaili Rimfeld, Eivind Ystrøm, Ted Reichborn-Kiennerud, Alexandra Havdahl.

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## Evocative Gene–Environment Correlation and the Education PGI/Educational Attainment Association: Genes Related to Educational Attainment, Presentation of Self, and the Perceptions of Others

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**Abstract**
Polygenic scores are broadly misconstrued as reflecting direct genetic effects on their respective phenotype. While this assumption might be accurate for some anthropometric traits like height, more complex traits such as educational attainment show very large indirect effects that stem from many sources. One unexplored source of confounding is the possibility of evocative gene–environment correlation (rGE). Using data from the National Longitudinal Study of Adolescent to Adult Health, we examine the relationship between interviewer assessments of respondent appearance as a function of respondent genotype. Results indicate that genes related to educational attainment also predict interviewer assessment of respondent's attractiveness and personal grooming. Importantly, these factors collectively explain nine percent of the relationship between education PGS and years of education with perceived grooming explaining most of this mediation. These results highlight the importance of social–behavioral mechanisms that may link specific genotypes to successful transitions through high school and college and continue to bridge research from the social and biological sciences.

**Keywords** Evocative rGE, Social–behavioral mechanisms

## The Association Between Metabolite Concentrations and Wellbeing in Adults

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**Abstract**
The biological complexity of wellbeing is studied from various perspectives, including genetics and epigenetics. However, there is a knowledge gap concerning other layers, such as metabolomics, which is dynamic and changes throughout life. This study explores the association between metabolites and wellbeing in a sample (N = 4748) drawn from the Netherlands Twin Register. A latent factor score for wellbeing was constructed based on: Quality of Life, Life Satisfaction, and Subjective Happiness. A total of 231 blood metabolites were analyzed using $^{1}$H NMR technique. Linear regression models were performed for each metabolite, while correcting for family clustering, relevant covariates, and multiple testing. None of the metabolites were significantly associated with wellbeing after multiple testing correction. Despite the lack of significant findings, the 34 metabolites with the lowest *p*-value (0.25) pointed to the same metabolic pathway: endogenous lipid metabolism. This pathway has previously been linked to wellbeing in a GWAS and associated with related phenotypes in other metabolomic studies. In conclusion, this study confirms the biological complexity of wellbeing and speculates on a potential role of lipids. Further research is needed to confirm these hypotheses.

**Keywords** Metabolomics, Wellbeing, NTR, Twin studies

**Grants**
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## Educational Genetics in Adaptive Learning Environments: Applying a Twin Model When Zygosity is Unknown

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**Abstract**
Given the significant role of education in society, thorough research into the drivers of school performance, including genetics, is essential. Edtech platforms like Oefenweb provide valuable, naturalistic data for such investigations. While Oefenweb's database includes over 400,000 children, it lacks genomic data and familial relationships, rendering traditional genetic designs inapplicable. To overcome this, we employed the finite mixture distribution twin model, which decomposes variance without known twin-pair zygosity by analyzing variance and covariance structures. We applied univariate and bivariate ACE models to six arithmetic abilities and login count. Model reliability was assessed through simulations, bootstrapping, and correlation analyses. Results indicated high heritability for arithmetic traits (85–98%), with low shared (0–13%) and unique (2–23%) environmental influences. For login count, we observed 51% additive genetic effects, 44% shared environmental effects, and 5% unique environmental effects. These findings align with similar studies (60–74% heritability in arithmetics), however positive kurtosis in twin difference scores likely biased these estimates, overestimating genetic effects and underestimating environmental effects. Additionally, twin selection errors could inflate genetic estimates. This study demonstrates the model's utility in differentiating traits influenced by genetics versus the shared environment, while emphasizing the need for careful interpretation due to potential biases. Future research should address kurtosis and employ additional sensitivity analyses to validate findings. Despite potential biases, this innovative approach provides valuable insights into the interplay of genetics and environment in educational outcomes. As data collection methods advance, this model could become a crucial tool for genetic analysis of previously inaccessible datasets.

**Keywords** Educational genetics, Unknown zygosity, Adaptive learning platform, Finite mixture distribution twin model

**Grants**
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## Like Parent, Like Child: A Meta-analysis of Parent–Offspring Resemblance in Reading Ability

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**Abstract**
**Background**
Dyslexia, a developmental reading disorder characterized by difficulties in decoding words, is influenced by an interplay of genetic and environmental factors, contributing to its familial nature. Children with an affected parent, designated as at family risk, are more susceptible to developing reading problems. This meta-analysis investigated parental reading skills in three groups: children with family risk and dyslexia, with family risk and without dyslexia, and controls. Additionally, we investigated the parent–offspring correlation of reading ability in unselected samples.
**Methods**
We extended Snowling and Melby-Lervåg (2016) by meta-analysing intergenerational transmission of reading skills in both family risk and unselected samples, employing novel methods that consider dependencies in the data.
**Results**
As by design, the parents of the children at family risk of dyslexia had worse reading skills than those of the children not at-risk (Cohen's d = − 1.22), with parents of dyslexic children (Cohen's d = − 2.02) typically performing worse than parents of non-dyslexic children (Cohen's d = − 1.58). Within the at-risk group, the parents dyslexic children generally had poorer reading skills than those of non-dyslexic children (Cohen's d = − 0.024). In studies with unselected samples, the parent–offspring correlation was 0.20.
**Conclusion**
These results demonstrate the intergenerational transmission of reading ability, and align with the notion that familial risk is not all or nothing but a continuum, as was concluded by Snowling and Melby-Lervåg (2016).

**Keywords** Dyslexia, Reading ability, Family risk, Parent–offspring resemblance

**Reference**
Snowling MJ, Melby-Lervåg M (2016) Oral language deficits in familial dyslexia: a meta-analysis and review. Psychol Bull 142(5):498–545

**Grants**
Grant support: Research Council of Norway, Project "GenEd", Grant No. 335634.

## Triangulating Evidence of Declining Heritability Across Educational Levels in Norway: Insights from Twin Models and Polygenic Scores

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**Abstract**
Although the genetic basis for educational attainment has been extensively studied, the progression from high school to doctoral levels remains less understood. Therefore this study employs a sequential hurdle model for analyses of twin models (N = 8910, cohorts 1965–2001) and polygenic scores (PGS, N = 114,699, cohorts 1965–1989) to explore the additive genetic effects (A) for these educational transitions across Norwegian cohorts. Findings indicate that genetic influence decreases with higher educational hurdles: for high school graduates, A = 0.6 [0.39, 0.76]; for bachelor's degree holders, A = 0.49 [0.36, 0.64]; for master's degree holders, A = 0.42 [0.23, 0.61]; and for doctorate holders, A = 0.3 [0.02, 0.6]. Similarly, McFadden's pseudo $R^2$ from PGS revealed decreasing genetic influence from high school (0.047 [0.044, 0.051]) to PhD (0.01 [0.007, 0.013]). The study also finds a selection effect, where individuals with higher PGS are more likely to surpass educational milestones, evidenced by the adjusted mean standardized PGS increasing almost linearly with educational level, from high school (− 0.29 [− 0.3, − 0.28]) to PhD (0.72 [0.67, 0.76]). This suggests that for each educational hurdle passed, the less genetic influences matter for passing the next hurdle, but that there is a selection process where individuals with higher PGS tend to more easily pass higher educational hurdles.

**Keywords** Educational transitions, Genetic heritability, Sequential hurdle model, Twin studies, Polygenic scores

## A Mendelian Randomization Analysis of ADHD on Education Performance Using Twins from the ABCD Study

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**Abstract**
Attention-deficit hyperactivity disorder (ADHD) is prevalent in children and associated with lower education performance. This association is often obtained from observational studies, which have limited ability in identifying causal relationships. Causes are useful, as they delineate intervention avenues more clearly, but causal inference in childhood is hindered by limitations in most common methods, with most results of possible causal effects known to date coming from adult data sets. The present study uses a twin design and polygenic risk scores (PRS) to investigate the causal relationship between ADHD symptoms and poor educational performance. The ABCD study data set is used and the Mendelian randomization (MR) extensions to twin design methodology is used to make causal inference. The models used are robust to horizontal pleiotropy, whose required absence is the most problematic assumption in MR. A causal effect of ADHD symptoms on educational performance was found. These findings highlight the complex interplay between ADHD and educational outcomes and emphasize the need for larger sample sizes in future research.

**Keywords** Mendelian randomization, Structural equation models, Direction of causation

**Grants**
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## Heritable Influences and Caregiving Environments: Predicting Effortful Control and Adiposity in Adolescents Using a Stress-Sensitivity Model

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**Abstract**
Stress exposure in early life can hinder children's development and lead to long-term health problems. Stress sensitivity can be assessed using hypothalamic–pituitary–adrenal (HPA) axis functioning via diurnal cortisol rhythms. Variations in HPA axis functioning stem from heritable factors and proximal caregiving environments, yet the relative contributions of each set of factors remain largely unknown. The present study leverages an adoption approach to examine the relative effects of birth mother life stress (BMLS) via child diurnal cortisol slopes (DCS) on effortful control (EC) and adiposity via BMI in early adolescence and whether adoptive parent stress modified this relationship. Participants were drawn from linked birth parents and adoptive families participating in the prospective, longitudinal, Early Growth and Development Study. The analytic sample included 561 adoptees, and their birth and adoptive parents. Structural equation models were used to examine the associations between BMLS, adoptive parent stress, child DCS, and adolescent EC and BMI. We found EC was predicted by both BMLS and adoptive parent stress. Specifically, BMLS was negatively associated with adolescent EC, such that BMLS predicted experiencing poorer overall EC in the adoptee. Adoptive parent stress did not modify the effect of BMLS or child DCS, but remained a significant predictor on the mediator, child DCS, such that children exposed to higher stress in the caregiving environment had flatter wake-to-bedtime cortisol slopes. By examining the relative contributions of heritable and environmental stress exposure on child development, this research can inform targeted resilience-focused interventions to help mitigate stress for vulnerable families.

**Keywords** Stress, HPA axis, Caregiving environment, Genetic, Resilience

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## Drawing Parallels Across Complex Trait Genomics and Social Science: A Framework to Combine Social and Genetic Risk Factors for Depression

**Katherine N. Thompson[1], Baptiste Couvy-Duchesne[2], Rafael Geurgas[1], Saul Newman[3], Robbee Wedow[1], Felix C. Tropf[1,3]**

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**Abstract**
Depression is one of the leading causes of disability worldwide. Research shows a multitude of social and genetic risk factors associated with depression, both at the clinical and symptom level. However, it is challenging to effectively combine a large number of social predictors alongside genetic risk to explain depression occurrence. The field of complex trait genomics has not only advanced our understanding of the biological underpinnings and mechanisms of depression, but has spent decades refining approaches to analyse big genetic data. Here, we use data from the National Longitudinal Study of Adolescent to Adult Health to demonstrate a new framework to model social risk factors for depression through drawing parallels across genomic and social data. First, we develop a multivariate mixed linear model that quantifies the variance in depression explained by all measured social data without reducing statistical power. We do this by repurposing the genetic-relatedness matrix used in genome-wide complex trait analysis (GCTA) to a social-relatedness matrix. GCTA uses a linear mixed model to fit millions of variables as random effects via a genetic relatedness matrix. We apply this technique using all social information about someone to estimate how socially-related each person is, or how similar each person is based on their social experiences. Second, we jointly model social and genetic predictors of depression by including both a social and genetic matrix. This allows us to estimate the variance in depression that is attributed to social and genetic factors and quantifies the contribution of gene–environment correlation.

**Keywords** Depression, Social-relatedness matrix, GCTA, Gene–environment interplay

## Metabolomics and Transcriptomic Analyses Highlight Differences in Gene-Expression of the TGF-B and WNT Signaling Pathways Between Monozygotic and Dizygotic Twins

**Nikki Hubers[1,2,3], Gabin Drouard[4], Rick Jansen[5], René Pool[1], Jouke-Jan Hottenga[1], Gonneke Willemsen[6], Jaakko Kaprio[4], Jenny van Dongen[1,2,3], Dorret I. Boomsma[1,2,3]**

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**Abstract**
High-throughput omics technologies allow for in-depth analyses of complex traits. Recent GWA studies identified genes associated with DZ twinning as well as hundreds of methylated sites in MZ twins. To gain further functional insight, we asked if these results are accompanied by differences in other omic layers. We compared 196 metabolites and 49,293 RNA transcripts of MZ and DZ twins from the Netherlands twin register (NTR), the FinnTwin12 (FT12) cohort and the old Finnish Twin Cohort (oldFTC). Meta-analyses of metabolite data in all three cohorts (N = 4837) did not show significant differences between MZ and DZ twins. When comparing RNA transcripts between MZ and DZ twins in the NTR (N = 2820) we identified one differentially expressed transcript of the protein-coding

Purine Rich Element Binding Protein G (PURG) gene. Replication analysis in the oldFTC (N = 367) of the top 5% most discriminant RNA transcripts found in NTR revealed 19 differentially expressed transcripts after correcting for multiple testing, none of which have been implicated for twinning before. Enrichment analyses for these 19 and the top 5% RNA transcripts showed differences in expression in the WNT-pathways and cell adhesion processes, which were previously indicated with MZ twinning, and the TGF-B pathway known to be associated with DZ twinning. Currently, we are following up on these findings by investigating whether the 19 differently expressed transcripts are associated with the polygenetic scores for MZ and DZ twinning.

**Keywords** Twinning, Metabolomics, Transcriptomics, Polygenic scores, Multi-omics

**Consortia**
BIOS Consortium, BBMRI-NL Metabolomics Consortium, Twinning Genetics Consortium.

**Grants**
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## Investigating the Relationship Between Sleep Phenotypes and Neurodevelopmental Conditions in the Norwegian Mother, Father, and Child Cohort Study (MoBa)

**Elizabeth C. Corfield[1,2], Laurie J. Hannigan[1,2,3], Ted Reichborn-Kjennerud[2,4], Helga Ask[2,5], Alexandra Havdahl[1,2,5]**

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**Abstract**
Investigations conducted in small clinical samples have shown that sleep problems are highly prevalent in children with neurodevelopmental conditions, such as autism and ADHD. However, there is limited knowledge about the patterns of co-occurrence in population-based samples or whether genetic predispositions to neurodevelopmental conditions correlate with genetic liabilities of sleep disturbances. In the Norwegian Mother, Father, and Child Cohort Study we examined the occurrence of sleep phenotypes in children and adolescents with neurodevelopmental conditions and investigated if genetic predispositions to neurodevelopmental conditions predict sleep phenotypes.

Children and adolescents diagnosed with autism, ADHD, or sleep problems were identified from health registries. Questionnaire data was used to examine mother-report (across childhood) and self-report (at age 14) sleep phenotypes (including sleep duration, difficulty falling asleep, and waking during the night). Sensitivity analyses will analyze groups of only ADHD and only autism, compare ADHD and autism, investigate results for children with and without intellectual disabilities, and examine interactions between time, diagnostic status, and polygenic scores.

Multivariate logistic regression modelling revealed that children and adolescents with neurodevelopmental conditions experience higher levels of both clinically diagnosed ($OR_{autism}$ = 5.58; $OR_{ADHD}$ = 4.79) and mother-reported ($OR_{autism}$ = 2.86; $OR_{ADHD}$ = 2.56) sleep problems. Polygenic score analyses showed genetic predispositions to autism and ADHD predict both clinically diagnosed ($OR_{autism}$ = 1.06; $OR_{ADHD}$ = 1.15) and mother-reported ($OR_{autism}$ = 1.04; $OR_{ADHD}$ = 1.08) sleep problems. Results from all sleep phenotypes will be presented at the meeting.

This study supports previous finding and reveals novel insights into the sleep of children and adolescents with neurodevelopmental conditions using a broad range of sleep phenotypes at multiple ages.

**Keywords** Sleep problems, Neurodevelopmental conditions, MoBa, Polygenic score

**Grants**
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## Prenatal Programming and Within Family Trajectories of Internalizing Symptoms, Obesity Risk, and Relative Pubertal Timing: A Discordant Sibling Design

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**Abstract**
Adolescent internalizing disorders and pediatric obesity rates are rising, and the age of puberty is declining which may be exacerbated by prenatal programming. Prior research is primarily based on non-genetically informed cross-sectional literature where individual differences are confounded. To accurately investigate whether these trends are connected, it is essential to account for genetic and environmental confounding between these phenotypes. Thus, we used a sub-sample of related individuals from the Adolescent Brain Cognitive Development (n = 1896 sibling pairs) study and developed a new model (Discordant Sibling Random Intercept Cross Lagged Panel Model) that was able to disaggregate within and between family variance over time in adolescence. Internalizing symptoms, body mass index (BMI), and pubertal development were collected at age 10, 11, 12, and 13. We additionally controlled for within-family birth weight and smoking during pregnancy as well as family-level demographics. There were sparse within-family associations between the phenotypes. However, there was an association between within-family birth weight and stable discordance factors, such that one

sibling having a higher birth weight relative to their sibling was associated with sibling BMIs being consistently different over time. There was also an association between having a lower birth weight compared to one's sibling was associated with sibling internalizing scores being consistently different over time. These associations persisted in the monozygotic twin subsample suggesting potential prenatal programming effects that lead to differences between siblings. Prior reported associations between internalizing symptoms, obesity risk, and pubertal timing may be partially due to confounded genetic and environmental influences.

**Keywords** Internalizing, Obesity, Puberty, Prenatal, Within-family

## Constellations of Genetic, Familial, and Environmental Risk for AUD: A Latent Class Analysis of the Collaborative Study on the Genetics of Alcoholism (COGA) Prospective Study

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**Abstract**
Multiple risk factors for alcohol use disorder (AUD) have been reliably identified. Better parsing heterogeneity in the constellation of risk factors among individuals with and without AUD will help elucidate the equifinality of AUD etiology. Data came from the Collaborative Study on the Genetics of Alcoholism (COGA) Prospective Study (n = 1935 and 918 individuals most similar to European and African genetic ancestry, respectively). Indicators included parent AUD; alcohol and other substance use and use disorder polygenic scores (PGS); early (before age 15) alcohol, cigarette, and cannabis initiation; peer alcohol and substance use; low parent education; assaultive and assaultive and non-assaultive trauma exposure. Data were randomly divided into 75% training and 25% test data, separately per ancestry group. Among participants most similar to European ancestry, a 4-latent class model fit the best, whereas a 2-class model best fit the data among individuals most similar to African ancestry. Peer alcohol and substance use distinguished classes the most, followed by early substance use initiation. Class membership was associated with the probability of AUD (e.g., a class with the highest proportions of peer use and early initiation had the highest AUD probability). These results, however, did not generalize to the test data. Latent classes of risk factors in a high-risk sample showed different degrees of risk across many indicators, notably with respect to peer substance use and early initiation. Results should be interpreted with caution given their failure to generalize out-of-sample. Future work should include larger sample sizes and additional theoretically justified risk indicators.

**Keywords** Alcohol use disorder, Polygenic risk, Latent class analysis, Risk factors

**Consortia**
COGA Collaborators.

**Grants**
SEP was supported by F31AA029934.

## Unraveling the Relationship Between Creativity and Mental Health Using a Genetically Informative Sample

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**Abstract**
There is a common mad-genius belief suggesting that desirable abilities, like creativity, come with the cost of mental health. Some studies have confirmed this belief by showing creativity to be positively correlated with certain mental complaints (e.g., schizophrenia and bipolar disorder), phenotypically and genetically. Nevertheless, inconsistent findings also exist and prompt us to verify this association while considering the heterogeneity in creativity measures and to examine if this association is causal. In Swedish twins (N = 14,660), we confirmed that two of the creativity measures, openness and artistic achievement, are positively associated with a wide range of psychological disorders (incl. major depressive, anxiety, bipolar, and stress related disorders) with $p < 0.001$, while divergent thinking and scientific achievement, are not. After we controlled for familial confounding using a co-twin control design in monozygotic (MZ) twins, most associations diminished and all previously significant associations became nonsignificant, implying familial confounding and no direct causation between creativity and mental health. However, confidence intervals of the co-twin control results were large due to low N of discordant MZ pairs. Additionally, polygenic risk scores of major depression, bipolar disorder, and schizophrenia can significantly positively predict creativity, further supporting the finding of genetic confounding in the creativity–mental health relationship. In summary, our study found that mental health problems are positively associated with creativity in general, but associations vary between different creativity measurements and are at least partly explained by shared genetic factors.

**Keywords** Creativity, Mental health, Genetics, Causality

## Using Coordinated Epistasis to Investigate Genetic Architecture of Psychiatric Comorbidity

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of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark. [8]Neurogenomics Division, The Translational Genomics Research Institute, Phoenix, AZ, USA. [9]Section of Genetic Medicine, University of Chicago, Chicago, IL, USA

**Abstract**
Cross-disorder analyses in psychiatry are centered around genetic correlation, which quantifies the average similarity of genetic effects across two traits. However, few studies have examined the genetic architecture of comorbidity or how it relates to the genetic architectures of the individual disorders involved. In the current study we develop a novel framework to model comorbidity based on the concept of Coordinated Epistasis (CE). Within this framework, we can distinguish synergy versus antagonism on comorbidity with respect to genetic effects on the underlying traits. In our proof of concept analysis, we use publicly available summary statistics from genome-wide association studies (GWAS) to create polygenic risk scores (PRS) for each of the six well-studied psychiatric disorders in the iPSYCH Consortium: schizophrenia (SCZ), bipolar disorder (BIP), autism spectrum disorder (AUT), attention deficit hyperactivity disorder (ADHD), anorexia (ANO) and major depressive disorder (MDD). Our approach partitions the genetic effects captured by these PRS by chromosomes and then evaluates their interaction effects onto each disorder. We find strong antagonistic CE between the PRS of SCZ and BIP in predicting BIP ($P = 3.3 \times 10^{-08}$) and SCZ ($P = 3.2 \times 10^{-02}$). This demonstrates that the CE framework indeed identifies interactions between distinct genetic pathways leading to SCZ and BIP when predicting individual disorders. Our on-going work examines whether the CE framework identifies interactions in individuals with comorbidity using simulations as well as real data from the iPSYCH Consortium and the UK Biobank.

**Keywords** Comorbidity, Psychiatric disorders, Coordinated epistasis, Genetic architecture

**Grants**
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## The Detection of Environmental Influences on Academic Achievement Appears to Depend on the Analytic Approach

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**Abstract**
One long-standing analytic approach in adoption studies is to examine correlations between features of adoptive homes and outcomes of adopted children (hereafter termed 'measured environment correlations') to illuminate environmental influences on those associations. Although results from such studies have almost uniformly suggested modest environmental influences on adopted children's academic achievement, other work has indicated that adopted children's achievement is routinely higher than that of their reared-apart family members, often substantially so. We sought to understand this discrepancy. We examined academic achievement and literacy-promotive features of the home in 424 yoked adoptive/biological families participating in the Early Growth and Development Study (EGDS; i.e., adopted children, adoptive mothers, birth mothers, and biological siblings of the adopted children remaining in the birth homes) using an exhaustive modeling approach. Results indicated that, as anticipated, adopted children scored up to a full standard deviation higher on standardized achievement tests relative to their birth mothers and reared-apart biological siblings. Moreover, these achievement differences were associated with differences in the literacy-promotive features of the adoptive and birth family homes, despite minimal measured environment correlations within adoptive families. A subsequent simulation study highlighted noise in measured environmental variables as an explanation for the decreased utility of measured environment correlations. We conclude that the field's heavy focus on measured environment correlations within adoptive families may have obscured detection of specific environmental effects on youth outcomes, and that future adoption studies should supplement their measured environment analyses with mean differences between reared-apart relatives.

**Keywords** Rearing environment, Academic achievement, Adoption study, Means and correlations

**Grants**
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## Why is Parental BMI Associated with Offspring Adiposity? A Mendelian Randomization Analysis of Genetic Trios

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**Abstract**
Children with obesity are more likely to have parents with obesity, too. Several environmental explanations have been proposed, such as foetal programming and parenting practices. However, the correlation could reflect direct genetic inheritance. Previous studies have found some evidence that mother's body mass index (BMI) associates with offspring BMI net of direct genetic effects, but the role of father's BMI or of diet as a mediating factor has not been examined. Therefore, we used Mendelian Randomization (MR) with genetic trio data from 2621 families in the Millennium Cohort Study, a UK birth cohort study of individuals born in 2001/2002, to examine the association between parental BMI ($kg/m^2$) and offspring BMI and diet measured from ages 3 to 17 years. Paternal and maternal BMI were

instrumented with polygenic indices (PGI) for BMI also conditioning upon offspring PGI. This allowed us to separate direct and indirect ("genetic nurture") genetic effects. We compared these results with associations obtained using phenotypic BMI data only, the standard research approach. Mother's and father's BMI was positively (phenotypically) associated with child's BMI. However, in MR analysis, associations with father's BMI were close to the null, while associations with mother's BMI remained consistent. There was also evidence of association between maternal BMI and offspring diet. Results were consistent with maternal genetic nurture effects on child's BMI but suggest that analyses of the association between father's and child's adiposity that do not account for genetic inheritance yield severely biased results.

**Keywords** Genetic nurture, Polygenic scores, Obesity, Mendelian Randomization

**Grants**
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## Disentangling Dyslexia: Towards Early Identification Through Biological and Behavioural Factors

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**Abstract**
**Background**
Dyslexia, a disorder marked by reading difficulties, significantly burdens individuals and society. For effective intervention, early identification is crucial.
**Methods**
Using FIOLA cohort data (van Bergen et al. 2015), this study examined early-life predictors of reading fluency in children (N = 217–285): behavioural factors in their parents (educational level and reading fluency) and the children's own biological predispositions [polygenic scores (PGSs) for dyslexia Doust et al. 2022 and educational attainment (EA) Lee et al. 2018, calculated with SBayesR]. PGSs were also used to predict arithmetic fluency for specificity. A hierarchical regression model included behavioural predictors first, then added biological predictors. For sensitivity, these analyses were repeated using Generalised Estimating Equations (GEE) including siblings.
**Results**
PGSs for EA and dyslexia significantly predicted children's reading ($\beta = 0.21, -0.25$) and arithmetic fluency ($\beta = 0.21, -0.19$). Adding both PGSs to the regression model significantly increased the explained variance by 7%. GEE results were very similar.
**Conclusion**
PGSs explain significant variance in children's reading ability beyond parental abilities, enhancing predictive power and facilitating early dyslexia identification. PGSs also open research avenues, including Mendelian Randomization, to better understand dyslexia aetiology.

**Keywords** Dyslexia, Polygenic scores, Parental skills, Intergenerational transmission

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**Grants**
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## Charting New Paths in Behavior Genetics: Developing a Comprehensive R Visualization Atlas

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**Abstract**
The adage 'A picture is worth a thousand words' is just as applicable to Behavior Genetics as it is to journalism. This talk briefly explored the importance of data visualization in our field, tracing advancements from Wright's (1920) classic path diagram of guinea pigs to modern applications. I presented a case study on the misappropriation and misinterpretation of a UMAP (Uniform Manifold Approximation and Projection) figure, highlighting the importance of intentional design.

To bolster the field's visualization toolkit, I introduced two R resources under development (Garrison 2024). The first, an atlas employing 'ggplot2' in R, guides researchers in selecting appropriate visualizations based on their data and provides source code for easy implementation. This atlas, accessible at r-computing-lab.github.io/-data-atlas/, includes various plot types and practical tips. The second resource is a proposed R package aimed at integrating OpenMx-compatible visualizations, streamlining the presentation of complex genetic analyses. Emphasizing practical advice, the talk covered effective uses of path diagrams, family trees, and interaction visualizations to improve the clarity and interpretability of genetic data. This initiative aims to democratize advanced data visualization tools, making them more accessible to the behavior genetics community and, in turn, amplifying the impact and reach of its research.

**Keywords** Data visualization, R, Graphics

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Wright S (1920) The relative importance of heredity and environment in determining the piebald pattern of guinea-pigs. Proc Natl Acad Sci USA 6(6):320–332. https://doi.org/https://doi.org/10.1073/pnas.6.6.320

## Association of Genetic Risk Scores of Schizophrenia-Specific and Highly Pleiotropic Variants for Neuropsychiatric Disorders with Brain Structure

**Lydia M. Federmann[1,2], Lisa Sindermann[2], Sabrina Primus[3], Frederico Raimondo[1,4], Konrad Oexle[3], Janik Goltermann[5], Juliane Winkelmann[3], Markus M Nöthen[2], Katrin Amunts[1,6], Thomas W. Mühleisen[1,6,7], Sven Cichon[1,7,8], Simon B. Eickhoff[1,4], Felix Hoffstaedter[1,4], Udo Dannlowski[5], Kaustubh R. Patil[1,4], Andreas J. Forstner[1,2]**

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**Abstract**
A genome-wide association study across eight neuropsychiatric disorders by Lee et al. (2019) identified 23 highly pleiotropic single-nucleotide polymorphisms (SNPs) that were associated with at least four disorders, as well as 22 SNPs that were predominantly associated with schizophrenia. Here, we compared their association with brain structure, which might shed light on their distinct underlying neurobiological processes. In data from the UK Biobank ($n$ = 28,952), we assessed the association between the genetic risk score of highly pleiotropic SNPs (GRS-pleiotropic) and likewise of predominantly schizophrenia-associated SNPs (GRS-SCZ) with 152 brain image-derived phenotypes of subcortical volume, cortical thickness, and surface area. Furthermore, we investigated the association of the GRSs with mental health outcomes. We found no significant association between the GRS-pleiotropic and brain structure after multiple testing correction, whereas the GRS-SCZ was significantly associated with, for example, increased putamen volume and increased surface area in the lateral orbitofrontal region. At the behavioral level, the GRS-pleiotropic was associated with irritability, fed-up feelings, and tense feelings, whereas the GRS-SCZ to worries, risk-taking, and sensitivity, among others. In conclusion, the GRS-pleiotropic was not associated with brain structure, whereas the GRS-SCZ was associated with the putamen, which plays a critical role in antipsychotic treatment. Both GRSs linked with a broad range of mental health outcomes. This study emphasizes that a better understanding of shared and disorder-specific effects is needed to improve clinical translation of GRSs in neuropsychiatric disorders.

**Keywords** Genomic imaging, Psychiatric disorders, Polygenic risk scores, Cross-disorder, Schizophrenia

**Reference**
Lee et al. (2019) Genomic relationships, novel loci, and pleiotropic mechanisms across eight psychiatric disorders. Cell 179(7): 1469–1482

**Grants**
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## Examining the Genetic Links Between Clusters of Immune-Mediated Diseases and Psychiatric Disorders

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**Abstract**
**Background**
Autoimmune and autoinflammatory diseases have been linked to psychiatric disorders in the phenotypic and genetic literature. However, a comprehensive model that investigates the association between all available psychiatric disorders and immune-mediated disease in a multivariate framework is lacking.
**Methods**
We utilize Genomic Structural Equation Modeling (Genomic SEM) to model the genetic correlations between five previously established factors of 13 major psychiatric disorders and five factors of 12 immune-mediated diseases that were determined based on a combination of exploratory and confirmatory factor analysis. The five immune factors represent rheumatic (RHEU), psoriatic (PSO), gastrointestinal (GI), gout (GOU) and celiac (CEL) diseases. The genetic correlations between these five factors were examined with five established factors of psychiatric disorders representing compulsive (COMP), schizophrenia/bipolar (SB), neurodevelopmental (NEURO), internalizing (INT), and substance use disorders (SUD).
**Results**
The five-factor model of immune-mediated diseases fit the data well (CFI = 0.93; SRMR = 0.07). Analyses revealed five FDR significant factor correlations between the immune and psychiatric factors, including RHEU with SUD, PSO with NEURO and SUD, and GI with SB and INT. Forthcoming work will investigate whether there are more specific trait-to-trait correlations that are driving the correlations across the factors.
**Conclusions**
Our results revealed genetic links between specific clusters of immune-mediated diseases and psychiatric disorders. The current analyses indicate that previously described relationships between specific psychiatric disorders and immune-mediated diseases capture broader pathways of risk sharing indexed by our genomic factors, yet are more specific than a general association across all psychiatric disorders and autoimmune diseases.

**Keywords** Genomic SEM, Immune-mediated disease, Multivariate genomics, Psychiatric genetics

## Sex Differences in the Association of Socioeconomic Factors and Cognitive Function with Family History of Alzheimer's Disease

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**Abstract**
**Introduction**
While higher educational attainment and cognitive function have been negatively associated with Alzheimer's disease (AD), recent evidence highlighted how these factors may have opposite effects when investigating AD family history (FHAD). To explore these dynamics, we investigated AD and FHAD with respect to socioeconomic factors and cognitive function, also testing differences between sexes.
**Methods**
Leveraging information available from the UK Biobank (UKB) and the All of Us Research Program (AoU), we applied a generalized linear regression model to test the association of educational attainment, household income, material deprivation, and cognitive function with AD and self-reported FHAD. To further investigate FHAD dynamics, polygenic risk score (PRS) and one-sample Mendelian randomization (MR) analyses were performed using UKB and AoU genetic data.
**Results**
In both sexes, we observed that high education, cognitive function, and income were associated with reduced AD and sibling-FHAD, while these factors were associated with increased mother- and father-FHAD. After controlling for parents' death and age of death, several of the latter associations were strongly attenuated. UKB-derived PRS for education, cognitive function, and income were also associated with increased mother- and father-FHAD in AoU. One-sample MR highlighted possible sex differences in the underlying dynamics with socioeconomic status positively associated with increased mother-FHAD in males and with increased father-FHAD in females.
**Conclusion**
The present study contributes to understanding the dynamics linking socioeconomic factors and cognitive factors with parents-FHAD, highlighting the possible effect of survival bias and sex-specific participation bias.

**Keywords** Family history of Alzheimer's disease, Socioeconomic status, Cognitive function, Polygenic risk score, Mendelian randomization

## Identifying Metabolomic Risk Factors for Psychiatric and Neurodegenerative Disorders: A Metabolome-Wide Mendelian Randomisation Study

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**Abstract**
Cross-trait genomic analyses have implicated shared risk loci and common causal pathways linking psychiatric and neurodegenerative disorders. Further identification of mutual and unique biological mechanisms may yield therapeutic interventions with effectiveness across multiple disorders or with disorder-specific effects. The metabolome offers a promising line of further investigation given metabolites modulate a diverse range of biological processes.

In this two-sample Mendelian randomisation study, we test the causal effect of ~ 1300 plasma metabolites ($N$ range = 8299–17,108) on anxiety, bipolar, depression, schizophrenia, amyotrophic lateral sclerosis, Alzheimer's disease, Parkinson's disease and multiple sclerosis ($N_{eff}$ range 89,308–825,105).

Out of 10,327 tests, 85 causal associations involving 77 unique metabolites passed FDR correction and sensitivity analyses (OR range 0.73–1.48; $p_{FDR} < 0.05$). No evidence of reverse causality was identified. Limited overlap was observed between metabolites significantly associated with psychiatric disorders and those associated with neurodegenerative disorders. Follow-up colocalisation identified evidence of a shared causal variant ($PP.H_4 \geq 0.8$) for 25 metabolite/psychiatric disorder trait-pairs at chr11:61293499-61854782—which contains the *FADS* gene cluster. These metabolites were all functionally related to linolenic/arachidonic acid. Additional colocalisation was identified between the ratio of histidine-to-glutamine, glutamine, and Alzheimer's disease at chr12:56615338-57115338. Gene expression levels of *SPRYD4* colocalised with histidine-to-glutamine, glutamine, and Alzheimer's disease in the same locus.

This study suggests that broadly unique metabolic pathways influence psychiatric and neurodegenerative disorder risk and provides evidence for the role of fatty acids across the psychiatric disorder spectrum, driven by shared causal variants on chromosome 11. Metabolites identified in this study may inform future therapeutic targets.

**Keywords** Metabolites, Psychiatric, Neurodegenerative, Mendelian randomisation

## Genomic Analyses of Bullying Victimisation in Children and Adolescents

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**Abstract**
**Background/Objectives**
Bullying victimisation is a phenomenon experienced by over 65% of children and adolescents globally. Quantitative methodical approaches have been used to identify profiles of bullying behaviours. However, these approaches require summary data from genome-wide association studies (GWAS's), which is currently unavailable for bullying victimisation. This paper aims to conduct a GWAS of bullying victimisation and perform downstream genetic analyses.
**Methods**
We use genotyped data from The Norwegian Mother, Father and Child Cohort Study (MoBa) (~ N = 43,000). Bullying victimisation is measured at ages 5, 8 and 14. The GWAS summary statistics of bullying victimisation is used to explore genetic correlations across 21 psychiatric conditions (e.g. schizophrenia and depression) and

associated traits (e.g. body mass index). Polygenic scores of the traits are generated and used to predict bullying victimisation. Mendelian randomisation analyses will check for causality.

**Results**

Significant genetic correlations (ranging 0.21–0.70) were identified between 15 traits and the exposure to childhood bullying. For adolescence bullying, high significant genetic correlations were identified in six traits (ranging 0.40–1.28). Genetic correlation and polygenic score analyses show consistent significant results between bullying victimisation and traits. Small causal effects are identified.

**Conclusion**

In line with evidence of pervasive pleiotropy between psychiatric and associated traits, bullying victimisation and mental health related phenotypes are genetically correlated. This study will contribute to our understanding of the implications pre-existing heritable mental health vulnerabilities have on exposure to bullying victimisation and allow for more robust methodologies to be used to examine the aetiology and consequences of bullying victimisation.

**Keywords** Bullying victimisation, Mental health, Genome-wide association study, Polygenic scores, Mendelian randomisation

**Grants**

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## The Fluid and Transactional Nature of Parent–Child Conflict from Childhood into Emerging Adulthood

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**Abstract**

The parent–child relationship is perhaps the most robust and consistent predictor of youth outcomes. Despite the large body of research establishing the sequelae of parenting, relatively little is known about its own developmental origins, or how to conceptualize genetic and environmental contributions to a phenotype that is inherently dyadic. Indeed, no study to date has leveraged both longitudinal and genetically informative methods to illuminate the etiology of parenting as a transactive, dynamic process unfolding across early development. The present study addressed this gap by examining the genetic and environmental origins of continuity and change in parent–child conflict from childhood through emerging adulthood. Participants ($N$ = 1030 twin pairs) were drawn from the Michigan State University Twin Registry and were assessed on up to five occasions. We first fitted a series of univariate and parallel process growth curve models to data from mothers and children, estimating latent intercept and slope factor scores for each participant. We subsequently estimated the genetic and environmental variance components of the intercept and slope, as well as the overlap in etiology across informants. Results indicated substantial environmental contributions to the intercept for both informants, whereas maternal rate of change in conflict was largely genetic in origin. Likewise, environmental factors explained the overlap between maternal and child intercepts, but the cross-informant correlation for the slope was primarily due to genetic influences. Such findings suggest that the transactional nature of parent–child conflict is increasingly driven by evocative genotype–environment correlation across adolescence.

**Keywords** Parenting, Developmental behavior genetics, Genotype–environment interplay, Trajectories, Transactional processes

**Grants**

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## Predicting Behavioural Traits from Polygenic Scores Between and Within Families

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**Abstract**

The ability to predict behavioral traits from polygenic scores (PGS) in unrelated individuals is strongest for cognitive traits, which include cognitive abilities, educational achievement, and years of schooling (educational attainment). PGS prediction of cognitive traits could be due to the emerging finding that cognitive traits, much more than psychopathology symptoms, include between-family (BF) effects that are not reflected within families (WF), such as population stratification, assortative mating, and environmentally mediated parental genetic effects (genetic nurture). The present study systematically compared BF effects for cognitive traits and psychopathology symptoms. We also triangulated these BF effects in twin study estimates of shared environment because twin studies detect genetic BF effects as shared environment.

Rather than relying on meta-analyses of diverse samples with diverse measures, we analyzed the same measures in the same twin sample (Twins Early Development Study) to assess BF effects for cognitive traits and psychopathology symptoms using 1200 DZ twin pairs, to assess genetic nurture using 1800 parents and their twins, and to estimate shared environmental influence using 3000 pairs of twins.

We found strong BF effects, modest effects of genetic nurture, and substantial shared environmental influence for cognitive traits but not for psychopathology symptoms.

We discuss the implications of our findings for the interpretation PGS prediction of behavioral traits within and between families.

**Keywords** Polygenic scores, Shared environment, Between families, Within families, Genetic nurture

**Grants**

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## Predicting First-Episode Psychosis Using Polygenic Risk and Longitudinal Child and Adolescent Phenotypes

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**Abstract**
Late adolescence and early adulthood are sensitive periods for the development of psychotic disorders. Psychosis first typically appears in early adulthood, but prodromal symptoms can emerge during adolescence, and developmental soft signs may appear in childhood. However, many individuals experiencing first-episode psychosis typically wait years for a diagnosis. Thus, identifying those at risk of transitioning to psychosis is essential. Our goal is to create a combined predictive model in a broadly defined high-risk group using genetic risk scores, prodromal symptoms, and childhood soft signs to predict conversion to psychosis. Genetic and diagnostic registry data were collected from children (total $n \sim 114{,}000$) in the Norwegian Mother, Father, and Child Cohort Study (MoBa). Our high-risk groups were defined by familial risk, prodromal symptoms, or genetic liability ($n \sim 4000$). Children from MoBa have been followed from gestation (mothers included between 1999 and 2008) and into their teens. We will use information on development and mental health collected at 6 and 18 months and 3, 5, 8, and 14 years and calculate polygenic risk scores (PRS) for schizophrenia (SZ). We then apply a machine learning algorithm (RUSBoost) using SZ PRS and selected diagnostic, developmental, and mental health variables to predict conversion. We will lastly perform feature selection, comparing the various predictors. We will present results from our prediction models and discuss the relevance of selected models in predicting conversion to psychosis. Hopefully, this project can pave the way for improved prediction of transition to psychosis that can be leveraged for early intervention and support.

**Keywords** First-episode psychosis, Polygenic risk, Prediction, Clinical high risk

**Grants**
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## Polygenic Analyses Show Important Differences Between MDD Symptoms Collected Using PHQ9 and CIDI-SF

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**Abstract**
Symptoms of Major Depressive Disorder (MDD) are commonly assessed using self-rating instruments like the Patient Health Questionnaire 9 (PHQ9, current symptoms), and the Composite International Diagnostic Interview Short-Form (CIDI-SF, worst-episode symptoms). In this project, we perform a systematic comparison between both sets of symptoms for their genetics and utility in investigating MDD heterogeneity using data from the UK Biobank (N = 41,948–109,417). We first assess their SNP heritability (h2) and genetic correlation (rG). We then compare their rG with non-MDD traits, and use Mendelian Randomization (MR) to assess if either set of symptoms have more genetic sharing with non-MDD traits. We further assess how specific each set of symptoms is to MDD, using the metric PRS Pleiotropy. Finally, we use Genomic SEM to identify factors explaining the genetic covariance between each set of symptoms. Results show that corresponding symptoms endorsed through PHQ9 and CIDI-SF have low to moderate genetic correlations (rG = 0.43–0.87), and this cannot be fully attributed to different severity thresholds or the use of a skip-structure in CIDI-SF. Both MR and PRS Pleiotropy analyses show that PHQ9 symptoms are more associated with traits which reflect general dysphoria, while the skip-structure in CIDI-SF allows for the identification of heterogeneity among likely MDD cases. Finally, the two sets of symptoms show different factor structures in Genomic SEM, reflective of their genetic differences. MDD symptoms assessed via the PHQ9 and CIDI-SF are not interchangeable: the former better indexes general dysphoria, while the latter is more informative of within-MDD heterogeneity.

**Keywords** Major Depressive Disorder (MDD), Genetic architecture, Polygenic analyses, Heterogeneity

**Grants**
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## Is Happiness Always a Personal(ity) Thing? A Quasi-replication and Extension of Previous Well-Being Studies

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**Abstract**
Neuroticism, Extraversion, and Conscientiousness are the most notable personality predictors of subjective well-being (SWB) in phenotypic and behavior genetic studies. Some of these links are intuitive. Neuroticism and Extraversion contain facets of emotional stability and tendencies towards positive affect, which reflect dimensions of SWB. Conscientiousness and Agreeableness might be related to behaviors that lead to outcomes positively associated with SWB. Based on the population-based German Twin Family Panel *TwinLife*, we aim to quasi-replicate and extend previous findings. We analyzed Big Five personality trait and life satisfaction data from 3080 twin pairs of three birth cohorts (born 2003/2004, 1997/1998, and 1990–1993). Prior research has repeatedly provided evidence that genetic variance in personality primarily or even completely explained genetic variance in SWB. Our results from Cholesky decomposition models replicated this pattern, with no significant unique genetic variance in life satisfaction. However, we found significant age cohort differences (e.g., Openness was only associated in ages 17–23, explaining 20% of genetic variance in Life Satisfaction). Using data from the *TwinSNPs* study, which genotyped $N = 5355$ individuals from twin families ($M_{\text{Age}} = 30.37$ [8–77]), we tested genetic personality–SWB associations based on polygenic scores (PGSs). PGSs were derived from genome-wide association studies on Big Five personality traits and dimensions of SWB, including life satisfaction. Neuroticism was the most robust correlate ($\beta = -0.205$; $p < 0.001$), while links with other personality traits showed significant differences between age groups (e.g., Conscientiousness only associated in ages 8–13 and 20–29).

**Keywords** Happiness, Subjective well-being, Big Five personality traits, Longitudinal twin study, Polygenic scores

**Grants**
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## Dutch Stakeholder Perspectives on Genomic Research in Autism: A Pilot Exploration

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**Abstract**
While genomic research in autism is growing rapidly, ethical implications should be carefully considered. The international autism community has raised concerns about privacy, transparency, data sharing, and even eugenics, and notes a disbalance between community research priorities and research funding. It is unclear to what degree these sentiments live in the Dutch autism community.

We explored the perspectives on genomic research of three Dutch autism community stakeholder groups recruited through the Netherlands Autism Register; autistic adults, parents of autistic children, and legal representatives. A total of 2366 individuals were included in our analyses, of whom 1767 autistic adults, 443 parents of autistic children, and 126 legal representatives. We compared perspectives between these groups through a repeated questionnaire, in 2018 and 2023.

We found that 94.8% of our sample considers it important knowledge that autism is highly heritable. As most important reasons for this were given: to be able to better diagnose autism, to increase general knowledge about autism, and to increase knowledge around co-occurring conditions. The three stakeholder groups did not differ on how important they found the heritability of autism, however, their reasons for finding it important were slightly different. We also found that 67.2% would like to know more about the heritability of autism. Findings did not change significantly over the 5-year period.

In conclusion, our preliminary analyses show that the Dutch autism community is generally positive towards genomic research in autism.

**Keywords** Autism, Stakeholders, Genetics

**Grants**
Authors MdW, TP and SB are supported by ZonMW (60-63600-98-834).

## Dissecting Genetic and Environmental Contribution on the Link Between Adolescent Gender Nonconformity and Trait Risk-Taking

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**Abstract**
The link between adolescent gender non-conformity and risk taking is normally and exclusively explained via unique social stressors faced by gender non-conforming youth. Empirical evidence focuses mainly on risky sexual behaviors rather than more general trait risk taking in gender non-conforming adolescents. It is worthy to explore both the genetic and environmental contribution to the correlation between adolescent gender-nonconformity and their trait risk taking. In this study, we aim to quantify the heritability to the link between the two phenotypes using Adolescent Brain Cognitive Development (ABCD) data which has a full sample size of about 12,000 youths aged 9–10 in the baseline with five follow-up years. Using Genetic Relatedness Matrix (GRM), we will first examine the univariate heritability of adolescent risk taking and gender non-conformity phenotypes respectively. Then we will further examine the bivariate heritability of the correlation between the two phenotypes. Based on the existing literature, we expect a positive link between gender non-conforming and risk taking in general. Our heritability results are expected to contribute to better understanding of the link between gender non-conformity and risk taking in adolescents.

**Keywords** Adolescents, Risk taking, Gender non-conformity, Gender identity, Heritability

## Illuminating the Developmental Etiology of Youth Resilience

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**Abstract**
Twin studies have highlighted the presence of both genetic and environmental contributions to resilience. To date, however, the specific mechanistic pathways remain unknown. One key possibility involves the biological embedding of stress via the methylation of genes. The present study seeks to investigate this possibility by exploring DNA methylation (DNAm) biomarkers of longitudinal profiles of resilience trajectories. We will leverage a longitudinal dataset (i.e., three waves: middle childhood, adolescence, and emerging adulthood) from the Michigan State University Twin Registry (MSUTR), restricting the sample to families residing in moderately-to-severely disadvantaged neighborhoods (N = 202 families). Psychological and social resilience will be assessed using the Child Behavior Checklist. Growth mixture models (GMMs) will first be conducted to identify longitudinal profiles of psychological and social resilience trajectories, respectively. We expect trajectories to vary both within (i.e., between persons) and across domains, such that youth will evidence two or more trajectory profiles unique to each domain. We will then perform Methylome-wide Association Studies for each domain, examining DNA methylation (across ~ 800,000 CpG sites) as a predictor of longitudinal resilience trajectory profile membership. Finally, to strengthen causal inferences, we will perform discordant monozygotic (MZ) twin difference analyses to circumvent genetic confounds. We expect to find several methylome-wide significant differentially methylated probes that predict resilience trajectory profile membership and are environmental in origin. Such findings would confirm the role of DNAm in resilience trajectories and provide a foundation for follow-up studies exploring DNAm as a mediator in the relationship between environmental promotive factors and resilience.

**Keywords** Resilience, Neighborhood adversity, DNA methylation, Development, Longitudinal

**Grants**

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## Wellbeing as a Catalyst: The Association Between Wellbeing at a Younger Age and Sociodemographic, Health, and Lifestyle-Related Factors Later in Life

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**Abstract**
Puberty into young adulthood is a turbulent period as it is characterized by, among other things, hormonal changes, physical growth, achieving a sense of purpose, and an increase in cognitive and intellectual capacities. Therefore, adolescence is a formative period in the upbringing of a person. Mental problems during adolescence have been associated with poorer economic, social, and health outcomes, however, research on wellbeing among adolescents and sociodemographic and health outcomes is lacking. In the current study we aim to study the hypothesis that wellbeing is a catalyst for positive life outcomes and that higher wellbeing during adolescence (14–16 years of age) is positively associated with sociodemographic, health and lifestyle-related factors later in life (20–25 and 30–35 years of age).

Wellbeing data, sociodemographic, health, and lifestyle-related factors are collected from participants of the Netherlands Twin Register. Multiple linear regression models will be performed to assess associations between wellbeing and sociodemographic, health, and lifestyle-related outcomes using a between-family and within-family approach. During the between-family approach, analyses will be corrected by twin status by applying generalized estimating equation models. For the within-family approach, instead, sibling fixed-effects will be introduced. Between and within-family analyses will be executed unadjusted and adjusted for age$^2$ at time of outcome, age window (i.e. the difference in age between the exposure and outcome), and gender.

Lastly, causal mediation analyses will be conducted for associations that were shown to be significant in the within-family analyses to test how wellbeing and outcomes are related through a panel of possible mediator variables.

**Keywords** Wellbeing, Health, Lifestyle, Sociodemographic, Twin studies

**Grants**
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## Phenome-Wide Associations of Sex Chromosome Aneuploidies Reveal Multi-system Health Effects

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**Abstract**
Sex chromosome aneuploidy (SCA) involves the loss or gain of one or more sex chromosomes (X or Y), leading to conditions such as Klinefelter syndrome (47, XXY), Jacobs syndrome (47, XYY), Triple X syndrome (47, XXX), or Turner syndrome (45, X). SCAs are among the most prevalent chromosomal aneuploidies observed in liveborn humans, associated with various medical and neuropsychiatric comorbidities encompassing learning disorders, fertility issues, and anthropometric features. However, existing research predominantly focuses on clinically ascertained cases, leaving the full spectrum of

health risks and comorbidities of SCA largely unknown. This study aimed to assess health outcomes linked with SCA utilizing health records via phenome-wide association studies (PheWAS) in the UK Biobank (application-ID 30782). ICD9/10 codes were mapped to phecodes, and the prevalence of these phecodes was compared between SCA cases and 1:5 matched controls. A total of N = 669 individuals with SCA were identified from N = 488,377 individuals with genotypic data. ICD codes mapped onto 1875 phecodes. Associated phecodes indicated a wide array of comorbidities spanning circulatory disease ($q = 4.7 \times 10^{-11}$), chronic obstructive pulmonary disease ($q = 1.2 \times 10^{-9}$), Type 2 diabetes ($q = 5.0 \times 10^{-7}$), and mood disorders ($q = 8.4 \times 10^{-6}$). Significantly enriched phecode categories encompassed multiple systems including genitourinary, digestive, neoplasms, and mental disorders. Further investigations are warranted to ascertain the etiopathogenesis of these comorbidities, including potential social and environmental determinants of health that might disproportionately affect individuals with SCAs. These findings suggest that SCAs may entail a broader spectrum of comorbidities than previously acknowledged. Understanding the comorbidities of SCAs is crucial for equipping clinicians, families, and individuals with the knowledge necessary for informed healthcare.

**Keywords** Sex chromosome aneuploidy, Phenome-wide association study, UK Biobank

**Grants**
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## Analyzing Novel Sexual Orientation and Gender Identity Phenotypes Using Genetic Relatedness Matrices

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**Abstract**
Gender nonconformity has previously been defined as a lack of adherence to social expectations in one's expression of gender according to their sex assigned at birth. Gender identity refers to the self-identified gender label of an individual (e.g., man, woman, nonbinary). The heritability of gender nonconformity and sexual orientation have previously been estimated using less comprehensive measures of gender expression and sexuality, which may have led to artificially inflated estimations. The Adolescent Brain Cognitive Development (ABCD) data repository, which contains nearly 12,000 adolescents ages 9–10 at baseline, includes unique survey data on various items regarding gender expression and sexual orientation. ABCD also contains full genotyping of all participants, allowing us to analyze family structures using sibling data and increase the accuracy of estimations of shared environmental influences on the phenotypes of interest. Using a unique measure of gender nonconformity, we will compare gender expression to gender identity for all participants. Estimations of heritability for both gender nonconformity and sexual orientation will be devised by conducting an analysis of bivariate heritability using genetic relatedness matrices. This study hopes to increase the accuracy of heritability estimations of variables related to sexual orientation and gender identity. Our results will advance the understanding of these phenotypes existing on a wide spectrum and the importance of recognizing the diversity and complexity of sexual orientation, gender identity, and gender expression in the general population.

**Keywords** Gender identity, Sexual orientation, Heritability, Adolescents

## Does the Multivariate Genetic Architecture of Negative Emotionality Differ by Trauma Exposure?

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**Abstract**
Lifetime trauma history is associated with elevations in a variety of forms of negative emotionality, ranging from depressed mood, to anxiety, to anger. These internalizing symptoms, while separable, are nevertheless correlated at both the phenotypic and genetic levels of analysis. Here we investigate whether lifetime trauma history is associated with differences in the multivariate genetic architecture of a multidimensional measure of neuroticism and its associations with major depressive disorder, a phenomenon which may result from gene-by-environment interactions that operate unevenly across different aspects of negative emotionality. We stratify item-level neuroticism data from UK Biobank by trauma exposure, and integrate summary GWAS data for depression from the Psychiatric Genomics Consortium. We compare multivariate genomic structure across groups by applying tests of Genomic Structural Invariance (GSI) within the Genomic Structural Equation Modeling (Genomic SEM) framework. Results have implications for theories of Gene-by-Environment interaction, and the generality vs. specificity at which such interactions may operate.

**Keywords** Neuroticism, Anxiety, Depression, Trauma, G × E

## Investigation of Age Effects in Polygenic Score-Based Models of Cultural Transmission

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**Abstract**
**Background**
With respect to cognitive abilities, it is well known from twin studies that shared environmental variance (C) wanes as children grow older. However, it is also known that C, as estimated in the classical twin design, is inflated by positive genotype (A)—shared environment covariance (covAC). This raises the question of whether waning C is due to decreasing shared environmental effects, (and) or to decreasing covAC.
**Objective**
Here, we focus on investigating cultural transmission, a case of covAC that arises when the parental genotype is associated with the child's phenotype by shaping their rearing environment. The present aim is to develop an age-moderated version of recently developed polygenic score (PGS)-based models of cultural transmission (Young et al. 2022).

**Methods**
We formulate the model as a structural equation model, and apply it to simulated data to study the statistical power to detect age-related changes in C and covAC, given variation in sample size, predictive power of the PGS and the magnitude of the effect of cultural transmission.
**Conclusion**
The power analyses will reveal under which circumstances we can test in empirical data whether waning C (e.g. for IQ) is due to decreasing C and/or to decreasing covAC.

**Keywords** Cultural transmission, Simulation study, Age effects, Polygenic scores
**Reference**
Young AI, Nehzati SM, Benonisdottir S, Okbay A, Jayashankar H, Lee C et al (2022) Mendelian imputation of parental genotypes improves estimates of direct genetic effects. Nat Genet 54(6):897–905

**Grants**
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## The Impact of Participant Behaviour and Study Design Characteristics on Findings Obtained from the UK Biobank

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**Abstract**
Large-scale volunteer-based studies such as the UK Biobank (UKBB) have become the cornerstone of genetic epidemiology. To ensure validity of findings obtained from biobank research, substantial efforts are made to eliminate potential sources of bias, such as population stratification, assortative mating, or indirect genetic effects. However, bias resulting from participant behaviour (e.g., study engagement, questionnaire completion) or study design features (e.g., sampling procedures, phenotype ascertainment) of volunteer samples are typically not considered in genetic studies.

UKBB participants typically show a better health profile than the population of interest. This talk will first present findings from research[1] evaluating the impact of this 'healthy-volunteer bias' on genome-wide findings. A number of empirical examples will illustrate how this bias can distort genetic associations and downstream analyses, particularly when studying behavioural and social outcomes. Second, many commonly studied phenotypes in the UKBB, notably sociodemographic data, health-related information, behavioral outcomes, and lifestyles, are ascertained using short self-report measures that are prone to errors resulting from misreporting. This talk will present recent evidence[2], including empirical and simulation data, that quantifies the extent of reporting error for self-report measures in the UKBB, assesses its properties and links with other participation behaviours, and proposes possible corrections for biases induced by reporting error and/or selective participation.

**Keywords** Participation bias, Biobank research, Complex traits, Self-report errors
**References**
[1]Schoeler T (2023) Participation bias in the UK Biobank distorts genetic associations and downstream analyses. Nat Hum Behav 7(7):1216–1227
[2]Schoeler T et al (2023) Self-report inaccuracy in the UK Biobank: impact on inference and interplay with selective participation. medRxiv

## Harnessing Epigenetics to Study the Shared Nature of Development in Childhood and Biological Aging in Adulthood

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**Abstract**
Humans change dramatically as they age. But what is aging? Is it solely characterized by declining function later in life? Or can it be considered as a process of change that occurs throughout the life course? Similarly, is development solely about growth in early life? Or is it a continuous process of change at any stage in the life span?

Recently, biological aging has been quantified in DNA-methylation samples of older adults and applied as so-called "methylation profile scores" (MPSs) in separate target samples, including samples of children. This nascent research indicates that (1) biological aging can be quantified early in the life course, decades before the onset of aging-related disease, (2) is affected by common environmental predictors of childhood development, and (3) shows overlap with "developmental processes" (e.g., puberty). Because the MPSs were computed using algorithms developed in adults, these studies indicate a molecular link between childhood environments, development, and adult biological aging.

Yet, if MPSs can be used to connect development and aging, previous research has only traveled one way, deriving MPSs developed in adults and applying them to samples of children. Researchers have not yet quantified epigenetic measures that reflect the pace of child development, and tested whether resulting MPSs are associated with physical and psychological aging. In this perspective I posit that combining measures of biological aging with new quantifications of child development has the power to address fundamental questions about life span: How are development and experience in childhood related to biological aging in adulthood?

**Keywords** Biological aging, Development, Epigenetics, Cognitive aging, Children, Social determinants of health

## Self-Injurious Behaviour During Early Childhood in the Norwegian Mother, Father and Child Cohort Study (MoBa)

**Anastasia Izotova[1,2,3], Line Indrevoll Stänicke[1,2], Becky Mars[4,5], Laurie J. Hannigan[1,3,4], Helga Ask[3,6], Kim Stene-Larsen[7], George Davey Smith[8], Anne Reneflot[7], Anne-Siri Øyen[1], Alexandra Havdahl[1,3,6], Robyn E. Wootton[1,3,9]**

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NHS Foundation Trust, Bristol, United Kingdom. [6]PROMENTA Research Center, Department of Psychology, University of Oslo, Oslo, Norway. [7]Department of Mental Health and Suicide, Norwegian Institute of Public Health, Oslo, Norway. [8]MRC Integrative Epidemiology Unit, University of Bristol, Bristol, United Kingdom. [9]School of Psychological Science, University of Bristol, Bristol, United Kingdom

**Abstract**
Self-harm is one of the strongest known risk factors for suicide and is a major public health concern. Most existing research is conducted with adolescents and adults and very little is known about the prevalence and risk factors for self-harm in younger children. This study uses population-based data from a large cohort study to address this important knowledge gap.

The Norwegian Mother, Father and Child Cohort Study (MoBa) included ~ 95,200 mothers during pregnancy weeks 15 to 20. Mothers reported on child self-injurious behaviour at ages 3 and 8 years. Questionnaire, registry, and genetic data is used to describe the biological, psychological, and social factors characterising self-harm at each time point, as well as its persistence over time. We generate neuropsychiatric polygenic scores in parents and children to explore genetic associations with childhood self-harm.

The preliminary results show that mothers reported self-harm behaviour for 7.4% of the children at the age of 3 years (n = 4264/57,965) and 1.8% of the children at 8 years (n = 751/42,982). Further analyses will include: (1) an estimate of persistence in self-harm between time points, (2) a descriptive comparison of characteristics of the self-harm groups compared to never-harmers using $\chi^2$/t-tests, (3) associations between parental as well as child neuropsychiatric polygenic scores and self-harm.

The results will help to close the knowledge gap on self-harm in young children. Identifying genetic and environmental risk factors for self-harm in young children will inform early prevention and intervention efforts and therefore contribute to effective suicide prevention.

**Keywords** Childhood self-harm, Polygenic scores, Cohort studies, MoBa

**Grants**
South-Eastern Norway Regional Health Authority (#2022039, #2020022) and the Research Council of Norway (#274611, #336085).

## Benefits of Nuclear Twin Family Models for Estimating Heritability: An Investigation of 93 Personality-Relevant Constructs Using JASP

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**Abstract**
To disentangle the genetic and environmental contributions to individual differences, classical twin models (CTMs) are predominantly used. Previous research revealed several conditions under which CTMs spawn biased heritability and environmentality estimates, including the simultaneous relevance of nonadditive genetic and shared environmental influences, the presence of assortative mating, and passive gene–environment covariance. Nuclear twin family models (NTFMs), adding information from parents of twins reared together, can handle these conditions. However, a recent overall picture on the actual impact of such conditions considering the empirical properties displayed by various personality-related phenotypes is lacking. The present study bridges this gap. We investigated differences between NTFMs and CTMs in terms of error-adjusted heritability and environmentality estimates across 93 personality-relevant constructs based on data from 343 monozygotic and 293 dizygotic German twin pairs and 320 parents gathered from the Study of Personality Architecture and Dynamics (SPeADy). Overall, NTFMs resulted in significantly lower heritability and higher environmentality estimates. Hence, broad-sense heritability and especially additive genetic variance were, on average, overestimated while environmentality was underestimated in CTMs. These trends pertained to about two-thirds of the investigated constructs, but reversely directed changes were detectable under specific conditions. We discuss the extent to which the assumptions underlying CTMs might bias conclusions regarding genetic and environmental contributions to personality differences, and demonstrate the benefits of NTFMs in estimating (broad-sense) heritability and environmentality.

**Keywords** Nuclear twin family model, Estimation biases, Individual differences, Heritability, Structural equation modeling

**Grants**
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## Within-Pair Associations Between Early Weight Catch-Up Growth and Body Mass Index in Adolescence: Findings from the Louisville Twin Study

**Sean R. Womack[1,2,3], Christopher R. Beam[4], Evan J. Giangrande[5,6,7], Sophie A. Bell[8], Deborah W. Davis[9,10], Deborah Finkel[11,12], Eric Turkheimer[8]**

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**Abstract**
On average, twins are born at low birth weight and display rapid weight catch-up growth across infancy (Womack et al. 2023). In singletons born at low birth weight, rapid weight catch-up growth is a risk factor for obesity and other cardiometabolic risk (Barker et al. 2005). Associations between weight catch-up and obesity have not been tested in twins, limiting our understanding of how individual differences in typical weight development are related to physical health at later developmental stages.

Using data from the Louisville Twin Study data (187 monozygotic pairs, 173 dizygotic pairs, 48.5% female), we tested biometric associations between early weight catch-up growth and body mass index at 15 years. Latent change score models were fit to model change in weight-for-age z-scores between birth and 24 months. BMI was regressed onto additive genetic, shared environmental, and nonshared environmental components of birth weight and the latent change score. There was a significant nonshared environmental association between weight catch-up and BMI, indicating that, within-pair, faster catch-up growth was associated with a higher BMI. Slowing the rate of weight catch-up growth among low-birth-weight infants may reduce obesity risk.

**Keywords** Catch-up growth, Low birth weight, Louisville Twin, Body mass index, Obesity

**References**

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**Grants**

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## Protective Genetic Associations Between Active Music Engagement in Mid- to Late-Life and Resilience to Mental Health

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**Abstract**

**Background**

Although music-based interventions may alleviate mental health symptoms, the relationship between active music engagement (i.e., playing an instrument or singing) and mental health is unclear. Heritability estimates of active music engagement range from 78% from twin models to 12% via SNP-based approaches. This study investigates the common genetic variation associated with active music engagement and its relationship to neurological function and mental health.

**Methods**

We conducted a GWAS in unrelated European samples (6,725,254 SNPS post-QC) of playing an instrument or singing in a choir several times per year or more (n = 3956 cases), compared to no engagement (n = 15,566 controls). SNP-based heritability was estimated using GREML in GCTA. We used FUMA for positional and eQTL mapping (GTEx v8). We examined genetic correlations (LDSC) with 11 GWAS from the Psychiatric Genomics Consortium.

**Results**

No loci surpassed genome-wide significance (5e−8). SNP-based heritability for active musical engagement was 8.41%, $p = 3.49 \times 10^{-7}$. However, at the suggestive threshold (5e−5), 31 genomic loci were identified. These loci mapped to 36 genes, including one eQTL mapping to TYW1 with 24 risk-increasing eQTLs in several brain regions and several positionally mapped genes related to neurobiological function: *CDH12* (rs10518914[7]), *ZNF536* (rs193058357), *LSAMP* (rs189907653), and PCDH9 (rs4493636). Genetic correlation analysis revealed significant negative correlations with several psychiatric diagnoses, including major depressive disorder, bipolar disorder, and schizophrenia.

**Conclusion**

Active music engagement in mid- to late-life is associated with resilience to mental illness. Future investigations are to examine the causality and directionality.

**Keywords** Music, Mental health, GWAS, Genetic correlations

**Grants**

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## A Systematic Review: The Reporting and Handling of Missing Data in Genetic Epidemiological Studies of Mental Health-Related Traits in Childhood and Adolescence Using Cohort Data

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**Abstract**

**Background**

Missing data are known to undermine the validity of research by introducing biased results, loss of statistical power and precision. The aim of this study is to assess how researchers report missing data, and the statistical methods used to address missingness in psychiatric genetic epidemiology studies, based on the strengthening and reporting of observational studies in epidemiology guidelines.

**Method**

A search was conducted in the OVID database host, which contains PubMed, EMBASE, and Psych-Info databases, for studies published between August 2012 and August 2023. Inclusion criteria were genetic epidemiological studies that used polygenic scores, genome-wide association studies, and Mendelian randomization, focusing on children and adolescents with mental health related outcomes, using population-based cohort data.

**Results**
A total of 97 studies were eligible for the review. Eighty-nine (91.75%) studies reported the number of complete cases in all waves while 56 (57.73%) reported the amount of missingness on key variables. To handle missingness, 29 studies applied advanced methods other than complete case analysis, where multiple imputation 15 (15.47%) being the most used, followed by FIML 7 (7.22%). Only 15 (15.46%) reported the assumed missing mechanism along with the missingness handling method. The reporting of the extent and handling of missingness at the item-level was very uncommon, 2 (2.27%) and 9 (10.23%) respectively.
**Conclusion**
Best practice recommendations for handling missing data emphasize the importance of understanding the type of missing mechanisms, and selecting an approach accordingly. In this review, proper handling and reporting of missing data in accordance with recommendations remains uncommon.

**Keywords** Missing data, Mental health, Children and adolescent, Polygenic Score, Genetics

**Grants**
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## Distinguishing Genetic Predictors of Dementia and Cognitive Decline

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**Abstract**
Dementia is characterized by declines in cognitive function to levels beyond which daily living is substantially impaired. This diagnostic threshold may be surpassed either because cognitive functioning has declined substantially, or because cognitive function has declined more modestly from a lower peak level, or some combination of the two (Tucker-Drob 2019). Although a range of genetic factors are associated with dementia risk, there is little understanding of the extent to which each of these factors act via child developmental pathways that give rise to adult peak levels of cognitive function, or adult pathways whereby high and low risk groups differ in their rate of cognitive decline. We use participants with genetic data from the Health and Retirement study to investigate the pathways by which predictors affect risk for dementia, particularly focusing on polygenic scores to elucidate potential developmental pathways of risk. Our findings provide insight on the extent to which genetic pathways of dementia risk operate through individual differences in trajectories of child cognitive development or through differences in trajectories of decline.

**Keywords** Cognitive aging, Dementia, Genetics

**Grants**
This research was supported by National Institutes of Health (NIH) Grants R01AG054628 and R01HD083613. E.M.T.-D. is a Member of the Population Research Center (PRC) and Center on Aging and Population Sciences (CAPS) at The University of Texas at Austin, which are supported by NIH Center Grants P2CHD042849 and P30AG066614, respectively.

## Using Mendelian Randomisation to Identify Predictors of Depression Trajectories from Adolescence to Early Adulthood

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**Abstract**
Depression is the leading cause of global disability, and prevalence is increasing, especially amongst adolescents and young adults. The average age of depression onset is before age 30, with a quarter of affected individuals experiencing symptoms before adulthood. Therefore, adolescence represents a key period for depression onset, where it is critically important to identify robust causal risk factors. Mendelian randomisation (MR) uses genetic variants as instruments to identify possible causal risk factors for an outcome of interest (e.g. depression). However, most MR estimates are not specific to developmental stages, instead representing average lifetime effects. In a novel approach, we will combine trajectories of depressive symptoms with MR to identify developmentally specific risk factors. First, we will construct trajectories of depressive symptoms using repeated assessments from the Avon Longitudinal Study of Parents and Children (ALSPAC) cohort. Depressive symptoms were measured using the short version of the Moods and Feelings Questionnaire (13 items), asked at 11 repeated assessments from ages 10 to 25 years (including both self-reports and parent-reports). We will use a repeated measures multi-level model to describe the trajectory of depressive symptoms through adolescence and into early adulthood. We will test whether non-linear models are a better fit to our data (e.g., multi-level linear splines models). Second, we will combine predicted depressive trajectories with MR to identify causal predictors of change in intercept and slope (allowing for covariance between intercept and slope). The modifiable risk factors we will focus on are: educational attainment, smoking, alcohol consumption, and body mass index.

**Keywords** Depression, Trajectories, Adolescence, Mendelian randomisation, ALSPAC

**Grants**
Supported by South-Eastern Norway Regional Health Authority Grants 2020024, 2018059, and 2020022.

## Genetic Pathways for Autism Spectrum Disorder Unique of ADHD at Multiple Levels of Biological Analysis

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**Abstract**
**Introduction**
Autism spectrum disorder (ASD) is a neurodevelopmental condition characterized by atypical patterns of social functioning and repetitive/

restricted behaviors. ASD commonly co-occurs with ADHD and, despite their clinical distinctiveness, the two share considerable genetic overlap. Given their shared genetic liability, it is unclear which genetic pathways confer unique risk for ASD independent of ADHD.

**Methods**

We applied Genomic Structural Equation Modeling (SEM) to GWAS summary statistics for ASD and childhood-diagnosed ADHD, decomposing the genetic signal for ASD into that which is unique to ASD (*uASD*) and that which is shared with childhood-diagnosed ADHD. We computed genetic correlations between *uASD* and 75 external traits to estimate genetic overlap between *uASD* and other relevant phenotypes. We go on to apply Stratified Genomic SEM to identify classes of genes enriched for *uASD*. Finally, we implement Transcriptome-Wide SEM (T-SEM) to explore patterns of gene-expression associated with *uASD*.

**Results**

We observed positive genetic correlations between *uASD* and several external traits, most notably those relating to cognitive/educational outcomes and internalizing disorders and symptoms (e.g., self-harm). Stratified Genomic SEM showed that heritability for *uASD* was significantly enriched in genes involved in evolutionarily conserved processes, as well as for a histone mark in the germinal matrix. T-SEM revealed 83 unique genes whose expression was associated with *uASD*.

**Discussion**

These findings delineate the unique biological underpinnings of ASD which exist independent of ADHD. This work also demonstrates the utility of Genomic SEM and its extensions for disambiguating shared and unique risk pathways for genetically overlapping traits.

**Keywords** Autism spectrum disorder, Genomic SEM, Neurodevelopmental genetics, Biological pathways, Cognition

**Grants**

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## Associations of DNA-Methylation Profile Scores of Cognition with Cognitive Development, Academic Performance, and Socioeconomic Attainments

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**Abstract**

Recently, general cognitive abilities (*g*) have been quantified in blood DNA-methylation samples of adults that can be applied as methylation profile scores in separate target samples (Epigenetic-*g*). Here, we evaluate whether Epigenetic-*g* in children, adolescents, and adults is associated with cognitive development, academic performance, and socioeconomic attainments. Our preregistered analyses include n = 1830, 8–18-year-olds from the US Texas Twin Project, n = 5432, 5–32-year-olds from the German Twin Family Panel Study (TwinLife), and n = 2262, 0–72-year-olds from the German SOEP-Gene cohort. In the Texas Twin Project, which provided saliva DNA-methylation samples, Epigenetic-*g* is associated with levels of general cognitive ability, longitudinal gains in *g*, and math and reading school grades. For instance, no child with below average Epigenetic-*g* attended advanced math classes. In contrast, in German SOEP-Gene, which provided buccal DNA-methylation samples, quantifications of Epigenetic-*g* were not associated with cognitive performance, in line with previous reports of low blood-to-buccal cross-tissue correspondence. These results suggest that Epigenetic-*g* quantified in saliva samples, but not buccal tissue, is sensitive to individual differences in levels and developmental gains in *g* as well as real-world academic outcomes in children and adolescents. Similar to widely used measures of biological aging, DNA-methylation quantifications of psychological phenotypes may be useful tools to study the etiology of individual differences across the lifespan.

**Keywords** Cognitive development, Epigenetics, DNA methylation, Educational attainments

## Causally-Informative Analyses of the Effect of Job Displacement on All-Cause and Specific-Cause Mortality: A Finnish Register and Twin Study

**Stephanie Zellers[1], Terhi Maczulskij[2], Elissar Al Kazzi[1], Antti Latvala[1,3], Jaakko Kaprio[1]**

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**Abstract**

Finland suffered a severe economic recession in the early 1990s with unemployment reaching almost 25%. At the individual and macroeconomic level, unemployment is related to mortality, particularly suicide mortality. Studies conducted at the individual level offer significant strengths with respect to generalizability but are limited in that most rely on self-report data. Macroeconomic studies can provide incredibly well-powered and objective data on how rates of various phenomena have changed but cannot easily generalize to the individual level. With both objective data about recession-driven exogenous employment shock (job displacement in the private sector) and mortality on the individual level, we evaluate the relationship between these variables using large and well-powered longitudinal register data from Finland between 1989 and 2019 (N = 590,823 individuals, N = 1,561,232 person-years, N = 85,977 total deaths by 2019). We find that job displacement is associated with increased risk of all-cause mortality through the end of follow-up even after accounting for sex, age, marital status, and education (HR = 1.09 [1.07, 1.11], $p < 2 \times 10^{-16}$). Furthermore, the risk of mortality varies by cause of death. Risk of death by suicide, violence, alcohol, accidents, and any disease are higher for displaced individuals at all follow-up periods examined (2000, 2005, 2010, 2019). Risk of death from cancer and ischaemic heart disease are higher for displaced individuals only when evaluating later follow up periods (e.g., 2010 and 2019). Lastly, we find some evidence for familial confounding of job displacement (MZ tetrachoric correlation = 0.42, same-sex DZ tetrachoric correlation = 0.29). Additional analyses incorporating twin data and lifestyle factors are ongoing.

**Keywords** Causality, Longitudinal, Survival analysis

**Grants**
We acknowledge our Grant Support from the Biology of Trauma Initiative at the Broad Institute of MIT and Harvard.

## Comparing Factor and Network Models of Cognitive Abilities Using Twin Data

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**Abstract**
Network models have become a popular alternative to factor models for analyzing the phenotypic relationships among cognitive abilities. Studies have begun to compare these models directly to one another using cognitive ability data, although such a comparison has so far not extended to genetics. Here we will present comparisons between factor and network models of cognitive abilities at both a phenotypic and genetic level. We analyzed data from the Twins Early Development Study that were collected using 14 cognitive ability measures from 11,290 twins in the UK aged 12 years old. We conducted phenotypic and genetic analyses in which numerous factor and network models were tested, including a novel network twin model. Factor and network models both provided useful representations of the phenotypic and genetic relationships among cognitive abilities. Surprisingly, several relationships among cognitive abilities within the genetic networks were negative, suggesting that these cognitive abilities might share some genetic variants with inverse effects. We discuss implications for future genomic research.

**Keywords** Cognitive ability, Intelligence, Factor analysis, Network analysis, Twin study

**Grants**
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## How Do Lifestyle Factors Modify the Association Between Genetic Predisposition and Obesity-Related Phenotypes? A 4-Way Decomposition Analysis Using UK Biobank

**Mengrong Zhang[1], Frederick Ho[1], Joey Ward[1], Rona Strawbridge[1], Donald Lyall[1], Jill Pell[1], Carlos Celis[1]**

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**Abstract**
**Background**
Obesity and central obesity are multifactorial conditions with genetic and non-genetic (lifestyle and environmental) contributions. There is incomplete understanding of whether lifestyle modifies the translation from respective genetic risks into phenotypic obesity and central obesity, and to what extent genetic predisposition to obesity and central obesity is mediated via lifestyle factors.
**Methods**
This is a cross-sectional study of 201,466 (out of approximately 502,000) European participants from UK Biobank and tested for interactions and mediation role of lifestyle factors (diet quality; physical activity levels; total energy intake; sleep duration, and smoking and alcohol intake) between genetic risk for obesity and central obesity. BMI-PRS and WHR-PRS are exposures and obesity and central obesity are outcome.
**Results**
Overall, 42.8% of the association between genetic predisposition to obesity vs. phenotypic obesity was explained by lifestyle: 0·9% by mediation and 41.9% by effect modification. A significant difference between men and women were found in central obesity, the figures were 42.1% (association explained by lifestyle), 1.4% (by mediation), and 40.7% (by modification) in women and 69.6% (association explained by lifestyle), 3.0% (by mediation), and 66·6% (by modification) in men.
**Conclusions**
A substantial proportion of the association between genetic predisposition to obesity/central obesity and phenotypic obesity/central obesity was explained by lifestyles. Future studies with repeated measures of obesity and lifestyle would be needed to clarify causation.
**Keywords** Obesity, Central obesity, BMI-PRS, WHR-PRS, Total lifestyle factor

## Does Socioeconomic Advantage Interrupt the Intergenerational Transmission of Psychopathology?

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**Abstract**
**Background**
Children with a parent with depression or anxiety have triple the risk of developing a mental health problem. This intergenerational transmission of common mental health problems is partly due to genetics, but largely due to environmental pathways. Understanding how to interrupt the intergenerational transmission of mental health problems could help relieve a large public health burden.
**Aims**
Our primary aim is to test whether the intergenerational transmission of psychopathology is interrupted by socioeconomic advantage (e.g., high income, parental education, and neighbourhood advantage). Our secondary aim is to examine whether socioeconomic advantage interrupts environmental and/or genetic transmission of psychopathology.
**Sample**
We will use data from the Norwegian Mother, Father and Child Cohort Study (MoBa), a population-based pregnancy cohort. We will focus on extended family units comprising parent twins, parent siblings, or parent cousins, and their children (estimated N = 34,000 parents, 42,000 children).
**Measures**
Maternal depression and anxiety will be assessed through mothers' reports on the Hopkins Symptom Checklist at child ages 1.5, 3, and 5 years. Child internalising and externalising problems will be measured using the Child Behavior Checklist at ages 1.5, 3, and 5 years.

Socioeconomic factors (family income, parent education, and neighbourhood socioeconomic position) will be assessed through Norwegian registry data.

**Analyses**

First, we will test whether socioeconomic factors moderate the phenotypic associations between maternal depression/anxiety with child internalising and externalising problems. Second, we will test whether socioeconomic factors influence environmental and/or genetic routes of intergenerational transmission, using the Multiple Children of Twins and Siblings design.

**Keywords** Intergenerational transmission, Mental health, Socioeconomic position, Children of twins design

## Toxic Environments: Examining Metal and Air Pollution Exposure as an Etiologic Moderator of Children's Externalizing Behaviors

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**Abstract**

Heavy metal exposures (e.g., lead, mercury, manganese) and air pollution cluster in highly disadvantaged neighborhoods and have been associated with externalizing behaviors in early childhood and adolescence. The mechanisms underlying these associations remain unclear, however. One possibility is that toxicant exposures alter the magnitude of genetic and environmental influences on child externalizing behaviors. The current study aimed to evaluate this possibility, examining whether and how toxicant exposure moderated the etiology of children's externalizing behaviors. Participants included 756 twins from the Twin Study of Behavioral and Emotional Development in Children (TBED-C), half of whom were oversampled to reside in disadvantaged neighborhood contexts. Recorded toxicant exposure levels in the children's neighborhoods were pulled from the EPA's national database of facility reporting regarding toxic chemical and pollutant releases (Toxics Release Inventory; TRI). Childhood externalizing behaviors were assessed by mother reports. Preliminary results revealed that toxicant exposure moderated shared environmental influences on children's externalizing behaviors, such that as toxicant exposure increased, the amount of variance in children's externalizing behavior accounted for by shared environmental factors also increased—the same pattern observed for neighborhood disadvantage. Next steps will be to clarify whether toxicant exposures account for the effects of neighborhood, or vice versa.

**Keywords** Toxicants, Externalizing behaviors, Genotype–environment interaction

**Grants**

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## Diurnal Cortisol Functioning and Negative Affectivity as Predictors of Child Depressive Symptoms; a Genetically Informed Adoption Design

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**Abstract**

Childhood depressive symptoms are transmitted through genetic and environmental pathways influenced by various factors, including diurnal cortisol functioning (e.g., diurnal cortisol slope and variability), negative affectivity, and prenatal distress. Further, the development of negative affectivity, diurnal cortisol functioning, and depressive symptoms are buffered by higher parental warmth. Thus, we test a conceptual model wherein negative affectivity and diurnal cortisol functioning at age 6 years mediate earlier genetic and prenatal distress influences on depressive symptoms at age 8 years, and whether these trajectories of risk are buffered by parental warmth using cohort I of the Early Growth and Development Study (N = 361), a US-based sample of children adopted into non-related families at birth. Because age 8 depressive symptoms were zero-inflated we used a negative binomial hurdle model to predict any depressive symptoms as well as severity of depressive symptoms if any were endorsed. Implications for the development of depressive symptoms in childhood will be discussed. The results of the study did not support the specific hypotheses of the current study. However, higher parental warmth ($B = -0.012$, $SE = 0.005$, $p = 0.030$) predicted fewer depressive symptoms among children who endorsed any depressive symptoms.

**Keywords** Heritable risk, Prenatal risk, Diurnal cortisol functioning, Negative affectivity, Depressive symptoms

**Grants**

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## On the Importance of Modelling Indirect Effects in Genetic Epidemiological Studies

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**Abstract**

Indirect genetic effects occur when the genotype of an individual affects the phenotype of a related individual. Failure to correctly

account for the presence of indirect genetic effects may bias estimates of direct effects, heritability and genetic correlations, and lead to spurious results in downstream polygenic score and Mendelian randomization studies. Conversely, estimating indirect genetic effects accurately offers the opportunity to shed light on critical biological, social and environmental processes that may vary across population groups. In this session, I discuss some of the latest breakthroughs in the development and application of statistical methods that explicitly model such effects in large-scale genetic studies of common complex traits and diseases. This includes how correct modelling of indirect and direct genetic effects permits (i) Mendelian randomization studies free from dynastic effects, assortative mating and population stratification, (ii) increased power to detect loci in GWAS meta-analyses of traits influenced by direct and indirect genetic effects, (iii) Mendelian randomization studies that correctly model the long-term causal effects of intrauterine exposures on future risk of disease in offspring, and (iv) finally, in certain situations, what we think might be pleiotropy robust approaches to causal inference.

**Keywords** Indirect genetic effects, Direct genetic effects, Mendelian randomization, Genomic structural equation modelling

## Gene–Environment Interplay isn't Just Moderation: Simultaneously Testing Moderation and Mediation in Gene–Environment Models for Externalizing Outcomes

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**Abstract**

Studies investigating gene–environment interplay often focus on just one form of interplay—moderation (i.e., G × E)—while neglecting the possibility of environmental mediation (Li and He 2021). The current study leveraged polygenic scores (PGSs) generated from a well-powered genome-wide association study on externalizing outcomes (Karlsson Linnér et al. 2021) to explore both forms of gene–environment interplay underlying the development of EXT behaviors using longitudinal data from the National Longitudinal Study of Adolescent to Adult Health ($n$ = 9974). Separate structural equation models were fit for the two dimensions of EXT: disinhibition (e.g., substance use) and antagonism (e.g., aggression). Moderation and mediation by supportive parenting, delinquent peer affiliation, school connectedness, and neighborhood connectedness were modeled in parallel. Results indicated that the effects of EXT PGS were partially mediated by delinquent peer affiliation for both dimensions of EXT, while there was no evidence of PGS × E effects. PGSs may incorporate environmental signals via gene–environment correlations and genetic nurture (Kong et al. 2018), which suggests the prominence of environmental mediation in genetic effects and the importance of modeling mediation and moderation simultaneously.

**Keywords** Externalizing behaviors, Gene–environment interplay, Psychosocial environment, Latent-class trajectory, Dimensional psychopathology

**References**

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Li JJ, He Q (2021) Polygenic scores for ADHD: a meta-analysis. Res Child Adolesc Psychopathol 49:297–310. https://doi.org/https://doi.org/10.1007/s10802-021-00774-4

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## From Simulation to Empirical Data: Building a Pipeline for SEM PGS

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**Abstract**

Offspring exhibit similarities to their parents due to both genetic and environmental factors. Traditional twin studies, which rely on comparing phenotypic covariances to infer unmeasured genetic and environmental factors, face challenges in disentangling these factors. Balbona et al. (2021) developed a model which correlates offspring phenotypes with the polygenic scores (PGS) of parents' transmitted and non-transmitted alleles, using structural equation modeling (SEM). SEM PGS provides the estimate of (1) additive genetic influences, (2) the familial environment arising from vertical transmission, and (3) the correlation between them.

Here we present for the first time the semi automated pipeline we built to apply the SEM PGS model onto empirical data, using the Norwegian Mother, Father and Child Cohort Study (MoBa) as our target sample. Our result so far consist of an in depth quality control of both target and discovery samples, calculation of weights using LDpred2, full trio selection, an in house script splitting the MoBa parental genotypes into transmitted and non transmitted alleles, another to matrix multiply weights from LDpred2 with each parental allele to obtain haplotypic polygenic scores (hPGS) and formatting of the data to input for the model which runs in OpenMx.

**Keywords** SEM PGS, Genetic nurture, Methods, Pipeline

**Grants**

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## The Heritability of Beliefs About Heritability: An Adoption Study of Lay Intuitions About 21 Human Traits

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**Abstract**
Genetic literacy and education continue to be salient topics as genomic technologies become more affordable and accessible to the general public. Despite this importance, there has been very little research on the structure and etiology of lay beliefs about the heritability of human traits. In a sample of 617 biological, adoptive, and mixed Minnesotan families, we administered the Lay Estimates of Genetic Influence on Traits ("LEGIT") questionnaire, which asks participants to rate, on a 1 to 5 scale, the strength of genetic influence on a variety of human traits for which robust heritability estimates are available. We replicated our previous finding (Willoughby et al. 2019) that the association between layperson ratings and published estimates is surprisingly high ($r = 0.76$; $N = 1424$).

We additionally conducted biometric decomposition of beliefs about LEGIT traits by adapting familial relationship equations from Keller et al. (2009) to include adoptive parent and sibling relationships. To our surprise, we found nonzero heritability estimates only for beliefs about breast cancer, colorblindness, and sexual orientation, while beliefs about intelligence, ADHD, and musical talent showed significant influence of the shared sibling environment. We further examine the structure of these beliefs by investigating differences between adoptive and biological offspring, as well as various predictors of accuracy. In sum, we find that LEGIT accuracy is significantly greater for individuals that are female, have children, and are older (all $p < 0.005$), optimistically suggesting the role of experience, rather than genetics, in developing accurate perceptions about the heritability of human traits.

**Keywords** Heritability, Genetics education, Lay beliefs, Adoption study

**Grants**
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## Health and Wellbeing Among Close Family Members of Children with Comorbid Health Conditions: Triangulating the Evidence on Spillover Effects

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**Abstract**
Multimorbidity—the co-occurrence of two or more concurrent health conditions—has been identified as a global priority for health research in recent years. Though well-studied in older age groups, multimorbidity among children and adolescents remains under-investigated. Given that caregivers and other family members of individuals with chronic illness are known to be at risk of health spill-over effects, establishing the implications of multimorbidity for parents and siblings in the same household is important. Yet shared genetic and environmental risk factors among family members complicate this task. In this study, we leverage detailed phenotypic and genetic data from longitudinal birth cohort data (primarily the Norwegian Mother, Father, and Child cohort study, "MoBa") to triangulate evidence on the question of spillover effects associated with early-life multimorbidity. First, we compare the extent of change in symptoms of anxiety and depression among parents and siblings of children with, respectively, one, more than one, or no clinically diagnosed health conditions. Second, we compare the health and wellbeing of parental siblings differentially exposed to child health problems. Third, we explore the relationship between child health status and sibling/parental health, progressively adjusting potential confounders in the form of measured environmental factors and health-related polygenic scores. The results of these analyses can collectively inform about the nature and extent of spillover effects associated with early-life multimorbidity, which are important to quantify and include in health economic evaluations of prospective interventions.

**Keywords** Multimorbidity, Comorbidity, Sibling control, Polygenic scores, Spillover effects

**Grants**
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## Why Does Educational Inequality Run in Families? 170,000 Extended Families Show Larger Role for Genetics than for Nuclear- and Extended-Family Environments

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**Abstract**
**Purpose**
Children's academic achievement is predicted by parents' educational attainment. This could be because lower-educated parents tend to offer fewer resources and less academic support. Importantly, parent–child associations have been interpreted as a causal effect of parenting, but parents provide their children not only with a rearing environment, but also genes. We can only discover causal effects of children's natural environment if we control for genetics.
**Methods**
Our study is based on the population register of Norway and includes ~ 570,000 children born between 1994 and 2011 who took

part in national standardized reading-comprehension (L1), English (L2), and maths tests in 5th, 8th and/or 9th grade. We analysed a combined score. In addition, we used register information on parental education and on kinship (~ 2000 pairs of monozygotic twins, ~ 2000 pairs of dizygotic twins, and ~ 170,000 pairs of full siblings). Transmission from parents to children was analysed in behaviour genetic models that divided intergenerational resemblance into genetic and environmental modes of transmission.

**Results**

The correlation between parental educational attainment and children's educational performance was 0.31. Of this association, 68% was due to genetic transmission, 12% was due to parental transmission, and 20% was due to extended-family transmission.

**Conclusion**

Parental education predicts children's education mainly because of genetic transmission. This underscores the importance for social scientists to consider the role of genetics in studying children's environments. Additionally, we demonstrate the environmental effects of parental educational attainment, and previously undocumented environmental effects of extended family members.

**Keywords** Intergenerational transmission, Education, Children-of-twin model, Register data, Gene–environment interplay

**Grants**

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**LT-6D**

## Factors Shaping Resilient Functioning in Children Facing Genetic Vulnerability and Adverse Home Environments

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**Abstract**

Genetic vulnerability and adverse home environments are powerful and robust indicators of child behavioral problems, yet many children with these risk factors continue to thrive. One explanation is that genetically- and environmentally-based assets mitigate the deleterious effects of heritable risk and home adversity. We examined if positive heritable characteristics (i.e., emotion regulation) and rearing environments (i.e., parental warmth) account for more positive child outcomes within the context of heritable (i.e., psychopathology) and environmental [i.e., adoptive parents' (AP) hostility and depressive symptoms] risks using the "residual method."

Participants included 561 adopted children and their birth parents (BP) and APs. Genetic influences were indexed by BP psychopathology and emotion regulation. AP-reported parenting and depressive symptoms were assessed at 27 months. AP-reported child behavior problems were assessed at 54 months.

AP hostility and depressive symptoms, but not BP psychopathology, predicted more child behavior problems ($R^2 = 0.10$). Next, residual scores were extracted from the model for subsequent analyses as they reflect how much better or worse individuals are functioning when compared to others with similar levels of genetic and environmental risk. Thus, these residual scores indicate individual degree of "vulnerable to resilient" functioning, with higher scores reflecting better functioning relative to the level of risk. Results from subsequent analysis suggested that BP emotion regulation, but not AP warmth, was related to resilient functioning. The protective effect of BP emotion regulation in predicting resilient functioning sheds light on the importance of addressing emotional regulation skills in parents and children.

**Keywords** Adoption design, Genetic influences, Resilience, Environmental influences, Children, Behavioral problems

**Grants**

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## Beyond Trio: Genetic Effects in the Extended Family and Proximal Environments

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**Abstract**

After the goldrush to estimate parental indirect genetic effects, it is time to focus on the social world beyond. Indeed, several studies now show that indirect genetics effects capture more than just effects of parents. Could we leverage the association of someone's genotype with the phenotype of their child, sibling, niece, neighbor, or classmate to further understand genetic effects of heritable traits acting through the environment?

By using GWASs summary statistics performed in a mix of related and unrelated individuals, we fit genomic structural equation models to estimate the direct, indirect genetic effects, and their covariance on complex traits. First, we investigate the performance of this method using simulations. Second, using genotyped individuals in the Norwegian MoBa cohort and outcomes for their relatives and peers recorded in the national registries, we perform GWASs of birthweight and educational achievement. Mimicking TrioGCTA using GWAS, we replicate child and parental effects found in previous studies for birthweight ($\sigma^2_{child} = 0.12$, $\sigma^2_{mother} = 0.03$, $\sigma^2_{father} = 0.005$). Using educational achievement as outcomes, we will further estimate genetic effects of extended family members and peers.

This method gives promise for not only estimating correlations between a range of indirect genetic effects, but also allowing for estimating parental effects while adjusting for extended family effects and population stratification. It will also allow for creating GWAS summary statistics for these indirect genetic effects.

**Keywords** Indirect genetic effects, Education, Birthweight, Peer effects, Genetic nurture, Methods

**Grants**

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## Understanding Etiological Heterogeneity: Employing Mixture Models and Instrumental Variables

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**Abstract**

Modeling etiological heterogeneity presents challenges due to multiple sub-forms with subtle or overlapping phenotypic characteristics. To address this, we utilize finite mixture models to reveal specific heterogeneity in causal relationships. Particularly critical for prediction and treatment are cases where comorbidity arises from one disorder causing another, or vice versa. Our approach develops mixture models, mixDoC and mixCLPM, based on widely-used causal models such as the Direction of Causation twin model and the Cross-Lagged Panel model, with the mixture being modeled with two classes: one representing phenotypic measure X causing Y and the other representing phenotypic measure Y causing X.

To enhance causal inference, we integrate polygenic risk scores as instrumental variables (IVs) and employ moderated IV frameworks to examine how the effect of exposure varies across different levels of the moderator. This approach captures context-specific effects, making our model more robust for real-world data. By doing so, we aim to unravel complex causal pathways underlying psychiatric traits and structural brain measures, identifying subgroups with distinct etiological mechanisms.

Through model refinement and simulation-based validation, our framework elucidates diverse etiological pathways contributing to psychiatric disorders and brain structure variations, providing insights for potential interventions. Notably, simulations illustrate that it is feasible to detect underlying heterogeneity when alternative models predict different mean levels for subgroups, achievable with measurements taken on only three occasions using mixCLPM. We therefore hope our work will provide a robust methodological framework for unraveling causal heterogeneity, thus enhancing the potential for targeted interventions.

**Keywords** Causal inference, Instrumental variables, Mixture modeling

**Grants**

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## Genome-Wide Association Study of Age at Onset of Walking

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**Abstract**

Bipedal ambulation is a key characteristic of our species. Age at onset of walking can be achieved within a relatively wide developmental period, typically between ages 8 and 18 months. A recent meta-analysis of infant twin studies on psychologically-relevant phenotypes reported 59% pooled heritability for infant psychomotor functions (Austerberry et al. 2022). Here we report new findings from a genome-wide association meta-analysis of age at onset of walking from four cohorts (total N = 70,560 European-ancestry infants). Age at onset of walking in months was asked in all cohorts, and was a normally distributed phenotype. Genome-wide association studies were conducted using fastGWA, accounting for family structure using a sparse genetic relatedness matrix, and plink. Birth year, sex, ten ancestry principal components and cohort-specific technical variables such as batch were included as covariates. SNP-based heritability was 24.1% (SE = 1.2%) and 90% of the SNP-based heritability was explained by $\sim$ 15.4K variants, suggesting high polygenicity. Ten independent genome-wide significant loci were identified. Age at onset of walking was significantly genetically correlated with phenotypes including ADHD and educational attainment. The polygenic score showed out of sample prediction of 2.7–5.6%; within and between sib–pair polygenic associations were not significantly different ($p < 0.05$), suggesting largely direct genetic effects. In a genotyped sample of newborns with MRI data, the polygenic score associated with brain regions involved in motor learning.

**Keywords** GWAS, Infancy, Neurodevelopment, Motor

**Reference**

Austerberry C, Mateen M, Fearon P, Ronald A (2022) Heritability of psychological traits and developmental milestones in infancy: a systematic review and meta-analysis of twin studies. JAMA Netw Open 5(8):e2227887

**Consortia**

Corfield, E., Odintsova, V.V., Hottenga, J.J., Pool, R., Wong, A., Cullen, H., Wilson, S., Warrier, V., Eilertsen, E., Andreassen, O.A., Middeldorp, C., St Pourcain, B., Bartels, M., Boomsma, D.I., Hartman, C.A., Robinson, E.B., Edwards, D.

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## Controlling for the Genetic Relatedness Matrix in Spatial Regression Models

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**Abstract**

Social science genetics aims to control for all genetic effects in regression models to minimize confounding when estimating social effects. The state-of-the-art approach to control for genes is using polygenic scores. However, such scores do not capture the entire genetic variance of measured variants due to the imprecision of the results from genetic discovery studies on which their construction is

based. It has been shown that a genetic distance matrix can capture the genetic variance explained by all measured SNPs. Here, we develop a new method, a whole-genome control regression model (WGCR). We combine the whole-genome matrix-based measure with spatial regression control for all genetic effects without additional assumptions or a priori knowledge from genetic discovery about the genetic architecture of the outcomes under study. The only precondition is the availability of molecular genetic information on a set of respondents. Spatial regression models typically use high-dimensional connectivity matrices to control for the pairwise "proximity" of individuals to correct for omitted variable bias arising from common environmental shocks that individuals share. This is either because they live close to each other (geographic proximity) or biases arising from behavioural influences individuals exert on one another as a result of social relationships (friendship). The methodological innovation will be the redefinition of space in these models: we use a spatial regression model with a matrix encoding pairwise genetic distance, instead of a matrix encoding physical distance or social relationships. We demonstrate the properties of this approach based on simulations and real-world examples.

**Keywords** Genetic confounding, Educational attainment, Spatial regression, Genetic relationship matrix, Polygenic scores

## Dissecting the Contribution of Common Variants to Risk of Rare Neurodevelopmental Conditions

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**Abstract**

Although rare neurodevelopmental conditions have a large Mendelian component, common genetic variants also contribute to risk. However, little is known about how this polygenic risk is distributed among patients with these conditions and their parents, its interplay with rare variants, and whether parents' polygenic background contributes to their children's risk beyond the direct effect of variants transmitted to the child. Here, we addressed these questions using genetic data from 11,573 patients with rare neurodevelopmental conditions, 9128 of their parents and 26,869 controls. Common variants explained $\sim$ 10% of variance in overall risk. Patients with a monogenic diagnosis had significantly less polygenic risk than those without ($p < 1.2 \times 10^{-3}$), supporting a liability threshold model. In a trio-based model, using a polygenic score for neurodevelopmental conditions, the transmitted but not the non-transmitted parental alleles were associated with risk, indicating a direct genetic effect. In contrast, we observed no direct genetic effect of polygenic scores for educational attainment and cognitive performance, but saw a significant correlation ($p < 1.1 \times 10^{-7}$) between the child's risk and non-transmitted alleles in the parents, potentially due to indirect genetic effects and/or parental assortment for these traits. Indeed, as expected under parental assortment, we show that common variant predisposition for neurodevelopmental conditions is correlated with the rare variant component of risk. Our findings thus suggest that future studies should investigate the possible role and nature of indirect genetic effects on rare neurodevelopmental conditions, and consider the contribution of common and rare variants simultaneously when studying cognition-related phenotypes.

**Keywords** Neurodevelopmental disorders, Educational attainment, Cognition

**Grants**

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## Genomic Insights into the Shared and Distinct Genetic Architecture of Cognitive Function and Schizophrenia

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**Abstract**

Cognitive impairment is a major determinant of functional outcomes in schizophrenia. Previous studies have suggested genetic overlap between global cognitive ability and schizophrenia, however further work is needed to delineate the shared genetic architecture. Here, we apply genomic structural equation modelling to identify latent cognitive factors capturing genetic liabilities to 12 cognitive traits measured in the UK Biobank (UKB). We explore the overlap between latent cognitive factors, schizophrenia, and schizophrenia symptom dimensions using a complementary set of statistical approaches, applied to data from the latest schizophrenia genome-wide association study (Ncase = 53,386, Ncontrol = 77,258) and the Thematically Organised Psychosis study (Ncase = 306, Ncontrol = 1060). We identified three broad factors (visuo-spatial, verbal analytic and decision/reaction time) that underly the genetic correlations between the UKB cognitive tests. Global genetic correlations showed a significant moderate negative genetic correlation between each cognitive factor and schizophrenia, and patterns of local genetic correlations were consistent with this result. We found evidence of substantial polygenic overlap between each cognitive factor and schizophrenia but show that most loci shared between the latent cognitive factors and schizophrenia have unique patterns of association with the cognitive factors. Biological annotation of the shared loci implicated gene-sets related to neurodevelopment and neuronal function. Lastly, we find that the common genetic determinants of the latent cognitive factors are not predictive of schizophrenia symptom dimensions. Overall, these findings inform our understanding of cognitive function

in schizophrenia by demonstrating important differences in the shared genetic architecture of schizophrenia and cognitive abilities.

**Keywords** Schizophrenia, Cognitive function, Genome-wide association study, Factor analysis, Multivariate analysis

**Grants**

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## GenomicSEM Modelling of Diverse Executive Function GWAS Improves Gene Discovery

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**Abstract**

Previous research has supported the use of factor scores as the gold-standard in measuring executive function. However, for logistical reasons genome-wide association studies (GWAS) of executive function have largely eschewed factor scores in favour of singular task measures. As low correlations have traditionally been found between individual executive function (EF) tests, it is unclear whether these GWAS have truly been measuring the same construct. In this study, we addressed this question by performing a factor analysis on summary statistics from 11 GWAS of EF taken from 5 studies, using GenomicSEM. Models demonstrated a bifactor structure consistent with previous research, with factors capturing common EF and working memory-specific variance. Furthermore, the GWAS performed on this model identified 25 new genomic risk loci for common EF and 7 for working memory reaching genome-wide significance beyond what was found in the constituent GWAS, together resulting in 32 newly mapped EF genes. These results help to clarify the underlying genetic structure of EF and support the idea that EF GWAS are capable of measuring genetic variance related to latent EF constructs even when not using factor scores. Furthermore, they demonstrate that GenomicSEM can combine GWAS with divergent and non-ideal measures of the same phenotype to improve statistical power.

**Keywords** Executive function, GWAS, GenomicSEM, Genetic correlations

**Consortia**

CHARGE Consortium.

## Genome-Wide Association Meta-analysis of Infant Fussiness in the First Year

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**Abstract**

A recent meta-analysis of twin studies reported significant pooled heritability of emotional traits in infancy. Our research builds on this further by identifying specific common genetic variants associated with infant fussiness in the first year.

This project employs data from community and population-based cohorts with parent-report measures of temperament for infants aged 0–12 months. The total meta-Genome-Wide Association Study (GWAS) sample size will be ~ 80,000; the largest of the samples being the Norwegian Mother, Father and Child Cohort Study (MoBa, N ~ 55,000).

Fussiness phenotype was measured in MoBa using the Infant Characteristics Questionnaire 'Fussy/difficultness' scale. Other samples included used different questionnaires to measure fussiness, such as the Carey Infant Temperament Scale's 'Mood' subscale or the Infant Behaviour Questionnaire's 'Negative Affectivity' subscale.

GWAS in each sample was performed using GCTA fastGWA with a genetic relatedness matrix to account for relatedness. Covariates included age at data collection, year of birth, sex, the first 10 ancestry principal components, and relevant technical variables (e.g., genotyping batch). Individual sample GWAS results were meta-analysed using METAL.

In MoBa, fussiness at 6 months had an estimated SNP heritability of 10.2% (SE = 0.9). Analyses are ongoing for some smaller samples. Complete meta-analysis results will be presented alongside secondary analyses interrogating any genomic regions of interest and genetic correlations with mental health and neurodevelopmental outcomes.

Infancy remains largely unstudied in relation to common genetic variation on behaviour. Our study demonstrates that common genetic variation plays a significant part in explaining individual differences in temperament before a child's first birthday.

**Keywords** Infant, Temperament, GWAS, Development

**Consortia**

EArly Genetics and Lifecourse Epidemiology (EAGLE) Consortium.

**Grants**

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## The Moralistic Fallacy

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**Abstract**

The "moralistic fallacy" is frequently wielded as a rebuttal against critics of race science, asserting that calls to censor such research stem from an illogical attempt to derive an "is" from an "ought" (Davis 1978). Frequently championed by proponents advocating

unrestricted academic inquiry, this notion is particularly prevalent in contentious discussions surrounding race science (Rushton and Jensen 2005, 2008). Originating as a philosophical concept and intended as an inversion of G.E. Moore's 'naturalistic fallacy' (1903), the moralistic fallacy was conceived by a microbiologist and has not undergone rigorous analytical scrutiny. With the recent resurgence of debates on race and IQ, propelled in part by advancements in genomics and data, the validity of the moralistic fallacy demands examination: is it truly a logical fallacy? This paper undertakes a systematic, analytical evaluation of the moralistic fallacy, delineating three plausible interpretations of the argument. However, each interpretation presents its own set of challenges, ultimately leading to the conclusion that the moralistic fallacy itself is inherently fallacious.

**Keywords** Race, Intelligence, Values, Controversy

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**Grants**

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## A Twin Study of Early-Life Temperament and Accelerated Biological Aging in Midlife

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**Abstract**

Childhood psychosocial risk factors, such as adverse childhood experiences and internalizing symptoms, have received attention with regard to possible associations with accelerated biological aging. Whether and how early-life temperament correlates with biological aging, however, remains unexplored. Using 395 twins ($M_{Age}$ = 49 years, SD = 9.58), including 91 and 62 complete monozygotic and dizygotic pairs, respectively, from the Louisville Twin Study, we examined associations between three temperament dimensions at age 30 months (activity, affect-extraversion, task orientation) and three DNA methylation (DNAm) age acceleration biomarkers in midlife (PhenoAge, GrimAge, and DunedinPACE). Phenotypic analyses revealed task orientation was correlated with GrimAge Acceleration after adjusting for smoking ($r = -0.15$ [0.95 CI − 0.26, − 0.04]). When subsetting our sample to those 40 years old and younger, an additional correlation was observed between Activity and GrimAge Acceleration (r = 0.39 [0.95 CI 0.25, 0.54]). Among those older than 40, a *negative* correlation was observed between activity and PhenoAge Acceleration (r = − 0.23 [0.95 CI − 0.36, − 0.09]). Mediation models indicated this correlation was driven by a negative effect of activity on PhenoAge Acceleration (b = − 0.10 [0.95 CI − 0.17, − 0.03]), coupled with a positive indirect effect (b = 0.04 [0.95 CI 0.01, 0.08]) mediated smoking, body mass index, self-rated health, and physical activity. Biometric models indicated the association between activity and GrimAge was driven by additive genetic variance, whereas the association between activity and PhenoAge was driven by nonshared environmental variance. Findings implicate temperamental activity as a correlate of DNAm age acceleration, with nonshared environmental factors explaining part of this association.

**Keywords** Temperament, Toddlerhood, Aging, Midlife, DNA methylation

**Grants**

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## New Game+: Video Gaming Correlates with Cognition, a Co-sibling Control Design

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**Abstract**

Engaging in cognitive and social leisure activities, like playing video games, is theorized to benefit cognitive functioning past young adulthood. Employing a co-sibling approach, we investigated whether genetic and/or earlier life experiences underlie gameplay–IQ correlations using data from Colorado Adoption/Twin Study of Lifespan behavioral development and cognitive aging (CATSLife; N = 897 [52.4% F]; $\bar{x}_{age}$ = 32.2 years, SD = 4.5). Cognitive functioning was indexed by WAIS-III full-scale, performance, and verbal IQ. Video game engagement was measured by the average cognitive and social demand (Video Game Demand Scale; Bowman et al. 2023) from reported genres; non-gamers (58%) were coded as 0. Gameplay was correlated with IQ ($r_{verbal}$ = 0.09 to $r_{performance}$ = 0.18), after covariate (i.e., sociodemographics and screen time) adjustment. Based on covariate- and sibling-adjusted generalized estimating equations (GEE) models, cognitive demand was associated with IQ at the individual and between-sibling level. For performance IQ, we found a significant adoptee within-effect and fully attenuated biological and twin sibling effect, suggesting genetic confounding. Genetic rather than familial/shared environmental factors may explain these associations, because of the overlapping cognitive abilities involved in performance IQ (e.g., visual spatial and processing speed) and skills challenged by gameplay mechanics, potentially highlighting selection processes. With growing public health interest in understanding links between gaming and functioning, future work will reveal whether gaming-IQ differences influence later cognitive maintenance and aging.

**Keywords** Video gaming, Cognition, Co-sibling approach

**Reference**

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**Grants**

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## Characterizing Behavioral Manifestations of a Genetic Liability Towards Self-Regulation Deficits Across Development

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**Abstract**

Behaviors and disorders such as ADHD, conduct disorder, and substance use disorders, aggression, risk taking, and sensation seeking are characterized by impulsivity and self-regulation deficits. These behaviors and disorders are highly correlated and are often referred to as *externalizing* phenotypes.

In the current study, we used polygenic scores (PGS), derived from a multivariate GWAS identifying genes that impact addiction outcomes via self-regulation (Karlsson Linnér et al. 2021), to estimate associations of genetic risk with relevant behavioral phenotypes within and across developmental periods, ranging from toddlerhood to early adulthood. We used data from five population-based and epidemiological datasets spanning three countries.

We found significant associations between the PGS and a variety of phenotypes from toddlerhood to early adulthood. In toddlerhood and early childhood, children with higher PGS were rated as more active, more social, and less shy. Higher PGS was consistently associated with measures of impulsivity from early adolescence to early adulthood. Individuals with higher PGS were more likely to experience symptoms of conduct disorder, oppositional defiant disorder, and attention deficit hyperactivity disorder, as well as greater alcohol use and problems, and nicotine, cannabis, and other drug use starting in early adolescence.

Overall, findings show wide ranging manifestations of genetic risk for self-regulation deficits across a variety of developmental periods. Results from this study will inform our understanding of how we can incorporate genetic information into targeted early intervention and prevention efforts for substance use and related disorders.

**Keywords** Externalizing, Substance use, Developmental

## Resilience Factors and Adolescent Externalizing Symptomology: Genetic and Environmental Contributions

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**Abstract**

Previous twin studies find that adverse life events and adolescent externalizing psychopathology are linked primarily through environmental covariance, supporting the assumption that adversity influences psychopathology through environmental mechanisms. A similar assumption underlies resilience factors for psychopathology. We tested this assumption by examining genetic and environmental contributions to the relations between two resilience factors (self-compassion, parental support) and adolescent externalizing symptoms. In a subsample ($N = 491$) of racially diverse adolescents participating in the Arizona Twin Project ($M_{age} = 13.62$, $SD = 0.95$; 32.4% MZ, 34.8% ssDZ, 32.8% osDZ; 49.6% female; 59.7% White, 25.8% Hispanic), twins reported on self-compassion and parental support, and parents reported on twins' externalizing symptoms (MacArthur HBQ). Using biometric twin modeling covarying age, sex, and family socioeconomic status, we found that additive (A) genetic, shared (C), and unique (E) environmental components influenced self-compassion (A = 18.8%, C = 11.1%, E = 70.1%), whereas dominant (D) genetic factors influenced parental support (A = 32.1%, D = 31.4%, E = 36.5%). Self-compassion and parental support were negatively associated with externalizing ($r = -0.178$ and $-0.275$ respectively, $ps < 0.001$). For self-compassion, the bivariate model revealed covariance attributable to genetic and environmental mechanisms (A = 43.6%, C = 34.7%, E = 21.7%; $r$A = 0.05, $r$C = 0.15, $r$E = 0.02). For parental support, a small, positive nonshared environmental covariance parameter was dropped, $\chi^2(1) = 0.65$, $p = 0.42$. The reduced model revealed strong genetic contributions (A = 88.2%, D = 11.8%; $r$A = $-0.52$, $r$D = $-1.0$). Such results suggest that self-compassion and externalizing symptoms are linked through both genetic and environmental covariance in adolescence, while parental support is heritable and linked to externalizing via genetic covariance. These findings have implications for the assumptions underlying traditional resilience models.

**Keywords** Resilience, Self-compassion, Parental support, Externalizing, Twins

**Grants**

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## Using Extended Twin Family Designs to Differentiate Multiple Forms of Intergenerational Transmission

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**Abstract**

Parent–offspring similarity is observed in numerous traits. Hence, modeling intergenerational transmission paths from parents to offspring has been an important topic in behavioral genetic models. By including parents of monozygotic and dizygotic twins, the nuclear twin family design (NTFD) can estimate the "phenotypic transmission" path from parents to offspring while controlling for the effects of shared genes. As an extension of the NTFD, the extended twin family design (ETFD) includes twins and their siblings, parents, spouses and children to allow the estimation of additional parameters and thus reduces biases when such parameters are non-zero. The Cascade model further extends traditional ETFD models by relaxing the assumption that both assortative mating and intergenerational

transmission are primarily phenotypic. While the ability to differentiate various forms of assortative mating (e.g., phenotypic assortment, social homogamy, genetic homogamy) with the Cascade model has been tested, similar studies have not been done to relax the assumption of phenotypic transmission. In this study, we introduce the Cascade model that can flexibly model different forms of intergenerational transmission (including phenotypic transmission, environmental transmission, or a mixture of different transmission forms). For example, environmental transmission occurs when only environmental influences on the parent phenotype influence the child phenotype, rather than the parent phenotype itself directly influences the child phenotype. We test whether our Cascade model provides unbiased parameter estimates and can reliably differentiate various forms of intergenerational transmission. Applying this version of the Cascade model to empirical data will provide important insights into the nature of intergenerational transmission processes.

**Keywords** Intergenerational transmission, Extended twin family design, Behavior genetics, Parent–offspring similarity

**Grants**

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## Cross-Sectional Causal Modeling with Polygenic Scores

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**Abstract**

In the standard instrumental variable (IV) regression model, the association between the exposure (X) and the outcome (Y) is decomposed into the association attributable to the causal effect of X on Y and the association attributable to background covariance (confounding). This decomposition is possible due to the presence of the IV of X, e.g., a polygenic score of X ($PGS_X$). This model may be extended to include the PGS of Y ($PGS_Y$), which allows one to consider the bidirectional causal relationship between X and Y, in the presence of confounding. Here, we demonstrate that it may be beneficial to include, in the unidirectional model, $PGS_Y$ as a covariate of Y, not an IV of Y. Specifically, if $PGS_X$ and $PGS_Y$ are correlated, leaving out $PGS_Y$ results in a biased estimate of the causal effect of X on Y. The correlation between the components (single-nucleotide polymorphisms; SNPs) of $PGS_X$ and the components (SNPs) of $PGS_Y$ may arise from many sources, including, among other things, pleiotropy, linkage disequilibrium, cross-trait assortative mating, and uncorrected population structure. Therefore, given possibly correlated $PGS_X$ and $PGS_Y$, regressing Y on $PGS_Y$ helps reduce potential biases in the estimated effect of X on Y. We used simulated and empirical data (drinks/week and lifetime smoking in the Netherlands Twin Register) to compare the causal estimates in the standard and extended IV regression models. The model further lends itself to bidirectional causal modeling, providing a model for data from unrelated individuals, analogous to the MR-DoC2 model for twin data.

**Keywords** Polygenic scores, Instrumental variables, Causal inference

**Grants**

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## Parsing the Genetic Architectures of Smoking Initiation and Tobacco Use Disorder Using Genomic Causal-Common-Contingent Modeling

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**Abstract**

Smoking initiation (SmkInit) and tobacco use disorder (TUD) are distinct stages of tobacco use progression. Large-scale genome-wide association studies (GWAS) have identified hundreds of genomic loci associated with these phenotypes. However, any observed genetic associations of TUD may be partly or fully mediated via SmkInit. Notably, all 88 loci associated with TUD are also associated with SmkInit. Prior twin/family-based studies have used the causal-common-contingent (CCC) model to tease apart (unobserved) genetic factors unique to nicotine dependence (ND; based on Fagerström test questions closely related to TUD) and have found $\sim$ 70% of ND's twin-heritability to be mediated by SmkInit. Here, we apply genomic structural equation modeling (gSEM) to fit the CCC model to European ancestry-specific summary statistics from the current largest GWAS meta-analyses of SmkInit (N = 2,669,029) and TUD derived from electronic health records, but not screened for SmkInit (N = 495,005). The Genomic CCC model suggested that $\sim$ 32% of TUD's SNP-heritability is unique beyond SmkInit—consistent with the estimates from twin studies. Subsequent CCC GWAS identified 19 loci associated with TUD beyond SmkInit, with the top loci

located in/near nicotinic receptor genes. Interestingly, the Genomic CCC model is equivalent to a Cholesky decomposition model ("GWAS-by-Subtraction"), with both models equivalent to linear regression (TUD regressed on SmkInit) in gSEM. Further methodological work is imperative to enable these models across ancestral populations for more inclusive and equitable research. This study represents a critical step towards disentangling the genetic architecture of multistage complex traits like tobacco use, with potential implications for stage-specific treatment strategies.

**Keywords** Causal-common-contingent model, Smoking, Tobacco use disorder, Genomic SEM, Cross-ancestry meta-analysis

**Grants**

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## Evocative Gene–Environment Correlation Between Polygenic Scores for Externalizing Behaviors and Supportive Parenting: A Within-Family Analysis

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**Abstract**

Polygenic scores (PGSs) that predict externalizing problems are also associated with parenting behaviors. One explanation for this association is evocative gene–environment correlation (rGE). However, testing evocative rGE is notoriously difficult due to the confounding effects of parental genotypes. Recent studies have either utilized an adoption design instead of PGSs or constrained their analysis to a smaller sample for which parental PGSs were available as covariates (Shewark et al. 2021; Kretschmer et al. 2022). Combining within-family design and PGS in studying evocative rGE may afford (1) more robust statistical control and (2) higher statistical power. Capitalizing on the family sample from the National Longitudinal Study of Adolescence to Adult Health (n = 959), we fit a multilevel structural equation model where parent-report supportive parenting mediated the effect of PGS on adolescent externalizing behaviors. The indirect effects of PGS were statistically significant both between- and within-family, mediating 12.3% and 34.7% of the PGS total effect, respectively. Our results provide robust evidence for evocative rGE effect on externalizing behaviors related to parenting and highlight the importance of considering environmental mediation as an important mechanism in the interplay between PGS for EXT behaviors and the environment.

**Keywords** Externalizing behaviors, Evocative gene–environment correlation, Parenting, Twins and siblings, Environmental mediation

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## Adolescent Psychopathology and Cognitive/Academic Functioning: Impact of Comorbidity Using a Genetically Sensitive Design

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**Abstract**

**Objectives**

Determine the: (i) nature of the associations between three domains of psychopathology (depressive, hyperactivity/impulsivity and conduct symptoms) and cognitive/academic performance among adolescents i.e., causal mechanisms and/or shared genetic effects; (ii) extent to which these associations vary by comorbidity.

**Methods**

The sample comprised participants in the UK Twins Early Development Study ($n \approx 12{,}000$ individuals). Psychopathology domains were assessed using standardised questionnaires, and Cognitive and academic performance using Standard Progressive Matrices and GCSE scores respectively. Comorbidity was derived as a count of borderline/high psychopathology present per individual. Twin modelling (including the bidirectional MRDoC model—MRDoC2) was used to investigate phenotypic correlations, direction of causation and moderation effects.

**Results**

There were small to moderate negative correlations between adolescent psychopathology domains and cognitive performance ($-0.15 \leq r \leq -0.01$) and academic performance ($-0.23 \leq r \leq -0.04$). Hyperactivity/impulsivity causally influenced cognitive/academic performance while poorer cognitive performance causally influenced conduct symptoms. Comorbidity significantly moderated the phenotypic associations between hyperactivity/impulsivity symptoms and cognitive performance ($\beta_{mod} = 0.07$, 95% CI 0.02, 0.12); and between conduct symptoms and academic performance ($\beta_{mod} = -0.05$, 95% CI $-0.08$, $-0.02$) whereby this association became more negative with increased comorbidity.

**Conclusion**

These findings indicate differential mechanisms for the links between different psychopathology domains and impaired functioning and the need to recognise comorbidities during assessment and treatment planning to promote optimal functioning among adolescents.

However, more methodological work is required to ascertain validity of moderation of these models.

**Keywords** Adolescent psychopathology, Comorbidity, Cognitive functioning, Academic functioning, Classical twin design

**Grants**
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## Shared Genetic Liability Between Nausea and Vomiting During Pregnancy and Mental Health Disorders

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**Abstract**
Severe nausea and vomiting (NVP) is experienced in ∼ 14% of pregnancies and progresses to hyperemesis gravidarum (HG) in 1–3% of cases. The degree of severity of NVP is partially due to genetic factors (heritability estimate 53%, 95% CIs = 38–65%). There is a phenotypic association between NVP/HG and depression and anxiety during pregnancy and postpartum, but also between pre-pregnancy psychiatric diagnoses and HG. We aimed to explore the genetic architecture of NVP and HG and to investigate whether there is an overlap between the genetic risk for severity of NVP and psychiatric disorders.

We conducted Multi-Trait Analysis of GWAS summary data (MTAG) using the summary statistics of GWAS meta-analyses of NVP severity (N = 77,649) and HG (N cases = 5598, N controls = 390,942) from 10 cohorts. We detected 10 novel genomic loci surpassing the genome-threshold and replicated 8 of the genomic loci already reported for HG. There were significant genetic correlations between the MTAG results and major depressive disorder, anxiety, PTSD and ADHD, ranging from 0.25 to 0.33. Using Mendelian Randomisation we found support for a causal inference (versus only correlation relationship) between PTSD and NVP severity but not for MDD. These results survived multiple testing correction due to other traits of interest (e.g. reproductive, metabolic) being analysed.

Our study increases our knowledge about the genetic factors underlying NVP severity/HG and suggests that some of the reported associations between NVP severity and mental health at an epidemiological level are partially due to shared genetics factors.

**Keywords** Nausea and vomiting during pregnancy, Mental health, Genetics

**Consortia**
NVP Genetics Consortium: Ky'Era Actkins, Karina Banasik, Thomas Folkmann Hansen, Ada Johansson, Yi Lu, Juan J. Madrid-Valero, Catherine M. Olsen, Lavinia Paternoster, Nirmala Pandeya, Juan F. Sánchez-Romera, Peter Straub, David Westergaard, José J. Morosoli, Kerrie McAloney, Simone Cross, Richard Parker, Lea Davis, Patrick Jern, Triin Laisk-Podar, Michelle Lupton, Nicholas G Martin, Mette Nyegaard, Dale R. Nyholt, Juan R. Ordoñana, Valgerður Steinþórsdóttir, David C. Whiteman DBDS Genomic Consortium FinnGen 23andMe.

**Grants**
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## Disentangling Sources of Gene–Environment Correlation: A Simulation Study on Detecting Cultural Transmission and Sibling Interaction Using Polygenic Scores

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**Abstract**
**Background**
In the classical twin design, unmodeled AC covariance (covAC) leads to misestimation of the C component. One way to address this is the incorporation of polygenic scores (PGSs). Previous approaches focused on two sources of covAC: cultural transmission, when the parents' genotype contributes to the effective environment of the child, and sibling interaction, when the genotype of one sibling contributes to the effective environment of another. One method disentangles these two sources of covAC through integrating PGSs into one regression model (Young et al. 2022). Here, we present a simulation study that tests this model.
**Method**
We generated data using exact simulation in OpenMx and varied the parameters for the two sources of covAC. We tested power using regression models that assume only one source of covAC, and a combined model with both sources.
**Results**
We present power calculations and regression coefficients. Fitting a regression model that assumes one source of covAC when another is present leads to falsely detecting an effect and drawing wrong conclusions about covAC mechanisms. The combined model offers an advantage over models that only assume one source of covAC.
**Conclusion**
This study elucidates how the regression model may be used to detect different covAC sources. We demonstrate that unmodeled sibling interactions can look like cultural transmission, and vice versa.

**Keywords** Gene–environment interplay, Simulation study, Classical twin design, Polygenic scores

**Reference**
Young AI, Nehzati SM, Benonisdottir S, Okbay A, Jayashankar H, Lee C et al (2022) Mendelian imputation of parental genotypes improves estimates of direct genetic effects. Nat Genet 54(6):897–905

**Grants**
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## Epigenome-Wide Association Study Meta-analysis of Wellbeing

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**Abstract**
Wellbeing is associated with both behavioral phenotypes as well as several key life outcomes, such as health, employment and coping with stressful events. These phenotypes associated with wellbeing could be potential indicators of differential epigenetic patterns between individuals that differ in their levels of wellbeing. We performed the largest epigenome-wide (EWAS) meta-analysis of wellbeing to date by combining data from 13 cohorts from Europe, Australia, and the USA ($N$ = 10,757 participants). We tested whether a weighted methylation score (MS) based on the EWAS meta-analysis summary statistics significantly predicted wellbeing over and above a polygenic score (PGS) for wellbeing in an independent cohort. After correcting for smoking and BMI, no genome-wide significant methylation sites were identified. The MS was significantly associated with wellbeing independently of the PGS (MS: variance explained = 0.90%, $p$ = 0.011, PGS: variance explained = 0.58%, $p$ = 0.002). Lastly, we compared methylation levels in 16 pairs of monozygotic twins discordant for wellbeing. These analyses revealed no significant within-twin methylation differences at the top-sites from the meta-analysis or at the genome-wide level. Our results suggest that larger EWAS meta-analyses with better phenotype assessment are required to identify methylation sites associated with wellbeing.

**Keywords** Wellbeing, Epigenome-wide, MZ discordant, Methylation score

**Grants**
This work was supported by the NWO Talent Programme—VICI Scheme Nr. VI.C.211.054 'The Power of Wellbeing' (PI Bartels) and an ERC Consolidator Grant (771057, WELL-BEING, PI Bartels).

## Distinguishing Specific from Broad Genetic Associations Between External Correlates and Common Factors

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**Abstract**
Within the Genomic SEM framework, common factors are often used to index shared genetic architecture across constellations of GWAS phenotypes. A standard common pathway model, in which a genetic association is estimated between an external GWAS phenotype and a common factor, assumes that all genetic associations between the external GWAS phenotype and the individual indicator phenotypes are mediated through the factor. This assumption is typically tested using the $Q_{Trait}$ statistic, which compares the *common pathway* model to an *independent pathways* model that allows for direct genetic associations between the external GWAS phenotype and the individual indicator phenotypes. Here, we expand upon the $Q_{Trait}$ approach by describing an effect size index that quantifies the degree to which the common pathways model is violated, and providing a systematic approach for identifying the specific indicator traits for which direct pathways to add to the standard common pathways model may be called for. Our method comprises a series of omnibus tests and outlier detection methods indexing the heterogeneity of associations between the genetic component of external traits and the individual indicators of common factors of shared genetic architecture. We provide a set of automated functions which we apply to investigate the patterns of associations across a set of external correlates with respect to a genetic general intelligence factor drawn from seven cognitive traits, and an Alzheimer's disease factor drawn from case–control and proxy GWAS of Alzheimer's disease.

**Keywords** Genomic Structural Equation Modeling, Genetic architecture, External correlates

**Grants**
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## The Effect of Polygenic Liability for Psychiatric Disorders on COVID-19 Outcomes in People with Depression: The Mediating Role of Anxiety

**Anna Monistrol-Mula[1,2,3], Sarah E. Medland[4,5], Lucía Colodro-Conde[4,5], Brittany L. Mitchell[4]**

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**Abstract**
**Background**
Genetics contribute to how individuals experience stressful life events. For instance, genetic vulnerability to psychiatric disorders has been associated with various clinical and psychological outcomes associated with the COVID-19 pandemic. Here, we explored whether genetic risk for several psychiatric disorders predicted poorer COVID-19 outcomes in people with pre-existing depression.
**Methods**
Data from three assessments of the Australian Genetics of Depression Study (N = 4417; 52.2 years ± 14.9; 76.2% females) were analysed. Outcomes included COVID-19 clinical outcomes (SARS-CoV-2 infection, long COVID) and COVID-19 psychological outcomes (COVID-related stress, COVID-19 burnout). Predictors included the polygenic risk scores (PRS) for depression, bipolar disorder (BD), schizophrenia and anxiety. Associations between PRS and outcome variables was assessed using linear/logistic/multinomial regressions. We further explored the potential influence of anxiety symptoms and resilience on the identified associations between the PRSs and COVID-19 psychological outcomes using mediation and moderation analyses.
**Results**
The selected PRS did not predict SARS-CoV-2 infection or long COVID. PRS for depression predicted higher levels of COVID-19

burnout and COVID-related stress, although the later only at a nominal level. PRS for BD predicted lower levels of COVID-19 burnout albeit only at a nominal level. Anxiety symptoms fully mediated the association between PRS for depression and COVID-19 burnout. Resilience did not moderate this association.

**Conclusions**

A higher genetic risk for depression predicted higher COVID-19 burnout and this association was fully mediated by anxiety symptoms. Interventions targeting anxiety symptoms might be particularly effective in mitigating the psychological effects of a pandemic among people with depression.

**Keywords** Mental health, Polygenic risk scores, Anxiety, COVID-19 pandemic

**Grants**

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## Variation in Corpus Callosum Volume in Autism

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**Abstract**

Autism spectrum disorder is a complex neurodevelopmental disorder characterized by social and communication difficulties and a high rate of co-occurring problems. Despite the phenotypic heterogeneity, significant progress has been made in characterizing the neuroanatomical underpinnings. In terms of brain structure, autism is associated with a reduced corpus callosum (Hardan et al. 2009).

To investigate whether similar associations are present in typically developing children, we used a longitudinal sample of twins (N = 229) and older siblings (N = 104), with brain MRI and information on autistic traits, at baseline (twins 10.1 ± 0.5 years; siblings 12.2 ± 1.2 years) and approximately 2.5 and 6.5 years of follow-up.

Volumes of the corpus callosum, divided into five subregions from posterior to anterior, were derived from the MRI scans. Autistic traits were based on mothers' ratings of 10 items from the Child Behavior Checklist.

Cross-sectional analysis at the three measurement times revealed no significant associations between the autistic trait scores and corpus callosum volumes. Furthermore, in a longitudinal test, we also could not find relationships between changes in autistic traits and corpus callosum volumes from the first to the second or from the second to the third measurement.

The results of this longitudinal study indicate no support for an association between autism behavior and corpus callosum volume. If more autistic traits are associated with a smaller corpus callosum in typically developing children, the association is likely small.

**Keywords** Autism, Twins, MRI, Corpus callosum, Subcortical volumes

**Reference**

Hardan AY et al (2009) Corpus callosum volume in children with autism. Psychiatry Res 174:57–61

## Trajectories of Conduct Problems from Early Childhood to Early Adulthood in the Twins Early Development Study: Risk Factors and Outcomes

**Elisavet Palaiologou[1], Joanna Bright[1], Gerome Breen[1], Elham Assary[1], Matthew Hotopf[2,3], Celestine Lockhart[1], Megan Skelton[1], Argyris Stringaris[4], Thomas McGregor[1], Essi Viding[5], Ewan Carr[1,6], Thalia C. Eley[1]**

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**Abstract**

Conduct problems are highly heterogeneous across development. Studies identify distinct symptom trajectory classes within single populations that follow unique patterns of onset, stability and severity. These trajectories are differently associated with risk factors and outcomes. To date, research primarily focuses on phenotypic risk factors and outcomes in adolescence or the early thirties. The role of genetic factors in conduct problem trajectories and associations between trajectories and outcomes in emerging adulthood are understudied. Moreover, evidence on the aetiological bases of the trajectories is unclear, and no studies have explored the prediction accuracy of trajectory class membership from genetic and phenotypic risk factors.

With data from the Twins Early Development Study (N = ∼ 15,000), we will use growth mixture models to identify conduct problem trajectories from ages 4 to 26. We will then examine associations between trajectories and genetic and early-life phenotypic risk factors and outcomes at age 26. We will use twin analyses to explore the aetiological bases of the identified trajectories and penalised regression models to assess the predictive accuracy of trajectory membership based on genetic and early-life phenotypic risk factors.

We expect to identify at least four conduct problem trajectories. We anticipate that genetic and phenotypic risk factors will be associated with more severe trajectories, which in turn will be associated with worse outcomes. We expect that all trajectories will be moderately heritable and that predictive accuracy will be increased by the inclusion of both genetic and phenotypic variables. Our study will have the potential to inform early prevention and identification processes.

**Keywords** Trajectories, Conduct problems, Genetics, Environment, Machine learning

**Grants**

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## A Genetically Informed Investigation into the Bidirectional Links Between Parenting Style and Child Externalizing Behaviours

**Joanna K. Bright[1], Yasmin I. Ahmadzadeh[1], Olakunle Oginni[1,2], Essi Viding[3], Eivind Ystrøm[4,5], Tom A. McAdams[1,4]**

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**Abstract**

Parenting style is frequently associated with child conduct problems. There are multiple potential reasons for such associations, and family-based designs offer an opportunity to investigate such explanations. Children of twin models compare monozygotic and dizygotic avuncular correlations to parent–child associations to decompose parent–child associations into genetic effects, shared environmental and non-shared environmental effects. Known environmental influences can be controlled for as confounds in analyses and family data can control for unknown environmental influences.

This project aims to account for genetic confounding and estimate the direction of effects between parenting and child externalising using a cross-sectional 'multiple children of twin and siblings' (MCoTS) sample, extracted from the Norwegian Mother Father and Child Cohort Study (MoBa; Magnus et al. 2016). MoBa has a total sample of over 95,000 mothers, 75,000 fathers and 114,500 children. We have extracted extended families from MoBa, comprising twins, siblings, half-siblings, and cousins in both the parent and child generation. The MCoTS model can control for passive gene–environment correlation by capturing genetic effects shared by parent and child phenotypes, and can be adapted to address direction-of-causation. We address the association between aggressive and rule breaking behaviour in children and inconsistent discipline and positive parenting in parents when the child was 5 years old.

Preliminary analyses suggest observed association between inconsistent parenting and child externalising remains after controlling for shared genes, whereas the association between positive parenting and child externalising is wholly explained by shared genes. Analyses of the direction of the significant environmental association are underway.

**Keywords** Family-based design, Conduct problems, Parenting

**Grants**

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## Shared Genetic Architecture and Causality Between Autism Spectrum Disorder and Irritable Bowel Syndrome, Pain, and Fatigue

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**Abstract**

Autism spectrum disorder (ASD) often co-occurs with functional somatic syndromes (FSS), such as irritable bowel syndrome (IBS), musculoskeletal pain, and fatigue. However, the underlying genetic mechanisms and causality have not been well studied. Using large-scale genome-wide association study (GWAS) data, we investigated the shared genetic architecture and causality between ASD and FSS. Specifically, we first estimated genetic correlations and then conducted a multi-trait analysis of GWAS (MTAG) to detect potential novel genetic variants for single traits. Afterwards, polygenic risk scores (PRS) of ASD were derived from GWAS and MTAG to examine the associations with phenotypes in the large Dutch Lifelines cohort. Finally, we performed Mendelian randomization (MR) to evaluate the causality. We observed positive genetic correlations between ASD and FSS (IBS: $r_g = 0.27$, adjusted $p = 2.04 \times 10^{-7}$; musculoskeletal pain: $r_g = 0.13$, adjusted $p = 1.10 \times 10^{-3}$; fatigue: $r_g = 0.33$, adjusted $p = 5.21 \times 10^{-9}$). Leveraging these genetic correlations, we identified four novel genome-wide significant independent loci for ASD by conducting MTAG, including *NEDD4L*, *MFHAS1*, *RP11-10A14.4*, and *C8orf74*. PRS of ASD derived from both GWAS and MTAG were associated with ASD and FSS symptoms in Lifelines, and MTAG-derived PRS showed a bigger effect size, larger explained variance, and smaller *p*-values. We did not observe significant causality using MR. Our study provided new evidence of shared genetic architecture between ASD and FSS, specifically with IBS, musculoskeletal pain, and fatigue. The findings confirm the genetic associations between ASD and FSS, and advance our understanding of the mechanisms underlying co-occurrence.

**Keywords** Autism spectrum disorder, Irritable bowel syndrome, Musculoskeletal pain, Fatigue, Genetic correlation, Polygenic risk score

## Genetics of SSRI Antidepressant Use in UK Biobank and the US Million Veteran Program

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Department of Veterans Affairs, San Diego, USA. [7]Psychiatry, University of California San Diego, New Haven, USA

**Abstract**
Antidepressants are among the most-prescribed drugs worldwide, especially selective serotonin reuptake inhibitors (SSRIs). To increase our understanding of biological relationships between SSRI use and psychiatric traits, we conducted genomewide association study (GWAS) meta-analysis using two large datasets: the US Million Veteran Program (MVP) and the UK Biobank (UKBB) across 22 autosomes and the X chromosome in 777,952 individuals of European ancestry (191,800 SSRI users, 586,152 controls) and 112,526 individuals of African ancestry (53,499 SSRI users, 59,027 controls). We identified 40 genomewide-significant (GWS) loci, 38 autosomal and two on the X chromosome. Analysis investigating the genetic overlap of SSRI use from the larger MVP-only portion of the summary statistics with UKBB-only traits using linkage disequilibrium score regression (LDSC) detected strong correlations with specific SSRIs in UKBB (fluoxetine rg = 0.82 and citalopram rg = 0.89) as well as with headaches (rg = 0.80), major depressive disorder (rg = 0.77), and spondylosis (rg = 0.84). A comparative genetic-correlation analysis found statistically significant differences between major depressive disorder from a prior GWAS and our current SSRI use data, most notably for educational attainment (rG with SSRI = − 0.38, rG with MDD = − 0.26) and cognitive performance (rG with SSRI = − 0.31, rG with MDD = − 0.15). SSRI use presents a potential proxy trait for major depressive disorder, genetics of SSRI use accordingly also presents significant differences that may prove useful to disentangle MDD from other traits that use similar pharmacological treatment and to understand medication treatment response and adverse effects.

**Keywords** GWAS, SSRI, MDD, Genetics, X chromosome

**Consortia**
VA Million Veteran Program.

**Grants**
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## Extracting Stability of Generalized Anxiety Increases Heritability Estimates: A Longitudinal Twin Study in Young Adults

**Julia S. Funk[1,2], Aliyah Kassam[1], Celestine Lockhart[1], Elisavet Palaiologou[1], Megan Skelton[1], Genevieve Morneau-Vaillancourt[1,3], Thalia C. Eley[1]**

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**Abstract**
Rates of generalised anxiety symptoms are rising dramatically amongst young adults, but twin analyses of anxiety during this life stage are scarce. Longitudinal twin studies of adolescent and adult anxiety typically find that genetic influences largely affect stability, with increased heritability for stable anxiety. We tested whether this also applies for the current generation of young adults. We used Twins Early Development study (TEDS) data to examine genetic and environmental contributions to continuity and change in generalised anxiety symptoms in young adulthood. We then calculated the heritability of a latent factor reflecting anxiety stability over this period. The sample comprised 6429 monozygotic and dizygotic twin pairs (10,836 individuals). Generalised anxiety was assessed six times between the ages of 23 and 28 years. A Cholesky Decomposition indicated that genetic factors largely accounted for continuity, with environmental factors accounting for both stability and change. The heritability of stable generalised anxiety (60%) was substantially higher than that at any single time point (39–46%). As such, extracting stability may provide optimal power to identify specific genetic influences on generalised anxiety. We are using this finding to improve power in a GWAS of dimensional anxiety, from which we will share preliminary findings if available. Our finding that some environmental influences had a persistent effect on generalised anxiety in this age-range may reflect the greater precarity of the environment for young adults today. We will end by describing specific factors identified in recent focus groups that young adults consider relevant to anxiety in their generation.

**Keywords** Anxiety, Stability, Contemporary adulthood

**Grants**
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## Validation of the Short-Form PTSD Checklist for Use in Large Genetically Informative Cohorts

**Jacob Knyspel[1], Jonathan Coleman[1]**

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**Abstract**
Measuring PTSD validly is essential for ensuring high quality research on the genetics of the condition. In several genetically informative cohort studies, such as the Genetic Links to Anxiety and Depression Study (GLAD) and Twins Early Development Study (TEDS), PTSD has been measured using a short-form version of the civilian PTSD checklist (PCL-c-SF). Although this measure has been validated in primary care contexts, it has so far not been validated for the purpose of broader cohort-based research. Here we discuss a current validation study in which we assessed the screening concordance between the PCL-c-SF and its long-form equivalent (PCL-5) in a subsample of GLAD. We found strong concordance between the two measures, with a PCL-c-SF cutoff of 18 providing the optimal balance of sensitivity and specificity. Using summary statistics from the most recent genome-wide association study of PTSD by the Psychiatric Genomics Consortium, we calculated polygenic risk scores which explained just over 1% of the variance in PTSD case/control status based on this cutoff. We conclude that the PCL-c-SF has suitable validity for use in large genetically informative cohorts. Limitations and implications for future research are discussed.

**Keywords** PTSD, Genomics, Psychometrics, Polygenic scores

**Consortia**
Thalia C. Eley, Patrizia Pezzoli, Brittany Mitchell, Anna Monistrol Mula, Patrizia Pezzoli, Celestine Lockhart.

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## The Development of Reappraisal and Suppression Strategies in Adolescents: A Twin Study

**Katherine N. Alexander[1], Amanda M. Ramos[1]**

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**Abstract**
Although work has examined how social and developmental processes influence adolescent emotion regulation, less work has used genetically informed designs to investigate its potential heritable and environmental influences. Two specific emotion regulation strategies used by adolescents are emotional suppression, which includes inhibiting expression of specific emotions, and reappraisal, which includes trying to shift cognitions about events to influence emotional states. The aim of the current research project was to examine how genetic and environmental influences on reappraisal and suppression strategies vary across time in an adolescent twin sample. Data were obtained from The Adolescent Brain Cognitive Development SM Study (ABCD Study®), including 1916 twins at wave 3 and 931 twins at wave 4. Adolescents reported on their emotion regulation strategies using a shortened version of the Emotion Regulation Questionnaire for Children and Adolescents (ERQ-CA) at ages 12.5 and 13.5 years. The results from a multivariate Cholesky Decomposition indicate genetic and non-shared environmental factors that explain common variance in the emotion regulation strategies across time as well as unique genetic and non-shared environmental influences that explain variance in emotion regulation at 13.5 years old. These results support the importance of heritable, environmental (e.g., peers, differential parenting) and unique developmental pathways in shaping specific emotion regulation strategies, which may have important implications for adolescent mental health and associated interventions.

**Keywords** Reappraisal, Suppression, Adolescence, Emotion regulation

## From Genetic Disposition to Observed School Achievement: Investigating the Mediating Role of Noncognitive Skills Across Development

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**Abstract**
While it is well established that academic achievement is heritable, the environmental, developmental, and psychological processes that translate genetic potential into observable variation in academic achievement remain less understood. This study investigates the mediating role of noncognitive skills—such as personality, motivation, and self-regulation—in the relationship between genetic dispositions and academic achievement. Utilizing data from 5016 children in the Twins Early Development Study, assessed at ages 7, 9, 12, and 16. We found that noncognitive skills significantly mediate the effects of both educational attainment, cognitive and noncognitive PGS on academic achievement. Longitudinal analyses confirmed the enduring influence of noncognitive skills, while within-family analyses highlighted their impact independent of shared genetic, environmental, and demographic effects. The mediation effects persisted after controlling for general cognitive ability and socioeconomic status. These findings underscore the importance of noncognitive skills and highlight the potential for targeted interventions to enhance academic outcomes.

**Keywords** Gene–environment correlation, Noncognitive skills, Socio-emotional skills, Academic achievement, Polygenic score prediction, Mediation

**Grants**
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## A Contextual Genomic Perspective on Physical Activity and Its Relationship to Health and Illness

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**Abstract**

Physical activity (PA) has pervasive health benefits and is genetically influenced. There is also a wide range of interindividual differences in requirements for PA in daily life. Some jobs require much more physical exertion than others; some individuals must exert more effort in their households than others; some individuals experience most of their PA while at leisure. PA related to work and home-time tend to differ phenomenologically from exercise taken at leisure, occurring more in concentrated bursts in the first two cases, and with the potential to be more sustained in the latter case. Using data from the Million Veteran Program (MVP), we conducted genetic analyses of leisure, work, and home-time PA. For leisure, for individuals of European (EUR) ancestry, n = 189,812 and SNP-based heritability ($h^2$) = 0.083 ± 0.005; for African ancestry, n = 27,044; $h^2$ = 0.034 ± 0.017; and for Latin American ancestry, n = 10,263; $h^2$ = 0.083 ± 0.036. EUR and cross-ancestry meta-analyses with UK Biobank identified 67 and 70 lead variants, respectively. Leisure-time PA was genetically distinct from PA at home or work, with the latter two showing weaker associations with health benefit on pathologies and lifespan. Mendelian randomization analyses showed protective effects of leisure-time PA on cardiovascular and respiratory system diseases, metabolic traits, aging, and other traits. There was also a protective role of leisure-time PA against COVID-19 hospitalization. Further analyses helped to delineate the relationships between PA and a range of psychiatric traits. These findings provide new insights into the biology of PA, showing the causal health benefits of leisure-time PA.

**Keywords** Genomics, GWAS, Physical activity

**Consortia**
VA Million Veteran Program.

**Grants**
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## Epigenome-Wide Analyses of Pubertal Development in Adolescence: Connections with Biological Aging

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**Abstract**
Puberty is highly heritable and responsive to environmental exposures and signals. Earlier and faster puberty is related to early life adversity and worse health across the lifecourse. Novel DNA-methylation methods provide the opportunity to increase biological insights about the etiology of differences in puberty. We leverage salivary DNA-methylation samples from two diverse population cohorts of adolescents to (1) identify genomic regions that are associated with pubertal development and (2) generate a novel DNA-methylation biomarker of pubertal development. We conduct epigenome-wide association studies (EWAS) of precocious puberty, puberty, and pace of puberty in the Texas Twin Project (TTP; n ∼ 1200) and the Future of Families Child Wellbeing Study (FFCWS; n ∼ 2000). We employ these results to create a novel elastic-net algorithm of puberty in TTP and test its performance in other developmental cohorts. We explore the relationship between puberty-related differences in DNA-methylation and insights from the study of biological aging in midlife and older adulthood using targeted enrichment analysis and by relating our puberty methylation profile score (MPS) to accelerated biological aging and pace of biological aging MPSs. DNA-methylation measurements of puberty have the potential to improve health forecasting and accelerate research on development and aging.

**Keywords** DNA-methylation, Epigenetics, Puberty, Development, Aging

**Grants**
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## Parent–Offspring Stress to Externalizing Pathways: A Test of Genetic Mediation Through Negative Emotionality

**Veronica Oro[1], Jenae M. Neiderhiser[2], Misaki N. Natsuaki[3], Gordon T. Harold[4], Jody M. Ganiban[5], Daniel S. Shaw[6], Leslie D. Leve[1,7]**

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**Abstract**
Negative emotionality has been found to underlie continuity in stress and psychopathology, but less is known about whether it drives intergenerational transmission. Using data from a parent–offspring adoption study, we sought to elucidate genetic mediation of parent–offspring stress to externalizing pathways through negative emotionality and whether parent emotional and verbal responsivity mitigated genetic risk. Specifically, this study examined whether the association between birth mother life stress and adoptee externalizing at age 11 was mediated by birth mother negative emotionality ($n$ = 561 children and their birth and adoptive parents) and whether adoptive parent responsivity moderated paths from birth mother life stress and negative emotionality to adoptee externalizing. Birth mother life stress was positively associated with her negative emotionality (β = 0.15, $p$ = 0.010), and birth mother negative emotionality was positively associated with adoptee externalizing (β = 2.38, $p$ = 0.001). The total effect of birth mother life stress was

significant ($\beta = 1.77$, $p = 0.035$). After adjusting for birth mother negative emotionality, the direct effect of birth mother life stress was nonsignificant ($\beta = 1.42$, $p = 0.096$), indicating full mediation. Finally, the mediated effect was significant ($\beta = 0.36$, $p = 0.040$). Adoptive parent responsivity was not found to moderate either path from birth mother life stress or negative emotionality, though supplementary analyses indicated that it did attenuate the association between birth mother life stress and adoptee internalizing. Findings suggest that parent–offspring stress to externalizing pathways are genetically mediated and that negative emotionality is a plausible mechanism.

**Keywords** Stress, Trauma, Intergenerational transmission, Negative emotionality, Externalizing

**Grants**

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## A Qualitative Analysis on Perceptions of Sexual Orientation and Gender Identity Genetics Research

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**Abstract**

Historically, genetics research has been a highly debated topic, as there are concerns with gene editing, testing, and various therapies. These concerns are especially prevalent for LGBTQ+ groups. The present study assesses the perceptions of sexual orientation and gender identities (SOGI) genetic research via qualitative analysis in both LGBTQ+ and non-LGBTQ+ populations. Qualitative data were collected via a survey posted on Reddit, an online forum for a variety of communities to connect, on a thread specifically for collecting survey data. Participants answered seven free-response questions regarding their perceptions of SOGI genetic research. Data was analyzed using thematic analysis procedures outlined by Braun and Clarke (2022). We identified several themes that allow for a better understanding of the general populations' perceptions on SOGI genetic research. This research allows us to better understand what both LGBTQ+ and non-LGBTQ+ individuals see as concerns regarding genetic research. We anticipate our research findings can be used to work towards educating the general public on genetics. Additionally, our results may allow us to better guide future research practices in an attempt to eliminate ethical concerns.

**Keywords** Bioethics, Sexual orientation, Gender minority, Public perception, Qualitative

## Predicting Political Beliefs with Polygenic Scores for Cognitive Performance and Educational Attainment

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**Abstract**

Intelligence is correlated with a range of left-wing and liberal political beliefs. This may suggest intelligence directly alters our political views. Alternatively, the association may be confounded or mediated by socioeconomic and environmental factors. We studied the effect of intelligence within a sample of over 300 biological and adoptive families, using both measured IQ and polygenic scores for cognitive performance and educational attainment. We found both IQ and polygenic scores significantly predicted all six of our political scales. Polygenic scores predicted social liberalism and lower authoritarianism, within-families. Intelligence was able to significantly predict social liberalism and lower authoritarianism, within families, even after controlling for socioeconomic variables. Our findings may provide the strongest causal inference to date of intelligence directly affecting political beliefs.

**Keywords** Causal inference, Genetics, Intelligence, Political belief, Polygenic score

**Grants**

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## The Pathway to Effortful Control in Adolescence: The Role of Parental Personality, Family Environment, and Genetic Factors

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**Abstract**

Effortful control (EC) is the ability to intentionally inhibit a dominant response in service of goals. During adolescence, poor EC is associated with mental health problems. We used a longitudinal adoption design to investigate the contributions of heritable and theoretically relevant family factors to EC from childhood to adolescence. Participants included 495 families from the Early Growth and Development Study. Children's heritable predisposition for EC was measured by Birth Mothers' (BM) self-reported EC at 5 months postpartum. Family factors relevant to EC were assessed by primary and secondary Adoptive Parents' (AP1, AP2) reports when children were 4.5 years old, including: perceptions of home order/structure and the AP's own EC. Child EC was assessed via AP1s' and AP2s' reports when children were 4.5 years, and by adolescent self-report at 11 years. AP1s' and AP2s' EC were positively associated with home chaos/stress at 4.5 years. BMs', AP1s', and AP2s' EC were not related to child EC at 4.5 or 11 years. The associations between AP1s' EC and child EC were mediated by home chaos/stress at 4.5 years: higher AP1 EC was associated with higher home order/structure, and higher home order/structure predicted higher child EC at 4.5 and 11 years. While these findings do not rule out genetic

influences on child/adolescent EC, they affirm the important role of the home environment in the development of EC.

**Keywords** Effortful control, Home environment, Heritability, Parental personality, Adolescence

**Grants**

This work was supported by R01 HD042608 from the Eunice Kennedy Shriver National Institute of Child Health and Human Development (PI Years 1–5: David Reiss; PI Years 6–10: Leslie D. Leve), R01 DA020585 from National Institute on Drug Abuse, NIH, U.S. PHS (PI: Jenae M. Neiderhiser), R56 HD042608 from the Eunice Kennedy Shriver National Institute of Child Health and Human Development, NIH, U.S. PHS (PI: Leslie Leve, PhD), and Grant UH3 OD023389 from Environmental influences on Child Health Outcomes (ECHO) Program, The Office of the Director, NIH, U.S., PHS (MPIs: Leslie D. Leve, Jenae M. Neiderhiser, and Jody M. Ganiban).

## Externalizing 2.0: Boosting Discovery of Genetic Variants Associated with Externalizing Behaviors with Larger and More Diverse Samples

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**Abstract**

The Externalizing Consortium previously modelled the shared genetic etiology of seven externalizing behaviors and disorders, including substance use and risk-taking, in 1.5M individuals. While over 500 independent genetic loci were associated with a general externalizing factor (EXT), those results were specific to European ancestry populations. Here, we extend such analyses to additional ancestral groups to more equitably advance genomic discovery for externalizing traits. We conducted genomic structural equation modeling and multivariate GWAS analyses in participants with European-like genomes (N ~ 3,176,000) and African-like genomes (N ~ 245,700) while accounting for differences in public and restricted (e.g., 23andMe, Inc.) data availability. We first examined the robustness and consistency of the EXT factor model with data available only for a subset of indicators, as is the case for African samples. We then compared the fit and parameter estimates of a harmonized EXT model across ancestries. If the factor models are comparable across ancestries, we will conduct a meta-analysis of the ancestry-specific externalizing GWASs using LOG-TRAM, a method that leverages local genetic architecture for trans-ancestry association mapping. Finally, we will present preliminary validation results, including the quasi-replication of identified genomic loci and tests of polygenic prediction.

**Keywords** Externalizing, African ancestries, Genome-wide association study, Polygenic index

**Grants**

The Externalizing Consortium is supported by a Grant from the National Institute on Drug Abuse (R01DA050721). Additional funding for investigator effort has been provided by N.S.C.-K. by T33KT6694.

## Measuring Alcohol Behaviors from Adolescence to Adulthood: An Analysis of Twin Data from the Minnesota Center for Twin and Family Research

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**Abstract**

Debates persist about the relationship between alcohol use and dependence. Namely, the DSM-V does not include specific criteria for the frequency or quantity of use in diagnosing Alcohol Use Disorder (AUD). This study investigates the latent structure and genetic and environmental contributions to alcohol use and dependence from adolescence to middle adulthood using factor and twin modeling. Participants from the Minnesota Center for Twin and Family Research (MZ twin pairs = 1205, DZ twin pairs = 676; 51.9% female) were assessed at 7 timepoints from ages 14 to 45. Alcohol use frequency and quantity were self-reported, while AUD symptoms were measured based on DSM-III-R and DSM-IV criteria. An ACE decomposed common pathway model was used to model a latent factor at each wave of assessment. Mean alcohol use and dependence peaked between late adolescence and early adulthood and declined thereafter. Heritability accounted for 50% of variance in the latent alcohol behavior factor at age 14, increasing to 73% by age 45. Within-age factor loadings could not be constrained to be equal during adolescence (change in AIC = + 1145.5), suggesting that measures of alcohol use and dependence are distinct. However, at older ages, measures of use and dependence loaded similarly onto the latent factor (change in AIC = − 3.9). Therefore, at older ages, measures of alcohol use and dependence appear to similarly assess the alcohol behavior construct, indicating that measures of use may be used as a proxy for dependence. Heritability of the latent alcohol behavior factor increased with age.

**Keywords** Alcoholism, Externalizing, Development, Behavior genetics, Lifespan

**Grants**

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## Examining Links Between Temperament and Emotion Expression in Middle Childhood: A Twin Analysis

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**Abstract**
Learning to recognize and interpret others' emotions is an important developmental task of child development and is essential for successful social interactions. Little is known about children's ability to accurately express emotions (EE), although some evidence suggests a relationship between EE and victimization. This study of 344 twins aged 6–11 years old (136 MZ, 208 DZ) used untrained coders to rate children's photographed expressions of happy, sad, fearful, or angry facial emotions to create scores for EE biases (being rated as expressing a particular emotion but not trying to) and errors (attempting to express a particular emotion but not being judged to show it). Genetic analyses showed genetic effects for angry errors (A = 0.53, E = 0.84), fearful errors (D = 0.47, E = 0.88), happy biases (A = 0.56, E = 0.83), and sad biases (D = 0.47, E = 0.88). In turn, these EE skills were related to components of parent-rated temperamental surgency/extraversion, specifically activity and intensity. Multilevel modeling analyses showed that the twin rated higher on intensity was less likely to show angry EE errors ($beta = -0.45$, $p = 0.027$), fearful errors ($beta = -0.54$, $p = 0.019$), sad biases ($beta = -0.89$, $p < 0.001$), or happy biases ($beta = -0.40$, $p = 0.011$), with mostly similar results shown for children's activity. These results suggest possible causal relations from temperament to EE skills, but there was no evidence for shared genes between temperament and EE skills. As EE skills may be important for children's social development, understanding how they relate to temperament may be valuable for individual training and interventions.

**Keywords** Temperament, Children, Emotion expression, Development, Surgency/extraversion

**Grants**
SIU School of Medicine RSG Award; SIUC Special Research Project Award (both to 1st author); Psi Chi Graduate Research Grant (to 2nd author).

## Increasing Exercise Intensity: A Study of Behavioral Substitution on Health Biomarkers

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**Abstract**
Engaging in exercise is a well-established counter to a sedentary lifestyle, although responsiveness to compensatory exercise remains unclear including genetic and environmental contributions. Our study leverages a natural experiment using twins to expand understanding between exercise and health. We utilize behavioral and biomarker data in an established adulthood sample from the Colorado Adoption/Twin Study of Lifespan behavioral development and cognitive aging (CATSLife1; N = 1327, *Mage* = 33.16 years, *SD* = 4.93). We applied an expanded co-twin control approach examining exercise intensity and several health metrics, further testing discordant monozygotic (MZ) pairs to evaluate increases in moderate or vigorous physical activity. Phenotypic results for cardiovascular and metabolic biomarkers indicate that vigorous exercise outweighs moderate exercise contributions beyond the 2:1 ratio expected from US guidelines which is further elaborated in the discordant MZ pair analyses. To illustrate, co-twin control models suggested environmental contributions between exercise and cardiac burden, indexed by the resting heart rate (RHR). Moreover, among discordant MZ pairs those performing more vigorous exercise had additional improvements in health biomarkers even at levels of 10 min of additional vigorous exercise per day. Our results suggest nuanced associations between exercise, intensity of engagement, and offsetting sedentary lifestyle which has implications for premature aging health risks and public health recommendations.

**Keywords** Exercise, Cardiovascular health, Metabolic health

**Grants**
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## Exploring the Direction of Effects Between Parenting Behaviours and Offspring Depressive Symptoms: An Intergenerational MR-DoC Study

**Meredith Han[1,2], Joanna Bright[1], Christopher Rayner[1], Olakunle Oginni[3], Eivind Ystrom[4,5], Tom McAdams[1,4]**

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**Abstract**
Parents play a role in offspring emotional development. Warm and sensitive parenting is associated with greater offspring socio-emotional competence while children experiencing corporal punishment and inconsistent discipline show increased internalising behaviours (Cooke et al. 2022; Yap and Jorm 2015).

However, this association may also capture effects of shared genetics and environment between mother and offspring (Cheesman et al. 2020). We extend the Mendelian Randomization-Direction of Causation (MR-DOC) model for use on intergenerational pedigree data (multiple children of twins, siblings, and cousins) (Oginni and von Stumm 2022). The MR-DOC model combines Mendelian Randomization (MR), using genetic variants as instrumental variables, with the direction of causation twin model, providing insights into likely directions of effects. This method controls for the effects of shared genetic and shared environment as well as relaxes the MR

assumptions by modelling unmediated pleiotropy, providing an unbiased estimate of the effects of parenting on child depressive symptoms.

Data were from the Norwegian Mother, Father, and Child Cohort Study (MoBa, n = 15,000 mother–child dyads). Mothers completed the Alabama Parenting Questionnaire (APQ), consisting of three subscales: positive parenting, inconsistent discipline, poor supervision, and the Short Mood and Feelings Questionnaire (SMFQ) as a measure of offspring depressive symptoms at 8.5 years. We calculate PGS using SNPs associated with maternal parenting behaviours using GWAS from general psychopathology (i.e., depression/education attainment). DOC models are constructed using the degree of genetic relatedness between family members. Using MR-DOC, we will explore whether the three parenting behaviours show potential causal effects on child depressive symptoms or vice versa.

**Keywords** Parenting, Depression, Cohort studies, Mendelian randomization analysis

## Detaching from the Nine-to-Five: How Retirement Has an Effect on the Genetic Influences on Chronotype

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**Abstract**

Our sleep–wake rhythm and chronotype are primarily determined by how our society is designed, i.e. our work lives. However, after retirement, these societal influences might decrease, making room for genetic influences to increase. In this study, we looked at wake-up times and chronotypes during employment and during retirement. Given that one can freely choose their wake-up time during retirement, we hypothesize that the variance of chronotype explained by genetics is larger in retirees compared to employees.

We used data from 20.189 participants from the Netherlands Twin Register. We first tested four assumptions about the association between wake-up times, chronotypes and employment status. The final hypothesis was tested by using polygenic scores for Chronotype to predict the variance in Chronotype in employees and retirees.

Aging was associated with earlier chronotypes ($p < 0.000$). For both early and late risers, the wake-up time depended on whether it's a workday or free day ($b = 0.76$, $p < 0.000$, $b = 2.32$, $p < 0.000$, respectively). Variance in chronotype was not predicted by employment status ($p > 0.05$). However, wake-up time was depended on employment status: the employees had an earlier mean wake-up time (06:50 h) than the retirees (07:30 h, $p < 0.000$). The results of the polygenic score predictions of Chronotype in the employees and retirees will follow.

This study enhances our understanding of how both our society and genetics influence our chronotype, shedding light on the gene–environment interplay in the circadian rhythm.

**Keywords** Chronobiology, Genetics, Polygenic scores, Employment status, Gene–environment interplay

**Grants**

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## Heritability of Plasma Biomarkers for Alzheimer's Disease: A Nuclear Twin Family Design

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**Abstract**

**Background**

Plasma biomarkers amyloid-β-42/40 ratio (Aβ-42/40), neurofilament-light (NfL), and glial fibrillary acidic protein (GFAP) may be a non-invasive tool for diagnosing Alzheimer's Disease (AD). We aimed to determine the genetic and environmental contributions to individual differences in the levels of these plasma biomarkers.

**Methods**

Twins, their parents and siblings were included (n = 6516). Biomarker levels were measured on the Simoa and adjusted for age. A twin-only 5-zygosities model was used to examine sex differences in genetic and environmental effects. A nuclear twin family design (NTFD) was used to estimate the proportional contributions of genetic, shared and unshared environment. Shared environmental factors consisted of environment shared by siblings, by twins only, or by parents and offspring.

**Results**

No sex differences in contributions to biomarker levels were found. The NTFD showed that 16% of the variance in Aβ-42/40 was explained by genetic effects and 84% by unshared environment. For NfL, 30% of the variance was explained by genetic effects, 27% by sibling-shared environment, and 43% by unshared environment. For GFAP, 45% of the variance was explained by genetic effects, 17% by sibling-shared environment, and 38% by unshared environment.

**Conclusion**

The low heritability of Aβ-42/40 is a mismatch with the high heritability of AD (60–80%), suggesting Aβ-42/40 ratio might be susceptible to environmental confounding. The sibling-shared environment for NfL and GFAP could be a reflection of generational effects. These findings are not definitive proof of confounding, but are a first step in identifying possible confounders.

**Keywords** Nuclear twin family design, Alzheimer's disease, Heritability, Blood biomarkers

**Grants**

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## Longitudinal Genetic Analysis of Aggression Across the Life Span

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**Abstract**
Aggressive behavior on average increases in adolescence and decreases in early adulthood, and hardly changes in level from thereon (e.g., Eron and Huesmann 1990). Longitudinal twin studies of aggression have mostly focused on childhood and adolescence. These studies demonstrated that genetic factors contribute to stability of aggression across childhood and adolescence (Tuvblad and Baker 2011). In this study we analyze genetic and environmental contributions to stability and changes in aggressive behavior across the life span. We analyze aggressive behavior of over 30.000 twin pairs participating in the Netherlands Twin Register, from age 3 to 90+. Aggression was measured by age-appropriate ASEBA aggressive behavior subscales. We fitted a genetic simplex model to the data to determine the contributions of genetic and environmental sources of influence to stability and change in aggression levels across the lifespan. We interpret the results in the light of differences in questionnaire versions, raters, and measurement invariance across questionnaire version and age.

**Keywords** Aggression, Genetic simplex model, Genetic covariance structure modeling, Longitudinal modeling, Development

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## Can Household Chaos Explain the Discrepancy Between Within- and Between-Family Polygenic Score Predictions of Children's School Achievement?

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**Abstract**
Polygenic score (PGS) predictions for educational attainment account for about 15% of variation in children's school achievement, yet this prediction effect is reduced within families because population-based estimates are biased by family-level confounders. The difference between within- and between-family predictions has been previously attributed to socioeconomic status (SES), a distal proxy measure that does not allow identifying specific environmental conditions that affect children's educational achievement. By contrast, the household chaos, which refers to the degree of structure and predictability of the home environment, is a proximal, likely modifiable environmental factor, whose role in within- and between-family PGS predictions for school achievement we will test here.

We conducted multi-level models in up to $N = 1889$ DZ twins, drawn from TEDS (UK; 9, 12, and 16 years) and TwinLife (Germany; 11 years), to discern within- and between-family predictions of school achievement from PGS and household chaos. Parent- and child-reported household chaos predicted school achievement phenotypically across development, yet these effects became non-significant after accounting for family-shared genetic and environmental confounders. Household chaos did not explain the within-family attenuation of the PGS predictions relative to the between-family level. The discrepancy between within- and between-family level PGS predictions was instead accounted for by SES, confirming prior research. We concluded that household chaos is likely not a causal factor of children's differences in school achievement, because their associations are mainly driven by passive gene–environment correlation.

**Keywords** Educational achievement, Within-family, Twins, Household chaos, Educational inequality

**Grants**
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## Incorporating Genetic and Clinical Predictors for Antidepressant Side Effects in the Genetic Link of Anxiety and Depression Study

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**Abstract**
Antidepressants are the most common treatment for moderate or severe depression. Side effects are crucial indicators for antidepressants, but their expression varies widely among individuals. In this study, we leveraged genetic and phenotypic data from self-reported questionnaires in the Genetic Link of Anxiety and Depression (GLAD) study to predict side effects and discontinuation across three antidepressant classes [SSRI, SNRI, tricyclic antidepressants (TCA)] at the first and the last (most recent) year of prescription. About 260 predictors spanning genetic, clinical, comorbidity, demographic, and antidepressant information were included. XGBoost, random forest, cubist, elastic net, and support vector machine (with RBF and polynomial kernel) were trained and their performance was compared.

The final dataset comprised 5358 individuals, with 4354 in the first and 3414 in the last year of prescription. The average prevalence of side effects and discontinuation was 74.1% and 28.7%, respectively. In the initial year, the best AUROC for predicting SSRI discontinuation and side effects were 0.66 and 0.60. In the last year of SSRI prescription, the highest AUROC reached 0.72 for discontinuation and 0.87 for side effects. Models for predicting discontinuation and side effects of SNRI and TCA showed comparable performance. The history of side effects and discontinuation of antidepressant use were the most influential predictors of the outcomes in the last year. Our findings demonstrate the feasibility of predicting antidepressant side effects using self-reported questionnaires, particularly for the last prescription. These results contribute valuable insights for the development of clinical decisions aimed at optimizing treatment selection with enhanced tolerability.

**Keywords** Antidepressants, Side effects, Machine learning

**Grants**

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## Investigating Generality and Specificity in the Association Between Genetic Risk for Psychopathology and Cognitive Development

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**Abstract**

Despite a well-established association between cognition and psychopathology, cognitive impairments are a critical dimension of psychopathology that remains under-investigated if compared to affective and emotional symptoms. Cognitive deficits often cut across traditional diagnostic boundaries in the same way that core affective and emotional symptoms are shared across multiple conditions. This overlap between cognitive and affective symptoms across psychiatric disorders makes investigating specific relationships between cognition and psychopathology, as well as their origins, difficult. Specific aspects of mental illness will likely impact cognitive skills differently, and vice versa, specific cognitive impairments will lead to different psychopathological profiles. Accounting for transdiagnostic effects provides a way to investigate specificity and obtain a clearer picture of the development of the associations between cognition and psychopathology.

This study investigates generality and specificity in the association between genetic disposition for psychopathology and cognition over development. Specifically, we investigate whether, after removing transdiagnostic genetic effects, polygenic scores (PGS) for psychopathology reveal differential associations with cognitive profiles (e.g., general and specific cognitive abilities) over development, from age 4 to age 21, potentially pinpointing critical developmental times. We constructed PGS in the Twin Early Development Study sample based on a recent genome-wide association study that isolated transdiagnostic effects from 11 major psychiatric disorders.

The results suggest that the transdiagnostic factor does not drive the association between psychiatric disorders and cognition. Instead, these associations are specific to each disorder and different cognitive abilities. This highlights the importance of considering specificity when studying cognition and psychopathology.

**Keywords** Psychopathology, Cognitive development, General cognitive ability, Specific cognitive abilities, Behavioural genetics, Polygenic scores

**Grants**

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We gratefully acknowledge the ongoing contribution of the participants in the Twins Early Development Study (TEDS) and their families. TEDS is supported by the UK Medical Research Council (MR/V012878/1 and previously MR/M021475/1).

## Revisiting the Fractionable Autistic Triad Hypothesis: How Do the Same and Different Teacher Versus Parent Perspectives Add to Our Understanding?

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**Abstract**

The fractionable autistic triad hypothesis proposes that the social, communication, and rigid/repetitive behaviour/interests (RRBI) components of autism traits are aetiologically largely independent. The hypothesis is supported by parent-report data. As a complement to the parental perspective, teacher-report data is used in the present study to revisit the fractionated triad hypothesis.

Participants (N = up to ~ 7800 twin pairs) were from the Twins Early Development Study (TEDS). Teachers and parents assessed the twins' social, communication, and nonsocial/RRBI autism traits at ages 7, 9, and 12 years. DSM-IV items were used at age 7 and the Childhood Autism Spectrum Test (CAST) was used at ages 9 and 12. Teachers were divided into two groups: those who evaluated both twins in a pair, and those who evaluated only one twin. For each type of rater (i.e., same teachers, different teachers, and parents), phenotypic correlations were first calculated; genetic and environmental contributions to the phenotypic correlations and the genetic correlations were then calculated using twin analyses. In addition, we calculated the correlations separately for males and females and conducted bivariate DeFries–Fulker extremes analyses for the extreme autism traits.

The results consistently supported the fractionation of the autism components in that phenotypic and genetic correlations between social, communication and nonsocial/RRBI traits were modest for teacher as well as parent ratings, for the high extremes and for the entire distribution, at all three ages and for both sexes. These findings indicate distinct genetic and environmental causes underlying each autism component.

**Keywords** Autism traits, Twins, Genetics, Teacher, Rater effects

**Grants**

We gratefully acknowledge the ongoing contribution of the families in the Twins Early Development Study (TEDS). TEDS is supported by the UK Medical Research Council (MR/V012878/1 and previously MR/M021475/1). Y.L. is supported by a KCL-CSC Joint Scholarship awarded by King's College London and China Scholarship Council. F.H. is part-funded by the NIHR Maudsley Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London. K.R. is supported by a Sir Henry Wellcome Postdoctoral Fellowship (213514/Z/18/Z). M.M. is supported by a Starting Grant from the School of Biological and Behavioural Sciences, Queen Mary University of London.

## ♫ Notes from a Genome: What Beethoven's DNA Tells Us About Genetics

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**Abstract**

Rapid advances over the last decade in DNA sequencing and statistical genetics enable us to investigate the genomic makeup of individuals throughout history. In a recent study, Ludwig van Beethoven's hair strands were used for genome sequencing and the authors explored genetic predispositions for some of his documented medical issues. They did not explore the trait that made him an iconic figure in Western culture, likely because there has not yet been a sufficiently informative genome-wide-association study (GWAS) of composer or musical talent. However, a GWAS involving 606,825 individuals with European ancestry, associated 69 genetic loci with variation in self-reported beat synchronization ability assessed with the question "Can you clap in time to the beat? Yes/No" and a follow-up family study showed that the polygenic scores (PGIs) tapped into broader genetic musical propensity. Therefore, we calculated this PGI for Beethoven and compared it against two population-based datasets of modern-day individuals for whom we have musical achievement data, namely 5648 individuals from the population-based Swedish Twin Registry's STAGE cohort and 6150 individuals from the population-based repository Vanderbilt's BioVU cohort. As outlined in our preregistration, we approached this analysis without any hypotheses regarding Beethoven's PGI placement, aiming to underscore the complexities involved in making individual-level genetic predictions for an individual that lived over 200 years ago.

**Keywords** Musicality, Genomics, Polygenic index

**Grants**

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## Age-Associated Genetic and Environmental Contributions to Epigenetic Aging Across Adolescence and Emerging Adulthood

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**Abstract**

Epigenetic aging estimators commonly track chronological and biological aging, quantifying its accumulation or its speed. These estimates reflect a combination of inherent biological programming and the impact of environmental factors, which are suggested to vary at different stages of life. The transition from adolescence to adulthood is an important period in this regard, marked by an increasing and, then, stabilizing epigenetic aging variance.

This study delves into understanding the age-associated genetic and environmental contributions to differences both between and within individuals across these developmental stages. For this purpose, we analyzed four differently developed epigenetic aging estimators, which were collected from 976 twins aged 8 to 31 years ($M$ = 16.0, $SD$ = 6.05) measured twice, 2 years apart.

Approximately half (33–76%) of the between-individual differences in epigenetic aging and most (70–99%) of the variance in within-individual changes across the 2 years were attributable to unique environmental factors. The contribution of these factors to the

variance of epigenetic estimators trained on chronological age notably increased with time. Genetic contributions were more pronounced in the between-individual differences than in the variance of within-individual changes. For epigenetic aging estimators trained on chronological age (Horvath Acceleration and PedBE Acceleration), both additive genetic factors (8–39%) and shared environmental influences (13–49%) contributed to the variance. For epigenetic estimators trained on biological indicators, either additive (GrimAge Acceleration) or additive and non-additive genetic (DunedinPACE) factors were relevant. The variance of three estimators, initially developed in adult samples, demonstrated increasing contributions of genetic factors to the variance across adolescence and emerging adulthood.

**Keywords** Epigenetic aging, Age-moderation, Adolescence, Gene–environment interaction

**Grants**

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## Gene–Environment Correlation: The Role of Family Environment in Academic Development

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**Abstract**

Academic achievement is partly heritable and highly polygenic. However, genetic effects on academic achievement are not independent of environmental processes. We investigated whether aspects of the family environment mediated genetic effects on academic achievement across development. Our sample included 5151 children who participated in the Twins Early Development Study, as well as their parents and teachers. Data on academic achievement and family environments were available at ages 7, 9, 12, and 16. We computed educational attainment polygenic scores (PGS) and further separated genetic effects into cognitive and noncognitive PGS. Three core findings emerged. First, aspects of the family environment, but not the wider neighbourhood context, consistently mediated the PGS effects on achievement across development, accounting for up to 34.3% of the total effect. Family characteristics mattered beyond socio-economic status. Second, family environments were more robustly linked to noncognitive PGS effects on academic achievement than cognitive PGS effects. Third, when we investigated whether environmental mediation effects could also be observed when considering differences between siblings, adjusting for family fixed effects, we found that environmental mediation was nearly exclusively observed between families. This is consistent with the proposition that family environmental contexts contribute to academic development via passive gene–environment correlation processes. Our results show how parents shape environments that foster their children's academic development partly based on their genetic disposition, particularly towards noncognitive skills.

**Keywords** Gene–environment correlation, Polygenic scores, Academic achievement, Family environment, Noncognitive skills

**Grants**

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## Linked Emergence of Racial Disparities in Mental Health and Epigenetic Biological Aging Across Childhood and Adolescence

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**Abstract**

Marginalization due to structural racism may confer an increased risk for aging-related diseases—in part—via insults on people's mental health. Here we leverage a prospective birth cohort study to examine whether the emergence of racial disparities in mental health and DNA-methylation measures of biological aging (i.e., DunedinPACE, GrimAge, PhenoAge) are linked across childhood and adolescence. We further consider to what extent racial disparities are statistically accounted for by perinatal and postnatal factors in preregistered analyses of N = 4898 participants from the Future of Families and Child Wellbeing Study, of which N = 2039 had repeated saliva DNA methylation at ages 9 and 15 years. We find that racially marginalized children had higher levels and diverging longitudinal slopes of both externalizing and internalizing behaviors. Black compared to White identifying children, children living in more racially segregated neighborhoods, and marginalized children with darker skin tones tended to have higher age-9 levels of biological aging and more biological age acceleration over adolescence. Longitudinal increases in internalizing and externalizing problems were correlated with longitudinal increases in biological aging. While racial and ethnic disparities in mental health were largely statistically accounted for by socioeconomic variables, racial differences in biological aging were often still visible beyond covariate controls. Our findings indicate that racial disparities in mental health and biological aging are linked and emerge early in life. Programs promoting racial health equity must address the psychological and physical insults of structural racism in children and adolescents. Comprehensive and diverse measures of racism are lacking in current population cohorts.

**Keywords** Racial and ethnic disparities, Epigenetics, Biological aging, Mental health, Childhood

## Rosetta: A Method for Integration of Genetically Informative Studies of Reading Comprehension

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**Abstract**
An increasingly necessary practice in statistical genetics is combining data sets across multiple studies to increase power and better understand genetic and environmental effects. However, studies that examine the same construct (e.g. reading comprehension) frequently use different measures of that construct, resulting in difficulty harmonizing or comparing studies for research and clinical applications. The current study employs Rosetta, a new methodological approach which allows for the concurrent analysis of related but incompletely overlapping data. We apply this approach to combine the extant Western Reserve Reading Project and Colorado Longitudinal Twin Study samples, expanding upon a study conducted by Soden et al. (2015) which compared the two. The combined Rosetta dataset was fit to a genetic simplex model to understand the genetic and environmental contributions to the longitudinal stability in reading comprehension from first through ninth grade. Reading comprehension was assessed across grades 1, 2, 3, 4, 5, and 9, and across three measures of reading comprehension ability. Results demonstrate expected heritability, shared environmental, and nonshared environmental estimates for reading comprehension development, but with vastly decreased standard error estimates. Reading comprehension also shows significant genetic and environmental longitudinal stability. We demonstrate the usefulness of Rosetta for developing more robust results on the development of reading comprehension skills to develop more precise intervention strategies. Rosetta code can be found on GitHub (https://github.com/cwbartlett/rosettaR) to be used by the research community.

**Keywords** Quantitative genetics, Data harmonization, Simplex modeling, Reading comprehension

**Grants**
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## Familial Transmission of Personality Traits and Life Satisfaction Are Far Higher than Typical Single-Method Studies Show

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**Abstract**
We argue that familial similarity of personality traits has been strongly underestimated in typical self-report studies. This is because a large fraction of trait score variance is rater-specific, reflecting rating biases and idiosyncratic item interpretations, besides random error and raters' unique views/information about themselves. Therefore, even self-ratings of identical siblings should not correlate more than ratings of two people who rate the same target they both know well (usually, $r \sim 0.50$). Using a large number of parent–offspring, sibling and second-degree relative pairs from the Estonian Biobank, we first combined self-ratings with informant-ratings to estimate the true correlations between relatives' Big Five personality traits and life satisfaction (i.e., correlations between the traits' variance shared among self- and informant-ratings). We then fit structural equation models to estimate additive genetic effects on these valid trait scores and the genetic correlations between personality traits and life satisfaction. Hypothesizing that there may be more potentially trait-relevant environmental variance across than within generations, we also compared the similarity of the second-degree relatives from the same and different generations to see if heritability fades with time/generation. As expected, we found relatives' true trait similarity to be about twice as high as typically seen in single-method studies, while the additive genetic effects typically exceeded 0.60. We also found some evidence that the share of genetic transmission in trait variance may fade with time/generation.

**Keywords** Siblings, Psychometrics, Parent–offspring

**Grants**
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## Genetic and Environmental Links Between the Big Five Personality Traits and Socio-political Attitudes

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**Abstract**
Right-Wing Authoritarianism (RWA) and Social Dominance Orientation (SDO) are two constructs that describe important individual differences in socio-political attitudes. Data from the German Twin-Life project were analyzed to estimate the extent to which variance in RWA and SDO is accounted for by genetic and environmental differences in core personality traits, operationalized as Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness. Nuclear twin family modelling was used to analyze 687 monozygotic twin pairs, 753 same-sex dizygotic twin pairs, 435 non-twin full siblings, 1173 mothers of twins, and 843 fathers of twins. Significant additive genetic effects were observed for all constructs, and non-additive genetic influences for personality traits only. Twin-specific environmental effects were found for RWA, SDO, Neuroticism, Openness, and Conscientiousness. Multivariate Cholesky decomposition model analyses based on the data from twins and non-twin siblings showed that additive genetic differences in Openness and Agreeableness accounted for significant genetic variation in RWA, while nonadditive genetic factors were not significantly linked. Additive genetic differences in Openness, Agreeableness, and

Neuroticism accounted for significant genetic variation in SDO, with nonadditive genetic influences having no impact. Individual-specific nonshared environmental differences in Conscientiousness accounted for significant nonshared environmental differences in RWA, while individual-specific nonshared environmental differences in Conscientiousness and Agreeableness accounted for significant nonshared environmental differences in SDO. No further significant links between the genetic and environmental sources of personality traits and the two socio-political attitudes were found. The results of this study are used to shed light on how and why personality traits and socio-political attitudes are often correlated.

**Keywords** Right-Wing Authoritarianism, Social Dominance Orientation, Multivariate Cholesky decomposition modeling, Personality traits

**Grants**

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## The Potential Mediating Role of Brain Volume in Associations Between Smoking and Psychiatric Disorders

**Margot P. van de Weijer[1,2], Shu Liu[1], Anaïs B. Thijssen[1], Robyn E. Wootton[3,4], Jentien M. Vermeulen[1], Guido A. van Wingen[1,5], Dirk J. A. Smit[1], Marcus R. Munafò[3], Karin J. H. Verweij[1], Jorien L. Treur[1]**

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**Abstract**

Smoking is one of the leading preventable risk factors for morbidity and mortality. It is much more common among individuals with a psychiatric disorder, such as depression, schizophrenia, or bipolar disorder, who—because of this—have a lower life expectancy. With respect to causality, existing evidence suggests that there are bi-directional, causal effects between smoking and these psychiatric disorders. However, there is a surprising lack of insight into what mechanisms might underlie these causal associations. With respect to causal effects of smoking on psychiatric disease risk, one potential mechanism is that smoking might lead to changes in brain volume in specific parts of the brain, which may in turn lead to an increased susceptibility for psychiatric disease. However, to date, there is no research examining the potential mediating role of brain volume in the smoking–psychiatric disease association. For this study, we examine the potential role of hippocampal volume, amygdala volume, lateral and medial orbitofrontal cortex volume, superior frontal cortex volume, and more generally whole cortex surface area and thickness, in (genetic) associations between smoking measures and major depressive disorder, bipolar disorder, and schizophrenia. We do so using three different study designs: mediation models in genomic structural equation modelling (genomic SEM), mediation models in local analysis of (co)variant association (LAVA), and multivariable Mendelian Randomization (MR).

**Keywords** Smoking, Psychiatry, Brain volume, GenomicSEM, Mendelian randomization

**Grants**

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## Examining the Interplay Between Polygenic Risk and Family Processes on Adolescent Externalizing Behaviors

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**Abstract**

Externalizing behaviors are influenced by a complex interplay between genes and environment. Genetic influences have been characterized using genome-wide polygenic scores (PRS) based on genome-wide association studies (GWAS). Most GWAS used adult samples, and whether adult-based PRS predict childhood/adolescent traits beyond child-based PRS is unclear. This study examined the interplay between PRS based on GWAS of childhood aggression (Agg-PRS) and adult externalizing behaviors (Ext-PRS) and family processes in predicting adolescent externalizing behaviors. Data were from the Adolescent Brain Cognitive Development study (White n = 5924, Black/African American n = 1697, Hispanic/Latinx n = 2151; age 9/10 years). Both Agg-PRS and Ext-PRS predicted higher externalizing behaviors among White and Latinx youth; Agg-PRS (not Ext-PRS) predicted higher externalizing behaviors among Black youth. Among White youth, there were gene–environment interactions (G × E) such that the association between family conflict and externalizing was stronger whereas the association between parental acceptance and externalizing was weaker among those higher in Ext-PRS. No interaction effects were found among Black or Latinx youth. Among White youth, gene–environment correlations (rGE) were found with both Agg-PRS and Ext-PRS associated with higher family conflict and lower parental acceptance, which were associated with more externalizing behaviors. Among Latinx youth, Agg-PRS was associated with lower parental acceptance, which was associated with more externalizing behaviors. No rGE was found for Black youth. Findings suggest that both child- and adult-based PRS are useful in predicting adolescent externalizing behaviors. Family processes are important in influencing externalizing behaviors through both G × E and rGE processes.

**Keywords** Polygenic, Externalizing, Family process, Gene–environment interplay, ABCD study

**Grants**

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## Adolescent Reward and Cognitive Control Constructs Independently Relate to Polygenic Scores for Adult Psychopathology

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**Abstract**
Reward sensitivity and cognitive control constructs are individually phenotypically related to internalizing and externalizing disorders. However, less is known about whether these constructs are independently associated with psychopathology at the genetic level, particularly during early adolescence. We used polygenic scores (PGSs) in a European ancestry subset (n = 6667, ages 9–10) of the Adolescent Brain Cognitive Development (ABCD) Study to evaluate how adolescent reward sensitivity and cognitive control are related to genetic risk for psychopathology. We used exploratory structural equation modeling (ESEM) on the full baseline cohort (n = 11,875) to build a phenotypic model of reward sensitivity and cognitive control; seven factors were extracted capturing youth- and parent-reported reward sensitivity (e.g. items from Behavioral Activation Scale), youth- and parent-reported control (e.g., inhibitory control items from Early Adolescent Temperament Questionnaire), and executive function as measured by tasks (e.g. Stop-signal task). We then ran a model where Addiction and Internalizing factor PGSs (based on prior genomic structural equation model GWASs) were allowed to predict psychopathology latent factors mediated by reward and control factors. Results suggest cognitive control as a transdiagnostic marker: the cognitive control ESEM factors were significantly related to both PGSs (βs =|0.03–0.08|) and significantly mediated the relationship to both psychopathology latent factors after multiple testing correction. The Addiction PGS was significantly, positively related to Reward Seeking (β = 0.04) whereas the Internalizing PGS was not (β = − 0.004). These findings suggest that individual differences in multiple aspects of adolescent reward sensitivity and cognitive control are independently related to genetic risk for addiction and psychopathology.

**Keywords** Internalizing disorders, Externalizing disorders, Cognitive control, Executive function, Reward sensitivity

**Grants**
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## The Genetic Architecture of Personality Traits: A GWAS Update from the Revived Genomics of Personality Consortium

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**Abstract**
Personality traits are fundamentally important to behavioral genetics. However, surprisingly little is known about the genetic architecture of personality; the most recently published comprehensive personality GWAS is 7 years old (Lo et al. 2017). We present results of a new GWAS of the Big Five personality traits, overhauling our understanding of personality genomics. We meta-analyze 46 European-ancestry samples, incorporating 520,000–1,100,000 participants per trait, and 7 African-ancestry samples, with 35,000–45,000 participants per trait, to uncover approximately four times the number of genome-wide significant SNPs as past published and preprinted efforts. SNP heritability ranges from 7 to 10% per trait. Follow-up comparisons reveal that personality's genetic architecture is robustly similar across different Big Five questionnaires, age groups, rater perspectives, and western countries. Whereas past personality GWAS have struggled to produce predictive PGIs, our substantially larger discovery samples allow us to surmount this issue. European-ancestry PGI analyses indicate substantial out-of-sample polygenic prediction, accounting for between 2 and 5% incremental variance per trait. For extraversion, agreeableness, and conscientiousness, these are the first ever significantly predictive PGIs. Comparison of between-family and within-sibship PGI estimates indicate that, unlike for other social traits, the GWAS of personality is minimally confounded by residual population stratification, gene-family correlations, and assortative mating. We close by discussing how these results and data will be publicly disseminated, facilitating research that emphasizes the "behavior" in "behavioral genetics."

**Keywords** GWAS, Personality, Big five, PGIs, Consortium
**Consortia**
Revived Genomics of Personality Consortium.

**Grants**
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## Genetic and Environmental Influences on Flanker Task Inhibitory Control Across Middle to Late Childhood

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**Abstract**
The Flanker Task assesses inhibitory control performance and is scored in myriad ways, making cross-study comparisons difficult, potentially explaining why heritability estimates are inconsistent across studies. Identifying optimal task scoring methods are necessary to accurately assess the etiology of executive functioning across development. We examined the genetic and environmental covariance of the Flanker Task across middle to late childhood, comparing

traditional [Inverse Efficiency Test (IET)] and alternative (bifactor model) scoring approaches. Participants were drawn from the sociodemographically diverse Arizona Twin Project (23% Hispanic/Latino; 30% near/below the poverty line) and completed the Flanker Task at ages 9 (n = 590 twins; $M_{age}$ = 9.72 years), 10 (n = 528 twins; $M_{age}$ = 10.90 years), and 11 (n = 388 twins; $M_{age}$ = 11.81 years) during home visits. For the IET score, mean reaction time (RT) was divided by the proportion of correct trials, and scores were mean-composited and reverse-scored. The bifactor approach estimated latent Flanker performance using trial-level accuracy (dichotomous) and RT (continuous) indicators. Univariate models suggested ACE models fit best at ages 9 and 10, with CE influences accounting for variation at age 11. For the IET approach, C influences could be dropped at each wave, and additive genetic influences accounted for longitudinal stability in performance. The bifactor approach also showed some genetic covariance longitudinally, however, this approach demonstrated the substantial role of the nonshared environment. Findings suggest that the IET scoring approach may be more appropriate for twin modeling. This work has implications for how childhood executive functioning performance is assessed, particularly when large assessment batteries are not available.

**Keywords** Executive functioning, Accuracy, RT, Task scoring, Twin

**Grants**
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## Tools for Biometric Modeling in Large Population Databases

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**Abstract**
Large population databases commonly include complicated family structures that are well-suited to modern biometric methods, yet much of the latent potential of these resources remains untapped due to numerous complexities that arise in their analysis. Three challenges in large population databases are (1) finding independent extended families within larger population databases, (2) determining coefficients of relatedness among all pairs of individuals within those extended families, and (3) fitting biometric models to the resulting set of large, complicated, extended families. We present a set of openly available tools in the BGmisc package in R that address each of these problems. To illustrate the application of these tools to solve common problems in large population databases, we apply them to small, fictitious examples before demonstrating them in more realistic scenarios. With a solution to this set of problems, researchers are able to make much greater use of the large, population databases already collected.

**Keywords** Path tracing, Biometric models, Population databases

**Grants**
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## Is Music Listening Associated with Mental Health? A Genetically Informative Approach

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**Abstract**
While some people actively play musical instruments, most primarily engage with music by listening to it. Building upon our prior research on music playing and mental health, I will present our recent findings on the association between music listening and mental health. In this study, we analyzed weekly music listening habits in a large cohort of Swedish twins collected across two waves in 2011 (~ 10,500 twins) and 2022 (~ 9000 twins), respectively, with roughly 5500 twins participating in both waves. We tested whether the amount of music listening was associated with registry based mental health diagnoses (i.e., major depression, anxiety, schizophrenia, bipolar or stress-related disorders) and self-reported mental health problems such as depressive symptoms, burnout and well-being using multivariate (Cox) regression analyses. For a 5500-participant subset, we generated polygenic scores for several mental health conditions to explore associations with music listening frequency. The amount of music listening correlated 0.49 (95% CI 0.47–0.51) across the two timepoints. Initial findings revealed that frequent listeners, across both years, were more likely to receive a stress-related diagnosis (HR 1.08). In 2022, frequent music listening was also associated with a higher risk for anxiety diagnoses (HR 1.08). Correspondingly, individuals genetically predisposed to major depression, depressive symptoms, and neuroticism listened more frequently to music (betas ranging from 0.05 to 0.06, $p < 0.001$). No link was found between music listening and the self-reported mental health measures. Longitudinal, co-twin-control, and gene–environment interaction analyses will be added to explore gene–environment interplay underlying the relationship between mental health and music listening.

**Keywords** Well-being, Music, Depression, Anxiety

## Exploring the Genetics of Stuttering Persistence

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**Abstract**
**Background**
Stuttering is a neurodevelopmental disorder characterized by disfluencies in speech, including blocks, prolongations and repetitions. It develops early in childhood in 5–8% of children and for 0.8% of children persists into adulthood. Twin studies estimated the heritability of stuttering at 40% to 80%. It however remains to be explored whether and how genetic factors may be involved in stuttering persistence.
**Methods**
We analysed twin heritability of parent/self-reported stuttering present at 5, 7, 10, 12, 14 and 16 years of age in data of 47,216 twins of the Netherlands Twin Registry. Next, we used GREML to estimate SNP and pedigree heritability for self-reported resolved and persistent stuttering in genotyping data of 700 cases and 15,625 controls from the Lifelines cohort. Polygenic scores for ADHD, neuroticism, anxiety/tension and worry/vulnerability, factors previously associated with stuttering and its persistence, are being used to predict stuttering and its persistence in Lifelines, and in 687 cases and 649 controls of the MPI Erasmus Genetics of Stuttering study.
**Preliminary Results**
Twin heritability estimates are higher for stuttering present at 7 to 16 years of age (77.3, 95% CI = 70.0–83.3) than for stuttering present at 5 years of age (35.8%, 95% CI = 26.0–45.7). Correspondingly, higher estimates for pedigree heritability were found for persistent stuttering (35.0%, se = 4.6%) than for resolved stuttering (4.7%, se = 3.8%).
**Conclusions**
Higher twin and pedigree heritability estimates for long/persistent stuttering indicates that genetic factors are involved in stuttering persistence. Polygenic score analysis will be applied to further characterize genetic factors involved in stuttering development and persistence.

**Keywords** Stuttering, Heritability, Polygenic scores

**Grants**
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## Genomic Links Between Symptoms of Eating Disorders and Suicidal Ideation

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**Abstract**
Eating disorders (EDs), encompassing anorexia nervosa, bulimia nervosa and binge eating disorder, are psychiatric conditions associated with high mortality rates, particularly in young individuals. Even though EDs have been linked to suicidal ideation, attempts and death, the relationship between these conditions remains poorly understood. Despite evidence of substantial genetic influence on each of these conditions, the common genetic factors contributing to their frequent co-occurrence have not been extensively explored. The majority of genome-wide association analyses have focused on individual phenotypes, limiting the ability to differentiate between variance common to a set of traits and variance specific to each. Utilizing a multivariate approach, we aimed to examine the shared genome-wide architecture of ED symptoms, measured using the ED100K questionnaire, and suicidal thoughts and behaviours in ~ 20,000 individuals ascertained for lifetime depression and/or anxiety, enrolled in the COVID-19 Psychiatry and Neurological Genetics (COPING) study. We employed a structural equation modelling framework to explore the factor structure underlying symptoms of EDs and suicidal ideation, followed by genome-wide association studies of the extracted factor scores and genetic correlation analyses. We observed distinct patterns of genome-wide association between latent structures of EDs and suicidal ideation. We modelled the general factor of susceptibility to EDs and suicidal ideation that was as strongly genetically influenced as both conditions, with mean SNP heritability of 7%. We found that factors related to EDs and suicidal ideation are strongly genetically correlated (mean rG = 0.45), but also show a unique residual component.

**Keywords** Eating disorders, Self-harm, Suicide, Genome-wide structural equation modelling

**Grants**
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## Biological Pathways of Cognitive Health, Major Depressive Disorder, and Schizophrenia in Diverse Ancestry Populations

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**Abstract**
Cognitive functioning (CF) is integral to mental health, with cognitive impairment phenotypically and genetically correlated with major depressive disorder (MDD) and schizophrenia (SCZ). Educational attainment (EA) is negatively correlated with MDD. Paradoxically, a modest positive genetic correlation between EA and schizophrenia is found, despite a strong positive correlation between EA and general cognitive ability. Due to the summative nature of genome-wide correlations, separating out a mixture of concordant and discordant genetic effects from the absence of an effect is not possible. The aim of this research is to leverage genome-wide association studies of CF, EA, SCZ, and MDD to parse the biological mechanisms underlying these relationships. Linkage-disequilibrium score (LDSC) regressions were used to calculate pairwise genetic correlations between MDD, CF, and EA, generating sample overlap estimates required for Local Analysis of (Co)variance Association (LAVA). Resulting genomic regions of interest shared between MDD, CF, and EA were examined for potential causal variants using fine-mapping and colocalisation tools. LDSC found a negative genetic correlation between both MDD and CF (− 0.1, $p = 7.8 \times 10^{-7}$) and EA (− 0.22, $p = 7.8 \times 10^{-19}$), whereas CF had a positive genetic correlation with EA (0.74, $p = 5.4 \times 10^{-28}$). LAVA found 13 genomic regions of interest for MDD–EA and six for MDD–CF. Two genomic regions of biological relevance for the relationship between MDD, CF, and EA were

identified through colocalisation and fine-mapping methods. This work will be expanded to SCZ using Asian populations. Genomic regions with pleiotropic patterns will undergo interrogation through rich annotations and gene set analysis.

**Keywords** Statistical genetics, GWAS, Psychiatric disorders, Cognition, Diverse ancestry

## Musical Rhythm Abilities and Risk for Developmental Speech–Language Problems and Disorders: Epidemiological and Polygenic Associations

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**Abstract**
Speech-, language-, and communication-related disabilities have tremendous health impacts, including increased risk for mental health disorders; poorer physical health; long-term issues with health-related quality of life; and poorer educational and socioeconomic outcomes. Identifying risk factors for these disorders is therefore paramount for understanding, identifying, and treating them. Impaired musical rhythm abilities have been identified as a potential risk factor for atypical development of speech, language, and communication abilities. Prior work examining their relationship has primarily used small samples; here, we studied associations at population-scale by conducting the largest systematic epidemiological investigation to date (total $N = 39{,}092$). Based on existing theoretical frameworks, we predicted that rhythm impairment would be a significant risk factor for speech–language disorders in the general adult population. Findings were consistent across multiple independent datasets and rhythm subskills (including beat synchronization and rhythm discrimination), and aggregate meta-analyzed data showed that rhythm impairment is a modest but consistent risk factor for developmental speech, language, and reading disorders (OR = 1.32 [1.14–1.49]; $p < 0.0001$). Further, cross-trait polygenic score analyses indicate shared genetic architecture between musical rhythm and reading abilities, providing evidence for genetic pleiotropy between rhythm and language-related phenotypes. Our results can inform future development and validation of more equitable and inclusive screening and intervention tools for individuals at-risk for/living with speech–language disabilities. For example, rhythm-based screeners may be more cost-effective, and not require separate assessments and norms to be developed for multilinguals or speakers of different languages (unlike language-based screeners), making them more accessible for global health.

**Keywords** Musicality genomics, Speech and language disorders, Genetic pleiotropy, Rhythm abilities, Epidemiology

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## Gene–Environment Interplay in Adolescent Developmental Psychopathology

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**Abstract**
A combination of genetic (G) and environmental (E) influences working in complex interplay are thought to underlie differences in symptoms of psychopathology between adolescents. However, studies that have investigated gene–environment interaction (G × E) in isolated aspects of developmental psychopathology are characterized by lack of robust effects, suggesting the need for a more comprehensive approach. We adopted a multivariable approach to investigate G × E in developmental psychopathology. Our sample included 4000 16-year-olds enrolled in the Twins Early Development Study. Adolescents and their parents provided data on externalizing and internalizing symptoms of psychopathology. We estimated G by combining polygenic scores for neurodevelopmental disorders and psychopathology. We measured E by combining home environments and life events, assessed during childhood and adolescence. We used elastic net regularization to examine the main effects of G and E on symptoms of psychopathology, their joint effects (G + E) and their interaction (G × E). Polygenic scores jointly accounted for less than 3% of the variance in developmental psychopathology. The prediction was stronger for externalizing symptoms, compared to internalising. Parent-rated psychopathology symptoms at age 16 were best predicted by parent-rated environments measured during childhood and early adolescence (R = 10.4%), while self-rated symptoms at age 16

were best predicted by self-reported contemporaneous environmental experiences ($R^2$ = 11%). We observed small and isolated G × E effects accounting for < 1% of the variance and weak correlations between G and E. The findings point to the challenges of investigating G × E in developmental psychopathology and highlight the role of specific family environments.

**Keywords** Developmental psychopathology, Behaviour problems, Gene–environment interaction, Gene–environment correlation, Adolescence

**Grants**

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## Triangulating Approaches to Evaluate the Genetic and Environmental Composition of Socioeconomic Status

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**Abstract**

Indicators of socioeconomic position are similar across generations, which is due to interplay of genetic and environmental modes of transmission.

The estimated contributions of genetic and social factors to socioeconomic status (SES) vary depending on (i) the definition and use of socioeconomic indicators, (ii) the statistical methods applied, (iii) the life stage, demographic composition and geographical contexts of the sample.

Here we show how four key SES measures and four distinct heritability methods compare and relate in the same Norwegian sample consisting solely of parents at the age of parenthood (age 35–45). The results show a range of pedigree heritability estimates ($h^2$ped = 36–69%) and molecular heritability estimates ($h^2$mol = 6.6–23.4%) highlighting advantages and limitations of each method. The consistent pattern of estimates across the four main socioeconomic indicators education, occupation, income, and wealth reveals a first principal component accounting for 52% of the phenotypic, 73.5–89.1% of the genetic, 100% of the shared environmental, and 31.1–32.4% of the non-shared environmental variance. The explained variance and loadings suggest common genetic and shared environment causes for the four SES indicators, while pointing towards split influences on education, occupation, income, and wealth in the non-shared environment.

Our results help to build an intuition for comparing methods and measures, and yield insight into the relationship between the subcomponents of social and economic position of parents in a Norwegian context. This is the first time multiple heritability estimates of four main socioeconomic status indicators and their principal components have been shown in the same sample of parents, thereby triangulating the basis of childhood socioeconomic status.

**Keywords** SES, Heritability methods, Dimensionality

**Grants**

ERC.

## Multi-polygenic Scores in Psychopathology: A Systematic Review

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**Abstract**

Polygenic scores (PGS) provide individual estimates of genetic propensity relative to others and hold potential for prediction in psychopathology. However, predictive power remains limited. To address this, multi-polygenic scores (MPS) that leverage shared genetic influences across disorders have been developed that integrate multiple PGSs into a unified statistical framework. We conducted a systematic literature review of MPS studies of psychopathological outcomes to evaluate methodological diversity, replicability, and study quality.

Following PRISMA guidelines, we systematically searched PubMed, Embase, and PsycINFO for peer-reviewed studies of MPS applications in psychopathology. Of 206 articles screened, 8 studies were included. Results indicate MPS improves prediction beyond single PGS models, yet significant methodological inconsistencies exist across studies. These include a lack of consensus on which PGSs to include and diversity in approaches used to construct prediction models. Studies almost exclusively focus on European genetic ancestry populations and provide minimal consideration of ethical implications beyond research ethics.

We highlight a critical lack of standardized methods in MPS implementation, making it challenging to reach a consensus on best MPS approaches and limiting the generalizability of findings. We underscore the need to develop standardized guidelines for MPS construction that will enable more robust and reproducible research. Importantly, the development of genomic prediction models presents significant ethical challenges, particularly given the bias in current PGSs. The need to embed ethical thinking from the beginning of PGS research is emphasized.

**Keywords** Multi-polygenic scores, Polygenic index, Systematic review, Psychopathology

## Genetic and Environmental Differences in Right-Wing Authoritarianism and Social Dominance Orientation from Adolescence to Adulthood: A Multi-cohort Twin Family Study

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**Abstract**

Past research found genetic and environmental contributions to Right-Wing Authoritarianism (RWA) and Social Dominance Orientation (SDO)—two core dimensions that describe individual differences in socio-political attitudes. However, less is known regarding potential age differences in the sources of variance in RWA and SDO. Based on data from 687 monozygotic twin pairs, 753 same-sex dizygotic twin pairs, 435 other siblings of twins, 1173 mothers of twins, and 843 fathers of twins from the German TwinLife project, multi-cohort nuclear twin family modelling (MC-NTFM) was used to estimate the extent to which genetic and environmental contributions to the variance in RWA and SDO vary across three age cohorts ($M_{\text{ages}} \approx 15, 21,$

and 27 years). In line with previous twin studies on several personality-relevant phenotypes, we expected genetic variance to increase with age, reflecting more active genotype–environment transactions as the twins matured, and declining trends for passive genotype–environment covariation, which can be analyzed directly in a MC-NTFM. Consistent with our expectations, twin similarity increased from adolescence to young adulthood. However, the MC-NTFM results were only partially consistent with our expectations regarding genotype–environment covariance trends. For RWA, both passive genotype–environment covariance and additive genetic components declined with age, while twin-specific shared environmental differences increased. For SDO, passive genotype–environment covariance were generally small and not statistically significant, while additive genetic variance components increased from adolescence to young adulthood. Counterintuitively, influences of twin-specific shared environmental factors on RWA and SDO were larger in adulthood than adolescence. The results were discussed against the background of human development theories.

**Keywords** Right-Wing Authoritarianism, Social Dominance Orientation, Multi-cohort nuclear twin family model, Genotype–environment covariation

**Grants**
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## New Ways to Share, Validate, Reward, and Interact with Research in Behavioral Genetics

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**Abstract**
Similar to other fields, scientists in behavioral genetics are struggling with reproducibility, replicability, access to data and code, and a slow and often arbitrary peer-review process that allows publishers to monetize the pro-bono work of scientists. Furthermore, the current system lacks explicit incentives for best scientific practices, such as sharing data and code. Emerging technologies for "decentralized science" (DeSci) are promising solutions to these problems. Using the example of a recent large-scale GWAS meta-analysis on income, this talk will showcase how already existing DeSci solutions help address these problems and provide an outlook for the future.

**Keywords** GWAS, Open science, Innovation

## The Association Between Frequency of Social Media Use, Wellbeing, and Depressive Symptoms: Disentangling Genetic and Environmental Factors

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**Abstract**
Meta-analyses report small associations (around $r = -0.10$) between wellbeing (WB) and social media use (SMU) and between anxious-depressive symptoms (ADS) and SMU (around $r = 0.10$). This study employs the classical twin design, utilizing data from 6492 individuals from the Netherlands Twin Register, including 3369 MZ twins (893 complete twin pairs, 1583 incomplete twin pairs) and 3123 DZ twins (445 complete, 2233 incomplete) to provide insights into the sources of overlap between WB/ADS and SMU.

In this study, both hedonic and eudaimonic WB scales were used. SMU was measured by (1) the time spent on different social media platforms ($SMU_t$), (2) the frequency of posting on social media ($SMU_f$), and (3) the number of social media accounts individuals have ($SMU_n$). Our results confirmed the low phenotypic correlations between WB/ADS and SMU (absolute correlations, between $r = 0.03$–$0.10$). For SMU, heritability estimates as high as 72% were obtained.

The small but significant phenotypic correlations between WB/ADS and SMU were mainly determined by genetic factors (in the range of 80–90%). For WB and SMU, genetic correlations were between $-0.10$ and $-0.0$, and for ADS and SMU genetic correlations were between 0.10 and 0.23. These genetic correlations implied limited but statistically significant sets of genes that affect WB/ADS and SMU levels. Overall, the results underline that the small associations between WB/ADS and SMU are partly driven by overlapping genetic influences. We encourage the use of more personalised approaches when considering the association between WB and SMU.

**Keywords** Classical twin design, Genetically informed designs, Natural language processing, Machine learning, Social media use

**Grants**
This work is supported by a European Research Council Consolidator Grant (WELL-BEING 771057, PI Bartels).

## Joint Multi-family History and Multi-polygenic Score Prediction of Major Depressive Disorder

**Rujia Wang[1,2], Helena Davies[1,2], Sang-Hyuck Lee[1,2], Jonathan R. I. Coleman[1,2], Raquel Iniesta[3], Thalia C. Eley[1,2], Gerome Breen[1,2]**

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**Abstract**
**Background**
Major depressive disorder (MDD) is a complex psychiatric disorder influenced by genetic, social, and environmental factors. Family history and polygenic risk scores of MDD and related psychiatric disorders are strong predictors for MDD, while childhood trauma (CT) also plays a crucial role. This study aimed to jointly model the predictive effect of multi-family history (mFH), multi-PRS (mPRS), and childhood trauma on the development of MDD and the number of MDD episodes experienced. We aimed to develop a predictive model for stratifying treatment plans and interventions effectively.
**Methods**
Data were obtained from the NIHR BioResource Genetic Links to Anxiety and Depression (GLAD) study and UK Biobank (UKB). MDD diagnosis followed DSM-V criteria using online questionnaire data. Family history (Yes/No) was reported for up to 22 psychiatric disorders. MegaPRS was used to calculate PRSs. Reported childhood trauma was identified via the 5-item childhood trauma screener questionnaire. Elastic net regression with nested cross-validation was applied.

**Results**
In GLAD (9927 MDD cases, 4452 controls), mFH explained 22.66% of MDD variance, followed by CT (14.53%), demographics (13.47%), and mPRS (13.22%). All predictors explained 44.52% of MDD variance (AUC = 0.84). In UKB (40,667 MDD cases, 70,755 controls), mFH explained 13.56% of MDD variance, followed by demographics (9.42%), CT (5.87%), and mPRS (3.63%). Together, all predictors explained 25.54% of variance (AUC = 0.75). As the number of MDD episodes increased, all predictors explained a larger variance and resulted in a higher AUC.
**Conclusions**
Integrating multi-family history, multi-PRS, childhood trauma, and demographics enhances MDD prediction, particularly regarding recurrent MDD episodes.

**Keywords** Major depressive disorder, Family history, Polygenic score, Machine learning, Prediction

## Investigating the Associations Between Childhood Trauma, Intimate Partner Violence, and Posttraumatic Stress Disorder Using Genetically Informed Methods

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**Abstract**
**Background**
Individuals with a history of childhood trauma (CT) face an increased risk of experiencing intimate partner violence (IPV) and posttraumatic stress disorder (PTSD) compared to their peers. This study investigated the associations between CT, IPV, and PTSD using genetically informed methods.
**Methods**
We estimated twin- and DNA-based heritability and genetic correlations between CT, IPV, and PTSD using multivariate twin models in 8980 26-year-old participants from the UK Twins Early Development Study (TEDS) and using linkage disequilibrium score regression based on recent GWAS summary statistics of CT, IPV, and PTSD. Additionally, we examined the potential causal effect of CT on IPV using direction of causation models.
**Findings**
TEDS participants exposed to CT alone faced a fourfold increased risk of IPV and PTSD, whereas those exposed to both CT and IPV faced an eightfold increased risk of PTSD. Across methods, genetic correlations were significant and substantial between CT and IPV ($r_g$ = 0.47–0.84) as well as between both exposures and PTSD ($r_g$ = 0.35–0.76). Direction of causation models suggested a potential causal effect of CT on IPV beyond their shared etiology ($\beta$ = 0.21).
**Conclusion**
Results support the presence of a genetic susceptibility to CT and IPV, a shared etiology between these exposures and PTSD, and a potential causal effect of CT on IPV. These findings underscore the need for interventions aimed at promoting safe intimate relationships following CT, considering genetically influenced traits, behaviors, and cognitive processes associated with vulnerability to repeated trauma exposure.

**Keywords** Trauma, PTSD, Childhood maltreatment, Intimate partner violence, Causal inference

**Grants**
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## Social Origins and Socioeconomic Outcomes: A Combined Twin and Adoption Study

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**Abstract**
Who gets ahead? The answer given by several theories and empirical studies in the social sciences to this question has typically involved some kind of privilege or resource: "The rich", "The educated", "The elite", "The well connected", and so on. Essentially, most of this research ascribes an important role to social origins, the rearing family, and parents' investments and aid, essentially emphasizing social environments for intergenerational persistence in socioeconomic status (SES). Yet, most of these studies do not account for genetic transmission, making it impossible to separate social from genetic mechanisms in intergenerational transmissions. Accumulating evidence suggests genetics play an important role in the transmission of SES from parents to children, but estimates of the importance of such factors differ substantially across measures and methods. Using three different research designs that account for potential genetic confounding and high-quality data from Norway, we estimate the strength of social environments in the intergenerational transmission of a range of SES indicators (i.e., education, occupational prestige, income and wealth). In particular, we use an adoption design where we compare intergenerational transmission for adopted and biological children, a multiple children-of-twin design and a nuclear-twin-family approach to obtain estimates of the social transmission of status, while accounting for genetics. By triangulating different data and designs, we obtain estimates that are more robust to idiosyncratic modeling assumptions. While we find strong intergenerational transmission of status, social origin measures are moderate predictors of socioeconomic outcomes and genetic mechanisms are substantial pathways of transmission that should be accounted for.

**Keywords** Intergenerational transmission, Socioeconomic status, Adoption design, Children-of-Twin design, Nuclear-Twin-Family-Design

**Grants**
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## Understanding the Genetic Architecture in the Tails of Complex Traits from Population Genetic Simulations

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**Abstract**

Understanding the genetic architecture of human traits is fundamental to our efforts of identifying causal variants and performing downstream analyses, such as genetic risk prediction. Prediction in relation to extreme values of trait distributions (i.e., the tails) is particularly important because it is here that behavioural and psychiatric disorders present. Historical processes, such as population bottlenecks and evolutionary selection, are known to shape trait genetic architecture but so far there has been little investigation of how these may impact genetic architecture across the distribution of a trait and especially within the tails. Here, we specifically focus on the role of selection and build a Wright–Fisher forward-simulation model of a human polygenic quantitative phenotype in a population to investigate its impact on tail genetic architecture. In validating our model across many simulations, we show how patterns of genetic variation in our simulation match to real-world data with respect to the distribution of allele frequencies and linkage disequilibrium, as well as the degree of genetic diversity in the population. Under stabilising selection, we find that large effect variants are maintained at low-frequency and contribute disproportionately more to the phenotypic variance at the tails of the phenotypic distribution than under neutral evolution. Stabilising selection is a form of selection that has been hypothesized to play a key role in shaping the architecture of present-day complex traits. Our work helps understand how evolutionary processes impact the tail genetic architecture of human phenotypes, which has implications for our understanding of behavioural and neuropsychiatric traits.

**Keywords** Tails, Genetic architecture, Stabilising selection, Forward-simulation, Rare variants

## Direct and Indirect Genetic Effects on Early Neurodevelopmental Traits

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**Abstract**

**Background**

Neurodevelopmental conditions are highly heritable. SNP heritability estimates can be confounded by genetic effects mediated via the environment (indirect genetic effects). The relative importance of direct versus indirect genetic effects on early manifestations of neurodevelopmental conditions is unknown.

**Methods**

The sample included up to 24,692 parent–offspring trios from the Norwegian Mother, Father, and Child Cohort Study cohort. We use Trio-GCTA to estimate latent direct and indirect genetic effects on mother-reported neurodevelopmental traits at age 3 years. Further, we investigate to what extent direct and indirect effects are attributable to common genetic variants associated with neurodevelopmental conditions and associated traits using polygenic scores (PGS) in regression modeling.

**Results**

Direct and indirect latent genetic effects contributed to attention (direct: explaining 4.8% of variance, indirect: 6.7%), hyperactivity (direct: 1.3%, indirect: 9.6%), and restricted and repetitive behaviors (direct effects: 0.8%, indirect effects: 7.3%). Variation in social and communication, language, and motor development was best explained by direct effects (5.1–5.7%). Direct genetic effects on attention were captured by PGS for ADHD, autism, educational attainment, and cognitive ability, whereas direct genetic effects on language development were captured by cognitive ability and autism PGS. Indirect genetic effects were primarily captured by educational attainment and/or cognitive ability PGS across all outcomes.

**Conclusions**

Results were consistent with differential contributions to neurodevelopmental traits in early childhood by direct and indirect genetic effects. Indirect effects were particularly important for hyperactivity and restricted and repetitive behaviors and interests and may be linked to genetic variation associated with cognition and educational attainment.

**Keywords** Neurodevelopment, Intergenerational transmission, Autism, ADHD, Genetic architecture

**Grants**

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## Examining ADHD Diagnosis and Cannabis Use Frequency: The Role of Family Environment in a Nationally-Representative Sample

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**Abstract**

The relationship between cannabis use and Attention-Deficit/Hyperactivity Disorder (ADHD) involves complex genetic and environmental factors. Research indicates parenting characteristics

and the home environment influence cannabis use (Mulligan et al. 2013). However, measures like the Home Observation for Measurement of the Environment (HOME) capture more than just the "shared environment", absorbing other between-family variance. Extending our prior work (Tyson et al. 2024), we incorporated the HOME measure to examine the ADHD–cannabis use relationship. Using the nationally-representative Children of the National Longitudinal Survey of Youth (NLSYC), we explored associations at both between- and within-family levels utilizing siblings and cousins. Without accounting for family structure, the classic approach found an association between ADHD and higher cannabis use, even after controlling for the HOME. At the between-family level, only dyads where at least one sibling had ADHD were associated with higher cannabis use. No significant associations were found at the within-family level. Using the HOME across childhood, we lost a lot of nuance which could be reflected by no significant associations. Regardless, our results show that cannabis use is not caused by an ADHD diagnosis or the home environment.

**Keywords** ADHD, Sibling comparisons, Cannabis, HOME

**References**

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Tyson KH, Webster EE, Garrison SM (2023) Unpacking the link between ADHD and cannabis use: a sibling comparison approach. Behav Genet 53(6)

## Multi-variant Fine-Mapping to Identify Putative Causal Variants from Genome-Wide Association Studies of Major Depressive Disorder

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**Abstract**

**Background**

Major depressive disorder (MDD) is a significant global contributor to chronic disability. Genome-wide association studies (GWAS) have implicated over 500 genomic regions as associated with MDD. However, these associations are largely restricted to statistical correlations and do not directly provide biological insight. Statistical and functional fine-mapping can enable the identification of putative causal genetic variants.

**Methods**

We conducted GWAS and meta-analysed linkage disequilibrium (LD) matrices from 65 European ancestry cohorts (N = 1,932,632) included in the Psychiatric Genomics Consortium (PGC) MDD Working Group. We conducted sample LD-accurate multi-variant fine-mapping of 1,088,850 common variants (MAF $> 0.001$) across 564 genomic regions genome-wide significant in the latest PGC MDD GWAS. We used PolyFun + SuSiE supported by additional analyses using PolyFun + FINEMAP. We defined credible sets of variants with cumulative posterior inclusion probabilities (PIP) $\geq 95\%$. We contrasted these results with parallel analyses using population-accurate LD from the Haplotype Reference Consortium.

**Findings**

Analyses suggest that 62% of loci contain a single causal signal. We identify 146 credible sets across 138 regions, including 8 single-variant sets and 60 sets with $< 10$ variants. Analyses using population-accurate LD show high concordance in high-confidence, multi-signal regions (correlation of PIPs = 0.95), although inconsistent results are seen in some specific regions. Annotation of credible sets to putative genes and molecular mechanisms is ongoing.

**Conclusions**

We present evidence for specific causal sets across genomic regions associated with major depressive disorder, and show that these can largely be recapitulated using reference LD panels.

**Keywords** Major depressive disorder, Fine-mapping, Causality, Genome-wide association study, Linkage disequilibrium

**Consortia**

Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium.

**Grants**

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## Do Chronic Pain Polygenic Scores Derived in Older Adults in the U.K. Biobank (UKB) Predict Adolescent Pain Reports in the Adolescent Brain Cognitive Development (ABCD) Study?

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**Abstract**

Chronic pain in adolescence is associated with worse outcomes in adulthood, including heightened incidence of new chronic pain. Adolescence is a developmentally active period, and it is unclear whether similar mechanisms underlie adolescent and adult chronic pain. We use two chronic pain polygenic scores (PGSs) computed in an older adult sample to predict young adolescent pain reports.

The sample consisted of 6387 children (mean ages 12.03–12.92; 52% boys, 48% girls; all European ancestry) from the longitudinal Adolescent Brain Cognitive Development (ABCD) study. We calculated PGSs from two previously reported genetic chronic pain factors (a general factor capturing genetic variance shared among 24 different chronic pain conditions and a specific factor capturing genetic variance shared among musculoskeletal chronic pain conditions) derived in the U.K. Biobank. We used the 2 pain factor PGSs to

predict self-reported pain, persistent pain (i.e., pain reported at 2+ timepoints), pain severity, and level of limitation in the ABCD.

The general chronic pain PGS predicted pain presence (log odds = 0.08, p = 0.004), pain severity (b = 0.16, p < 0.001), limitation from pain (b = 0.09, p = 0.039), and pain persistence across 2+ timepoints (log odds = 0.09, p = 0.033). The musculoskeletal pain PGS did not significantly predict any pain report.

Polygenic scores developed to predict risk of pain in older adulthood generally captured risk for self-reported adolescent pain. These results inform how we should conceptualize adolescent pain, specifically that pain reports in early adolescence seem indicative of pain risk throughout the lifespan.

**Keywords** Chronic pain, Adolescent, Polygenic score, ABCD, UKB

**Grants**

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## Preliminary Results from the ClozaGene Study Examining the Experiences of 1021 Australians Prescribed Clozapine

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**Abstract**

The ClozaGene Study is a nation-wide cohort of adults who have been treated with clozapine. While clozapine is indicated in the management of treatment-resistant schizophrenia, it is associated with a considerable adverse drug reaction (ADR) burden, and not all patients achieve adequate symptomatic response. This retrospective cohort study aimed to examine the environmental and genetic factors that influence response to clozapine treatment and the risk of experiencing ADRs. A total of 1021 participants (41.0% female; aged 46.2 ± 10.6 years [range 18–66]) were recruited via a mail-out based on prescriptions for clozapine. Participants completed a self-report questionnaire. A total of 771 (75.5%) participants provided a saliva sample for genotyping and 781 (76.5%) consented to a review of their pathology test records. Most participants (90.1%, n = 912) were living with schizophrenia while 41.5% reported a lifetime diagnosis of depression, 15.6% bipolar disorder, and 8.1% schizoaffective disorder. Clozapine was currently prescribed to 944 (92.5%) participants and 41.4% of participants were currently prescribed one or more antipsychotic medications. Nearly three quarters of participants living with schizophrenia reported that clozapine helped control their schizophrenia symptoms moderately to very well. The most commonly reported ADRs were sialorrhea (80.5%), weight gain (71.2%), constipation (57.1%), and sedation (52.9%). Our findings highlight the high rate of psychotic and metabolic symptoms and ADRs among adults prescribed clozapine in the general Australian population. Genomic analyses focus on identifying genetic variants influencing clozapine treatment response and side-effects are underway.

**Keywords** Treatment-resistant schizophrenia, Metabolic disorders, Adverse drug reactions, Effectiveness

**Grants**

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## gSEM: Common and Independent Genetic Factors Underlie Substance Use, Psychotic, Mood, and Anxiety Disorders

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**Abstract**

Substance use disorders (SUDs) commonly co-occur with mood, anxiety, and psychotic disorders. Genome-wide association studies (GWAS) have shown these disorders to be highly polygenic. These studies also provide evidence of pleiotropy among these and other psychiatric traits, which partially accounts for their co-occurrence. We used genomic structural equation modeling (gSEM) and GWAS-by-subtraction to examine transdiagnostic and disorder-specific genetic risk across substance use, psychotic, mood, and anxiety disorders in African (AFR) and European (EUR) ancestry individuals. In AFR individuals, transdiagnostic genetic factors represented SUDs [1 lead single nucleotide polymorphism (SNP)] and psychiatric disorders (no lead SNPs). In EUR individuals, genetic factors represented SUDs (143 lead SNPs), psychotic disorders (162 lead SNPs), and mood/anxiety disorders (112 lead SNPs). Genetic factors also accounted for shared variance between SUDs and psychotic disorders (76 lead SNPs) and SUDs and mood/anxiety disorders (63 lead SNPs). Genetic correlations with phenotypes differed for shared and disorder-specific components. Phenome-wide association studies (PheWAS) of polygenic risk scores (PRS) for shared and disorder-specific components also differed in their phenotypic associations. For example, the component of genetic risk for TUD that operates through the SUD factor was associated with other psychiatric disorders, whereas the component that is independent of the SUD factor was associated with cardiometabolic traits. Combining transdiagnostic and disorder-specific genetic approaches provides insights into the structure and classification of co-occurring conditions, refining our understanding of comorbidity, and informing nosology.

**Keywords** Genomic SEM, Substance use disorders, Mood and anxiety disorders, Psychotic disorders, Electronic health records

## Quantifying Opinions of Sexual Orientation and Gender Identity Genetics Research

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**Abstract**
Opinions of behavioral genetics research vary considerably across people and populations. This is especially true given the various ethical concerns that surround behavioral genetics research and the differential impact that genetics research can have on marginalized communities. In recent years, research has been done seeking out genetic associations and heritability estimates of genes related to sexual orientation and gender identities, prompting increased concerns about the uses and ethics of these findings. This study looks at opinions regarding sexual orientation and gender identity genetics research using a quantitative questionnaire designed to look at opinions on sexual orientation and gender identity research. The scale was developed by using prior qualitative research on the opinions of sexual orientation and gender identity genetics research from LGBTQ+ populations. We will compare LGBTQ+ and non-LGBTQ+ participants to see if there is variation in opinions and awareness of possible concerns and benefits associated with conducting sexual orientation and gender identity genetics research. We expect to find variation in opinions on sexual orientation and gender identity genetics research both within and between LGBTQ+ and non-LGBTQ+ participants. Findings will be informative as guidance for future sexual orientation and gender identity genetics research, as it will help to better understand concerns held by both those who are directly affected and the general population.

**Keywords** Bioethics, Public perception, Sexual orientation, ELSI genomics, Gender identity

## Assessing Causal Links Between Age at Menarche and Adolescent Mental Health: Triangulation of Evidence from a Registered Report

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**Abstract**
The timing of puberty may influence adolescent mental health. Earlier age at menarche has been associated with depression in adolescents. Previous research suggests that this relationship may be causal, but replication and an investigation of whether this effect extends to other mental health domains is warranted. In this Registered Report, we used a new wave of data (N = 13,398) from the Norwegian Mother, Father, and Child Cohort Study. We combined multiple regression, one- and two-sample Mendelian randomisation (MR), and negative control analyses (using pre-pubertal symptoms as outcomes) to assess causal links between age at menarche and adolescent mental health. As hypothesised, earlier age at menarche was associated with elevated depressive symptoms in early adolescence based on multiple regression ($\beta = -0.11$, 95% CI [$-0.12$, $-0.09$], $p_{one\text{-}tailed} < 0.01$). One-sample MR analyses suggested that this relationship may be causal ($\beta = -0.07$, 95% CI [$-0.13$, 0.00], $p_{one\text{-}tailed} = 0.03$). There was also some evidence of a causal relationship with depression diagnoses during adolescence based on one-sample MR (OR = 0.74, 95% CI [0.54, 1.01], $p_{one\text{-}tailed} = 0.03$), corresponding to a 29% increase in the odds of receiving a depression diagnosis with each earlier year of menarche. Negative control and two-sample MR sensitivity analyses were broadly consistent with this pattern of results. Meanwhile, we found little evidence of effects on other domains of mental health after accounting for co-occurring depression and other confounding. Age at menarche affected adolescent depression, but not other domains of mental health. Our findings suggest that earlier age at menarche is linked to problems in specific domains rather than adolescent mental health in general.

**Keywords** Age at menarche, Depression, Adolescent mental health, Mendelian randomisation, MoBa

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## Testing for Polygenic Adaptation in Complex Human Traits Using Cross-Ancestry and Within-Sibling GWAS Results

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**Abstract**
Genome-wide association studies (GWAS) have identified multiple genetic variants associated with externalizing behaviors and other complex traits. Researchers are interested in applying GWAS results to investigate polygenic selection, i.e., evolutionary changes in a phenotype that arise through slight changes in allele frequencies across many trait-associated sites. However, GWAS results can also reflect various biases (e.g. population stratification, assortative mating) which confound signals of polygenic selection (Sohail et al. 2019). Two approaches to this problem have been proposed: (1) using

GWAS results from samples that differ in ancestry than the samples in which selection is being tested, and (2) using results from within-sibling GWAS (Le et al. 2022). Here, we investigate the comparability of results from tests of polygenic selection using these two methods. We use **public and restricted (23andMe, Inc.)** GWAS summary statistics from six phenotypes, including anthropometric traits (height, BMI), externalizing-related substance use behaviors (cigarettes per day, problematic drinking), and differences in reproductive lifespan (age at menarche, age at menopause), estimated in three populations (Japanese ancestry individuals, African ancestry individuals, and European siblings). These summary statistics will be used to test for polygenic selection in the Allen Ancient DNA Resource (Mallick et al. 2023), comprising 1291 individuals from Europe who lived up to 10,000 years ago. Results will be informative regarding whether researchers can triangulate results across complementary methods and samples to investigate polygenic selection in complex traits.

**Keywords** Polygenic selection, Ancient DNA, Genome-wide association study, Externalizing, Complex traits

## Genetic and Environmental Influences on Brain Age Measures at Age 29 and Their Associations with Early Life Cognition

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**Abstract**

Predicted BrainAGE difference (PBAD), an aging biomarker, quantifies the gap between one's chronological age and predicted age based on structural brain indices. Despite recognition that differences in rates of biological aging depend on activities and experiences shaped partly by genetics and genetically influenced environmental influences, most studies of BrainAGE have focused on cross-sectional estimates in either developing adolescents or late–middle-aged to elderly (age 60+) samples. Almost no work has examined BrainAGE in established adulthood, prior to the manifestation of significant age-related declines. In 605 young adult twins, we examined genetic and environmental influences on PBADs based on brain imaging at age 29 years. PBADs were significantly heritable for surface area (A = 58%) and subcortical volume (A = 48%), but not cortical thickness (A = 4%), and area and subcortical PBADs were only moderately genetically correlated (rG = 0.46). Shared environmental influences were largest but nonsignificant for cortical thickness PBAD (C = 41%) vs. area and subcortical volume PBADs (C = 0–15%). Controlling for intracranial volume, PBADs, particularly subcortical volume PBAD, were significantly related to general cognitive ability measured as early as age 7 (partial rs = − 0.12 to − 0.17), and the relationships between subcortical PBAD and ages 29 and 16 cognition were explained by significant genetic correlations (rGs = − 0.39 to − 0.43). Results suggest that prior to midlife, adult PBAD estimates for different brain indices capture distinct genetic and environmental variance and relate to neurodevelopmental factors such as childhood cognitive ability, consistent with the hypothesis that individual differences in brain aging have roots in earlier life when cognitive reserves are formed.

**Keywords** Brain age, Neuroimaging, Aging, Twins

**Grants**

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## Externalizing Behavior Predicts the Willingness to Donate DNA for Science

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**Abstract**

The accumulation of large genetic data is crucial for the scientific advancement of genetic research and precision medicine, but various participation biases threaten the validity of genetic research data. To better understand the decision to participate and its relationship with preferences and socioeconomic characteristics, we studied the stated willingness to donate DNA for science in a representative Dutch household panel. There were two randomized conditions, varying (i) the information material, and (ii) the intended financial incentive. The first condition had little effect, suggesting insensitivity to the information material. Proposing a higher incentive had a significant but modest effect, suggesting that higher incentives are not cost-effective. Preregistered association analyses across behavioral and psychological domains show that risk willingness strongly predicted the willingness to participate, which reverberates with stated reasons not to participate that concentrated on personal risks, e.g., privacy violations and exploitation. There was also some evidence of association with other externalizing traits, such as smoking, drug use, and heavy drinking. Theoretically, the willingness to take risks, impulsivity, reward sensitivity, or traits of self-control all have plausible roles in the decision to participate. The study replicated previous findings, e.g., with health, some psychological scales (trust and reciprocity), and confidence in science or societal institutions. However, the study found conflicting evidence for some previously reported associations, possibly because of our representative sample. We conclude by proposing strategies to encourage participation and discuss ways to prevent selection bias on certain personal characteristics.

The article is currently under review for journal publication.

**Keywords** Data collection, Genetic research data, Non-response bias, Participation bias, Selection bias, Volunteer bias

**Grants**

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## Crops and Cradles: Analyzing the Impact of Agricultural Transition on Reproductive Preference

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**Abstract**
This study investigates the impact of the Neolithic Revolution on reproductive preferences in modern and archaic populations. I test the hypothesis that the origin of agriculture influenced the evolution of reproductive ability and demographic trajectories across different countries. Utilizing recent advancements in human genetics and the Allen Ancient DNA Resource (AADR), I construct an individual-level polygenetic score for age at first birth (AFB) as a proxy for reproductive capability. It empirically assesses the relationship between genetically determined reproductive predispositions and the transition to agriculture among both ancient (110,450 BP to 60 BP; N = 10,076) and contemporary (N = 10,504) populations globally. I find robust evidence supporting the hypothesis that the shift to agricultural lifestyles has positively influenced reproductive abilities in humans, observable across 151 countries. This phenomenon is attributed to natural selection processes. Among diverse agricultural practices, paddy rice farming emerges as the most significant predictor of variations in reproductive preferences worldwide. I further extend the analysis to contemporary China using population-based 1990 census data (N = 3,562,834), in which paddy rice farming history not only helps to explain the total number of children, but also the sex ratio at the household level, presenting comprehensive insights into the legacy of agricultural practices on human reproductive preference. The results underscore the profound influence of agricultural transition on demographic trends, contributing to a deeper understanding of the interplay between historical societal shifts and genetic evolution in shaping human fertility patterns.

**Keywords** Agriculture, Reproduction, Neolithic revolution, Polygenic score, Sex ratio

**Grants**
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## Genomic SEM Factors of Substance Use Related Psychopathology, Internalizing, and Externalizing Predicts Adolescent Drug Use in the ABCD Sample

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**Abstract**
There are several pathways that confer risk for developing a substance use disorder. Among those pathways include problems in externalizing (aggressive behavior, rule breaking, hyperactivity, inattention and impulsivity, and conduct problems) and internalizing behaviors (anxiety, depression, withdrawal, somatic complaints, and social isolation). Adolescence serves as a unique developmental period in which externalizing and internalizing traits play a significant role in the early initiation and escalation to problematic substance use. Recent advances in GWAS have enabled the identification of genetic factors common and specific to substance use, externalizing, and internalizing behaviors, but models of their predictive utility are limited. Using data from the Adolescent Brain Cognitive Development (ABCD) study (release 5.0), we evaluated whether genetic factors underlying substance use, internalizing, and externalizing traits were associated with substance use in white adolescents < 16 years of age. Three polygenic scores (PGSs) were developed using PRS-CS based on summary statistics from three genomicSEM factors. These factors represented genomic influences shared across internalizing, externalizing, and substance use behaviors (i.e., SU-Psychopathology), as well as those specific to internalizing and externalizing traits after accounting for what was shared across the three domains (INT-specific, and EXT-specific respectively). In a model including all 3 PGSs, the SU-Psychopathology PGS was associated with alcohol use in adolescents < 16 years of age (but not nicotine or marijuana use), as well as externalizing traits. Finally, the EXT-specific PGSs were significantly associated with externalizing traits (as measured by the CBCL) while INT-specific PGSs were associated with both externalizing as well as internalizing traits.

**Keywords** Adolescence, Polygenic scores, Substance use, Externalizing, Internalizing

**Grants**
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## Multivariate SEM–PGS Model: Using Polygenic Scores to Investigate Cross-Trait Genetic Nurture and Assortative Mating

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**Abstract**
Recent research has uncovered genetic nurture effects across a wide range of human behaviors (Kong et al. 2018), with a limitation of primarily capturing same-trait genetic nurture effects. However, cross-trait genetic nurture also exists as a consequence of cross-trait vertical transmission. The current study proposed multivariate SEM–PGS, a novel tool extending the univariate SEM–PGS model to a multivariate framework (Balbona et al. 2021; Kim et al. 2021) to account for familial similarities arising from cross-trait genetic nurture. Multivariate SEM–PGS uses polygenic scores in an SEM model, offering more insights into the gene–environment interplay in human development. We formulated mathematical expectations for estimates of direct genetic effects, same-trait and cross-trait genetic nurture, vertical transmission, and assortative mating effect. The validity of our mathematical derivations is supported by Monte Carlo simulations. By fitting the model with simulated data, we will demonstrate the precision of the estimates under varying effect sizes and sample sizes. Additionally, we will evaluate potential biases of the estimates introduced when the model's underlying assumptions are violated. Further promising model modifications to capture a broader spectrum of familial gene–environment interplay will be discussed.

**Keywords** Genetic nurture, Vertical transmission, Structural equation modelling, Polygenic scores, Assortative mating

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## Changes in BMI and Cognitive Functioning from Childhood to Midlife Predict Midlife Physical Function: The Louisville Twin Study

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**Abstract**

Physical decline in late life is the culmination of processes already underway in mid-adulthood and even childhood. Although age changes may not become apparent until after age 65, many risk factors for late-life decline are already well-established in midlife and before. Earlier identification of individuals at risk for accelerated physical decline provides opportunities for earlier intervention. The relationship of changes in measures of body mass (BMI) and general cognitive functioning (GCF) from age 15 years to midlife with midlife physical functioning were examined and genetic and environmental contributions to the associations were investigated. Participants were 184 twins from the Louisville Twin Study who participated in the ongoing midlife assessment (mean age = 51.75, SD = 7.1). The Functional Aging Index (FAI) uses functional biomarkers accessible from most epidemiological studies of aging to capture age changes in functioning of four body systems: sensory (vision and hearing), pulmonary (lung function), strength (grip strength), and movement/balance (gait). Comparisons of data at age 15 and in midlife indicated significantly greater increases in BMI and decreases in GCF for the highest quartile of FAI. Application of moderator models indicated that higher risk at both age 15 and midlife (higher BMI or lower GCF) was associated with greater genetic variance in FAI. Results suggest that interventions in childhood may affect midlife and later functioning, and that both early physical and cognitive functioning play a role in midlife physical functioning. In other words, early intervention in one domain may affect outcomes across domains of aging.

**Keywords** BMI, Cognitive functioning, Functional aging, Midlife, Moderation

**Grants**

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## External Validation of a Population-Based Multimodal Machine Learning Prediction Model for Wellbeing in a Depression and Elderly Cohort

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**Abstract**

We recently developed a machine learning prediction model for adult wellbeing based on psychosocial, genetic, and external exposome data in the Netherlands Twin Register. We included survey-based psychosocial features (e.g., personality, lifestyle, socio-economic status), 60 polygenic scores, and neighborhood level exposures (e.g., air pollution, urbanicity) linked to participants' postal codes. The psychosocial predictors were highly predictive of wellbeing ($R^2$ = 0.63), while the genetic and environmental exposures were not, independently nor incrementally.

In this study, we will externally validate our prediction models in two other Dutch cohorts. The Netherlands Study of Depression and Anxiety (NESDA) is a longitudinal, multi-site, case–control cohort study of 3348 baseline participants (70% cases) followed from 2004 in 7 waves. The Longitudinal Aging Study Amsterdam (LASA) is a longitudinal cohort study of older adults (initial ages 55–84) followed from 1992 in > 10 waves.

We use wave 1 (2004–2007), 5 (2010–2013), and 7 (2019–2023) of NESDA, wave F (2005–2006), H (2011–2012), and J (2018–2019) of LASA, and wave 6 (2002–2004), 8 (2009–2012), and 14 (2019–2022) of NTR. Outcomes will be happiness (NESDA) and life satisfaction (LASA and NTR). Following our prior work, we create stacked ensemble models by training three machine learning algorithms (extreme gradient boosting, support vector machines, and random forest), the predictions of which feed into a final meta-model. Both cross-sectional and longitudinal models—predicting over 3 to 20 years—will be trained on each data modality separately and in combination. Performance will be assessed by $R^2$ and root mean squared error (RMSE) values.

**Keywords** Wellbeing, Exposome, Polygenic scores, Machine learning prediction, External validation

**Grants**

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## The Genetic Specificity of Specific Cognitive Abilities After Controlling for General Cognitive Ability

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**Abstract**
Diverse cognitive abilities correlate about 0.30 phenotypically and about 0.60 genetically. Their phenotypic overlap defines general cognitive ability (g), driven largely by genetic overlap. Consequently, much of our understanding of the genetic landscape of specific cognitive abilities (SCA) likely reflects g rather than the SCA themselves. Removing this g-associated genetic variance will sharpen research on SCA. Here, we use Genomic Structural Equation Modelling (Genomic SEM) to remove the shared genetic variance among 12 diverse SCAs that capture verbal and nonverbal cognitive domains. We applied Genomic SEM to summary statistics from the largest genome-wide association studies of verbal SCA (GenLang Consortium, 5 tests) and nonverbal SCA (UK Biobank, 7 tests) to chart the genetic landscape of the 12 SCA independent of g (SCA.g) as compared to uncorrected SCA. We found that SNP heritabilities were nearly as high for SCA.g as for SCA: the average SNP heritability was 0.16 (SE = 0.02) for SCA and 0.13 (SE = 0.02) for SCA.g. Despite this, we found that the genetic landscape of SCA is transformed after controlling for genomic g. The matrix of positive genetic correlations for SCA (average genetic correlation of 0.45 among our 12 tests) disappeared for SCA.g (average genetic correlation of − 0.07) and included some strong negative correlations; for instance, Memory and Word (− 0.72), Fluid and Symbol (− 0.72), and Tower and Spelling (− 0.79). The summary statistics of SCA.g can be used by researchers to create polygenic scores that focus on the specificity of specific cognitive abilities.

**Keywords** Specific cognitive ability, General cognitive ability, Genomic-SEM, Polygenic scores

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## Migration, Assortative Mating, and Educational Attainment

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**Abstract**
Assortative mating on educational attainment can arise through phenotypic assortment and social homogamy. What happens when individuals change their levels of social homogamy through migration? As higher educated individuals are more likely to migrate for work or school, does this enhance social homogamy and lead to a stronger degree of assortment? This study investigates whether migration influences assortative mating on educational attainment.

We investigated investigated this in 33,972 spouse pairs within the UK Biobank. Migration was quantified by migration distance, a binary variable based on a distance threshold, and a binary variable indicating whether individuals moved out of their census region. Multiple regression analyses revealed that migration distance significantly predicted the difference in years of education between spouses. Moreover, spouse correlations in years of education were significantly higher among couples where both spouses migrated compared to those where neither did. Sensitivity checks confirmed these findings, suggesting that migration is associated with an increase in assortative mating on educational attainment.

These results align with previous research on assortative mating and extend the literature by highlighting migration's role. This has implications for genetics research, as both migration and assortative mating can bias genetic effect estimates, and their interaction may introduce additional biases. Furthermore, this study underscores broader social implications, as assortative mating may contribute to increasing social inequality over generations.

**Keywords** Migration, Assortative mating, Educational attainment, Polygenic scores

## How Schools and Residential Areas Moderate the Effects of Polygenic Scores for Complex Behavioral Traits on Educational Performance in Norway

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**Abstract**
A previous study has shown that schools moderate the association between students' ADHD polygenic index (PGI) and educational achievement. However, it remains interesting whether the interactions occur on other heritable mental disorders (i.e. autism and depression) as well as personality traits at different levels. In this study, we are engaged in disentangling how schools and students' residential areas interact with the relationship between mental disorders, personality traits, and educational performance. Using the data (n = 25,000) from genotyped trios in the Norwegian Mother, Father, and Child Cohort Study (MoBa) and the national administrative registry, we quantify students' PGI for each disorder and their educational performance in GPA. School identifiers and geographical codes at the level of municipality, district, and neighborhood are obtained from the Norwegian Statistics Bureau database. Multilevel models with random intercepts and slopes are applied to analyze the mental disorder effects on educational performance across schools and residential

areas. To exclude the effect from the parents' selections of schools and residential areas, we adjust for the parents' PGI. Results from this study may contribute to an in-depth understanding of the differences between individuals in the genetical effect of mental disorders and personality traits on educational performance due to the various environmental contexts.

**Keywords** Schools, Gene–environment interaction, Polygenic index, Within family

**Grants**
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## Accelerated Biological Aging During the COVID-19 Pandemic: A TwinLife Epigenetic Change Satellite Study

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**Abstract**
Socioeconomic hardship and psychosocial stress are associated with accelerated biological aging. As a global stressor, the Coronavirus Disease 2019 (COVID-19) pandemic was accompanied by severe socio-economic and health-related burden. However, the extent to which the perceived hardship of the COVID-19 pandemic affected aging-related epigenetic regulation remains unknown. We investigated whether accelerated epigenetic aging (AgeAccel) changed during the COVID-19 pandemic, whether this change can, in part, be explained by COVID-19-related perceived threat (e.g., for education/work) and burden (e.g., social isolation), and whether the association with change was dependent on sex or socioeconomic status (SES). We used saliva DNA samples collected pre-pandemic in 2018–2020 and follow-up in Sep–Dec 2021 from 533 monozygotic and 513 dizygotic twins of the TwinLife Epigenetic Change Satellite (TECS) project (360 children, 393 adolescents, 293 adults). Epigenetic aging was computed using the Pediatric-Buccal-Epigenetic (PedBE) clock, the Horvath clock, the GrimAge2 clock, and DunedinPACE. Analyses were carried out using the coxme R-package for full sample analyses and the lme4 R-package for monozygotic co-twin control analyses with raw clock estimate as dependent variable, and age at sampling, timepoint, cell type, and sex as covariates. Epigenetic aging increased in PedBE, GrimAge, and DunedinPACE but was not associated with COVID-19-related threat or burden in the full sample or co-twin control analyses. SES was negatively associated with GrimAge AgeAccel and DunedinPACE. Adolescent females showed faster DunedinPACE than males. Further analyses will include examining the effect of COVID-19 experiences in vulnerable subgroups based on genetic and epigenetic sensitivity to glucocorticoid (stress hormone) exposure.

**Keywords** Biological aging, Stress sensitivity, Epigenetic clocks, COVID-19, Stressful life events

## The Associations Between Educational Experiences and Mental Health from Childhood to Young Adulthood

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**Abstract**
Lifetime trajectories of psychopathology tend to start early in development, with most cases emerging by adolescence. The Children's Society, a UK charity, found that more children report unhappiness related to school than other aspects of life, including friends and family. Despite this, studies exploring the relationship between the school environment and young people's mental health remain scarce. Our previous work, using the Twins Early Development Study (TEDS; N = ~ 6500, ages 7–21), showed that subjective parent-, teacher-, and twin-reported educational experiences (e.g., classroom environment) were substantially heritable (~ 50%) and associated with mental health additively, explaining up to 49% of the variance. These associations remained after controlling for early psychopathology symptoms and adult psychiatric polygenic scores.

Here, we extend our investigation to the objective educational experiences that are largely shared between twins (e.g. school quality) using TEDS (N = 6398, ages 7–26) and UK-wide registers (National Pupil Database, Ofsted). Using a cross-lagged panel model, we will assess longitudinal associations between objective experiences and mental health. Multiple regression and prediction modelling (elastic net) will also explore how objective and subjective educational experiences jointly predict mental health, contemporaneously, and over time. In both analyses, we will control for early mental health, 30 adult psychiatric polygenic scores, and socioeconomic status. The plan has been pre-registered (https://osf.io/3c5rk/), and analyses are ongoing.

We aim to clarify how both objective and subjective educational experiences shape mental health trajectories in young people. These insights could have important implications for the strategies used to prioritise student wellbeing in schools.

**Keywords** Schools, Mental health, Gene–environment interplay

**Grants**
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(FP7/2007–2013), Grant Agreement No. 602768. RF is supported by a Royal Holloway, University of London College Studentship.

## The Impact of Climate Conditions on the Aetiology of Psychological Distress: Insights from a Genetically Informative Design

**Juan J. Madrid-Valero[1], Federico J. Blanco[1], José M. Martínez-Selva[1], Juan F. Sánchez-Romera[1], Juan R. Ordoñana[1]**

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**Abstract**
Reports of seasonal variation in psychological distress are common in the literature. Seasonal variation is overtly related to environmental factors, such as daylight hours and temperature. Hence, many studies have tried to analyse the possible association between those variables and psychological distress, but results have been inconclusive. While some studies have found a worsening of depressive symptoms in winter, others have found this pattern during summer. Geographical, cultural, and methodological variations could be responsible, in part, for these inconsistencies. To shed light over this question, this study aimed to examine the effect of temperature and daylight hours (photoperiod) on psychological distress in a Mediterranean adult population. The sample comprised 2143 participants from the Murcia Twin Registry (mean age 53.66; SD 7.4). Psychological distress was measured using the Spanish version of the anxiety/depression subscale of the EQ-5D questionnaire. Polygenic Scores (PGS) for depression were calculated using LDpred2. Data on temperature and daylight hours of the day of the interview were obtained from the Spanish Meteorological Agency. In order to test the association between environmental factors and psychological distress, a series of regression and moderation models were fitted while including genetic vulnerability for depression and other related variables. PGS for depression ($P = 0.02$) and temperature ($P < 0.001$) were both significant independent predictors of psychological distress. These variables did not show a significant interaction but were affected by other variables such as sleep quality. There is a strong association between environmental temperature and psychological distress but there is no evidence of a causal path between them moderated by genetic vulnerability.

**Keywords** Depression, Environmental temperature, Photoperiod, Psychological distress

**Grants**
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## Negative Life Events and Epigenetic Ageing: a Study in the Netherlands Twin Register

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**Abstract**
Understanding the long-term impact of negative life events, especially on health outcomes and premature mortality, is vital. This study aimed to enrich the existing literature by examining a broad spectrum of negative life events, ranging from crime victimization and financial troubles to losing a loved one in 1808 adults (average age 27.69 years, range 17.60–79.60, 29.70% male) from the Netherlands Twin Register. Epigenetic aging was assessed using five biomarkers: Hannum, Horvath, PhenoAge, GrimAge, and DunedinPACE, with DNA methylation measured using the Illumina 450k array in blood. Two primary analyses were conducted: a population-level analysis using generalized estimating equations and a discordant monozygotic twin design using fixed effects regression ($N = 274$ pairs), applying Bonferroni correction for multiple testing (adjusted alpha 0.002). Our findings indicated a significant association between experiencing a higher number of life events and accelerated epigenetic aging, with the GrimAge and DunedinPACE biomarkers. Financial adversities, sexual crimes, and job loss emerged as significant predictors of biological aging. The association between the number of life events and financial problems and epigenetic age acceleration measured by the GrimAge biomarker persisted, even after controlling for BMI, smoking, and white blood cell counts. However, no strong relationship was found in the MZ twin analysis after controlling for familial factors. This study highlights the complex interplay between environmental stressors and biological aging, underlining the importance of considering both genetic and environmental factors in studies of epigenetic aging and emphasizing the need for more detailed, longitudinal research to further unravel these relationships.

**Keywords** Discordant twin analyses, DNA methylation, Epigenetic biomarkers, Biological aging, Stressful life events

**Grants**
This study was supported by the Dutch Victim Support Fund. Data collection in the Netherlands Twin Register was funded by the Netherlands Organization for Scientific Research (NWO), Grant Numbers 31160008, 904-61-193 and 985-10-002. The Netherlands Organization for Scientific Research (NWO): Biobanking and Biomolecular Research Infrastructure (BBMRI-NL, NWO 184.033.111) and the BBRMI-NL-financed BIOS Consortium (NWO 184.021.007), NWO Large Scale infrastructures X-Omics (184.034.019), Genotype/Phenotype Database for Behaviour Genetic and Genetic Epidemiological Studies (ZonMw Middelgroot 911-09-032); Netherlands Twin Registry Repository: Researching the Interplay Between Genome and Environment (NWO-Groot 480-15-001/674); epigenetic data were generated at the Human Genomics Facility (HuGe-F) at ErasmusMC Rotterdam.

## The Effects of Housing-, Health- and Crime-Related Life Events on Social Withdrawal in Young Adults: A Longitudinal Co-twin Control Study

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**Abstract**
Most research has focused on the detrimental consequences of social withdrawal for life outcomes, with little insight into how life experiences affect social withdrawal. Because social withdrawal reportedly decreases during the transition from late adolescence to emerging adulthood, these life phases of self-focus and exploration offer the opportunity to study the causes of individual changes in social withdrawal. The present study investigates the effect of life events on individual differences in social withdrawal in early adulthood. Analyses will be based on data of up to $n = 2600$ and 3500 twins (aged 16–18 at T1 and 18–26 at T2) from the Netherlands Twin Register and involve a longitudinal co-twin control design. Two measurement points (i.e., before and after the life event) will be considered to control for effects of social withdrawal on the likelihood of experiencing the life event. Social withdrawal will be assessed via five items of the withdrawn(/depressed) scale of the Youth Self-Report, respectively Adult Self-Report. The living situation is categorically measured, differentiating diverse forms of housing circumstances. The considered life events are (a) leaving the parental home, (b) a serious illness or injury, (c) a traffic accident, (d) theft, burglary, or vandalism, (e) a violent crime (robbery, physical violence), and (f) a sexual crime (rape, sexual assault), with (b) to (f) having experienced less than 1 year ago. We will discuss the implication for interventions that aim to counteract social withdrawal.

**Keywords** Co-twin control design, Life experiences, Social withdrawal, Young adulthood

## Investigating Genetic Nurture Effects in Internalizing Disorders

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**Abstract**
Internalizing disorders (IDs), marked by anxiety and depression, run in families due to the direct inheritance of genetic risk factors from parents to offspring. Beyond direct transmission, parental genes can also indirectly impact offspring through the rearing environment, which is known as genetic nurture (Kong et al. 2018). The role of this pathway in IDs remains unclear. This study investigates the potential impact of genetic nurture effects in IDs, with a focus on current and lifetime diagnoses and symptoms of major depression, dysthymia, generalized anxiety disorder, social phobia, agoraphobia, and panic disorder. We computed polygenic scores (PGS) based on the largest major depression GWAS for parental haplotypes that were transmitted (T) and non-transmitted (NT) to offspring. We used mixed-effect models to examine the associations of both PGS-T and PGS-NT with outcomes in offspring. A positive association of the PGS-NT with offspring outcomes would imply an environmentally mediated effect of parental genes. Our preliminary analyses in a large sample (N = 4435–17,198) of Dutch adults with at least one genotyped parent resulted in significant associations between PGS-T and IDs in offspring, as expected. The absence of significant associations between PGS-NT and internalizing outcomes in offspring tentatively suggests a limited role of genetic nurture. Disentangling the interplay between genetic and environmental risk factors is valuable for understanding underlying mechanisms and identifying targets for prevention of IDs.

**Keywords** Internalizing disorders, Anxiety, Depression, Genetic nurture, Indirect genetic effects

**Reference**

Kong A, Thorleifsson G, Frigge ML et al (2017) The nature of nurture: effects of parental genotypes. Science 359(6374):424–428

## Testing for Treatment Effect Heterogeneity: Educational Reform, Genetic Endowments, and Family Background

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**Abstract**
We study the heterogeneous effects of a Swedish educational reform that increased compulsory schooling from 7 to 9 years. Following a pre-registered Analysis Plan, we examine how the reform differentially affected 20 outcomes in various domains (educational, labor market, reproductive, substance use, and health) for individuals with different genetic endowments. To estimate the effects of the reform, we leverage the gradual roll-out of the reform with a differences-in-differences estimator that compares municipalities and birth cohorts before and after the reform. Among 105 pre-registered tests of heterogeneous effects of the reform as a function of genetic endowments (across outcomes, sexes, and family SES background), we find two significant interactions after correction for multiple hypothesis testing. One of these implies that, among females from high-SES families, the reform had a relatively larger effect on earnings for those with a lower genetic propensity for educational attainment. Our results suggest that sizeable heterogeneous effects of educational reforms may not be ubiquitous but do exist, and highlight how careful study designs with preregistration can help uncover them.

**Keywords** Treatment effect of schooling, Education reforms, Gene–environment interactions

## Socioeconomic Position and Somatic Comorbidity in Mental Disorders

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**Abstract**
**Background**
Individuals with mental disorders have been shown to be at increased risk for somatic non-communicable diseases (NCD), such as cardiovascular diseases (CVDs), cancers, chronic obstructive pulmonary disease (COPD) and diabetes. Low socioeconomic position (SEP) has been associated with increased risk of mental disorders and somatic NCDs. However, it is unknown how SEP modifies the associations between mental disorders and somatic NCDs. In this study, we estimated how associations between mental disorders and somatic NCDs vary by SEP indicators.
**Methods**
Data on all Norwegians residents from 2008 to 2010 were linked to nationwide registers on primary and secondary health care, dispensed prescriptions, causes of death, and sociodemographic variables. We assessed four categories of mental disorders (psychotic, bipolar, depressive and anxiety disorders) and four categories of somatic NCDs diagnosed during follow-up 2010–2020. Educational attainment and household income for index persons and their parents were used as SEP indicators. Cox regressions were used to calculate hazard ratios for pairs of mental disorders and somatic NCDs, with and without SEP adjustment and stratification.
**Results**
All groups of mental disorders were associated with increased risk of CVD, COPD and diabetes, but not with cancers. Low SEP measures were associated with mental disorders and somatic NCDs but did not exacerbate this association.
**Conclusion**
We found an increased risk of most somatic NCDs in individuals with mental disorders, but SEP measures did not modify this risk. These patterns of associations will be followed by quantitative genetic analyses using the pedigree structure of the entire Norwegian population.

**Keywords** Comorbidity, Mental disorders, Non-communicable diseases, Socioeconomic position

## Genomic and Non-genomic Contributions to Children's Cognition, Language, and Social Skills: Evidence for Independent Phenotypic Domains with Strong Gene–Environment Correlations

**Fenja Schlag[1], Lucía de Hoyos[1], Ellen Verhoef[1], Simone van den Bedem[1], Brad Verhulst[2], Simon E. Fisher[1,3], Beate St Pourcain[1,3,4]**

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**Abstract**
Cognitive, language and social abilities are complex phenotypes crucially shaping children's development. However, the underlying aetiological factors structures, including genetic and environmental influences, gene–environment correlations and phenotypic crosslinks, are only partially understood.

Here, we model multivariate genomic and residual factors and their interplay ($r_{GE}$), contributing to cognition and language performance and social behavioural/communication difficulties (8–13 years) assessed in two independent, population-based cohorts: the UK ALSPAC study ($N \leq 6543$) and the US ABCD study ($N \leq 4412$). We model genomic covariance in unrelated children adopting a data-driven genetic–relationship–matrix structural equation modelling (GRM-SEM) approach to identify the best-fitting genetic and residual structure, including $r_{GE}$.

In ALSPAC, an independent pathway model fitted the data best. We identified two independent genomic and two independent residual factors. The first genomic and residual factors captured each cognitive/language ability, with the largest loading for verbal IQ. The second genomic and residual factors explained each social ability, with the largest loading for social communication skills. Across these phenotypic domains, we identified domain-specific $r_{GE}$ with different effect directions: cognition/language $r_{GE} = 0.89$ (SE = 0.18), social $r_{GE} = -0.62$ (SE = 0.17). Within ABCD, genetic contributions to the selected phenotypes were low, limiting the interpretation of shared structures. However, we confirmed similarities between the residual structure in ABCD and the genetic and residual structures in ALSPAC by fitting a hybrid Cholesky (genetic part)/independent pathway (residual part) model.

By mid-childhood, cognition/language phenotypes versus social difficulties are aetiologically distinct phenotypes, as shown by domain-specific genomic and non-genomic contributions and gene–environment correlations, at the population level.

**Keywords** Cognition, Language, Social behaviour, GRM-SEM, Gene–environment correlation

**Grants**
Max Planck Society.

## Trio Genome-Wide Association Studies

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**Abstract**
**Background**
There is increasing evidence that population-based GWAS estimates for some phenotypes capture associations relating to population stratification, assortative mating and indirect genetic effects. Recent studies employing within-family designs have highlighted substantial attenuation in SNP–phenotype association estimates, highlighting the importance of accounting for different sources of association.
**Objectives**
This study aims to investigate the extent to which differences between SNP–phenotype associations derived from population-based and within-family designs account for familial and demographic factors.
**Methods**
We will conduct a consortium-based parent–offspring trio GWAS to compare SNP–phenotype association estimates against population-based and within-sibship estimates. Our primary analysis will test for shrinkage in SNP–offspring associations, controlling for parental genotype across genome-wide significant SNPs identified in previous GWAS. We will employ a joint shrinkage test across all known SNPs and validate findings through genetic correlation analyses between within-family and population-based estimates, focusing on direct and indirect genetic effects. We will meta-analyse results across all studies.
**Conclusion**
By disentangling the direct and indirect genetic effects on phenotypic traits, this study aims to provide a clearer understanding of the genetic architecture underlying complex traits. Our findings are expected to refine the interpretation of GWAS estimates, promoting a more

nuanced approach to genetic research in the context of familial and demographic dynamics.

**Keywords** GWAS, Population stratification, Indirect genetic effects, Within-family GWAS, Parent–offspring trios

## Self-Reported Health, Basic Psychological Needs and Life Satisfaction: A Twin Study

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**Abstract**
It has been found in previous studies that self-reported health is positively associated with life satisfaction in 32 European countries, including Croatia (Kööts-Ausmees and Realo 2015), and that satisfaction of basic psychological needs is positively associated with life satisfaction for both emerging and middle-aged adults in Croatia (Butkovic et al. 2020). Previous studies have also shown that self-reported health (Silventoinen et al. 2007), psychological needs (Bratko et al. 2022) and life satisfaction (Stubbe et al. 2005) are heritable, but genetic and environmental contributions to their associations have not yet been investigated. A sample of 334 Croatian twin pairs, with 103 monozygotic (MZ) and 231 dizygotic (DZ) twin pairs (85 male pairs, 136 female pairs, 113 opposite-sex pairs, *M*age = 18.63, *SD*age = 2.31) filled in Satisfaction with Life Scale (Diener et al. 1985), Basic Psychological Need Satisfaction Scale (Deci and Ryan 2000; Gagné 2003) and single item assessing self-reported health. As in previous studies, self-reported health, basic psychological needs and life satisfaction were heritable. Using Cholesky decomposition, significant genetic and environmental overlap was found between self-reported health, basic psychological needs and life satisfaction, with 58% of the genetic and 19% of the environmental influences to individual differences in life satisfaction shared with self-reported health and basic psychological needs.

**Keywords** Self-reported health, Basic psychological needs, Life satisfaction, Twin study

## The World as I See It: Genetic and Environmental Influences on World Beliefs and Their Relationship with Well-Being, Loneliness, and Social Isolation

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**Abstract**
Positive beliefs about the world, known as primals (e.g., believing that the world is a safe place) are associated with numerous psychological and social outcomes, for instance with personality traits, higher life satisfaction, and more social connections. We demonstrated phenotypically that primals and the big five personality traits are distinct, yet related, concepts with low to moderate absolute correlations ranging from 0.04 to 0.34. Moreover, we showed that genetic factors contribute significantly to people's beliefs about the world with heritability estimates ranging between 20 and 48% (Perizonius et al. in press). Here, we investigate genetic and environmental influences on primals, and their associations with well-being, loneliness, and social isolation, utilizing bivariate twin models in a large Swedish twin sample ($N \sim 9000$). Replicating previous work, we found positive world beliefs to phenotypically relate to higher well-being, lower feelings of loneliness, and less social isolation ($r = 0.11$ to $0.40$, $-0.09$ to $-0.22$, $-0.07$ to $-0.27$, respectively). Co-twin control designs indicated familial confounding in these associations, though indications for potential causal relations were additionally shown. Bivariate twin modeling showed significant moderate genetic correlations between primals and well-being, loneliness, or social isolation, implying that overlapping genes play a role in the associations. To conclude, we found genetic influences on positive beliefs about the world and on their relationship with well-being, loneliness, or social isolation. Importantly, these relationships are not fully explained by familial factors, suggesting potential causal relations.

**Keywords** Positive psychology, Heritability, Personality, Genetics, Well-being, Social isolation, Loneliness

**Grants**
The Project was partly supported by Templeton Religion Trust (TRT0298). Furthermore, we would like to acknowledge Jer(emy) Clifton and Erik Pettersson for helping with the translations of the original PI-99, and Klaus Frieler for his consultation regarding the factor analysis statistics. In the end, we would like to express our gratitude towards all of the Swedish twins that participated and towards the Swedish Twin Registry for providing the data.

## Impact of Internal Migrations on the Genetic Structure in Estonia

**Ivan A. Kuznetsov[1], Mait Metspalu[1], Uku Vainik[1,2,3], Luca Pagani[1,4], Francesco Montinaro[1,5], Vasili Pankratov[1]**

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**Abstract**
Past demographic events and isolation-by-distance are considered to be the main sources of spatial genetic structure in human populations. However, the role of recent fine-scale migrations in changing this structure remains understudied. In this study, we utilize a sample of more than 200,000 individuals to explore the genetic correlates and consequences of contemporary migration in Estonia.

We use the Estonian Biobank, which is a volunteer-based cohort including the genetic and phenotypic data on approximately 20% of the Estonian resident adult population. Polygenic scores (PGS) for 169 and 24 diverse traits were calculated using population-based and within-sibship GWAS summary statistics, correspondingly. Individual migration events were tracked with the information on the regions of birth and residence.

Despite the homogenizing effect of migration on the genome-wide genetic structure, it intensifies inter-regional differences in PGS for certain traits, derived from both population and within-sibship GWA studies. The strongest effect is observed for educational attainment

(EA). Spatial differentiation of PGS for EA is mainly driven by the migration of individuals with relatively high PGS to the two largest cities from the rest of the country starting not later than the middle of the twentieth century. PGS for EA is not only associated with migration to the cities but also with migration to the more distant of the two cities or between the cities.

Our findings contribute to the understanding of the interplay between genetics and geographic mobility, offering a more profound insight into the contemporary processes of population structure change.

**Keywords** Migration behaviour, Population structure, Educational attainment, Urbanization

**Consortia**

Estonian Biobank Research Team: Andres Metspalu, Lili Milani, Tõnu Esko, Reedik Mägi, Mari Nelis, Georgi Hudjashov.

## Does the 'word gap' Cause the Intergenerational Transmission of Education?

**Sophie von Stumm[1], Anna Brown[1], Alexandra Starr[1]**

[1]Education, University of York, York, United Kingdom

**Abstract**

The 'word gap', which refers to differences in children's early life exposure to language learning opportunities, is thought to be a key driver of the intergenerational transmission of educational inequality. To discern causal and non-causal pathways through which mothers' lexicon may mediate the association between mothers' and children's educational attainment, we analysed data from E-Risk, a genetically sensitive, longitudinal cohort study of 1116 families from England and Wales.

Mothers' lexicon was quantified from interview recordings (N = 894). Twins' (age 18) and mothers' (at twins' age 5) highest educational attainment were recorded. Polygenic scores from GWAS for years spent in education (EA3) were used to index genetic propensities for education and lexicon.

Genetic nurture and nurture (effects independent of genetics) pathways that can be interpreted as causal accounted for 65.8% of the transmission of educational attainment from mothers to children, while genetic confounding and gene–environment correlations explained 34.2% of the remaining variance. Mothers' lexicon drove 16.7% of the genetic nurture and 56.6% of the nurture effects. The findings substantiate that mothers' lexicon are likely effective targets for interventions that seek to disrupt the perpetuation of family background inequality in education.

**Keywords** Polygenic scores, Genetic nurture, Intergenerational transmission, Education, Twin cohort

**Grants**

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## The NIH Environmental Influences on Child Health Outcomes (ECHO) Program: Publicly Available Data from the ECHO Cohort

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**Abstract**

The purpose of this poster is to detail the wealth of de-identified data from the ECHO Cohort that is publicly available and is of potential interest to the BGA community. The goal of the ECHO Cohort is to understand the effects of a broad range of early environmental influences on child health and development. ECHO is dedicated to both learning what factors affect child health and to finding ways to enhance it. ECHO studies five outcome areas: pre-, peri-, and post-natal; upper and lower airways; obesity; neurodevelopment; and positive health. The ECHO Cohort began as a synthetic cohort of 69 existing pediatric cohorts. There are more than 100,000 participants, including more than 65,000 children. More than 88,000 biospecimens have been collected. ECHO is making de-identified data publicly available, including data on early development, demographics, environmental data (i.e. physical, chemical, psychosocial, natural and built environments), pregnancy and birth, COVID-19, and a broad range of health conditions. Biospecimens and genetic data will be available in the future. De-identified data can be accessed through the Data and Specimen Hub (DASH) on the *Eunice Kennedy Shriver* National Institute of Child Health and Development (NICHD) website. More information is available on the ECHO website.

**Keywords** Data, Development, Child health

## Co-development of ADHD and Emotional Problems Trajectories from Childhood to Early Adulthood: Predictors and Outcomes

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**Abstract**

Comorbidity between ADHD and emotional problems (EP) has been reported in many previous studies, with 25–42% of children diagnosed with inattentive ADHD subtype and 15–36% of children diagnosed with hyperactivity-impulsivity ADHD subtype also meeting diagnostic criteria for anxiety and/or depression. In comparison to children diagnosed with only ADHD and no comorbidities, those with ADHD and comorbidities are more likely to experience persistent ADHD symptoms into adulthood.

The current study aims to utilize a joint trajectory-based approach to explore the heterogeneity of concurrent inattention/hyperactivity-impulsivity and emotional problems from childhood to adulthood. To predict the latent groups of these trajectories, the study will incorporate Polygenic Risk Scores (PRS) for psychopathologies and family environmental factors. Additionally, adult mental health outcomes and educational attainment variables will be considered as outcomes of the joint trajectories.

We anticipate that problems with inattention will tend to persist throughout development, whereas there will be a remission of hyperactivity–impulsivity symptoms with increasing age. At least four classes are expected in the results of ADHD trajectories: an unaffected class, a persistently high class, a class with high inattention/hyperactivity symptoms but low emotional problems, and a class with high emotional problems but low inattention/hyperactivity symptoms.

We also hypothesize that the membership of joint developmental trajectory classes will be predicted by family factors, including genetics and the family environment. Furthermore, memberships in joint developmental trajectory classes will serve as predictors for life outcomes, such as mental health status and educational attainment in adulthood.

**Keywords** ADHD, Emotional problems, Trajectory

**Grants**

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## Measuring the Associations Between Brain Morphometry and Polygenic Risk Scores for Substance Use Disorders in Cannabis Naive Adolescents

**Sydney Kramer[1], Mei-Hsin Su[1], Roxann Roberson-Nay[2], Nathan Gillespie[2]**

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**Abstract**

A number of studies have shown that substance use is associated with individual differences in adult brain morphology. However, it is unclear if individual differences in brain morphology precede or follow substance use. We explore the association between a polygenic risk score (PGS) for cannabis use disorder (CUD) and general substance use liability and brain morphology based on structural MRI data assessed in an adolescent, drug naive population. We analyzed 221 monozygotic (MZ), 327 dizygotic (DZ) twin pairs and 385 regular sibling pairs (mean = 10 years, range = 9–11 years) of European descent from the Adolescent Brain Cognitive Development Study (ABCD Study®). Individual level PGSs for CUD and brain regions based on PGC and CHARGE consortium GWAS summary statistics were calculated. We fitted multivariate twin models in OpenMx in R to measure the genetic contribution of the CUD and brain region PGSs to six cortical and two subcortical phenotypes. All data were adjusted for the effects of age, sex, total intracranial volume and the top 10 genetic principal components. Our results suggest that variation in brain morphometry in drug naive subjects is unrelated to genetic risk of CUD. Results regarding the other SUD PRS to be discussed. Broadly, our results are consistent with the hypothesis that associations between CUD PRS and brain morphometry seen in older samples likely arises closer in time to or after drug exposure.

**Keywords** Brain, Substance use, Polygenic risk, Twin, Cerebral cortex

## Genetics of Environmental Sensitivity to Psychiatric and Neurodevelopmental Phenotypes: Evidence from GWAS of Monozygotic Twins

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**Abstract**

**Background**

Individuals differ in how sensitive they are to their environments. Individual sensitivity to environmental exposures may be genetically influenced. An implication of such interplay is changes in trait variance across genotype values. GWAS of monozygotic (MZ) twin phenotypic differences offers an alternative method to current population-based methods to investigate phenotypic variance due to gene–environment interactions.

**Methods**

Using genome-wide and phenotypic data from monozygotic twins, we conducted the largest GWAS meta-analysis of MZ differences to date (N from 3636 to 21,792 MZ individuals, depending on phenotype) to identify genetic variants associated with variance in psychiatric, neurodevelopmental and psychological phenotypes [symptoms of attention deficit hyperactivity disorder (ADHD), anxiety, autism spectrum disorder (ASD), depression, psychotic-like experiences, neuroticism and wellbeing]. The analyses were conducted in the largest available samples, as well as separately in developmentally stratified samples. The SNP heritability of phenotypic variance was also estimated.

**Results**

We identified several putative novel genetic factors associated with phenotypic variance and provided a first estimate of heritability of environmental sensitivity for ADHD (adults: 18% [− 4, 40]; children: 4% [− 8, 16]), depression (adults: 2% [− 10, 14]; children: 4% [− 12, 20]) and ASD (adults: 9% [− 20, 38]) symptoms. Monozygotic twins are an important new source of evidence about genetics of phenotypic variability and gene–environment interaction.

**Keywords** Monozygotic twins, MZ differences, Heritability, Environmental sensitivity

**Consortia**

The Within Family Consortium: Margot van de Weijer 10,11, Laurence Howe 3, Teemu Palviainen 5, Katrina Grasby 13, Rafael Ahlskog 14, Marianne Nygaard 15,16, Rosa Cheesman 17, Kai Lim 2, Chandra Reynolds 18, Juan Ordoñana 19, Lucia Colodro-Conde 13,20, Scott Gordon 13, Juan Madrid-Valero 21, Anbupalam Thalamuthu 22, Jouke-Jan Hottenga 10,11, Jonas Mengel-From 15,16, Nichola J. Armstrong 22,23, Perminder S. Sachdev 22, Teresa Lee 22, Henry Brodaty 22, Julian Trollor 22,24, Margaret Wright 25, David Ames 22, Vibeke Catts 26, Antti Latvala 5, Eero Vuoksimaa 5, Travis Mallard 27,28, Kathryn P. Harden 29, Elliot Tucker-Drob 29, Sven Oskarsson 14, Christopher Hammond 30, Kaare Christensen 15,16,31, Mark Taylor 32, Nancy L. Pedersen 32, Karen Mather 22, Sarah Medlands 13, Nicholas Martin 13, Meike Bartels 10,11, Paul Lichtenstein 32 10 Department of Biological Psychology, Vrije Universiteit Amsterdam, The Netherlands 11 Amsterdam Public Health Institute, Amsterdam University Medical Centres, The Netherlands. 12 Institute for Molecular Medicine FIMM, University of Helsinki, Finland 13 QIMR Berghofer Medical Research Institute, Australia. 14 Department of Government, Uppsala University, Norway. 15 The Danish Twin Registry, Department of Public Health,

University of Southern Denmark, Denmark 16 Department of Clinical Biochemistry, Odense University Hospital, Denmark 17 PROMENTA Research Center, Department of Psychology, University of Oslo, Norway. 18 Department of Psychology and Neuroscience, University of Colorado Boulder, USA. 19 Department of Human Anatomy and Psychobiology, University of Murcia, Spain. 20 School of Psychology, The University of Queensland, Brisbane, Australia. 21 Department of Health Psychology. University of Alicante, Spain. 22 Centre for Healthy Brain Ageing, Discipline of Psychiatry and Mental Health, School of Clinical Medicine, Faculty of Medicine and Health, University of New South Wales, Australia. 23 School of Electrical Engineering, Computing and Mathematical Science, Curtin University, Australia. 24 Department of Developmental Disability Neuropsychiatry, School of Clinical Medicine, Faculty of Medicine and Health, University of New South Wales, Australia. 25 Queensland Brain Institute, University of Queensland, Brisbane, Australia. 26 University of Melbourne Academic Unit for Psychiatry of Old Age, St George's Hospital, Melbourne, Australia. 27 Department of Psychiatry, Harvard Medical School, USA. 28 Center for Precision Psychiatry, Department of Psychiatry, Massachusetts General Hospital, USA. 29 Department of Psychology, University of Texas at Austin, USA. 30 Department of Twin Research and Genetic Epidemiology, King's College London, UK. 31 Department of Clinical Genetics, Odense University Hospital, Denmark. 32 Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, Sweden.

## Tracking School-Related Stressors in Accelerated Epigenetic Aging

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**Abstract**

School-related stressors (e.g. bullying, exhaustion due to high pressure at school, problems with teachers) are essential determinants of mental and physical health throughout adolescence and into adulthood. Prolonged stress, and school stress in particular, affects multiple biological systems including DNA methylation (DNAm), one of the epigenetic markers responsible for the accessibility of the genetic material for its transcription. Addressed cumulatively, alterations in DNAm allow examination of the possible effect of stress in biomarkers of aging processes, called epigenetic aging acceleration and its pace.

The body of the literature suggests that the relationships between stress experience and epigenetic aging during adolescence are rather complex. While some studies report the association of accelerated epigenetic aging (e.g. being epigenetically older compared to the biological age) with peer struggles, severe childhood experiences like abuse or violence, and discrimination in ethnic groups, other research fail to confirm the link between reports of being stressed and epigenetic age. This controversy proposes that the effect can be nuanced and different stressors or their combination, possible confounds, and moderators should be accounted for. Following preregistration: https://osf.io/kdgu8, we will present the results of the analysis of the relationships between three school-related stressors (high school pressure, problems with teachers, and experience of bullying at school) with epigenetic aging estimators. Leveraging the rich pheno typic data in the German Twin Family Study, we will control for genetic and familial underpinnings, and investigate the extent to which age moderates the association between school stress and epi genetic age.

**Keywords** Adolescence, Epigenetic aging, School pressure, School-related stressors, Twin life

**Grants**

This work was supported by the German Research Foundation (DFG) as a part of the Project Immunotwin under Grant (Number 451773429), the TwinLife Epigenetic Change Satellite (TECS) Project Under Grant (Number 458609264), and the Long-Term Project TwinLife under Grant (Number 220286500).

## Family Socioeconomic Status Compensates and Triggers Genetic Propensity for Externalizing Behavior on Adverse High School Outcomes

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**Abstract**

This study investigates whether children's genetic propensity for externalizing behaviors influences their likelihood of experiencing adverse outcomes in high school and whether this relationship varies by family socioeconomic status (SES). According to behavioral genetics and social stratification theories, such as the diathesis-stress and compensatory advantage models, we test whether there is a stronger genetic association to adverse outcomes among low-SES children. Using data from the National Longitudinal Study of Adolescent to Adult Health (Add Health) from the US, we construct a polygenic index (PGI) for externalizing behaviors, and we look at different adverse high-school educational outcomes, such as school failure, suspension, expulsion, absence, and misbehavior. The findings reveal two key insights. First, a higher genetic propensity for externalizing behaviors increases the likelihood of experiencing adverse high-school outcomes. Second, the genetic propensity for externalizing behaviors is more predictive of adverse high-school educational outcomes among low-SES children compared to high-SES children. The robustness of these findings is confirmed through several checks, including replication with the British National Child Development Study (NCDS), alternative modeling, correcting for biases in polygenic scores, and using various socioeconomic status measures. Our results suggest that high-SES families compensate for their children's high genetic propensity for externalizing behaviors, whereas low-SES families trigger these genetic propensities in their children.

**Keywords** Family socio-economic status, Externalizing behaviours, Adverse school outcomes, Educational inequalities, Gene–environment interaction (G × E)

## Effect of APOE ε4 Status on Cognitive Ability Trajectories in the Louisville Twin Study

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**Abstract**
The APOE ε4 allele is the single strongest genetic risk factor of Alzheimer's disease and correlates with adult global cognitive functioning, executive functioning, and episodic memory (Small et al. 2004). APOE ε4 status also predicts differences in cognition in childhood and adolescence (Reynolds et al. 2019). Studies predicting cognitive developmental trajectories from APOE status are rare and thus require replication. Moreover, prior studies relied on sum or standardized scores that cannot model intraindividual cognitive growth. Using item-level data from the Wechsler Intelligence Scale for Children and the Wechsler Adult Intelligence Scale in 1344 individual twins (MZ = 306 families; DZ = 349 families) from the Louisville Twin Study, we tested effects of APOE status on cognitive ability trajectories of four subtests (vocabulary, similarities, information, arithmetic). A 2-parameter logistic item response theory model was used to estimate latent subtest scores from ages 7 through 15 and midlife. APOE ε4 did not predict differences in the intercept or slope of any of the subtests nor differences in their midlife subtest scores. In contrast to other twin studies of cognitive development, we did not observe pernicious effects of APOE ε4 on verbal or arithmetic ability development or ability in midlife.

**Keywords** Cognitive development, Longitudinal twin design, APOE, Item response theory

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**Grants**
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## Estimation of Causal Effects Across the Lifecourse Through Multivariable Mendelian Randomization

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**Abstract**
Mendelian Randomisation (MR) is a powerful tool that utilises genetic variants to estimate the causal effect of an exposure trait on an outcome trait. As genetic variants are fixed across an individual's lifetime, the effects obtained from MR studies are often interpreted as the lifetime effect of the exposure in question. However, the effect of genetic variants on a single behavioural trait may vary throughout an individuals lifetime. We discuss the implications of time-varying genetic effects for interpreting MR effect estimates for such traits. We then show how MR including measures of the same trait at multiple time points in an individuals lifetime as exposures can be used to estimate the relative importance of those timepoints, exploiting variation in the genetic effects on that trait at different timepoints. We illustrate this method and the assumptions required to estimate the causal effects of childhood and adulthood adiposity smoking behaviour, using data from the UK Biobank and a recent large GWAS of smoking behaviour. These results show a direct causal effect of adulthood adiposity on smoking behaviour [OR for smoking initiation per category change in adiposity 1.40, 95% confidence interval 1.27–1.55], but not childhood adiposity [OR 0.97, 95% CI 0.86–1.09]. As smoking initiation often occurs during adolescence or early adulthood, many years before the measure of adult adiposity, we hypothesise that the results are due to a stronger correlation in the genetic effects on adiposity between adolescence and adulthood than between childhood and adolescence.

**Keywords** Lifecourse, Mendelian randomization, Adiposity, Smoking behaviour

## Strong Stability of Cognitive Ability from Infancy Through Adulthood

**Daniel E. Gustavson[1], Giulia A. Borriello[2], Robin P. Corley[1], Sally-Ann Rhea[1], Sally J. Wadsworth[1], Naomi P. Friedman[1,3], Chandra A. Reynolds[1,3]**

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**Abstract**
Measures of intelligence are highly stable between adolescence through late life but measurement of general cognitive ability very early in life is difficult. DiLalla et al. (1990), for example, used data from the Twin Infant Project to validate measures of cognition at 7 and 9 months by evaluating which measures were best associated with mid-parent IQ scores. The current project resurrects this effort to examine how infant IQ scores relate to intelligence now measured in the same individuals at 3, 7, 16, and 29 years using standard instruments. Importantly, some of the same measures that best correlated with mid-parent IQ in infancy [e.g., the Fagan Test of Infant Intelligence (FTII); tester-rated task orientation at 7 or 9 months] continued to correlate with full scale IQ at most or all subsequent waves (e.g., $r$ = 0.01 to 0.16 for FTII, $r$ = 0.15 to 0.21 for task orientation). We also observed strong stability of standardized IQ measures between ages 3 and 29 ($r$s = 0.36 to 0.86) with a genetic Cholesky decomposition suggesting 55% of the variance in IQ at age 29 could be explained by genetic influences underlying IQ by age 3 and/or 7. Our findings from this unique, longitudinal and genetically-informative dataset were discussed based on current frameworks of cognitive development.

**Keywords** Infant cognition, Heritability, General cognitive ability,

Twin study, Longitudinal analysis

**Reference**
DiLalla LF, Thompson LA, Plomin R, Phillips K, Fagan III JF, Haith MM et al (1990) Infant predictors of preschool and adult IQ: a study of infant twins and their parents. Dev Psychol 26:759

**Grants**
National Institute on Aging: R01 AG046938.

## A Propensity Score Approach to Examining the Effects of Prenatal Substance Exposure on Development

**Ami S. Ikeda[1], Chelsie E. Benca-Bachman[1,2], Leslie A. Brick[2], Rameez A. Syed[1], Carla Aleman[1], Lejun Liu[1], Manjushri Karthikeyan[1], H. M. Sean Lee[1], Valerie S. Knopik[3], Rohan H. C. Palmer[1]**

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**Abstract**
Environmental perturbations such as prenatal substance exposure during critical periods may lead to disruptions in biological systems that have downstream consequences across development. However, the persistent and transient effects of prenatal substance exposure across development are not well understood. Data from the Adolescent Brain Cognitive Development (ABCD) study (release 5.0) was used to examine the effect of prenatal cannabis exposure on delays in developmental milestone attainment. A propensity score approach was used to control for familial (e.g., family history of psychopathology) and environmental characteristics (e.g., socioeconomic status) associated with the likelihood of being exposed to cannabis during pregnancy (i.e., confounding factors) while examining the effect of prenatal cannabis exposure (PCE) on several developmental outcomes. The familial and environmental characteristics robustly captured mothers who 'did use' versus 'did not use' cannabis during their pregnancy, with an area under the ROC curve of 0.90. The prospective effects of PCE on five developmental milestones were examined. PCE was not significantly associated with any delays in gross motor or language attainment but was positively associated with delays in bedwetting. However, after accounting for confounders via propensity score for PCE (pPCE), the PCE effect became non-significant. Finally, delays in language were associated with pPCE but not PCE, specifically negatively associated with familial history of substance use and birth weight, and positively associated with maternal ADHD symptoms. Future work will expand on these models to examine other behavioral outcomes that may confer risk for early substance initiation through a developmental lens.

**Keywords** Family design, Propensity score, Prenatal exposure, Developmental outcomes

**Grants**
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## The Development of Emotional Overeating: A longitudinal Twin Study from Toddlerhood to Early Adolescence

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**Abstract**
Emotional overeating (EOE) describes the tendency to overeat in response to stress and negative emotions, and has been associated with obesity and mental health outcomes. However, little is known about its development. Here we aimed to model individual differences at key stages of development—toddlerhood (16 months), early childhood (5 years), and early adolescence (12 years). Data were from the UK Gemini Twin Study, which includes 2402 twin pairs born in 2007. Parents reported on each child's EOE at 16 months (n = 3784), 5 years (n = 2064), and 12 years (n = 964), using the Emotional Overeating Scale of the Child Eating Behaviour Questionnaire (CEBQ). A Cholesky Decomposition twin model was used to quantify the additive genetic, shared and nonshared environmental influences on EOE at the three time points, partitioned into aetiological effects unique to each age and those carried over from one time point to the next. Heritability was minimal at 16 months and 5 years (9% and 7% respectively) but increased to 34% by 12 years. Shared environmental effects explained the majority of variance in EOE at all three time points, but significantly less at 12 years (41%) than earlier (> 81%). The longitudinal phenotypic associations (r = 0.23–0.43) were explained by the continuing shared environment. The shared environment plays a major role in the development of EOE in early life. Most aetiological influences on EOE were unique to each age, indicating the need for family-based interventions that are targeted to each developmental stage.

**Keywords** Gemini twin study, Longitudinal, Child development

## Genetic Risk and Prenatal Substance Exposure on Callous-Unemotional Behaviors and the Effect on Parenting: A Developmental Cascade Model

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**Abstract**
Callous-unemotional (CU) behaviors in children include shallow affect and low levels of empathy, guilt, and positive interpersonal emotions. These behaviors forecast severe and chronic antisocial behaviors over time. Although the influences of genetics and rearing environment on CU behaviors have been explored previously, the pathways through which these genetic influences may operate have

yet to be examined. We focus on one specific pathway involving prenatal substance exposure.

We tested a developmental cascade that places children on a path for antisocial behavior, specifically hypothesizing that genetic influences indirectly contribute to child CU behaviors at 27 months via prenatal substance exposure. In turn, child CU behaviors were postulated to be linked to lower parental warmth and higher hostility at 4.5 years old.

Participants were from the EGDS ($n$ = 561 families). Genetic risk was indexed by the biological mother's externalizing problems. Prenatal substance use was measured using biological mother self-report at 5 months post-partum. Adoptive parent reported on their children's CU behaviors at 27 months and their own warmth and hostility at age 4.5. Preliminary analyses using linear regression found birth mother externalizing problems predicted CU behaviors at 27 months (b = 0.15), and a significant association between CU behaviors at 27 months and hostile parenting at child age 4.5 years (b = 0.31), controlling for hostile parenting at 27 months. Additional analyses will use structural equation modeling to examine the direct and indirect effects of genetic and prenatal influences on CU behaviors and CU behaviors' effect on parenting.

**Keywords** Callous-unemotional behaviors, Parenting, Prenatal substance exposure, Genetic risks, Adoption design

## Establishing the Bidirectional Effects Between Body Mass Index and Manic and Depressive Symptoms Using Two-Sample Mendelian Randomisation

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**Abstract**

Depressive and bipolar disorders pose great disease burdens demonstrating the need to establish causative risk-factors for prevention and treatment. Higher body mass index (BMI) also creates disease burden through detrimental effects on physical but also mental illnesses. There is strong evidence for bidirectional effects between higher BMI and major depressive disorder (MDD) however the associations between BMI and bipolar disorder (BD) appear more inconsistent. One reason for this complexity may be differences in how BMI affects depressive and manic symptoms, which BD diagnosis alone may be insufficiently sensitive to capture. As manic and depressive symptomology appear genetically uncorrelated, disentangling genetic effects on the symptom-level may be informative. Mendelian Randomisation (MR)—which uses genetic variants as instruments to predict outcomes through exposures—is a useful tool to test for possible causal effects. We use a two-sample MR approach to assess the bidirectional causal effects between BMI and depressive symptoms or ever having manic symptoms in population samples, alongside BD and MDD case–control status. Alternatively, the inconsistent effects of BMI on BD may be explained by variations in the effect of BMI across age, especially when diagnosis often precedes adulthood where typical BMI measures are taken in genomic studies. Therefore, we will also compare the effects of childhood versus adult BMI on each of our psychiatric outcomes (MDD, BD, depressive symptoms, manic symptoms). We further aim to replicate our findings in clinical BD samples. Such findings may help elucidate potential reasons for unclear associations between BMI and BD.

**Keywords** Mendelian randomisation, Depression, Bipolar, Symptoms

## Genetic Architecture of Pubertal Traits and Sex-Steroids: Cross-Sex Factor Associations

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**Abstract**

Genome-Wide Association Studies (GWAS) of puberty-related traits have identified genetic loci in both sexes, and bivariate genetic correlations note areas of convergence and divergence within and across sex. However, systematic examination of the genetic inter-relatedness of these traits is lacking. The current project investigates the genetic architecture of pubertal timing, pubertal growth, and sex-steroid phenotypes in both males and females.

Sex-stratified GWAS summary statistics for age of menarche (AAM), relative age of facial hair (FH) and voice break (VB), pubertal growth (single height [females: age 10, H10F; males: age 12, H12M], standardized height difference age 8-adult [HDTotal] and age 14-adult [HDAdol]), and sex-steroids [testosterone, estradiol, and sex hormone binding globulin (SHBG)] were used to generate genetic and sampling covariance matrices. Factor structure was examined using Genomic Structural Equation Modeling.

The best fit model (CFI = 0.998, SRMR = 0.06) suggests a sex-steroid factor identified by SHBG ($\lambda_{FEMALE}$ = 0.90; $\lambda_{MALE}$ = 0.91), male–testosterone ($\lambda$ = 0.77), and female–estradiol ($\lambda$ = 0.20); a female-specific factor identified by AAM ($\lambda$ = 0.92) and all three pubertal growth indicators (H10F: $\lambda$ = − 0.35; HDTotal: $\lambda$ = 0.64; HDAdol: $\lambda$ = 0.74); and a male-specific factor identified by FH ($\lambda$ = 0.73), VB ($\lambda$ = 0.93), and HDAdol ($\lambda$ = 0.91), with a residual correlation between FH and VB (r = 0.19). The three factors were all significantly correlated ($r_{FEMALE–MALE}$ = 0.83; $r_{FEMALE–STEROID}$ = 0.13; $r_{MALE–STEROID}$ = 0.17).

These results reveal that female- and male-puberty factors index similar genetic signal, and are modestly associated with adult sex-steroids. Investigating the association between these factors and health traits may elucidate the degree to which sex-specific and cross-sex pubertal and sex-steroid genetic signals play a role in physical and behavioral health.

**Keywords** Puberty, Sex-steroids, Genomic SEM, Sex-differences

**Grants**

K08MH132903.

## Citizen Scientific Racism: Analyzing the "misappropriation" of Behavior Genetics

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**Abstract**
When the manifesto of a white nationalist gunman cited behavior genetics research as a justification for his murder of people in Buffalo, NY, acute attention was drawn among scientists to the problem of racist misappropriation of genetics research. This talk will present an overview of results from an ongoing research project about the rearticulation of scientific racism as a citizen science movement. I describe some of the practices that this movement employs in order to appropriate behavioral genetics and turn it toward their racist concerns. In the spirit of offering news behavioral geneticists can use to address this problem I present two issues. First, I challenge the "misappropriation" and "weaponization" framing of citizen scientific racist efforts by citing recent examples of behavior genetic research that closely link to their agenda. Second, I describe some of structural and cultural vulnerabilities that have made the field of human genetics, but especially behavioral genetics, vulnerable to appropriation by the citizen scientific racism movement.

**Keywords** Racism, Open science, Misappropriation, Sociology of science, Social impact

**Grants**
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## Rethinking Partner Similarity: Understanding Indirect Assortative Mating and Its Consequences for Educational Attainment

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**Abstract**
Partners tend to be similarly educated due to some form of indirect assortative mating, but earlier studies disagree on why. We attribute the inconsistencies to a simplistic understanding of indirect assortment and inconsistent use of relevant terms. In this paper, we develop a more coherent framework for understanding indirect assortative mating and provide updated definitions of key terms. We then develop a new partners-of-twins model that can estimate genetic and environmental influences on a latent sorting factor associated with the phenotype of interest (here, educational attainment). Using data on 575,902 individuals from Norwegian registry data (including 252,487 mate pairs and 3556 twin pairs from the Norwegian Twin Registry), we estimate that partners correlate 0.71 (95% CIs 0.69, 0.72) on a latent sorting factor, which we estimate to be about 51% (40%, 55%) heritable, 45% (33%, 55%) shared environment, and 4.7% (4.2%, 5.2%) unique environment. If accurate, then this would imply a genotypic correlation between partners of 0.36 (0.28, 0.45), which is somewhat larger than would be the case under direct assortment.

**Keywords** Assortative mating, Educational attainment, Twins

**Grants**
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## Untangling the Impact of the Home Environment on ADHD Symptoms: A Study on Cousins Raised Separately

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**Abstract**
The relationship between ADHD symptoms and the environment has been widely studied in literature, often yielding inconsistent results. Notably, although ADHD lacks a shared-environment component, factors such as increased household chaos have been implicated in exacerbating ADHD symptoms (Agnew-Blais et al. 2022), suggesting an environmental influence.

The present study builds upon this work by examining ADHD symptoms and HOME-SF using cousins raised separately. We will be implementing a genetically-informed design using the Children of the National Longitudinal Survey of Youth 1979. Specifically, we will use the discordant kinship model (Garrison and Rodgers 2016) on cousins raised in separate households. This model explicitly distinguishes within-family variance from between-family variance while controlling for residual variation from gene-and-environmental differences, which will allow us to untangle the impact of the environment. Preliminary results are nuanced, and suggest that the relationship is primarily at the between-family level.

**Keywords** Cognitive stimulation, Home environment, ADHD, Cousins

**References**

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## Shared Genetic Liability Across Clusters of Psychiatric and Non-psychiatric Medical Illness

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**Abstract**
**Background**
Individuals with one psychiatric disorder are at such an elevated risk for a second psychiatric diagnosis that psychiatric comorbidity is the norm rather than the exception. Prior large-scale epidemiological studies have found that pervasive comorbidity extends beyond the psychiatric space, with a psychiatric diagnosis associated with an

elevated risk of developing additional non-psychiatric medical illness. While pairwise patterns of comorbidity have been extensively described, our understanding of the broader, multivariate system of relationships underlying these patterns is poorly understood.
**Method**
Using Genomic SEM, we conduct the most comprehensive analysis of psychiatric and non-psychiatric medical comorbidity to-date, to evaluate the extent to which latent genomic factors for 144 medical traits across 8 domains of non-psychiatric medical illness (e.g., cardiac, pulmonary) evince genetic overlap with previously identified factors of shared psychiatric liability for 14 major psychiatric disorders. In planned follow-up work, we will utilize developed measures of heterogeneity ($Q_{Trait}$) to examine to what extent identified factor correlations are driven by more specific relationships between pairs of psychiatric and non-psychiatric conditions.
**Results**
We find that internalizing, neurodevelopmental, and substance use factors are broadly associated with increased genetic risk sharing across all medical factors. Conversely, we find that a compulsive factor is generally protective against adverse medical illness, whereas genetic risk sharing between medical factors and thought disorders was limited.
**Conclusion**
Through the most comprehensive analysis of psychiatric and medical comorbidity, our results elucidate shared liability previously observed phenotypically and reveal pervasive risk sharing between specific clusters of psychiatric and non-psychiatric medical disorders.

**Keywords** Genomic SEM, Psychiatric disorders, Medical disorders, Shared liability, Multivariate genetic analysis

## Parsing the Genetics of Obsessive–Compulsive Disorder: Phenotypic Characterisation of Genetic Subtypes

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**Abstract**
**Introduction**
Obsessive–compulsive disorder (OCD) has a high number of comorbid disorders, including Anorexia Nervosa (AN), Major Depression Disorder (MDD), Tourette's Syndrome (TS), and Bipolar Disorder (BD). This is mirrored by moderate to high genetic overlap of these disorders with OCD. Interestingly, the comorbid disorders shown no or limited genetic overlap between them. Genetic risk for OCD may therefore be separable in different sources of genetic variation that reflect different aetiological pathways. Although these different aetiological pathways all increase the risk for OCD, they are expected to have different characteristic symptoms and behaviours.
**Method**
We used the genetics of the most recent OCD GWAS. With Genomic SEM and the GWAS of TS, AN, MDD, and BD, we extracted genetic variation shared between OCD and each of the comorbid disorders (AN, TS, BD) and not shared with the other comorbid disorders. For each we created a novel GWAS. Next, we characterised the genetic subtypes of OCD by genetically correlating them with a wide range of behavioural variables, including personality and a substance use GWAS.
**Results**
The three shared risk GWAS (OCD + BD, OCD + AN, OCD + MDD) show many substantial differences in genetic correlations. OCD + BD showed increased correlations with risk-related behaviours (substance use, risk taking), extraversion, and lower correlations with diabetes. OCD + MDD showed strong personality differences (mostly in the neurotic spectrum), higher OCD + TS showed no significant differences in genetic correlations after corrections. Many more traits showed large differences and therefore characterise genetic subtypes.

**Keywords** OCD, Genomic factor analysis, Translational psychiatry

**Grants**
This research is supported by the 2023 Amsterdam Neuroscience PoC Grant to OvdH and DS.

## Using Negative Control and Sibling Designs to Test the Maternal Immune Activation Hypothesis

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**Abstract**
The maternal immune activation (MIA) hypothesis posits that the neurodevelopment of offspring is disrupted through the activation of inflammatory pathways in the mother during pregnancy (notably inflammatory cytokines such as IL-6 and IL-17A), potentially leading to neurodevelopmental disorder (NDD) such as autism spectrum disorder and ADHD. It is therefore suggested that maternal exposure to stresses such as mental illness or infection can affect placental functioning and epigenetic programming. Most of the evidence for this comes from mouse models or ecological epidemiological studies of famine or pandemics and it could be argued that neither give sufficient evidence of it playing a substantial role in the current prevalence of NDD's.

To test the MIA hypothesis we consider whether maternal infection or depression/anxiety in pregnancy causes offspring neurodevelopmental disorder (NDD) using a large primary care database (410,461 children) we conducted two negative-control studies and a sibling analysis. For the negative control, we used maternal exposure to either a period of depression/anxiety or infection in the year before pregnancy or the year after. The results suggest no substantial causal effect of either exposure on NDD. I will discuss the implications for the MIA and some strengths and weaknesses of using negative control and sibling designs using routine data.

**Keywords** Maternal immune activation, Neurodevelopmental disorder, Registry data, Sibling analysis, Negative control

## Parental Influence on Children's Educational Achievements: Analyzing Direct and Indirect Genetic Effects Through Trio-GCTA

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and Mental Health, Norwegian Institute of Public Health, Oslo, Norway

**Abstract**
The intergenerational transmission of education is an important subject in social science. There is substantial empirical evidence indicating that children of highly educated parents often reach higher levels of academic achievement than children of less educated parents. This pattern of inheritance is attributable not only to the direct genetic influences but also to the environmental conditions and opportunities that parents create for their children, which can be proxied by indirect genetic effects. In this study, we present new insights into the separation of direct and indirect genetic effects on educational achievements by employing genome-wide complex trait analysis within parent–offspring trios (Trio-GCTA). Using 23,457 genotyped parent–child trios from the Norwegian Mother, Father, and Child Cohort Study (MoBa), we compared direct and indirect genetic effects on educational outcomes at various stages of schooling (5th, 8th, 9th, and 10th years) and explored the differences between girls and boys. Our preliminary findings highlight the significant influence of direct genetic effects on children's educational outcomes while suggesting a notable impact of paternal and maternal indirect genetic effects at different time points of children's education. Understanding the relative contributions of direct and indirect genetic effects on children's educational achievements is crucial for comprehending intergenerational educational mobility and addressing educational inequality.

**Keywords** Indirect genetic effects, Education, Within-family, Trio-GCTA, MoBa

**Grants**
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## Using Polygenic Scores Corrected for the General Psychopathology Factor to Predict Specific Psychopathology

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**Abstract**
Although polygenic scores can be used to predict genetic liability to psychiatric problems, they often show transdiagnostic associations with multiple psychopathology traits, suggesting that these scores index genetic liability to general psychopathology in addition to trait-specific liability. One way to examine genetic specificity in predicting psychopathology is to create polygenic scores corrected for the general factor of psychopathology (p). In the proposed study, we will use polygenic scores constructed from our previous work on genome-wide association summary statistics of major psychiatric disorders that are uncorrected and corrected for p using GWAS-by-subtraction, to predict diverse psychopathology traits at age 26 in a sample of unrelated individuals from the Twins Early Development Study.

Our previous genomic analyses of genome-wide association summary statistics using Linkage Disequilibrium Score Regression led to three hypotheses for PGS prediction of individual differences in measured psychopathology traits corrected and uncorrected for p. First, we expect that PGS corrected for p will on average predict less variance in their target psychopathology traits than PGS uncorrected for p. Second, cross-trait associations of PGSs with other psychopathology traits will be lower for PGSs corrected for p than for PGSs uncorrected for p. Third, and most importantly, PGSs corrected for p will yield more specific patterns of predictions across traits as compared to standard PGSs uncorrected for p.

Although PGSs corrected for p predict less variance in phenotypes, we hope that research on specificity will foster greater understanding of the nature of psychopathology unclouded by the manifold influence of p.

**Keywords** Polygenic scores, Psychopathology, p-Factor, General psychopathology factor

**Grants**
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## Generational Differences in Genetic Effects on Alcohol and Tobacco Use Behaviors: A G × E Analysis in the UK Biobank

**Jordan D. Alexander[1], Gretchen R. B. Saunders[1], Xingyan Wang[2], Shuang Gao[3], Seon-Kyeong Jang[4], Dajiang J. Liu[3], Scott I. Vrieze[1]**

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**Abstract**
Alcohol and tobacco use are complex behaviors influenced by both genetic and environmental factors. Over the past 50 years, rates of both alcohol and tobacco use have changed substantially in many countries, partly reflecting changes in attitudes and policies targeting their use. Despite these changes, little research exists on whether genetic effects on alcohol or tobacco use have differed over time. To investigate whether genetic influences on alcohol or tobacco use are moderated by birth year, a proxy for temporal changes in environmental features affecting substance use, we conducted a GWAS of alcohol and cigarette smoking phenotypes (using a calibrated genomic-control value to reduce false-discovery rates) in the UK Biobank which included a SNP-by-birth-year interaction. We identified three genome-wide-significant ($p < 5 \times 10^{-8}$) SNP by birth year

interaction effects for cigarettes smoked per day ("CPD"; $N = 148{,}246$), two for alcoholic drinks per week ("DPW"; $N = 377{,}058$), and one for age of smoking initiation ("AI"; $N = 152{,}464$). None of the significantly interacting SNPs had significant main effects on their respective phenotypes and all but one were intronic. SNPs within 10,000 base pairs of significantly interacting SNPs have been identified in GWAS of traits such as asthma (CPD), alcohol dependence (DPW), tobacco use (DPW; AI), intelligence (DPW), body mass index (DPW; AI), and testosterone levels (CPD). Results are preliminary and will be meta analyzed with results from additional cohorts to obtain greater effect size precision and to assess robustness across ancestry and nationality.

**Keywords** Gene-by-environment interaction, Substance use, Secular change, GWAS, UK Biobank

**Grants**
T32DA050560; R01HG011035; R01DA044283.

## Polygenic Scores for Intelligence and Educational Attainment Predict Cognitive Measures Between Infancy and Adulthood

**Giulia A. Borriello[1], Daniel E. Gustavson[2,3], Robin P. Corley[2], Sally-Ann Rhea[2], Sally J. Wadsworth[2], Naomi P. Friedman[2,3], Chandra A. Reynolds[2,3]**

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**Abstract**
Genetic influences on intelligence are stable across the lifespan. However, most GWAS of cognitive ability or educational attainment have focused on a single timepoint and on studies of adults. The goal of this study is to evaluate whether polygenic scores (PGSs) for intelligence and educational attainment predict cognitive ability in individuals measured longitudinally from infancy through adulthood, as well as to determine which measures of infant cognitive ability best capture early genetic influences on intelligence. Up to 639 individuals from the Colorado Longitudinal Twin Study (with European ancestry) were assessed with measures of general intelligence (e.g., WISC/WAIS) at mean ages 3, 7, 16, and 29 years with additional measures of cognition assessed at 7 and 9 months (as part of the Twin Infant Project). Educational attainment was assessed at the final timepoint. PGSs were based on general cognitive ability, educational attainment, and child intelligence. Results demonstrated that PGSs for general cognitive function predicted educational attainment and IQ from age 7 onwards ($\beta = 0.09$ to 0.25) but not measures of infant cognition ($\beta < 0.08$). Findings for educational attainment PGSs were similar, but they also predicted infant vocalizations at 9 months. Child IQ PGSs were generally uncorrelated with all cognitive measures, but did predict vocalizations at 9 months and IQ at age 16. Findings suggest that genetic influences on intelligence predict task performance very early in life and that infant vocalizations are a potentially good index of early genetic influences on intelligence.

**Keywords** Polygenic index, General cognitive ability, Longitudinal analysis, Infant cognition

**Grants**
National Institute on Aging: R01 AG046938.

## Moving Beyond the Frailty Index as a Single Aggregate Measure: Genomic Analyses Reveal Clinically Relevant Subclusters Across Frailty Indicators

**Isabelle F. Foote[1], Jonny P. Flint[2,3,4], John D. Fisk[5], Tobias K. Karakach[6], Andrew Rutenberg[7], Nicholas G. Martin[8], Michelle K. Lupton[8], Simon R. Cox[2], Michelle Luciano[3], Kenneth Rockwood[5], Andrew D. Grotzinger[1,9]**

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**Abstract**
Frailty is a multifaceted clinical state associated with adverse outcomes in older adults. It is commonly measured using the Frailty Index, a proportional score based on the number of 'deficits' an individual presents with. Although this approach provides valid risk assessment in clinic, using this aggregate score of heterogeneous deficits in genetic analyses limits our understanding of frailty etiology. We used Genomic Structural Equation Modelling to perform a multivariate genome-wide association study (GWAS) of the latent genetic profile between 30 frailty deficits. Our model included a bifactor representing genetic overlap between all 30 deficits, plus 6 residual latent factors of genetic overlap between discrete subsets of deficits, which were uncorrelated with the bifactor. These subgroups can be defined as pathways reflecting social isolation, unhealthy lifestyle, multimorbidity, metabolic/respiratory status, cognition/perceived health, and disability. GWAS of the frailty factors identified 408 risk loci. Polygenic risk score analyses revealed that the latent factors predicted frailty in three external datasets. Finally, we estimated the genetic correlation between each frailty factor and 50 aging-related outcomes. We show that whilst some outcomes are associated with the whole frailty spectrum (e.g. earlier mortality and increased infection risk), other outcomes are associated with specific deficit factors (e.g. frailty pathways linked to cognition are positively genetically correlated with Alzheimer's disease). Our findings provide novel insights into the genetic basis of frailty and support the refinement of frailty as a multifactorial construct to help stratify frail individuals who are at risk of adverse health outcomes as they age.

**Keywords** Frailty, Aging, Genomic SEM, Multivariate GWAS

**Grants**
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## Transferability of Tobacco and Alcohol Use Polygenic Scores Across Populations

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**Abstract**
Polygenic risk scores (PRS) based on a genome-wide association study (GWAS) of 2.7m individuals have been found to explain up to 9.6% of the variance in alcohol and tobacco use. This level of predictive power, however, is only true for individuals of European (EUR) ancestries, the same ancestries upon which the discovery GWAS was based. In the current study we use summary statistics from this GWAS to describe the accuracy of EUR-based PRS across diverse ancestries using > 300,000 individuals from the All of Us cohort. We find that the relative accuracy of PRS for smoking behaviors is 43% for American (AMR), 35% for East Asian (EAS), and 15% for African (AFR) ancestries. Further, we find that the predictive accuracy of scores decline as a function of genetic distance from the discovery GWAS even within ancestry. Comparisons of predictive accuracy by allele frequency and recombination rates suggest that differences in frequencies and linkage disequilibrium explain much, but not all, of the reduced prediction (approximately 85% of the loss in AFR ancestries, 58% in EAS ancestries, and 25% in AMR ancestries). Work is ongoing to estimate the contribution of differences in causal effect sizes across ancestries to the reduced accuracy using admixed AFR and AMR ancestry individuals from All of Us ($N \approx 100k$). The results of this work will inform our understanding of the genetic architecture of complex traits across genetic ancestries and create more accurate PRS for individuals from ancestries that have largely been left out of genomic research.

**Keywords** Polygenic score, Alcohol, Tobacco, Genetic ancestry

**Grants**
K01 DA 057376; R01 DA 044283.

## Genetic and Environmental Influences on Reading Performance

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**Abstract**
Genetic factors are known to play a role in shaping reading abilities and the underlying cognitive processes, but understanding how genetic and environmental factors interact to influence reading outcomes remains largely unknown. By evaluating the interplay between genetic and environmental influences on reading performance, this study aims to provide insights into the complex mechanisms underlying reading abilities.

To this aim, we leveraging the longitudinal Adolescent Brain Cognitive Development dataset (approximately 4000 individuals of European ancestry and two timepoints: mean ages 10 and 12). Genetic influences are operationalized as polygenic scores (PGS) for multiple reading-related traits, including cognitive performance, educational attainment and dyslexia.

First, we establish the individual associations between selected reading-related PGSes and reading outcomes, observing consistent effects across the two timepoints (adjusted $R^2$ range 3.5–7%). Next, we confirm that these associations are attenuated (adjusted $R^2$: 1–5%) but significant after adjustment for socioeconomic (household income and parental education) and linguistic factors (bilingualism, dual context exposure), and evaluate potential interactions between the PGSes and these environmental variables. Finally, we integrate all PGSes into a multi-PGS analysis to elucidate their combined contribution beyond environmental variables and the extent to which they explain independent variance in reading. The resulting best model includes four PGS that additively account for up to 9% (adjusted $R^2$) of variance, indicating that genetic and environmental influences additively contribute to reading outcomes. Sensitivity analyses with other cognitive outcomes (vocabulary and fluid intelligence) highlight that the observed pattern is specific to reading.

**Keywords** Reading, Polygenic scores

## Reducing Racial Prejudice through Genetics Education and Curriculum: Teaching Perspectives and Results

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**Abstract**
Genetic essentialism is the belief that people of the same race share genes that make them physically, cognitively, and behaviorally uniform, and different from other races. Having strong beliefs in genetic essentialism have been found to significantly predict public opposition to modern policies that promote racial equality among white and non-white racial groups in the United States. This study examines the effects of the Humane Genomics Literacy (HGL) intervention, which is a genetics education intervention that is designed to reduce beliefs in genetic essentialism. It is believed to influence how students grasp and apply genetic evidence to refute genetic essentialist arguments. While the HGL curriculum has been taught in middle school and high school settings, this presentation will cover an implementation in a college-level setting. We will discuss some aspects of teaching and implementing this curriculum in a college setting. We will then discuss the results and impact of genetics education interventions at the college level on beliefs in genetic essentialism of race, employing a quasi-experimental design based on class assignment rules at Purdue University. Students were randomly assigned to one of the two curriculum intervention treatments (baseline or "epistemic") during the semester. Contents taught in each session were different in that one session included materials related to epistemic complexities of human genetics research and the other did not. The outcomes we will examine are measures for cultural essentialism, racial essentialism, colorblind racism, and racialization, which were measured at three timepoints throughout the course.

**Keywords** Genetics curriculum, Genetic essentialism, Educational intervention, Genetics education

**Grants**
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## Characterizing the Genetic Architecture of Impulsivity and Its Overlap with Psychopathology

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**Margaret L. Westwater[8], Camille M. Williams[9,10], Lea K. Davis[11,12,13], Armin Raznahan[14], Elliot M. Tucker-Drob[9,10], Karmel W. Choi[1,2,3], Tian Ge[1,2,3], Jordan W. Smoller[1,2,3], Abraham A. Palmer[4,15], Sandra Sanchez-Roige[4,11,15]**

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**Abstract**
Impulsivity is a complex psychological construct with a fundamental role in health and behavior, but measurement issues hinder our understanding of its etiology. Neuroscientific insights support 'splitting' impulsivity into distinct facets, but factor analytic research has shown that similar traits can be 'lumped' together at multiple levels. Here, we use multivariate methods to advance our understanding of impulsivity by testing the splitting versus lumping hypotheses across biological scales. Using the Genomic SEM framework, we analyzed eight impulsivity facets ($N$s $\sim$ 130k) with two models: an unstructured 'omnibus' model (splitting) and a structured 'common factor' model (lumping). We evaluated these models and their outputs across genomic, transcriptomic, and neuroanatomical levels. We then conducted polygenic score (PGS) analyses in two independent cohorts to characterize links with health and well-being. Results revealed pleiotropy among impulsivity facets (mean $r_g = 0.43$, range = $-0.09$ to 0.79) that could be approximated with a common factor model (CFI = 0.94, SRMR = 0.08). However, subsequent analyses revealed that SNP- and gene-level effects were largely inconsistent with this model of general liability. Many genetic relationships with psychopathology were also facet-specific, such as those with externalizing behavior and autism spectrum disorder ($Q_{\text{Trait}}$ $P$s $< 8.48e{-6}$). PGS results underscored the consequences of erroneous lumping, as 67 associations with medical outcomes were missed with a common factor PGS. Collectively, the results of this study generate new insights into the etiology of impulsivity, implicating novel links to neurodevelopment and health. Our results support multidimensional approaches as a more informative avenue for studying the heterogeneous biology of impulsive traits.

**Keywords** Impulsivity, Externalizing, Psychopathology, GWAS, Genomics

**Consortia**
23andMe Research Team, Sarah L. Elson, Pierre Fontanillas.

## Multivariate Modeling of Assortative Mating: Quantifying Variations of Within-Trait and Cross-Trait Homogamy in Spousal Selection

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**Abstract**
Marital resemblance may arise from multiple sources, e.g., population stratification, spousal selection (aka assortment or homogamy), and confluence due to shared experiences or mutual interaction. Though present for many human traits, the sources of marital resemblance are largely unknown. This lack of knowledge spans both within-trait and across-trait marital resemblances.

We implemented a model to estimate these potential sources of homogamy simulating data from siblings and their spouses based on existing extended pedigree SEM models (Keller et al. 2009). Homogamy can be modeled as a selection process that changes the covariance between spouses but leaves means and variances unchanged. This mechanism was simulated using a partial sort of generated spouse pair data to approximate potential manifestations of spousal homogamy. Implementation of the Pearson–Aitken selection formula within OpenMx correctly recovers the degree of changes to the overall model covariance among the twins and spouses that would result from homogamy. We evaluate competing hypotheses for multivariate spousal resemblance, where homogamy is purely within-trait, purely across-trait, and both. Extensions to the model including other sources of marital resemblance are considered.

**Keywords** Marital resemblance, Assortative mating, Quantitative methods, Structural equation modeling, Multivariate modeling

**Reference**
Keller MC, Medland SE, Duncan LE, Hatemi PK, Neale MC, Maes HH, Eaves LJ (2009) Modeling extended twin family data I: description of the Cascade model. Twin Res Hum Genet 12(1):8–18

## Parent-of-Origin Effects in Attention-Deficit/Hyperactivity Disorder

**Dinka Smajlagic[1], Elizabeth C. Corfield[2,3], Siobhan Connolly[4], Hakon Hakonarson[5,6], Irwin Waldman[7], Josephine Elia[8], Elizabeth Heron[4], Mona Bekkhus[1], Martin Tesli[9,10], Ted Reichborn-Kjennerud[2,11], Alexandra Havdahl[1,2,3], Jan Haavik[12,13], Stefan Johansson[14], Tetyana Zayats[1,15,16]**

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**Abstract**
Attention-deficit/hyperactivity disorder (ADHD) is the most common, highly heritable childhood-onset neuropsychiatric disorder, characterised by multifaceted genetics. To date, genetic studies of ADHD have predominantly focused on additive effects, explaining about a quarter of its heritability. Thus, we aimed to provide novel insight into the complex genetic architecture of ADHD by examining parent-of-origin effects (POE) with additive and indirect genetic effect.

We meta-analyzed the data collected through Psychiatric Genomics Consortium and Norwegian Mother, Father, and Child Cohort Study, consisting of 2574 ADHD cases. POE, additive, and indirect genetic effects were evaluated using multinomial modelling implemented in EMIM software. We explored these associations in the light of the largest ADHD genome-wide association study to date (38,691 cases and 186,843 controls). Gene- and pathway-based association analyses were performed using MAGMA software.

Although we did not discover any genome-wide significant associations, our results have potentially identified the presence of POE and indirect genetic effects in the development of ADHD. Our strongest signal was observed for the POE (effect indicative of imprinting; P = 3.70E−06), however, it was among the top three most significant associations in other two effects as well. *PGBD5 gene*, involved in brain development, was among the top three most significant genes across effects in gene-based analyses.

In conclusion, this is the first and the largest genome-wide parent/offspring study in ADHD, exploring POE, additive, and indirect genetic effects simultaneously. Our results indicate that transmitted alleles contribute to more than just additive effects. Larger samples are needed to confirm our findings.

**Keywords** POE, ADHD, Imprinting

**Grants**
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## Examining the Heritability of Objectively Assessed and Parent-Reported Sleep Duration in a Community Sample of Twins Across Infancy, Childhood, and Adolescence

**Christy Bui[1], Kathryn Lemery-Chalfant[1], Sierra Clifford[1], Mary C. Davis[1], Leah D. Doane[1]**

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**Abstract**
Meta-analytic evidence suggests that heritability of sleep duration (SD) increases throughout infancy and childhood, peaking in adolescence. Few genetically-informed studies examine SD across development, and estimates vary by reporter or assessment. Extending prior published heritability estimates of SD in the Arizona Twin Project from infancy and middle childhood, we examine new data from the 9-, 10-, 11-, and 13-year waves (*N*s = 368–636 twins; MZ = 29.6–32.5%, ssDZ = 32.3–36.3%, osDZ = 32.8–35.1%; female = 48.6–51.2%; non-Hispanic white = 56.5–61.3%, Hispanic = 20.9–25.8%). Caregivers reported on twin sleep at each wave. Twins wore wrist-based accelerometers for one week at 8-, 10-, and 13-years.

In univariate models, parent-reported SD was largely shared environmental in infancy ($C$ = 0.81–0.83), and moderately heritable at 5–13 years ($h^2$ = 0.31–0.55, $C$ = 0.36–0.59). In contrast, objectively measured SD was highly heritable at 8, 10, and 13 years ($h^2$ = 0.66–0.81, $C$ = 0.00, $E$ = 0.19–0.34). Bivariate Cholesky decomposition models were fit at ages 8-, 10-, and 13-years with parent-reported SD as the first phenotype, and objective SD as the second. At 8 years, objective SD shared significant genetic overlap with parent-report, with little independent genetic variance ($A21$ = 0.74, $A22$ = 0.00), however, at later ages, most genetic variance was independent ($A21$ = 0.16–0.29, $A22$ = 0.33–0.59).

The heritability of SD changed across development and was highest in adolescence, and parent-report but not objective SD showed significant shared environmental influences, supporting a multi-method approach to SD. Age-related differences in genetic covariance suggest that influences on parent-reported SD may differ across this developmental transition. Future studies should examine G × E interactions, including contextual differences in heritability.

**Keywords** Sleep duration, Heritability, Longitudinal, Childhood, Twins

**Grants**
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## Genetic and Environmental Contributions to Physical Activity and Dietary Intake in the Context of Adiposity: Insights from a Community Sample of Twin Children

**Eva M. Bartsch[1], Sierra Clifford[1], Gianna Rea-Sandin[1,2], Mary C. Davis[1], Leah D. Doane[1], Kathryn Lemery-Chalfant[1]**

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**Abstract**
According to the obesogenic environment theory, physical activity (PA) and dietary intake are related to adiposity, despite adiposity being highly heritable during childhood. To examine genetic and environmental contributions to moderate-to-vigorous activity (MVPA), sedentary behavior (SB), and diet, we assessed children's diet and PA over 1 week via daily food diaries and actigraphy, and body mass index (BMI) during a home visit in twins from the Arizona Twin Project ($n$ = 654; MZ = 31.1%, ssDZ = 37.9%, osDZ = 31%; 48.6% female; 58.5% White, 32.5% Hispanic; $M_{BMI}$ = 16.76), aged 8.44 ($SD$ = 0.68) years. Using OpenMx for biometric modeling, covarying age and sex, we found that variation in SB (A = 64%, C = 0%, E = 36%) and MVPA (A = 78%, C = 0%, E = 22%) was explained by genetic and unique environmental factors. Sugar,

vegetable, and fruit consumption were influenced by environmental factors (A = 0%, C = 80–87%, E = 13–20%), whereas protein, carbohydrate, fat, and fiber consumption were influenced by genetic and environmental factors (A = 21–58%, C = 22–69%, E = 9–20%). Average levels of health behaviors and BMI were not correlated highly enough to parse genetic and environmental covariance ($r < 0.2$). Findings suggest that PA and diet do not play a significant role in shaping adiposity in a community sample at this age. Genetic factors may influence children's requirements for certain sources of nutritional energy, whereas consumption of sugar, fruits, and vegetables may depend on availability. Future analyses, based on family systems theory, will assess the genetic and environmental covariance of BMI and daily and structural stress to investigate whether stress may be a viable target for prevention/intervention efforts for children at high genetic risk for high adiposity.

**Keywords** Physical activity, Dietary intake, Adiposity, Twin study

**Grants**
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## Education Reduces Dementia Prevalence and Mitigates the Genetic Risk of Dementia

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**Abstract**
The number of cases of Alzheimer's Disease and Related Dementias (ADRD) worldwide is growing exponentially. However, there is little evidence on causal factors that might protect individuals against ADRD risk. We study the causal effect of education on ADRD by combining a natural experiment with dementia outcomes measured from administrative health records. The natural experiment, a change in compulsory schooling laws in the UK in 1972, generated as-good-as-random variation in secondary education. ADRD was ascertained based on medical history, hospital records, and death registries, addressing concerns about selective sample attrition. We find that education reduces risk of ADRD. The evidence suggests education also delays the onset of ADRD. The reduction in ADRD risk is larger for individuals with greater risk of AD due to genetic factors, implying this genetic risk is not immutable and can be modified by social policy.

**Keywords** ADRD, Education, Natural experiment, Gene–environment interaction

**Grants**
We gratefully acknowledge the support of the following Grants: R00 AG062787 (PI Turley) and R01 AG078522 (PI Barcellos) from the National Institute of Aging (NIH), and 010623-00001 (PI Benjamin) from Open Philanthropy. This research has been conducted using the UK Biobank Resource under Application Number 15666.

## Biobank Participants' Perspectives on Five Polygenic Scores for Social Traits

**Jean Cadigan[1], Margaret Waltz[1], Courtney Canter[2], Anneliese Long[2], Anya E. R. Prince[3]**

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**Abstract**
Polygenic scores utilize data from biobank participants to measure associations between genomic variants and traits of interest. This methodology, originally used to develop risk scores for medical traits, has expanded to social and behavioral traits. Despite this expansion, research examining how biobank participants view the use of their data to develop polygenic scores for social and behavioral traits is lacking. To address this, we surveyed over 400 U.S.-based biobank participants. We asked their opinions about five polygenic scores for social traits: educational attainment, aggression, life satisfaction, religiosity, and obesity. We also explored participants' perceptions of the potential uses of these traits' polygenic scores outside of research in six possible scenarios (e.g., by elementary schools, life insurance companies, and in selecting embryos with IVF). Support for using the scores outside of research was low. Among the five polygenic scores for the traits we surveyed, participants were most likely to support using a score for educational attainment, particularly to identify elementary school children who need additional interventions and resources (46.6%). Participants were least likely to endorse using polygenic scores for religiosity and obesity outside of research. Among the potential uses presented, participants were least likely to support life insurance companies using polygenic scores to help determine insurance premiums. These findings have implications for how biobank administrators may approach fielding specimen requests for this research and consenting biobank participants. We will also discuss how acceptance of different polygenic scores for social and behavioral traits intersect with the spectrum of acceptable uses.

**Keywords** Biobanks, Polygenic scores, Consent, Public perceptions, Social traits

**Grants**
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## Polygenic Scores for Social and Behavioral Traits: From Survey Questions to Public Perceptions

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**Abstract**
From educational attainment to aggression to religiosity, polygenic scores are rapidly being developed for a range of social and behavioral traits. Leveraging data from biobanks, researchers develop

polygenic scores by studying the associations between biobank participants' genetic variants and survey responses. This interplay between survey and genetic data opens the door to the development of polygenic scores for a wide variety of social traits, but also limits the research based on how survey questions in existing datasets are phrased. This presentation reports on findings from our study examining current trends in the development and use of polygenic scores for social and behavioral traits in research. We coded 984 recent journal articles to document the development and use of emerging polygenic scores. In this presentation, we will report on over 40 recent polygenic scores for social and behavioral traits identified in the literature. Using three to five traits as models, we will first interrogate the survey questions utilized to develop the scores, highlighting the strengths and limitations that stem from the phrasing of the underlying survey questions. Drawing on rich qualitative data from our focus groups with biobank participants, we will then compare how the traits are scientifically measured and defined with public perceptions of the traits themselves. This pairing of the underlying trait development with public insight will help to illuminate disconnects that may hinder public understanding and use of polygenic scores as well as potential sources of ethical, legal, and social implications (ELSI) in the development of PGS.

**Keywords** Polygenic scores, Biobank participants, Focus groups, Social and behavioral traits, Survey trait measurement

**Grants**

This work was funded by the US National Human Genome Research Institute (NHGRI) under Award Number R01 HG012402 01.

## Positive Neighborhood Features Moderate the Etiology of Youth Resilience Trajectories

**Elizabeth A. Shewark[1], Alexandra Y. Vazquez[1], Sarah L. Carroll[1], Amy K. Nuttall[2], Amber L. Pearson[3], Laura E. Miller-Graff[4], Luke Hyde[5], Kelly L. Klump[1], S. Alexandra Burt[1]**

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**Abstract**

It remains unknown whether children's resilient outcomes are differentially stable over time and whether the underlying etiology of those trajectories differ. Thus, we examined the etiology of resilience development across multiple domains. We selected a subsample of the Twin Study of Behavioral and Emotional Development in Children experiencing particularly high levels of contextual adversity (i.e., family poverty, neighborhood poverty, and exposure to community violence; 852 twins in 426 families). Mothers reported on their child's psychiatric and social resilience on the Child Behavior Checklist. We used Latent Growth Curve Modeling to examine psychiatric and social resilience development across three assessments from middle childhood to emerging adulthood. Results demonstrated that psychiatric and social resilience increased over time. We extracted the latent slope to conduct univariate G × E analyses testing how neighborhood factors (i.e., informal social control, norms, and social cohesion from the neighborhood matters questionnaire) might moderate the etiology of the rate of change for each resilience domain. Preliminary G × E analyses showed no significant moderation of the rate of change for psychiatric resilience. However, informal social control significantly moderated the etiology of the rate of change for social resilience (AC moderation fitting the data best), such that as informal social control decreased, the amount of variance accounted for by shared environmental factors increased. Next, we will evaluate how these neighborhood features moderate the etiology of the development of additional resilience domains as well consider access to greenspace (e.g., parks) as a potential moderator. Implications will be discussed.

**Keywords** Neighborhoods, Resilience, Development, Genotype–environment interplay

**Grants**

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## What Are the Longitudinal Associations Between Maternal Nurturance and Youth Antisocial Behavior? A Cross-Lagged Twin Differences Study

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**Abstract**

Lower maternal nurturance is consistently associated with higher levels of youth antisocial behavior (ASB), but the genetic and environmental origins of their association remain unclear. In middle childhood, the association between parental nurturance and youth ASB was found to be genetically mediated and likely reflective of child-driven evocative gene–environment correlations. However, no studies have examined the etiology of this association over time, so it is uncertain whether or how the associations observed in childhood persist into adolescence. To fill this gap, we sought to illuminate prospective associations and potential changes in the etiology of the relationship between maternal nurturance and youth ASB across development using a longitudinal cross-lagged twin differences design. Participants were drawn from the Michigan State University Twin Registry and included 2060 twins ages 6–11 at Time 1 and 706 twins ages 7–19 reassessed 4–6 years later. As expected, differential nurturance in childhood did not predict differences in adolescent ASB. However, co-twin differences in the levels of ASB during childhood predicted moderate differences in parental nurturance during adolescence, whereby the monozygotic or dizygotic co-twin exhibiting higher levels of ASB in childhood received less maternal nurturance as an adolescent. Effect sizes did not differ across zygosity. These results strongly suggest a child-driven but environmental effect of early ASB on later maternal nurturance, such that higher levels of ASB in childhood evoke less parental nurturance in adolescence, but do so via environmental mechanisms. These findings have important implications for interventions and etiologic models of youth ASB across development.

**Keywords** Twin differences, Parenting, Antisocial behavior

**Grants**

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National Institute for Child Health and Human Development (NICHD, **1K99HD110604**; awarded to EAS).

## Expanded GWAS of > 60,000 Individuals with Attention-Deficit/Hyperactivity Disorder Identifies > 50 Significant Loci and Enables Profiling of Differential Effects Across Traits and Within Families

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**Abstract**

Attention-Deficit/Hyperactivity Disorder (ADHD) is a heritable neurodevelopmental disorder that affects 5% of individuals and has lifelong associations to health and behavior.

Here, we report > 50 loci significantly associated with ADHD in GWAS of > 60,000 individuals with ADHD and > 800,000 controls from > 45 international cohorts, replicating previous findings and doubling the existing number of identified loci for ADHD (Demontis et al. 2023).

These GWAS results appear consistent across study designs, including strong concordance family-based cohorts and case/control or population-based cohorts (genetic correlation = 0.82, se = 0.18). Top loci also show no evidence of heterogeneity across study designs (p = 0.59). The family-based cohorts additionally support previously reported genetic correlations with ADHD; for example, probands with ADHD strongly over-inherit polygenic risk for smoking (p = 1.8e−17).

Finally, using a newly-developed method (Elliott et al. 2024) we observe that contrary to simple causal models (e.g. Mendelian randomization) the observed cross-trait relationships across top GWAS loci are highly heterogeneous for many ADHD-related traits. For example, six ADHD-associated loci are associated with increased educational attainment despite a strong overall negative genetic correlation (heterogeneity p = 1.7e−44).

Together, these results present new insights into novel ADHD-associated loci and their action in different contexts.

**Keywords** GWAS, Attention-deficit/hyperactivity disorder, Within-family analysis, Mendelian randomization

**References**

Demontis D, Walters GB, Athanasiadis G et al (2023) Genome-wide analyses of ADHD identify 27 risk loci, refine the genetic architecture and implicate several cognitive domains. Nat Genet 55(2):198–208

Elliott A, Walters RK, Pirinen M et al (2024) Distinct and shared genetic architectures of gestational diabetes mellitus and type 2 diabetes. Nat Genet. https://doi.org/https://doi.org/10.1038/s41588-023-01607-4

**Consortia**

ADHD Working Group of the Psychiatric Genomics Consortium; iPSYCH-Broad Consortium.

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## The Number of Comorbidities in Depression Patients Correlates with Their MDD Polygenic Score

**Penelope A. Lind[1], Ian B. Hickie[2], Nicholas G. Martin[3], Sarah E. Medland[1]**

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**Abstract**

Major depressive disorder (MDD) is a common mental health disorder accompanied by considerable comorbidity and excess mortality. There are extensive research findings that individuals with depression experience high levels of comorbid chronic illness and reduced life expectancy. We examined multimorbidity data in the Australian Genetics of Depression Study (AGDS) using the validated Rx-Risk Comorbidity Index, a pharmacy-based measure of 46 morbidity categories.

**Methods**

Prescription data record linkage for 15,890 AGDS participants were available from 01/07/2013 to 31/12/2017 for all dispensed medications listed on the Australian Pharmaceutical Benefits Scheme (PBS) schedule. Forty-six mental and physical health morbidities were inferred by mapping the prescription data using Anatomical Therapeutic Chemical Classification System codes and PBS Item Codes. Prevalence rates of each comorbidity were then compared with an unselected 10% Australian representative population sample (10PCT) with PBS claims data available from 01/07/2010 to 31/12/2014.

**Results**

The average number of inferred comorbidities was higher among AGDS participants (4.6 ± 2.9) than 10PCT individuals (3.0 ± 3.0). Excluding depression, 89.1% of AGDS participants had one or more inferred comorbidity; most commonly pain (51.0%), inflammation/pain (40.3%), and anxiety (32.3%). In the AGDS, the number of inferred comorbidities was higher among women compared to men

($p < 0.001$) and increased with participant age, BMI, number of depressive episodes experienced. Most interestingly, there was a highly significant correlation between number of comorbidities and MDD polygenic risk score.
**Conclusion**
This study shows that in depression patients the number of comorbidities, independently estimated from prescription data, is correlated with the patient's MDD polygenic score.

**Keywords** Depression, Comorbidity, Mental health, Physical health

**Grants**
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## Using Polygenic Risk Scores to Characterise Antidepressant Treatment Response in the Australian Genetics of Depression Study

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**Abstract**
**Background**
One in five people will be diagnosed with depression in their lifetime, and approximately one third of those are refractory to treatment. The incorporation of genetic information alongside demographic and clinical predictors is a promising avenue to improve our understanding of antidepressant treatment response. In this study, we aim to test whether polygenic scores (PGS) can be used to characterise individuals with differential treatment response in the Australian Genetics of Depression Study (AGDS).
**Methods**
To stratify individuals based on prescription patterns, we used prescription record data from $\sim$ 16,000 AGDS participants. Participants with more than 20 prescriptions of the same antidepressant were allocated to SSRI or SNRI response groups and were contrasted to a treatment resistant groups based on self-report of electroconvulsive therapy (ECT) treatment, bipolar disorder (BD) diagnosis and prescription of lithium. Individuals in each group were characterised using demographic and self-reported course and treatment outcome data, PGS and *CYP2C19* and *CYP2D16* metaboliser profiles.
**Results**
The median number of lifetime depressive episodes was higher in the SNRI than SSRI response groups. However, the treatment-resistant groups reported the highest number of depressive episodes and were more likely to be male. This corresponded with a higher PGS for ADHD, depression and bipolar disorder being significantly associated with non-responders. Lastly, *CYP* gene metaboliser profiles did not differ significantly between response groups.
**Conclusion**
Understanding the genetic influences on long-term pharmacological treatment usage is a promising field of research to reduce or mitigate the risk of poor treatment outcomes for individuals with depression.

**Keywords** Depression, Polygenic risk scores, Heterogeneity, Stratification

## Genetic Pathways to Music-Evoked Emotions: From Twins to Genome-Wide Association Studies

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**Abstract**
Music is ubiquitous across human cultures, present and past. Nowadays, one of the major reasons that individuals listen to music is to express, release, and influence emotions. To better understand the origins of such music-evoked emotions, we collected data on the emotion–evocation facet of the Barcelona Music Reward Questionnaire from a sample of Swedish twins ($N = 9169$) and Dutch genotyped individuals ($N = 15{,}648$). SEM of Swedish twins' data indicated that emotion–evocation traits are heritable (range-$h^2_{twin}$ [0.28, 0.38]). Further multivariate analysis showed that an independent genetic pathway could substantially capture the genetic correlation across the items used to measure the emotion–evocation facet (range $r_{A\text{-}twin}$ [0.59, 0.83]). Triangulation by multiple-thresholds GREML in the Dutch sample confirmed substantial emotion–evocation trait-level heritability, with a partial contribution of both SNP (range-$h^2_{SNP}$ [0.08, 0.13], $se = 0.02$) and familiar-relatedness (range-$h^2_{\pi>0.05}$ = [0.15, 0.21], $se = 0.04$) to the overall observed genetic-relatedness-based heritability (range-$h^2$ = [0.25, 0.31], $se = 0.04$). Further genomic SEM analyses on derived item-level GWAS summary statistics confirmed a common emotion–evocation genetic factor. From twins to genome-wide association studies, these results reveal shared genetic influences on the ability to experience emotions from music and pave the way for future studies on the origins of an important aspect of human affect.

**Keywords** Music, Affect, Twin, GREML, Genomic-SEM

**Grants**
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## Investigating Non-shared Environmental Influences: A Rapid Review of Qualitative MZ Twin Differences Studies

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**Abstract**

**Background**

The environment is an important influence in the development of behaviours and health outcomes, both physical and psychological. One way to study environmental determinants of behaviour and health is through the MZ differences design, whereby environmental exposures are used to predict differences in outcomes in identical twin pairs. This research design aims to identify sources of discordance between twins. Qualitative methods, which focus on context and subjective experience, provide a means to study the origins of discordance among MZ twins in depth, with emphasis on lived experience, whilst controlling for genetics and exposure, but not perception, of environmental factors shared by family members.

**Methods**

To foster the development of the qualitative MZ differences design, we conducted a systematic review of the literature. We examined the methodological features of studies using qualitative data collection and analysis methods to investigate differences in monozygotic twins' experiences.

**Results**

We identified seven studies, covering a range of phenotypes (e.g., anxiety or smoking). Data was mostly collected through inductive, semi-structured interviews with each twin separately, through existing twin studies or registries.

**Conclusions**

Given the suitability of this research design to explore the environment within a genetically sensitive design, and the growing interest in this approach, we make five key recommendations on its use, which include more detail on the qualitative analysis process as well as an acknowledgement of theoretical considerations.

**Keywords** Non-shared environment, Twins, Qualitative

**Grants**

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## Effects of Polygenic Scores for a Common Executive Function Factor on the Course of Executive Function

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**Abstract**

Executive Functions (EFs) are meta-cognitive capabilities controlling and coordinating mental processes. It has been shown that these are mediated by a latent common EF (cEF) factor, which is highly heritable. We have previously researched contributions of cEF polygenic scores (PS) calculated from summary statistics of a large cEF UK Biobank GWAS (Hatoum et al. 2023) to various cognitive phenotypes. Consistently we found that cEF PS explained the highest proportion of variance in a phenotypic common EF score, derived from a confirmatory factor analysis of cross-sectional data. Here, we research longitudinal cognitive effects of cEF PS in the PsyCourse Study, a transdiagnostic deep phenotyping study (N = 1786) of the affective-to-psychotic continuum (n = 1320), including neurotypic individuals (n = 466). We model a latent phenotypic cEF factor at each measurement point. After demonstrating measurement invariance across time points, we predict factor scores using a second-order growth model and analyze these factor scores using a linear mixed model containing cEF PS. Results show significant main effects of both cEF PS and measurement point, and, importantly, a significant interaction of these variables. This result indicates that the facilitation of executive performance improvement over time appears to be a crucial aspect of the genetic basis of the cEF factor.

**Keywords** Cognition, Schizophrenia, Bipolar, Neuropsychology, PRS

**Reference**

Hatoum AS et al (2023) Genome-wide association study shows that executive functioning is influenced by GABAergic processes and is a neurocognitive genetic correlate of psychiatric disorders. Biol Psychiatry 93(1):59–70

**Grants**

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## Genetic and Environmental Influences on Self-Reported Treatment Outcomes for Anxiety and Depression: A Twin Study

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**Abstract**
In recent years there has been a marked increase in the number of young adults experiencing anxiety and depression. Despite the existence of relevant treatments, many individuals with a common mental health disorder do not receive help, and of those who do, a proportion do not recover. A better understanding of the factors that influence individual differences in treatment related outcomes is crucial for improving prognosis. However, there has been limited genetically sensitive research in this area. This study aims to investigate the genetic and environmental influences on self-reported lifetime diagnosis of anxiety and/or depression, treatment receipt, and treatment outcome. We will use data collected at age 26 years from 8297 (65.0% female) twins in the Twins Early Development Study (TEDS), a UK cohort born between 1994 and 1996. Diagnosis of an anxiety disorder (generalised anxiety disorder, specific phobia, social phobia, agoraphobia, and panic disorder) or depression, treatment receipt, and treatment outcome were measured using a self-report version of the Composite International Diagnostic Interview, Short Form (CIDI-SF) for each disorder. Univariate twin analyses revealed, for the first time, significant genetic influence on treatment receipt and treatment outcomes as well as on lifetime diagnosis. Shared environmental influences were not significant. Findings from a multivariate Causal Contingent Common-Pathway Model to estimate the genetic and environmental influences on the covariance between study variables will also be presented.

**Keywords** Anxiety, Depression, Treatment receipt, Treatment outcomes, Twin study

**Grants**
We gratefully acknowledge the ongoing contribution of the Twins Early Development Study (TEDS) participants and their families. TEDS is supported by a Program Grant from the UK Medical Research Council (MR/V012878/1 and previously MR/M021475/1), with additional support from the US National Institutes of Health (AG046938). This study presents independent research part-funded by the National Institute for Health Research (NIHR) Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London. The views expressed are those of the authors and not necessarily those of the NHS, the NIHR, or the Department of Health.

## Parental Effects on Child ADHD Outcomes via DNA Methylation at Birth: A Genetically Informed Prospective Cohort Study

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**Abstract**
**Background**
DNA methylation (DNAm) is an epigenetic process that has been associated with a variety of environmental exposures as well as with (mental) health conditions. There is mixed evidence for the mediating role of DNAm, or whether DNAm is rather a biomarker of exposures and/or outcomes. Here we investigate the role of DNAm at birth in mediating potential effects of parental factors on child attention deficit hyperactivity disorder (ADHD) outcomes using genetically informed approaches.
**Methods**
We calculated parental and child polygenic scores for education, smoking and ADHD to examine individual/familial genetic effects on ADHD via DNAm, indicated by methylation profile scores (MPS) for education, smoking during pregnancy and ADHD. Furthermore, we performed two-step within-family Mendelian randomisation to test causal parental effects on child ADHD via DNAm.
**Results**
Both parental and child polygenic scores were associated with child ADHD outcomes. Parental phenotypes (education, smoking during pregnancy) were associated with their respective MPS in cord blood. Results of the mediation analyses differed between observational regression analyses and genetically informed analyses, and we find little evidence for a causal direct effect of parental factors on ADHD.
**Conclusion**
Our results suggest that DNAm at birth is associated with parental factors and child ADHD outcomes, however, findings do not provide strong evidence for causal effects.

**Keywords** Epigenetics, Polygenic scores, Mendelian randomization, MoBa, GenR

**Grants**
This work is supported by the ERC under the European Union's Horizon 2020 Research and Innovation Programme (#863981).

## The Phenotypic and Genetic Architecture of a Neurodevelopmental Spectrum Across Childhood and Adolescence

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**Abstract**
Transdiagnostic dimensional approaches, such as models including general psychopathology (p) factor, and internalising and externalising spectra, are a useful alternative to categorical approaches for uncovering the aetiology of psychiatric conditions and informing their classification. Recent phenotypic research delineates an additional neurodevelopmental spectrum broadly capturing features of neurodevelopmental conditions and accounting for their high co-occurrence, but its aetiology remains to be clarified. In > 10,000 twins from the Twins Early Development Study (TEDS), we extracted a phenotypic neurodevelopmental factor through hierarchical exploratory factor analysis of neurodevelopmental and mental health assessments in childhood (age 7), early adolescence (age 12) and mid-adolescence (age 16). Linear multilevel models examined the association of this neurodevelopmental factor with polygenic scores for psychiatric disorders, cognitive and educational outcomes, and brain phenotypes. Results across development indicate that the neurodevelopmental factor is predicted by polygenic scores (PGS) for neurodevelopmental disorders (ADHD, Tourette Syndrome),

psychiatric disorders (particularly mood disorders), cognitive ability, and educational achievement. Most effects remained significant when controlling for other psychopathology dimensions (e.g., externalising and internalising factors). Multivariate analyses combining multiple PGS further confirm that the neurodevelopmental spectrum reflects transdiagnostic genetic liability for neurodevelopmental disorders, but also for related psychiatric, cognitive and educational phenotypes. These findings extend prior phenotypic work, support the validity of the neurodevelopmental spectrum, and further our understanding of its aetiology across development.

**Keywords** Neurodevelopmental, Transdiagnostic, Factor analysis, Polygenic score, Development

**Grants**

TEDS is funded by a Programme Grant (MR/V012878/1) to Professor Thalia C. Eley from the UK Medical Research Council (MRC) (previously MR/M021475/1 awarded to Professor Robert Plomin).

## Mapping Brain–Personality Associations with Genetic Correlations Reveals Polylocalised, and Replicable Brain Maps

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**Abstract**

Mapping normal personality trait variation to brain systems has been theoretically attractive for personality neuroscience, but empirically largely unsuccessful. Here, we used genetic correlations to relate genome-wide association studies of 13 brain structure modalities (N ∼ 36K) to the GWASes of Neuroticism (N ∼ 380K, UKB) and Big Five (N ∼ 75K, Estonian Biobank, EstBB).

We found several univariate associations, mostly with cortical surface area. Namely, UKB Neuroticism related negatively to occipital and orbitofrontal cortex. Conscientiousness related negatively to insular and inferior frontal cortex, and Openness positively to insular cortex. Further, the genetic correlation brain maps replicated—the UKB Neuroticism brain map was reproduced for EstBB Neuroticism across surface area and 12 other brain modalities (mean brain map r ∼ 0.56).

To interpret the personality brain maps, we used the Neuromaps toolbox that integrates dozens of brain maps from the literature. Namely, we related the association maps that personality had with brain surface area to six common brain system components. A fronto-parietal sensory-association brain map had positive associations with Openness and negative associations with Neuroticism. Synaptic density was positively related to Extraversion and negatively to Conscientiousness.

In summary, personality is highly polylocalised in the brain, just as complex traits are highly polylocalised in the genome. The brain patterns may be interpreted in reference to common brain systems, suggesting a potential way forward in understanding the brain roots of personality.

**Keywords** Genetic correlation, GWAS, Brain, Personality, Brain map

**Grants**

Uku Vainik has been funded by Estonian Research Council's Personal Research Funding Start-Up Grant PSG759.

## Genome-Wide Association Meta-analysis of Attention-Deficit/Hyperactivity Disorder Symptoms and Symptom Domains in Adults

**Elis Haan[1,2], Laura E. Hegemann[3,4], Natàlia Pujol Gualdo[1], Katri Pärna[1], Siim Kurvits[1], Alexandra Havdahl[3], Elizabeth C. Corfield[3,4], Triin Laisk[1], Kadri Kõiv[1], Helga Ask[3,5], Kelli Lehto[1]**

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**Abstract**

Attention-deficit/hyperactivity disorder (ADHD) is characterized by inattention (INA) and hyperactivity–impulsivity (IMP) symptom domains, but the underlying genetics of adult ADHD symptoms is largely unknown. We aimed to conduct genome-wide association study (GWAS) meta-analysis on adult ADHD symptoms and INA/IMP domains in two large population-based biobanks.

ADHD symptoms were assessed with the six-item Adult ADHD Self-Report Scale in the Estonian Biobank (n = 77,276) and the Norwegian Mother, Father and Child Cohort Study (n = 64,229). We performed GWASs in both cohorts on the continuous ADHD score, INA/IMP domains and for each item separately. Sex-stratified analyses were also conducted. Meta-analysis (n = 141,505) was followed by post-GWAS analyses, e.g. function annotation, SNP-based heritability, polygenic score (PGS) prediction.

Nine independent genetic loci were associated with ADHD score, IMP, and INA (four unique and five shared). Lead SNPs are located near genes H4C8, ARNTL/BMAL1, FNBP4, CADM1, BAIAP2, LYPD1, FOXP2, HTR1D and KCNN2, with five of them previously linked to ADHD. The SNP heritability estimates for ADHD score, INA/IMP and each item ranged between 0.04 and 0.08. Genetic correlations between men and women were 0.86, 0.86 and 1.0 for ASRS score, INA and IMP, respectively. The PGS of ADHD symptoms predicted ADHD diagnosis in a hold-out Estonian Biobank subsample.

This is the first genome-wide association analysis on adult ADHD symptoms that has identified associated genetic variants. Our preliminary results suggest differences in genetic architecture between ADHD symptoms, highly correlated sex-stratified results and significant PRS prediction of ADHD diagnosis. More detailed post-GWAS results will be presented at the meeting.

**Keywords** ADHD, Inattention, Impulsivity, GWAS, Adults

**Grants**

Estonian Research Council Grant No. PSG615.

**SY-7C [T]**

## Direct and Indirect Genetic Influences on Adolescent Mental Distress During the COVID-19 Lockdown

**Johanne H. Pettersen[1,2,3], Espen M. Eilertsen[4], Laura Hegemann[1,3,5], Laurie J. Hannigan[1,2,5,6], Kristin Gustavson[3,7],**

**Ingunn O. Lund[1,2,3], Pia Jensen[1,2,3], Ole A. Andreassen[8,9], Alexandra Havdahl[1,4,5], Eivind Ystrom[1,2,4], Ragnhild E. Brandlistuen[2], Helga Ask[1,2,4]**

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**Abstract**

**Background**

Indirect genetic influences are environmentally mediated effects of other family members' genotypes on an individual's phenotype, such as their reported level of mental distress. Such influences are estimated separately from the direct genetic influences (effects of an individual's genotype on mental distress). In this study, we will investigate direct and indirect genetic influences from mothers and fathers on mental distress levels reported by adolescent offspring during the COVID-19 lockdown period from March to May 2020. We also examine whether the indirect genetic effects changed over time in lockdown.

**Method**

Based on a preregistered analysis plan, we use data from the Norwegian Mother, Father, and Child Cohort Study (MoBa). Since March 2020, all adult MoBa participants and 16–17-year-old adolescents were invited to complete short bi-weekly COVID-19 surveys, including measures of mental distress at three timepoints (N = 10,375 adolescents). First, we use the Trio genome-wide complex trait analysis (Trio-GCTA) method to separate the direct and indirect genetic effects on adolescent's mental distress. Then, we use multivariate models to investigate the effects of specific risk factors through the use of polygenic scores for anxiety, depression, neuroticism, and ADHD. Lastly, we investigate whether these genetic effects (particularly indirect genetic effects) changed over time in lockdown.

**Results**

Results will be presented at the meeting.

**Discussion**

These results will provide new knowledge on how family members influence each other under stressful circumstances. The pattern and strength of direct and indirect genetic effects in families can inform future family interventions.

**Keywords** Adolescent mental health, COVID-19, Trio-GCTA, Polygenic risk, MoBa

**Grants**

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## The International Statistical Genetics Workshop

**Michel G. Nivard[1,2], Abdel Abdellaoui[3], David M. Evans[4], Matthew C. Keller[5], Jeff Lessem[5], Hermine H. Maes[6], Sarah E. Medland[7], Benjamin M. Neale[8]**

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**Abstract**

The International Statistical Genetics Workshop (ISGW), and its predecessors, originated in 1987 and have taken place in Boulder, Colorado since 1990 (Maes 2021). The ISGW has been an important gateway to the field of behavioral genetics.

Recent workshop innovations integrate two programs aimed at increasing diversity of the student body. For each ISGW ten diversity scholarships are available for US citizens. Additionally, the International Scholar and Cultural Exchange Program (ISCEP) provides funding for students from lower and middle income countries to participate.

Following the success of two online workshops during the pandemic, we switched to a triennial cadence with workshops focused on latent genetic models (2024, 2027), molecular genetics (2025, 2028) and online (2026). Historically materials used during the workshops are made available online. The website (https://www.colorado.edu/ibg/isgw-online) has been updated with an organized and indexed repository of video lectures developed during online years, allowing us to increase the reach of the workshop.

Changes to the curriculum more tightly integrate familial and molecular genetic analysis as a means to understand quantitative genetics. In years with an epidemiological focus we aim to unify twin/family methods with molecular data to better understand and quantify sources of variation for complex traits. The biologically focused years teach advances in genetic discovery and interpretation of those results. These changes ensure the ISGW evolves to cover methods at the cutting edge of our field and leverages the union of familial and genetic data to answer epidemiologically and biologically pertinent scientific questions, while increasing accessibility of the ISGW.

**Keywords** ISGW, Boulder, Training

**Consortia**

ISG Planning Committee.

**Grants**

Funding by the Regeneron Genetics Center and NIMH R25 MH019918.

## Leveraging Genetics to Examine the Causal Role of Adolescent Depression on Later Adult Outcomes: Evidence from One Sample and Two Sample Mendelian Randomization

**Alex S. F. Kwong[1], Robyn Wootton[2], Mark J. Adams[1], Amelia Edmondson-Stait[1], Tom Palmer[3], Christel Middeldorp[4], Andrew McIntosh[1]**

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**Abstract**
**Background**
Adolescent depression is common, and associated with later poorer social, psychological and behavioural outcomes. However, the majority of these findings are associational, and not necessarily causal. Recent advances in genomic research, and the use of longitudinal data to help improve precision and temporal sequences can allow for more causal inference methods to be tested.
**Methods**
We used prospective measures of self-reported depression, assessed across adolescence and a variety of early adulthood outcomes in individuals from the Avon Longitudinal Study of Parents and Children (ALSPAC; n = ∼ 4411). We examined associations between adolescent depression and outcomes through Poisson regression and one-sample MR. We then used summary outcome data from the UK Biobank to test causal associations between depression and outcomes using two-sample MR.
**Results**
Adolescent depression episodes were observationally associated with all of the downstream adult outcomes. Moreover, there was also causal evidence from one-sample MR that a greater number of adolescent depression episodes were associated with poorer general health, greater financial problems, daily smoking and lower physical activity. Two sample MR supported these results. There was no evidence of instrument confounding.
**Discussion**
Adolescent depression is likely to play significant role in shaping later adulthood outcomes, as shown by traditional regression analysis, and through more causal inference methods such as Mendelian randomisation. These findings suggest that preventing and successfully treating adolescent depression could mitigate some of the negative downstream effects on later outcomes. More research is needed to determine whether the timing or onset of adolescent depression is more important.

**Keywords** Mendelian randomization, Depression, Cohort studies, Longitudinal

**Consortia**
Psychiatric Genetics Consortium Major Depressive Disorder Working Group.

**Grants**
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## Phenotype Imputation Increases the Power of Rare Genetic Association Studies of Fluid Intelligence in UK Biobank

**Wei Huang[1], David van den Berg[2], Petr Danecek[1], Daniel S. Malawsky[1], Dirk J. A. Smit[2], Karin Verweij[2], Sarah J. Lindsay[1], Abdel Abdellaoui[2], Matthew E. Hurles[1], Hilary C. Martin[1]**

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**Abstract**
UK Biobank (UKB) is the largest study with measurements of cognitive ability and has greatly increased the power for association studies of this phenotype. However, fluid intelligence (a key measure of cognitive ability), is missing in nearly two-thirds of participants.

In this study, we use SoftImpute to impute missing fluid intelligence in UKB by using a set of 83 phenotypes that show high phenotypic and genetic correlations with measured fluid intelligence. The phenotypic correlation between imputed and real data was 0.57, similar to the test–retest correlation in people who took the test twice (r = 0.65). We found a high genetic correlation ($r_g$ = 0.94) between imputed fluid intelligence and a previous GWAS of intelligence (Savage et al. 2018). By combining imputed and measured fluid intelligence scores in a GWAS, we increased the number of genome-wide significant SNPs from 879 to 3635. Using the exome-sequence data, we found significant negative associations between rare deleterious variant burden and both measured (N = 103,128) and imputed (N = 213,440) fluid intelligence. The effect was slightly stronger in observed data than in imputed data (ß = − 0.026 vs. − 0.022), but more significant in the latter due to the larger sample size ($p$ = 1.21E−17 vs. 8.83E−26).

Our results suggest imputing missing fluid intelligence values can effectively increase sample size and imitate the phenotypic and genetic correlation of observed data simultaneously. We are now exploiting these imputed fluid intelligence measures to increase the power to find genes in which rare variants are associated with cognitive ability.

**Keywords** Phenotype imputation, Fluid intelligence, UK Biobank, Association studies

**Consortia**
Department of Human genetics, Wellcome Sanger Institute Department of Psychiatry, Amsterdam University Medical Center.

## Parent-Specific and Individual Genetic Effects of Education-Related Polygenic Scores on Behavioural Problems in Childhood: A Longitudinal Analysis of the Millennium Cohort Study

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**Abstract**
Parental educational attainment has consistently been associated with a variety of children's outcome, including conduct problems. The association between parental education and child behaviour is complex and involves a combination of genetic and environmental factors. New methods combining family designs with molecular data provide an opportunity to explore how parental heritable characteristics and children's behaviour could be the results of parental transmission of genetic risk or indirect genetic influences via the environment (i.e., genetic nurture).

Recent studies analysing mother–father–child trios (i.e., trios) have found significant genetic transmission effects on conduct problems but no genetic nurture effects for a wide range of risk factors for conduct problems, except for small genetic nurture effects of maternal polygenic scores for educational attainment and cognitive performance.

In this talk, we present the results from a longitudinal analysis on genetic transmission and genetic nurture effects of parental polygenic scores for educational attainment, and cognitive and non-cognitive

skills, on child's externalizing and internalizing problems. We analysed data from 3228 trios from the Millennium Cohort Study where parents reported on child internalizing and externalizing difficulties when children were 3, 5, 7 and 14 years old.

This study answers to the need to replicate recent studies on this topic, expanding previous research by including data from children at an earlier age (3 and 5 years old), evaluating internalizing as well as externalizing problems, and analysing the effect of polygenic scores for cognitive and non-cognitive skills separately along with educational attainment.

Detailed results and implications will be discussed.

**Keywords** Behavioural problems, Genetic nurture, Cognitive skills, Non-cognitive skills, Polygenic scores

**Grants**

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## Investigating Non-additive Genetic Effects in Education Outcomes Using Longitudinal Data

**Ralph Porneso[1], Espen Moen Eilertsen[1], Perline Demange[1], Ziada Ayorech[1], Nicola Barban[2], Alexandra Havdahl[3], Eivind Ystrøm[1]**

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**Abstract**

Genome-wide association studies (GWAS) on educational attainment and related outcomes have succeeded at identifying hundreds to thousands of genetic variants with small additive effects using measures collected at a single time point. Recently, GWAS leveraging measures collected at multiple time points have found novel genetic associations, suggesting the presence of previously undetected time-specific variants. We hypothesize that a subset of these variants may be involved in non-additive interactions, i.e. gene-by-gene or gene-by-environment. In the present study, we use trajGWAS to concurrently model longitudinal mean and variance effects on academic performance in Math and Reading at grades 5, 8 and 9 in $\sim$ 70,000 children from the Norwegian Mother Father and Child study (MoBa). We also perform both population and within-family analyses to differentiate direct and indirect effects on the phenotypic mean and variance. Summary statistics from our trajGWAS of academic performance will be used to construct a polygenic index based on variance effects (vPGI) in the ALSPAC cohort to investigate the cumulative effect of these variants on phenotypic dispersion. This study is a first step into detecting variants with non-additive effects, which might help us better understand gene-by-gene and gene-by-environment interactions, as well as partially bridge the missing heritability gap.

**Keywords** G $\times$ E, Longitudinal, Variance GWAS, vPGI, MoBa

**Grants**

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## Intergenerational Transmission of Comorbid Internalizing and Externalizing Psychopathology at Age 11: Evidence from an Adoption Design for General Transmission of Comorbidity Rather than Homotypic Transmission

**Kristine Marceau[1], Sohee Lee[1], Muskan Datta[1], Olivia C. Robertson[2], Daniel S. Shaw[3], Miksaki Natsuaki[4], Leslie D. Leve[5], Jody Ganiban[6], Jenae M. Neiderhiser[7]**

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**Abstract**

Psychopathology is intergenerationally transmitted through both genetic and environmental mechanisms via heterotypic (cross-domain), homotypic (domain-specific), and general (e.g., 'p-factor') pathways. The current study leveraged an adopted-at-birth design, the Early Growth and Development Study (57% male; 55.6% White, 19.3% Multiracial, 13% Black/African American, 10.9% Hispanic/Latino) to explore the relative influence of these pathways via associations between adoptive caregiver psychopathology (indexing potential environmental transmission) and birth parent psychopathology (indexing genetic transmission) with adolescent internalizing and externalizing symptoms. We included composite measures of adoptive and birth parent internalizing, externalizing, and substance use domains, and a general 'p-factor'. Age 11 adolescent internalizing and externalizing symptom scores were the average of adoptive parent reports on the Child Behavior Checklist (n = 407). Examining domains independently without addressing comorbidity can lead to incorrect interpretations of transmission mode. Therefore, we also examined symptom severity (like the 'p-factor') and an orthogonal symptom directionality score to more cleanly disentangle transmission modes. The pattern of correlations was consistent with mostly general transmission in families, with youth showing comorbid internalizing and externalizing symptoms, rather than homotypic transmission. Findings more strongly supported potential environmental or evocative mechanisms of intergenerational transmission than genetic transmission mechanisms. Parent specific effects will be discussed.

**Keywords** Heritable risk, Adoption design, Environmental risk, Intergenerational transmission, Psychopathology

**Grants**

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## Longitudinal Genetic Effects of Common and Rare Variation on Cognition and Behavioural Traits Across Development

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**Abstract**
Genome-wide common and rare protein-coding variants are known to play a role in risk of intellectual disability and contribute to the heritability of cognitive and behavioural traits in the general population. However, the contribution of these variants to cognition across development is unclear. Using two UK birth cohorts (> 6000 unrelated probands respectively) with SNP array and whole exome sequence data, we investigate the temporally dynamic genetic architectures of cognitive performance and behaviour. We perform mixed effects modelling with polygenic indices and deleterious rare variant burden (DRVB) scores, a measure of the exome-wide burden of deleterious variants that considers genic constraint, to assess associations with repeated measures of IQ and behavioural phenotypes estimated via the Strengths and Difficulties Questionnaire (SDQ) across early life. We find DRVB is negatively associated with IQ and positively associated with the conduct problems, hyperactivity, peer problems, and emotional symptoms SDQ subscales. The variance explained by DRVB decreases from childhood into adolescence for both IQ ($p < 10^{-4}$) and the hyperactivity SDQ subscale ($p = 0.048$). In contrast, polygenic indices for educational attainment and cognitive performance had a positive association with IQ and explained more variance as children aged ($p < 10^{-10}$). Polygenic indices had a negative association with the conduct problems, hyperactivity, and peer problems SDQ subscales. When conditioning on IQ, DRVB and polygenic index associations with SDQ measures were no longer significant ($p > 0.05$). Our results suggest common and rare exonic genetic variants have dynamic effects on cognition and related phenotypes across development.

**Keywords** Cognition, Rare variants, Polygenic indices, Constrained genes

**Grants**
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## An Exploration of the Genetic Etiology of MDD and Depressive Symptoms in the BIONIC Sample

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**Abstract**
Although broad phenotyping strategies have led to significant genetic discoveries for major depressive disorder (MDD), there is a need for uniform in-depth phenotyping that adheres to clinical criteria; both to replicate the findings thus far and to fine-tune the genetic signal of this heterogeneous disorder.

In the Netherlands we have established the Biobanks Netherlands Internet Collaboration (BIONIC) with the aim of facilitating Genome-wide Association Studies (GWAS) in a relatively homogeneous population. We collected data on DSM-5 lifetime MDD diagnosis, sociodemographic traits, depressive symptoms, episode characteristics and psychiatric history in 132,850 Dutch individuals with a standardized instrument; 66,684 individuals also have genome-wide SNP data available. We tested for genetic association in a series of GWASs of MDD case–control status and MDD symptom status, followed by genetic follow-up analyses.

Data from 16 cohorts were combined to create a dataset of N = 29,650 lifetime MDD cases (22%) meeting DSM-5 criteria and 94,300 screened controls. In the GWAS dataset of N = 66,684 Dutch individuals the prevalence of MDD was 25.3%. The GWA mega-analysis of lifetime MDD in the Netherlands identified one significant hit on chromosome 1. The SNP-based heritability for MDD was estimated at 17.7% on the liability scale, which we attribute to uniform adherence to clinical criteria and homogeneity of the sample population. The genetic correlation with the latest international MDD effort was 86.9%.

We initiated this project because the complex genetic basis of MDD requires large population-wide studies with uniform in-depth phenotyping, and show the feasibility of such an approach.

**Keywords** Depression, GWAS, Heterogeneity, Symptomatology

**Consortia**
The BIONIC Consortium.

**Grants**
Funding for the BIONIC Project was awarded to Dorret Boomsma and Brenda Penninx by the Biobanking and Biomolecular Resources Research Infrastructure (BBMRI-NL: 184.021.007; 184.033.111).

## Examining the Shared Genetic Liability of Prescription Opioid Misuse, Depression, and Anxiety

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**Abstract**
The opioid crisis is an urgent and growing public health concern. To date, genetic examinations of prescription opioid misuse (POM), defined as using a prescription opioid in ways other than as prescribed by a physician, have been limited. Depression and anxiety are among the most commonly observed psychiatric disorders and are also frequently comorbid with POM, however little is known about their common genetic risk. We examined the shared genetic vulnerability between depression, anxiety, and POM within The National Epidemiological Survey on Alcohol and Related Conditions-III (NESARC-III N = 11,744, depression N = 3383 [27.0%], anxiety N = 2567 [20.3%], POM N = 1718 [14.6%]). The SNP-heritability estimated within NESARC-III was 6% (SE = 0.03) for depression, 1% for anxiety (SE = 0.03), and 3% for POM (SE = 0.05). When examined, we did not detect significant co-heritability between POM and depression, and POM and anxiety. We also examined the predictive utility of depression and anxiety Polygenic Scores (PGS) to account for the variance in POM. PGSs were developed using two of the largest available meta-Genome Wide Associations Studies available. We found that POM was genetically correlated with the depression PGS ($r = 0.02$, $p = 9.00 \times 10^{-3}$); however, after accounting for covariates, this association no longer reached statistical significance ($\beta = 0.21$ [SE = 0.13], $p = 0.12$). POM was not associated with the anxiety PGS. This study provides some evidence for the shared genetic vulnerability between POM and depression, however larger samples are needed for replication.

**Keywords** Opioid, Depression, Anxiety, Comorbidity

**Grants**
This work was supported by a Grant from the National Institute on Drug Abuse (DA042103; awarded to RHCP).

## Modeling Intergenerational Parental and Extended Family Effects in the Presence of Indirect Assortative Mating

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**Abstract**
Kinship-based intergenerational models, such as the children-of-twins model, typically distinguish between passive genetic transmission and effects of parental traits. They typically do not model effects of the wider environment, which are evident in the similarities between extended family members (uncles, aunts, cousins). Recent studies indicate that extended family effects could be important in explaining parent–child associations.

Assortative mating is particularly strong for educational attainment. This necessitates the inclusion of mating patterns in studies on education and related traits. Existing studies accounting for assortative mating typically assume direct assortment, based on observed phenotypes, even though considerable evidence suggests that partner correlations in educational attainment are due to indirect assortment on correlated traits. We present a model that captures indirect assortative mating using extended family data and apply it to intergenerational modelling of twins, siblings, and their partners and children.

In children-of-twins models, effects of the parental phenotype depend on the correlations between parents and offspring being higher than the avuncular correlations. However, wide family effects increase both parent–offspring and avuncular correlations, making the difference between these less prominent. This leads to downwardly biased estimates of parental effects in the presence of indirect assortative mating. We present an intergenerational model accounting for indirect assortative mating. Using population-wide Norwegian register data, we present empirical results for selected phenotypes, including educational attainment.

**Keywords** Intergenerational transmission, Children of twins, Assortative mating

**Grants**
Norwegian Research Council (#334093, #300668).

## Does Emotional Distress in Pregnant Mothers Have a Detrimental, Causal Influence on Offspring Outcomes? A Systematic Review of Causal Inference Research

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**Abstract**
There is a vast literature documenting associations between mothers' mental health during pregnancy and child outcomes. Questions remain as to whether these associations represent causal pathways. It is important to determine whether possible confounding factors (e.g., mother–child genetic relatedness, or mothers' postnatal mental health) drive spurious associations between mothers' mental health during pregnancy and child outcomes, before assuming causality. Unravelling causal pathways in this area is important for providing parents with accurate and effective information and support during pregnancy.

We conduct a systematic review of empirical research involving causal inference methods to account for confounding in associations between mothers' emotional distress in pregnancy and any offspring outcome, at any age postnatally. Causal inference methods could include [A] Artificially occurring experiments that randomly manipulate the exposure (e.g., randomised controlled trials: which reduce mothers' emotional distress); [B] Naturally occurring quasi-experiments using random variation in the exposure (e.g., instrumental variable analyses, including random exposure to natural disasters: which induce mothers' emotional distress); [C] Naturally occurring quasi-experiments using naturally occurring adjustment methods (e.g., sibling comparison or extended family designs: which control for unmeasured confounding by design); or [D] Control-based statistical methods [e.g., doubly robust methods: which include an exposure model (e.g., propensity score) and an outcome model to adjust for confounding].

3114 Records have been screened against our criteria. We critically assess the retained records, summarising whether associations between mothers' emotional distress during pregnancy and offspring postnatal outcomes survive correction for potential sources of measured and unmeasured confounding addressed across different causal inference methods.

**Keywords** Pregnancy, Anxiety, Depression, Quasi-experimental, Causal inference, Child development

**Grants**
This work was supported by a Wellcome Trust Senior Research Fellowship awarded to TAM (220382/Z/20/Z).

## Isolating the Genetics of Mania through Genomic Structural Equation Modelling

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**Abstract**
Bipolar disorder, also known as manic-depression, affects 1–2% of the population and is a severe, often progressive disorder that features relapsing–remitting and cyclical episodes of mania and severe depression, frequently accompanied by psychosis. While there has been considerable success in understanding the genetics of depression and psychosis via successful studies of major depressive disorder

(MDD) and schizophrenia, there has been little study of the specific genetics of mania. This is because genetic studies of bipolar disorder include genetic effects associated with all three of its primary phenotypic dimensions: mania, depression, and psychosis. Understanding the genetic effects specific to mania could contribute to the design and discovery of more targeted therapeutic interventions for bipolar disorder.

We first processed and QC'd GWAS summary statistics via the GenomicSEM (genomic Structural Equation Modelling) R package, using its implementation of the LD Score Regression (LDSR) algorithm to estimate genetic covariances, which we then used in a gSEM model based on the GWAS-by-subtraction approach. We extended the traditional GWAS-by-subtraction to allow for dual phenotype subtraction and constructed a mania latent trait by subtracting the genetic effects of MDD and schizophrenia from the genetic effects associated with bipolar disorder. The SEM analysis revealed significant and variable associations between bipolar disorder and Depression, Psychosis, and Mania, with standardized estimates/weightings of 0.29, 0.58, and 0.67 respectively. Seven significant genomic loci associated with mania were identified. The genetic correlation pattern validates the execution of the subtraction.

**Keywords** Genetics, Behavioral, Biological psychiatry, Structural equation modelings, Human genetics, Genomic medicine

## Nature Meets Schooling: Why Boys Fall Behind in GPA Despite Equal Genetic Potential

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**Abstract**

Girls are now outperforming boys in school. In Norway, this is most noticeable in the middle school grade gap: out of a maximum of 6 points, the average GPA is 4.5 for girls, but only 4.1 for boys. As we found that middle school GPA in Norway is highly heritable ($h^2 \sim 0.77$), we conducted a genetically-informed study of whether the gender gap could be explained by differential gene–environment interaction ($G \times E$) across genders. We linked data on the GPA of Norwegian 10th-grade students and population-wide administrative data with molecular genetic data from the Norwegian Mother, Father and Child (MoBa) study which consists of > 27,000 genotyped parent–child trios. $G \times E$ was investigated by regressing GPA on polygenic indexes (PGI) for education. We found that within-family PGI for education predicted GPA similarly for both genders (girls: $\beta = 0.22$, SE = 0.01, boys: $\beta = 0.24$, SE = 0.01), but that this association varied more for girls across schools (SD = 0.06) compared to boys (SD = 0.03). To address limitations in the PGI approach, we will also extend our study to incorporate an extended family model which includes relations as distant as second cousins, who attend both same and different schools. This approach allows us to expand our sample size to 600,000 students and analyze over 2,000,000 genetic relationships. Combining PGIs and the extended family by school design will allow us to paint an unprecedented detailed picture of how gender-specific gene–environment interactions produce gender differences in GPA.

**Keywords** Extended family model, Polygenic indexes, Educational gender gap, $G \times E$

## Multivariate Path Analysis with the Reticular Action Model (RAM)

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**Abstract**

In 1984, in the journal *Genetic Epidemiology*, Vogler and Cockerham published an extension of the rules of path analysis to multivariate data collected from relatives. This method replaced latent and observed variables by vectors of variables, and the paths connecting them by matrices of paths. Although not widely used, the method has considerable potential for multivariate generalizations of path models, for both data from relatives and other situations such as longitudinal studies. Onyx software (onyx-sem.com) enables users to draw path diagrams, and have them translated to OpenMx code. Drawing multivariate diagrams with Onyx (or any other software) rapidly becomes tedious when the number of variables is greater than two. To circumvent this inconvenience, an algorithm to convert univariate-type Onyx models to the multivariate case is described. The benefits and drawbacks of utilizing the RAM formulation for multivariate models of familial resemblance are considered.

**Keywords** SEM, Relatives, Path analysis, Methodology

**Grants**

Dr. Neale is grateful for support from NIH Grant R01 DA049867 from the National Institute on Drug Abuse.

## Adolescent Multimorbidity and Health-Related Quality of Life: Genetic Insights from the Norwegian Mother, Father and Child Cohort Study (MoBa)

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**Abstract**

Multimorbidity, defined as the presence of two or more conditions at the same time, is a phenomenon that has received limited attention in the context of younger age groups. The intrinsic heterogeneity of multimorbidity calls for an approach to its study that broadens the focus from specific patterns to common underlying factors. Health-related quality of life (HRQOL) is a construct that includes physical, mental and social well-being dimensions. Studying the genetic

background of HRQOL in adolescence may help tracing trajectories of multimorbidity throughout life.

The aim of this study is to investigate the genetic architecture of three putative indices of HRQOL dimensions—pain, mental wellbeing and school functioning—measured in adolescence, and investigate links to psychiatric and somatic health conditions. Using a sample of over 83,000 genotyped 14-years-olds from the Norwegian Mother, Father and Child Cohort study (MoBa), we (1) investigate differences in the three HRQOL domains according to diagnosis and multimorbidity status; (2) estimate the polygenic prediction of adolescent HRQOL by polygenic scores (PGS) for 30 psychiatric, neurodevelopmental, and somatic conditions, testing whether associations are general or domain-specific; (3) estimate the SNP-wise associations and heritability of adolescent HRQOL domains via GWAS. Conditional on power, we also use genomic structural equation modelling to investigate how genetic factors contribute to the interplay of the three HRQOL domains, and the extent to which accounting for variation associated with educational attainment affects links to other health conditions.

**Keywords** Multimorbidity, Comorbidity, Health-related quality of life, Adolescence, School functioning

**Grants**

This work is supported by the South-Eastern Norway Regional Health Authority (Grant Number 2922083).

## Genetic Associations with Educational Field Choices in > 460,000 Nordic Individuals

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**Abstract**

Educational pathways are not only characterized by duration, but by field of interest. Yet little is known about the genetic architecture of educational fields as defined by the International Standard Classification of Education (ISCED). We conducted genome-wide association study (GWAS) meta-analyses of 10 broad fields of education using population-wide administrative data from Finland and Norway (n = 463,134). We identified 23 independent genome-wide significant loci associated with 7 of the 10 fields of education. SNP-based heritability estimates for educational fields (7% on average) remained significant after adjusting for educational attainment and were not significantly attenuated when controlling for birthplace and parents' fields, showing that the GWAS do not only capture influences on attainment or confounding by social stratification. We discovered two key components explaining the genetic associations with field choices, characterised by Technical versus Social tendencies (genetically correlated with Extraversion and Agreeableness), and Radical versus Normative tendencies (genetically correlated with Openness to Experience and Conscientiousness). Weaving together genetics, traits and social contexts, we create a fuller picture of the underpinnings of educational choices, shifting the focus of social science genomic research from the conventional hierarchy of attainments, towards multidimensional tastes.

**Keywords** Education, Genome-wide association study, Personality, Family, Geography

**Grants**

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## Political Stratification and Genetic Sorting: Evidence from Swedish Twins

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**Abstract**

Many democratic societies have witnessed an increasing degree of political polarization over the last half-century. This is manifested on the one hand as an ideological and affective divide, but also as a corresponding geographical divide: people are spatially segregated by their political affiliation. Considering existing evidence on the possible partial genetic underpinnings of political attitudes, this raises the question of to what extent political segregation is driven by genetic self-selection into neighborhoods, and whether this contributes to a restructuring of population stratification.

In this study, we use data on genotyped Swedish twins coupled with fine-grained data on election results at the neighborhood ("electoral district") level to investigate the genetic correlates of geographical segregation on political affiliation. Using within-family models, we are able to disentangle whether this gene–environment correlation reflects causal sorting patterns, but also (due to the long time-span covered by our data) whether such sorting has changed the character of pre-existing population stratification. Finally, thanks to detailed data on socioeconomic factors for the full population, we are also able to separate how much of these changes are rather driven by internal economic migration between regions that is merely correlated with ideological leanings.

**Keywords** Population stratification, Political polarization, Family design, Register data

**Grants**

Swedish Research Council (2023-01343; 2019-00244).

## Using Repeated Measures to Improve the Precision and Power of Genome-Wide Association Studies (GWAS)

**Alex S. F. Kwong[1], Mark J. Adams[1], Poppy Z. Grimes[1], Tim T. Morries[2], Gareth Griffith[2], Thalia C. Eley[3], Kate Tilling[2], Andrew McIntosh[1]**

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**Abstract**

**Background**

Genome Wide Association Studies (GWAS) have been vital to understanding the genetics of complex traits. However, the majority

of GWAS use data from one occasion, even in longitudinal studies with repeated assessments. Traits like depression are often subject to measurement error. Using a more precise and longitudinal phenotype of depression could reduce measurement error and increase power and precision in GWAS, further enhancing understanding of the genetics of depression.

**Methods**

We used data from the UK Biobank (max n = 462,566) on the PHQ-2, a measure of depressive symptoms assessed up to eight occasions. We tested a GWAS baseline model (a traditional cross-sectional GWAS) against four longitudinal GWAS models: (1) the mean of all the assessments, (2) a common factor model, (3) a precision-weighted shrinkage model and (4) a genomicSEM model. We conducted analysis across multiple ancestries.

**Results**

All longitudinal GWAS models outperformed the GWAS baseline model in European ancestries, with the most powerful model being the precision-weighted shrinkage model which had an increase of 34 more lead SNPs (121% increase), 107 more independent significant SNPs (173% increase), 50 more mapped genes (35% increase) and a greater SNP heritability (35% increase), compared to the baseline model. There were no GWAS lead SNPs identified in the South Asian baseline model, but two lead SNPs in the precision-weighted shrinkage methods.

**Discussion**

Leveraging repeated information within GWAS could help improve power and precision to detect novel biological underpinnings for traits like depression. Such methods could be useful for detecting novel loci in smaller studies.

**Keywords** GWAS, Depression, Longitudinal

## Schizophrenia Genetic Risk and Voxel-Wise Structural MRI: A Multiplex, Extended Pedigree Study

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**Abstract**

Despite schizophrenia's high heritability ($h^2 = 0.80$), the effects of its genetic risk on the brain remain poorly understood. To address this question, we employed a multiplex (two schizophrenia probands per family), extended pedigree (first to fourth degree relatives of probands) design that allows the estimation of genetic correlations between schizophrenia and brain measures in one of the largest such studies to date. A total of 1304 participants, including 789 relatives from 52 multiplex extended-pedigree families of 123 schizophrenia patients along with 517 unrelated controls were assessed for diagnoses and 506 participants provided quality MRI scans (230 relatives, 30 with schizophrenia, from 32 pedigrees and 276 unrelated controls). Here we present data from structural MRI scans to examine schizophrenia genetic risk effects on gray matter concentration at voxel-wise resolution. Analyses used the SOLAR-Eclipse program with False Discovery Rate (FDR) correction, $p < 0.05$. From approximately 2 million voxels, 210,152 (11.5%) were significantly heritable, primarily in subcortical structures. 24,101 (1.3%) Voxels were significantly genetically correlated with schizophrenia. Although distributed throughout the brain, voxels significantly genetically correlated with schizophrenia were especially concentrated in subcortical (including the brainstem) and frontal regions. Most (70%) of genetic correlations were negative, indicating lower gray matter concentration with schizophrenia risk. These results may suggest that developmentally earlier brain regions (subcortical compared to cortex) are most affected by schizophrenia genetic risk. These findings also suggest additional study of the brain stem, which is often not included in MRI studies of schizophrenia.

**Keywords** Schizophrenia, Structural MRI, Voxel-based morphometry, Extended pedigree

**Grants**

NIH: Collaborative R01s-MH42191, R01-MH63480, and R01-MH60722.

## Internalising and Cardiometabolic Multimorbidity in Individuals with Neurodevelopmental Copy Number Variants in UK Biobank

**Ioanna Katzourou[1], LINC Consortium[1], George Kirov[1], James Walters[1], Michael J. Owen[1,2], Peter Holmans[1], Marianne van den Bree[1,2]**

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**Abstract**

Physical and mental health multimorbidity is common, with one in two individuals with a mental disorder also reporting poor physical health. Multimorbidity is difficult to manage and results in reduced quality of life and increased healthcare costs. Individuals with a copy number variant (CNV) are at increased risk of developing either mental or physical ill health, however the effects of CNVs on multimorbidity are less well established. Moreover, the combined impact of common and rare genetic variation on multimorbidity is largely unexplored. In this study, we investigated the risk of developing internalising and cardiometabolic multimorbidity (ICM-MM) in individuals with a CNV associated with increased risk for neurodevelopmental disorders (NDD-CNV). We identified individuals with ICM-MM through electronic healthcare records (primary care records and hospital episode statistics) in UK Biobank. Three internalising (depression, anxiety and somatoform disorder) and five cardiometabolic conditions (hypertension, dyslipidaemia, obesity, type II diabetes and chronic kidney disease) were included in our definition of ICM-MM. The association of the presence of a NDD-CNV with ICM-MM was assessed. Individuals with a NDD-CNV were at higher risk of developing internalising disorder and each of the cardiometabolic disorders individually, as well as ICM-MM (OR = 1.25, $p = 3.26 \times 10^{-7}$). Ongoing work is exploring the effect of common genetic variation and environmental exposures on the risk of developing ICM-MM in individuals with a NDD-CNV.

**Keywords** Multimorbidity, Internalising, Cardiometabolic, Copy number variation, UK Biobank

**Consortia**

LINC Consortium.

## A Chip Off the Old Block? Genetics and the Intergenerational Transmission of Socioeconomic Status

**Sjoerd van Alten[1], Marina Aguiar Palma[2,3], Silvia Barcellos[4], Leandro Carvalho[4], Titus Galama[2,4]**

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**Abstract**
We study how one generation's genetics affect the economic prospects of the next generation, using molecular genetic data. To measure this, we merge genetic data of first- and second-generation individuals, available from the Dutch Lifelines Biobank, with socioeconomic outcomes of the third generation, available from administrative records. We use Mendelian imputation to study the causal relationship between the PGI for educational attainment on the next generation's educational attainment, income, and wealth. We find that parental genetic makeup substantially affects the SES-related outcomes of children: a one standard deviation increase in the parental PGI increases a child's years of schooling by 0.31 years, individual income by 2.06 percentiles, and net wealth by 1.12 percentiles. Although our current estimates identify a lower bound, their magnitudes are comparable to existing estimates of intergenerational effects obtained from studies of adoptees. Our estimates for wealth-related outcomes increase strongly with age, which highlights the importance of estimating genetic contributions of wealth late in the life cycle. Last, we show how the composite effect of parental genes on children's SES can be decomposed into three distinct mechanisms: (1) a genetically transmitted effect that arises because parental genes determine the genetic makeup of their children, (2) a non-genetically transmitted effect that arises because parental genes influence the family background of these children and (3) assortative mating. We conclude that the effects of parental genetic makeup on children's SES mostly operate through genetic transmission, but also find evidence of the existence of an effect operating through family background.

**Keywords** Socioeconomic status, Educational attainment, Intergenerational transmission, Inequality

**Grants**
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## The Impact of Stressful Life Events on Genetic Liability for Neuroticism

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**Abstract**
Stressful life events play key roles in the development of psychiatric disorders by directly increasing phenotypic risk and by amplifying the genetic risk for the phenotypes. While twin studies emphasize the central role of gene–environment interactions (G × E) for a broad range of phenotypes, there is limited G × E research in genome-wide data. Existing G × E studies have been unable to reliably identify genomic variants for neuroticism that are sensitive to differences in adversity. Using a novel methodological approach, we will expand recent work examining how stressful life events (i.e., biological sex, childhood maltreatment, abusive adult relationships, and socioeconomic status) interact with genetic risk variants for neuroticism. To do so, we will conduct moderated genome-wide association studies (i.e., G × E GWAS) and calculate genetic marginal effects for each SNP at each level of the four stressors in the UK Biobank. Our results will identify genetic associations for neuroticism that are magnified at specific levels of exposure to stressful life events. While we remain underpowered to identify genomic variants that significantly interact with stress, our analyses will identify loci that may only be significant if people experience adversity. Standard GWAS analyses will not identify these variants as the prevalence of adversity is relatively low, masking the genetic signal. Nevertheless, these loci significantly contribute to neuroticism. Following the moderated GWAS, we will calculate the SNP-heritability for each level of the four stressors to demonstrate that adversity amplifies the polygenic architecture of neuroticism. These results will quantify stress-specific mechanisms that magnify the genetic liability for neuroticism.

**Keywords** G × E, GWAS, Heritability, Neuroticism

## The Phenotypic and Genomic Structure of Normal and Psychopathological Personality in the GLAD + COPING Sample

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**Abstract**
We investigate the genetics of personality patterns in the Genetic Links to Anxiety and Depression or GLAD study and normal controls from other cohorts within the NIHR BioResource in the United Kingdom (GLAD+). We analysed normal and psychopathological personality measurements and their genomic associations alongside diagnostic psychopathological traits in the GLAD+ sample ($N > 23{,}000$). Personality was measured via the HEXACO normal personality scale and the Personality Inventory for DSM-5 (PID-5), designed to measure extreme personality relevant to psychiatric disorders. Trait heritabilities and genetic correlations were obtained using LD score (LDSC) regression and GCTA-GREML. The six HEXACO dimensions obtained LDSC heritabilities from 12% (Honesty Humility) to 3.8% (Emotionality). Heritabilities of the five factors in PID-5 were estimated as Negative Affect: 7.8% (0.029), Antagonism: 7.3% (0.027), Detachment: 8.5% (0.028), Disinhibition:

7.9% (0.025), and Psychoticism: 7.2% (0.028). LDSC genetic correlations ($r_g$) revealed strong relationships between PID-5 Negative Affect and normal range Neuroticism ($r_g$ = 0.76, s.e. = 0.32) while HEXACO Honesty–Humility showed significant, large negative $r_g$s with extreme personality including Negative Affect ($r_g$ = − 0.97, s.e. = 0.33) and Psychoticism ($r_g$ = − 0.82, s.e. = 0.32). We further explore the pattern of extreme and normal personality traits $r_g$s with psychiatric disorders, and whether previously reported patterns from other samples are replicated in the GLAD+ COPING cohort. Genome-wide association study (GWAS) summary statistics of our investigated traits will be made publicly available for further research.

**Keywords** Personality, Genomics, GWAS, GLAD

**Grants**

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## Associations Between Offspring and Parental Polygenic Scores for Depression and Offspring Clinically Diagnosed Mental Illness: Influences from Age at Onset and Parental Sex

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**Abstract**

**Background**

Depression is a moderately heritable mental illness affecting about 5% of individuals worldwide. Research on depression genetics is challenged by the heterogeneous presentation of this condition. We investigate potential sources of heterogeneity using polygenic scores (PGS) for depression (DEP-PGS), by estimating how offspring and parental DEP-PGS associate with clinically diagnosed mental illness in offspring.

**Methods**

We included data on 42,265 parent–offspring trios in the Norwegian Mother, Father and Child Cohort Study, linked to diagnostic information in the Norwegian primary and secondary care registers with up to 17 years of follow-up. Mental illness was indexed by a clinical diagnosis in primary or secondary care of depression, behavioral disorders, phobias, and developmental disorders. We calculated DEP-PGS for parents and their offspring, both for transmitted and untransmitted alleles, and estimated associations between DEP-PGS and mental illness in offspring using logistic regression, with and without adjustment for parental DEP-PGS.

**Results**

Both offspring DEP-PGS calculated using transmitted and untransmitted alleles associated positively with most mental illness diagnoses. The association between offspring DEP-PGS and a primary care depression diagnosis was estimated to odds-ratio (OR) = 1.38 (95% confidence interval = 1.33, 1.43), and to OR = 1.45 (1.37, 1.54) for a secondary care diagnosis. In models with parental PGS, the maternal DEP-PGS generally associated higher with offspring mental illness than paternal DEP-PGS, and was more pronounced for diagnoses in childhood.

**Conclusion**

DEP-PGS associates broadly with clinically diagnosed mental illness. Our results suggest that maternal polygenic liability towards depression predicts offspring mental health to a greater degree, compared to the paternal liability.

**Keywords** Polygenic scores, Mental illness, Gene–environment interplay

**Grants**

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## Developmental Timing of Pain Symptoms: Associations with Early Adulthood Pain and Anxiety in Two Longitudinal Studies

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**Abstract**

**Background**

Pain symptoms are highly common in childhood and adolescence and are frequently associated with co-occurring and subsequent symptoms of anxiety. Despite their high co-morbidity, few studies have investigated whether the timing of pain influences the development of anxiety in early adulthood.

**Aim**

This study will investigate the relationship between, and genetic and environmental influences of pain symptoms across childhood and adolescence and anxiety in early adulthood.

**Methods**

Data from the Twins Early Development Study (TEDS), and the Avon Longitudinal Study of Parents and Children (ALSPAC) will be used. Measures include somatic symptoms from the strength and difficulties questionnaire in TEDS (parent-report at ages 4, 7, 9, 12, 16 and 21 years; and self-report at ages 9, 12, 16 and 21), and ALSPAC (parent report at ages 4, 8, 9, 11, 13, and 16). Pain and anxiety symptoms at age 26 years were assessed in TEDS using the Patient Health Questionnaire-15, and at age 24 in ALSPAC using the Clinical Interview Schedule CIS-R questionnaire.

**Analysis**

Structural equation modelling will used to derive the following estimates: (i) the phenotypic longitudinal paths between latent factors of childhood and adolescent pain symptoms and early adulthood pain and anxiety, while adjusting for observed confounds (sex, age, and socioeconomic position TEDS and ALSPAC); (ii) a multivariate ACE model to estimate genetic and environmental influences of pain symptoms (TEDS); (iii) phenotypic informant discrepancies in parent and self-reported pain symptoms (TEDS). Findings will be discussed in light of the evidence of pain and anxiety in young people.

**Keywords** Mental health, Somatic symptoms, Genetic influences, Environmental influences

## The Association Between Physical Activity and Dementia Risk After Adjusting for Genetic and Environmental Confounds: A Multi-national Twin Study

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**Abstract**

Physical inactivity is an established risk factor in the epidemiological literature for dementia and cognitive decline. However, exercise interventions conducted in various populations have demonstrated mixed efficacy on mitigating cognitive decline which calls into question the causal nature of the association between physical activity and dementia. One possible explanation is that traditional population studies have not accounted adequately for genetic and environmental confounds in their association. In the current study, we tested the extent to which physical activity correlates with an estimate of dementia risk, statistically adjusting their association for overlapping genetic and shared-environmental variance. Using a large, multi-national sample of MZ and same-sex DZ twin pairs (families: $n_{MZ} = 940$, $n_{DZ} = 841$) drawn from twin studies in Sweden, Australia, and the United States, we fit a cross-sectional bivariate twin model to self-reported leisure time physical activity and latent dementia risk score data. We accounted for genetic and shared-environmental confounding by estimating additive genetic, shared-environmental, and nonshared-environmental associations between physical activity and dementia risk score. Our results show that physical activity significantly predicts dementia risk after adjusting for their genetic and shared-environmental correlations. Additionally, the nonshared environment accounts for 63% of the association between physical activity and dementia risk score suggesting minimal genetic confounding. The results of this study give insight into the physical activity and dementia association and provide evidence that suggests increasing leisure time activity levels may be one avenue to reduce dementia risk.

**Keywords** Twins, Subjective physical activity, Exercise, Dementia risk

## Comparative Penetrance and Pleiotropy of Neuropsychiatric Copy Number Variants and Polygenic Risk Scores in the United States Veteran Population

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**Abstract**
Rare copy number variants (CNVs) have established relevance for psychiatric and neurodevelopmental disorders, but recurrent events account for a small fraction of affected persons with diagnosed conditions. With the advent of large-scale biobanks linked to electronic health records (EHRs), the broader phenotypic consequences of CNVs can be explored. We undertook a comprehensive survey of CNVs in 662,681 participants in the Million Veteran Program (MVP) and Cooperative Studies Program (CSP) #572 studies. We observe comparable prevalences in MVP and the UK Biobank, including penetrant deletions of 22q11.2 and 3q29. Specific CNVs (e.g. 16p11.2 duplications, Klinefelter's) and total CNV burden (e.g. events spanning haploinsufficient/triplosensitive genes) were associated with schizophrenia, affective disorders, autism, and higher body mass index; these relationships were replicable in 124,000 African American and 50,000 Hispanic participants. Carriers had both a greater number of recognized diagnosed conditions and greater degree of phenotypic similarity across disease domains than age-, sex-, and ancestry-matched non-carriers. Phenome-wide association studies highlighted robust associations with laboratory tests, antipsychotic and antidepressant treatment, and self-reported personality traits and cognitive problems. Among 9300 participants with confirmed schizophrenia or bipolar I disorder with performance-based neurocognitive testing data, we observed carriers of deletions of 22q11.2 and 16p12.1, and duplications of 15q11.2 and 13q12.12, whose performance was 1.5 standard deviation below the mean of the aggregated sample of veterans. Ongoing analyses compare carriers of deleterious CNV and individuals in the top strata of multivariate polygenic risk, and highlight benefits of jointly modeling common and rare variants in large scale genetic studies.

**Keywords** Copy number variants (CNVs), Polygenic risk scores (PRS), Schizophrenia, Electronic health records (EHRs)

**Consortia**
Cooperative Studies Program (CSP) #572 Million Veteran Program (MVP) Consortium on the Genetics of Schizophrenia (COGS) Genomic Psychiatry Cohort (GPC) Investigators.

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## Associations Between Genetic Predisposition to Psychiatric Disorders and the Severity of Acute COVID-19 Illness: A Meta-analysis Across Five COVIDMENT Cohorts

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**Abstract**
**Background**
Psychiatric disorders have been associated with severity of COVID-19 illness, often measured by hospitalizations and mortality. To what extent shared genetic factors contribute to this association is poorly understood, especially in non-hospitalized general populations with milder COVID-19 illness course. We aimed to investigate the associations between polygenic liability to psychiatric disorders and acute COVID-19 illness severity using data from five population-based cohorts of the COVIDMENT consortium across Denmark, Estonia, Iceland, Norway, and the UK.
**Methods**
The total sample comprised 77,086 subjects (64% female, 41% SARS-CoV-2-positive) responding to questionnaires from May 2020 to March 2023. Acute COVID-19 severity was defined by (1) self-reported number of days bedridden (never, 1–6 days, > 7 days), and (2) symptom count. Polygenic risk scores (PRS) for nine psychiatric phenotypes were constructed using PRS-CS. Estimates from cohort-based stepwise adjusted regression analyses were meta-analyzed using random effects model.
**Results**
In the fully adjusted models, extended time bedridden was associated with the PRS of five psychiatric traits: anxiety (relative risk ratio 1.23 [1.10–1.38]), neuroticism (1.20 [1.07–1.35]), depression (1.19 [1.05–1.34]), schizophrenia (1.15 [1.03–1.30]), and bipolar disorder (1.07 [1.02–1.12]). Depression and neuroticism replicated using symptom count as the outcome. We additionally ran the models in COVID-19 negative individuals reporting flu-like symptoms, and seven out of nine psychiatric PRSs were associated with symptom count [incident rate ratio range 1.01–1.04].
**Conclusion**
These findings imply that individuals with a genetic predisposition to psychiatric conditions may be at risk of a more severe acute course of COVID-19 and other respiratory viral infections.

**Keywords** Psychiatric polygenic risk scores, Acute COVID-19 infection severity

**Grants**
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## Multi-generational Effects of Dementia Risk and Onset on Labour Market Outcomes and Financial Decision Making

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**Abstract**
This article uses random genetic variation and longitudinal economic data to better understand the economic costs of dementia. We construct a novel dataset by merging data from health records of dementia diagnosis and tax records available from administrative sources, with multi-generational genetic and survey-based data from the Dutch Lifelines Biobank. We explore the multi-generational structure of the Lifelines to estimate direct genetic effects of dementia risk not only on dementia diagnosis and cognition but also on measures of labour market participation, welfare use, income, net wealth, savings and healthcare expenditures. We analyse these outcomes for the affected individuals themselves but also for family members. We hypothesise that caretakers' labour market participation and financial decision-making will also be affected. We also explore the longitudinal nature of the dataset to look at the timing of the effects relative to the timing of dementia diagnosis. Finally, in terms of mechanisms we look at differential effects by gender, socio-economic status, age at onset of dementia and other known dementia risk factors such as hypertension and diabetes.

**Keywords** Dementia, Labour market, Net wealth, Direct genetic effects

**Grants**
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## Exploring Genetic Effects on the Variability of Childhood Behaviour in the MoBa Cohort

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**Abstract**
Gene–environment interaction is likely to contribute to human development, but effects are difficult to detect, especially for behavioural outcomes. Genetic variants associated with phenotypic variability are potential candidates for tests of interaction.

We tested for genome-wide association of mean (mGWA) and variance (vGWA) effects for 71 behavioural traits across childhood in the Norwegian Mother, Father and Child Birth Cohort Study (MoBa; NMEAN: 44,788). We estimated genetic effects on standardised factor scores (mean effects) and quantile integrated rank scores (variance effects). Independently associated loci were included in phenome-wide association analyses. We estimated the fraction of phenotypic variance accounted for by genome-wide mean and variance effects, as well as shared effects between behavioural traits. Summary statistics from vGWA were used to compute variance-based polygenic scores (vPGS) in an independent sample, the Twins Early Development Study.

Across the 71 behaviours, there were 17 independent genetic variants associated with phenotypic variability, and 44 associated with mean levels. Overall, mean and variance effects were uncorrelated, but this varied across phenotypes. Genome-wide variance effects explained up to 9.7% of the variance (mean = 1.8%), whereas mean effects explained up to 15.8% of the phenotypic variance (mean = 6.0%). Analyses including vPGS are ongoing.

This is the first study to comprehensively explore the role of variance effects on childhood behavioural traits. Overall, there were fewer variance effects than mean effects, likely due to smaller effect sizes and less statistical power. However, we detected several candidate SNPs for follow-up analyses and provide summary statistics for future meta-analyses.

**Keywords** Gene–environment interaction, G × E, vQTL, GWAS

**Grants**
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## The Role of Early-Life Positive Parent Attributes on the Genetic and Environmental Contributions to Mental Health Symptomology in Middle Childhood

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**Abstract**
**Objective**
Early-life experiences including caregiver attributes that shape the context in which children develop can impact a child's mental and physical health. However, less attention has focused on how early-life positive environmental factors influence child mental health outcomes. This study examined the moderating role of early-life positive parent personality on the heritability of mental health symptomology (internalizing, externalizing, ADHD symptoms) among twin children.
**Methods**
A subset of 399 families (31% MZ, 34% same-sex DZ, 34% other-sex DZ twins) from the Arizona Twin Project were included. Primary caregivers reported on multiple dimensions of their own personality (e.g., optimism, self-compassion) when the twins were 12 months. Approximately 8 years later (*Mean age* = 9.71), twins (51% female; 57% White, 27% Hispanic) and caregivers completed an assessment in which caregivers reported on child mental health symptomology.
**Results**
Univariate ACE models indicated genetic and shared environmental influences on internalizing (A = 0.35, C = 0.47, E = 0.18) and externalizing (A = 0.53, C = 0.34, E = 0.13) symptoms, whereas ADHD symptoms were largely influenced by additive genetics

(A = 0.81, C = 0.11, E = 0.08). As indicated by a worsening of fit when comparing full moderated to non-moderated models, early-life positive parent personality moderated the shared environmental contribution to both externalizing ($\Delta$-2LL = 3.72, $\Delta df = 2$, $p = 0.16$) and ADHD ($\Delta$-2LL = 3.25, $\Delta df = 2$, $p = 0.20$) symptoms.

**Conclusions**

At low levels of early-life positive parent personality there were greater shared environmental influences on child externalizing and ADHD symptoms. Findings highlight the contribution of early-life parent attributes to the environmental contributions on children's mental health symptomology and suggest targeting these parental factors to promote children's positive mental health outcomes.

**Keywords** Mental health, Moderated heritability, Positive parent attributes, Children, Twins

**Grants**

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## Maternal Health in Pregnancy and Autism Risk: Genetic and Non-genetic Mechanisms

**Vahe Khachadourian[1], Elias S. Arildskov[2], Jakob Grove[2], Paul F. O'Reilly[3], Joseph D. Buxbaum[1], Abraham Reichenberg[1], Lisa A. Croen[4], Diana Schendel[5], Sven Sandin[6], Stefan N. Hansen[7], Magdalena Janecka[8]**

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**Abstract**

Efforts to identify modifiable risk factors for autism have revealed multiple associations between maternal health in pregnancy and autism risk offspring. However, the mechanisms underlying these associations remain unknown, including the role of transmitted and non-transmitted genetic variation.

We used the Danish registry data (N = 1.13M), with 3 generation family linkage, and genotype data (N $\sim$ 130k, iPSYCH) to test the associations between all maternal diagnoses and autism. To triangulate evidence for genetic confounding, we implemented (1) family-based (paternal negative control; sibling comparison; risk in maternal cross and parallel cousins), and (2) polygenic risk score (PRS) analyses.

We identified 30 maternal diagnoses associated with autism risk, after accounting for their comorbidity and chronicity, familial correlations and potential confounders, and controlling for multiple testing. We observed evidence for genetic confounding for most of those associations, including through both transmitted (psychiatric and neurological disorders) and non-transmitted (obstetric complications) variation. For example, for epilepsy: (i) autism risk was the same irrespective of which parent was affected; (ii) children born to affected mothers had higher autism PRS, compared to children of unaffected women; and (iii) autism risk was the same in offspring of sisters and brothers of women with epilepsy, and for both higher than in the general population. We found no evidence for genetic confounding for very few associations, including predominantly medical codes indicating severe maternal injury in pregnancy.

In summary, associations between maternal health in pregnancy and autism risk are largely attributable to genetic factors, both transmitted and non-transmitted.

**Keywords** Autism, Familial confounding, Perinatal epidemiology, Neurodevelopment

## International Statistical Genetics Workshop Online Content and Resources

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**Abstract**

The material used each year to teach the International Statistical Genetics Workshop (ISGW) has been publicly available online since 1996. For every course held, a new release was created with the updated material used at the most recent course. The faculty of the ISGW have always intended the course material to be a common good to promote the learning and teaching of behavioral genetics. The COVID-19 pandemic meant the typically in-person course was held online for 2 years. These online courses led to the creation of over 100 videos which have been organized into a series of lectures and topics covering the broad topics of: Common and rare association, downstream genomic data analysis, theory and background, and twin and family. Each of those categories contains multiple lectures on topics such as multivariate and univariate twin models, Mendelian randomization, genomic SEM, polygenic scores, and many others. This resource is funded to be maintained and updated with additional content. The online ISGW content has already made an impact on the field. It has been viewed for over 11,800 h by students from more than 70 countries. Along with the lectures there are also practicals with example scripts, problems, and solutions. One of the primary goals of the online resources are to make behavioral genetics techniques accessible to people who are unable to travel to genetics workshops, and to supplement local expertise everywhere in the world. Entry to this resource is through the web at https://www.colorado.edu/ibg/isgw-online and the public discussion forum at https://isgw-forum.colorado.edu/.

**Keywords** Learning, Teaching, Genetics

**Grants**
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## Associations Between Cannabis Use and Suicidal Thoughts and Behaviors: A Clinical Longitudinal Sibling Study

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**Abstract**
Cannabis use has been associated with an increased risk for suicidality across the lifespan. However, there is a paucity of longitudinal research in this area, and the causal directionality of this association or the potential role of familial confounds remains unclear. Further, research addressing these questions has not been conducted in clinical samples with high-frequency cannabis use. The current study seeks to extend the present literature by examining the cross-sectional and prospective effects of cannabis use on suicidality using a co-sibling control analysis in a sample with earlier initiation and heavier use patterns than commonly reported. Participants were recruited from the Genetics of Adolescent Antisocial Drug Dependence study (N = 1261). Participants were primarily male (59.9%) and the sample was diverse (45.1% Non-Hispanic White). A sibling in each family was a clinical proband identified due to delinquent behaviors and/or substance use disorder symptoms. Semi-structured clinical interviews assessed substance use and suicidality. Covariates included alcohol use, tobacco use, age, and sex. Overall, results were not consistent with the hypothesis that cannabis use has a causal influence on suicidality (all $p$ values $> 0.05$) in this sample. Exploratory analyses suggest that the association between tobacco use and suicidality may be consistent with causality, but this finding should be interpreted with caution given that plausible alternative hypotheses were not examined. In sum, findings suggest that moderate-to-heavy adolescent cannabis use did not have adverse effects on suicidality in this high-risk clinical sample.

**Keywords** Cannabis, Suicide, Adolescence, Tobacco, Clinical

**Grants**
DA035804, DA053693, DA054087, DA057894, DA056408.

## Genetic and Environmental Influences on the Covariation Between Subjective Well-Being, Eudaimonic Well-Being, and Problematic Substance Use

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**Abstract**
In studying factors associated with problematic substance use, researchers have most often focused on characteristics that are associated with increased risk. However, it is also important to understand characteristics that are associated with resilience and reduced risk. Towards that end, there is a growing body of research that has examined the inverse association between well-being and substance use. There are two major traditions within the well-being literature: subjective (or hedonic) well-being, which deals with happiness, satisfaction with life, and the balance between positive and negative affect, and eudaimonic (or psychological) well-being, which deals with human potential and engagement with existential challenges of life. The present study aims to build on prior research on subjective well-being, eudaimonic well-being, and substance use by using a genetically informed sample to examine (a) the associations among subjective well-being (i.e., life satisfaction), psychological well-being, and substance use; and (b) the genetic and environmental influences on any observed associations obtained among these constructs. Analyses will involve data from the Colorado Adoption/Twin Study of Lifespan Behavioral Development and Cognitive Aging (CATSLife), which is a sample of (a) adoptive and nonadoptive probands and siblings, and (b) twins. Associations among measures of subjective well-being, psychological well-being, and substance use will be examined, and multivariate Cholesky decompositions will be conducted to examine the magnitude of the association between the two types of well-being and substance use explained by genetic, shared environmental, or nonshared environment influences.

**Keywords** Well-being, Substance use, Twin, Adoption

**Grants**
AG046938, DA054087, DA057894.

## Lifespan Predictors of Cognitive Aging in the Louisville Twin Study

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**Abstract**
Prevention and treatment of Alzheimer's disease and related dementias depends on understanding mechanisms that contribute to pathology decades before clinical symptoms of cognitive and functional decline are observed. In 2020, the Lancet Commission identified 12 potentially modifiable lifespan risk factors for dementia (Livingston et al. 2020). The Louisville Twin Study (LTS) is the first

twin study to utilize prospective childhood developmental data combined with comprehensive midlife assessment to investigate risk and protective factors for cognitive decline. We present results from 287 middle-aged twins (60 monozygotic pairs, 41 dizygotic pairs, mean age 51.9 years ± 7.0). All IQ scores are based on complete, in-person WISC and WAIS-IV assessments. Mean childhood IQ (average age of assessment 14.4 ± 1.9 years) was 102.7 ± 13.4 and mean midlife IQ was 105.4 ± 13.8. Our models identify risk and protective factors that predict midlife IQ after controlling for childhood IQ. Higher childhood socioeconomic status and midlife educational attainment were protective factors for cognitive decline, consistent with existing cognitive reserve hypotheses. Our models also suggest childhood trauma, smoking, psychopathology severity, physical functioning and mobility, and accelerated DNA methylation age in midlife are risk factors for cognitive decline. We combine identified predictors of cognitive aging to evaluate a unit-weighted index of cognitive decline. These results are followed up in "quasi-causal" twin analyses that identify phenotypic associations that are independent of between-family genetic and environmental confounds.

**Keywords** Midlife, Cognitive aging, Socioeconomic status, Epigenetic age, Louisville Twin Study

**Reference**

Livingston G, Huntley J, Sommerlad A et al (2020) Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. Lancet 396(10248):413–446

**Grants**

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## Associations Between Orexin-Enriched Sleep Polygenic Scores and Pain, Sleep Disturbances, and Alcohol Sips in Early Adolescence

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**Abstract**

Physical pain, sleep disturbances, and alcohol use are prevalent and often co-occur in adolescence, but few common genetic factors have been identified, especially in early adolescence when these issues start to emerge. The orexin system is integral to sleep–wake regulation, with recent evidence suggesting its role in reward system functioning and pain modulation, making it a candidate for a shared etiological factor.

We examined associations between orexin-enriched polygenic scores for sleep and pain, sleep disturbances, and alcohol sips in early adolescence. Data were from the Adolescent Brain Cognitive Development (ABCD) study [Black/African American (AA) n = 1784, White/European American (EA) n = 6180, Hispanic/Latinx (LX) n = 2409]. Number of pain sites, sleep disturbances, and alcohol sips were assessed via self-report at age 11/12. Polygenic scores (PGS) were created using PRS-CSx based on 27 genes enriched in the orexin system with risk alleles informed by GWAS on sleep issues in AA and EA samples (Austin-Zimmerman et al. 2023). Racially/ethnically aligned orexin-enriched PGS for sleep were used for the AA and EA subgroups and meta-analyzed scores for the LX subgroup. Pain, sleep disturbances, and alcohol sips were positively correlated in all subgroups. PGS were associated with pain, sleep disturbances, and alcohol sips only in the EA subgroup.

Findings suggest that during early adolescence pain, sleep disturbances, and alcohol sips may be partially driven by the orexin system in EA youth. Delineating the role of the orexin system may help inform its utility as a clinical target and in prevention efforts during early adolescence.

**Keywords** Polygenic, Orexin, Sleep, Pain, Substance use

## The Preprint Problem: Fringe, Genetically Informed Studies of Group Differences in Behavior Housed on Open Science Platforms

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**Abstract**

Preprint servers and open science platforms have revolutionized the scientific process. A fundamental feature of these platforms is a lack of peer review—virtually anyone with an internet connection can upload their research in a few clicks. Although this setup has facilitated rapid dissemination of results and open access to research, it has also enabled fringe researchers to post and share pseudoscientific, genetically informed studies of differences in behavior. Because preprint archives are now routinely used by mainstream academics, preprints grant a degree of legitimacy to fringe research that otherwise may have been relegated to a blog post or fringe publication. Previous studies have documented individual examples of pseudoscientific, genetic studies of behavioral differences among socially constructed groups being posted on preprint archives, but the scope of this problem remains unclear, making it difficult to formulate responses and potential solutions.

The present study aimed to quantify the number of pseudoscientific studies of group differences in behavior—including studies that used genetic methods—housed on popular preprint servers and open science collaboration platforms. Dozens of such preprints were identified. Text analysis was then used to comb these preprints for commonly used themes, terminology, and subject matter. Preprints tended to focus on controversial and politically charged topics, often comparing majority to minoritized groups. A subset of preprints was investigated in further depth to track their wider dissemination online. Challenges and possible solutions to managing fringe preprints will be discussed.

**Keywords** Open science, Misuse, Social implications, Text analysis, Preprints

## Stratifying Psychotic Disorder Severity and Genetic Risk Using Longitudinal Inpatient Hospitalization Data

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**Abstract**

**Background**

Despite successful efforts to identify risk variants, the genetic etiology of psychotic disorders remains poorly understood. Previous studies have often utilized cross-sectional, relatively superficial psychosis phenotypes. Longitudinal electronic health record and register data enable deeper phenotyping that appreciates heterogeneity in disease severity and course.

**Method**

Data were drawn from SUPER-Finland: over 10,000 participants with psychotic disorders were genotyped, sequenced, interviewed, and linked to population registers. We identified every inpatient hospitalization for which the primary reason for admission was a psychotic disorder. Analyses focused on SUPER participants for whom at least 20 years of register data following psychosis onset (n = 3637). Variables related to the frequency, duration, and temporal pattern of inpatient psychosis admissions were extracted, residualized on date of birth, and used for clustering. Clusters were compared on psychiatric and physical health comorbidities, other phenotypic and functional outcome measures, and polygenic risk profile.

**Results**

Results supported a two-cluster solution. One cluster was characterized by a more severe presentation including earlier age-of-onset, more total hospitalizations, and more total days in the hospital ($p$s $< 0.001$). This clustering did not recapitulate traditional diagnostic categories, supporting a transdiagnostic approach. Membership in the more severe cluster was associated with lower functioning, cognitive performance, and educational attainment; significantly higher average polygenic risk for schizophrenia ($p < 0.05$); and significantly lower EA PGS ($p < 0.01$). Clusters did not differ significantly on PGS for depression, bipolar disorder, or intelligence.

**Conclusions**

Longitudinal hospitalization data may help parse heterogeneity in severity and course of psychotic illness, facilitating genetic analyses.

**Keywords** Psychosis, Longitudinal, Polygenic scores, Population register, Electronic health record

**Consortia**

SUPER-Finland Researchers.

## When More Genetic Information Biases Decision-Making in Embryo Selection

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**Abstract**

Polygenic embryo screening (PES) is a technology that screens the DNA of embryos and uses polygenic scores (PGS) to estimate their likelihood of developing conditions and traits in the future. Based on these scores patients undergoing In Vitro Fertilization (IVF) can prioritize which embryos they select to transfer for implantation. In a randomized experiment with Reproductive Endocrinologists and Infertility specialists (REI Clinicians), we tested how the presentation of polygenic score information about embryos' risk of developing schizophrenia and Type 1 Diabetes affected their recommendation for embryo selection. Clinicians were shown two embryos (A and B) in one "Baseline" scenario. In another scenario, they were shown A, B, and C (a third embryo) with the highest risk for one of the two conditions but equal risk for the other, making it a "decoy" that is objectively at higher risk and would not be selected. Our results from 122 Clinicians demonstrate that when the decoy (embryo C) was introduced to target Embryo A, 52% of clinicians selected embryo B, but when the decoy embryo C was introduced to target Embryo B, 78% of clinicians selected embryo B. This suggests that over a quarter of clinicians switched their recommended selection due to the presence of the decoy. These findings have important implications for ethical considerations regarding the uses of PES and suggest that, in this context, more genetic information can lead to worse decision-making.

**Keywords** Polygenic embryo screening, Polygenic scores, Decoy effect, Decision making bias

**Grants**

This work was sponsored by the United States National Human Genome Research Institute of the National Institutes of Health (Grant #NIH R01MH011711).

## The Longitudinal Associations Between Adolescent Self-Esteem and Parental Warmth: A Twin Study in Early Adolescence

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[1]Psychology, Hebrew University of Jerusalem, Jerusalem, Israel

**Abstract**

There is ample evidence that the adolescent–parent relationship is associated with adolescent self-esteem. However, the majority of research and clinical practice has assumed that this association is unidirectional (i.e., parenting affects the child's self-esteem). A possible bidirectional association, in which adolescent self-esteem predicts parenting, has been rarely examined. We address this issue with a longitudinal study of twin adolescents. We examined data from a longitudinal study of Israeli twins (at ages 11 and 13, N = 1155) in which adolescents reported on their self-esteem and parents reported on warmth and involvement toward the child. A cross-lagged panel showed that self-esteem at age 11 predicted parental warmth for both mothers and fathers at age 13. The association was not bidirectional (i.e., parental warmth did not predict self-esteem). Differences in correlations between monozygotic and dizygotic twins were found for both self-esteem and parental warmth, indicating genetic contribution to both factors. We will expand the analysis and examine to what extent both self-esteem and parental warmth are accounted for by genetic factors, and whether the longitudinal association between self-

esteem at age 11 and parental warmth at age 13 can be accounted for by genetic factors.

**Keywords** Self-esteem, Parenting, Bidirectional effects

**Grants**
Israel Science Foundation.

## Genetic and Environmental Influences on Children's Temperament Dimensions and Types

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[1]Psychology, Arizona State University, Tempe, USA

**Abstract**
Temperament, or individual differences in emotional reactivity and self regulation, is frequently measured in childhood, moderately heritable, and forms the affective core of later personality. Theory suggests that temperament is an ideal moderator of development, such that the influence of environmental risk and promotive factors on development vary by temperament. The field largely relies on parent-reports along multiple dimensions of temperament. The current study compared heritability estimates obtained from (a) reported and observational assessments, and (b) dimensions (factor analysis) and types (latent profile analysis).

Samples included twins who were approximately 7–8 years of age participating in the Wisconsin Twin Project (WTP; 787 twin pairs, 49% female, 88% White), or the Arizona Twin Project (ATP; 359 twin pairs, 51% female, 30% Hispanic, 58% White).

With WTP, parent-reported temperament factors of negative affectivity ($a^2 = 0.79$), surgency ($a^2 = 0.49$), and effortful control ($a^2 = 0.54$) were heritable. Similar heritability estimates were obtained for microanalytically-coded observational Lab-TAB measures of temperament organized into Research Domain Criteria composites—frustrative nonreward ($a^2 = 0.78$, $c^2 = 0.17$), loss ($a^2 = 0.54$, $c^2 = 0.26$), fear ($a^2 = 0.44$, $c^2 = 0.55$), and reward responsiveness ($a^2 = 0.86$), with notable shared environment. Three replicated temperament types based on parent-report in ATP yield similar estimates, 'regulated/typically reactive' ($a^2 = 0.71$), 'dysregulated/negatively reactive' ($a^2 = 0.80$), and 'well-regulated/positively reactive' ($a^2 = 0.31$, $c^2 = 0.43$). Computer-assisted coding of facial negative affect ($a^2 = 0.54$) and positive affect ($c^2 = 0.46$), supports the pattern of higher shared environment with observational assessments.

Findings support the use of both reported and observational assessments, as well as dimensions and types as tapping into constitutionally-based individual differences that may moderate developmental processes.

**Keywords** Temperament, Childhood, Measurement, Latent profile analysis, Twins

**Grants**
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## Comparing Multivariate GWAS Results from Different Software Applications

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**Abstract**
The popularity of multivariate genome-wide association study (GWAS) methods is increasing rapidly, but the relative strengths and weaknesses of each approach remains unknown. In this project, we conducted genome-wide latent variable analyses of neuroticism in the European subsample of the UK Biobank in GW-SEM, TATES, a sum-score, and GenomicSEM. We find that GW-SEM identifies 530 genomic loci with significant associations with neuroticism, TATES identifies 98, the sum-score approach identifies 77, and GenomicSEM identifies 82. To rule out the possibility that the overabundance of associations from GW-SEM is due to statistical artifacts, we conducted a series of simulation studies to examine the Type I Error rate with combinations of simulated and permuted phenotypes and genotypes. We found no evidence of genomic inflation. Then, we conducted a series of GWAS simulations comparing the latent variable and sum score models (both raw data techniques), increasing the number of items from 3 to 10. As the number of items increased, the benefit of the latent variable model increased markedly (as indexed by an increasing difference in the z-scores). This is consistent with the magnitude of the inflated number of associations detected in GW-SEM relative to the other methods. When comparing GWAS results from the multivariate models to the individual item GWAS, the more consistent the associations across the items, the more likely we would observe a multivariate signal, with GW-SEM being more sensitive than the summary statistics methods.

**Keywords** Multivariate GWAS, Neuroticism, GW-SEM

## Unraveling Adolescent Behavioral Profiles: Polygenic Risk and Associations with Alcohol-Related Outcomes in the Adolescent Brain and Cognitive Development (ABCD) Study

**Belal Jamil[1], Angel D. Treviño[1], Jinni Su[1], Kit K. Elam[2], Kathryn Lemery-Chalfant[1], Josè M. Causadias[3], Eleanor K. Seaton[4], Kevin J. Grimm[1], Rick A. Cruz[1]**

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**Abstract**
Adolescent precursors to alcohol use, including internalizing, externalizing, and impulsive behaviors, have often been studied independently despite evidence that these characteristics are genetically and behaviorally interrelated. Polygenic risk likely influences profiles of substance-use related behaviors, however, there is limited research in this area, particularly among minoritized youth. Latent profile analysis (LPA) was employed with data from the ABCD study to delineate profiles based on externalizing and internalizing scores, and impulsivity dimensions (e.g., negative urgency). The sample consisted of Non-Hispanic White/European American ("White"; n = 6170), Hispanic/Latinx ("Latinx"; n = 2098), and non-Hispanic Black/African American ("Black"; n = 1777) adolescents at baseline (T1, 9/10 years old). Alcohol sips and perceived alcohol use harm at the 3-year follow-up (T4) were estimated across classes. Polygenic risk scores (PRS), calculated with PRS-CSx, for adult Major Depressive Disorder (MDD-PRS), externalizing behaviors (Ext-PRS), and Alcohol Use Disorder (AUD-PRS) were tested as predictors for adolescent class membership and T4 alcohol-related outcomes. LPA identified various classes across White (3 classes), Latinx (3 classes), and Black adolescents (4 classes). MDD-PRS predicted greater odds for higher-risk profiles among White and lower-risk profiles among

Black adolescents. AUD-PRS predicted greater odds for higher-risk profiles among Black adolescents. Ext-PRS predicted greater high-risk profiles odds across all three racial/ethnic groups. Ext-PRS predicted lower perceived harms of alcohol use and greater alcohol sips at T4 among White adolescents. These results highlight racial/ethnic variations in adolescent behavior profiles, utility of PRS, and demonstrate the importance of considering how various behavioral precursors of alcohol use can operate conjointly.

**Keywords** Polygenic risk, Latent profile analysis, Adolescents, Racial/ethnic minorities, Alcohol use

**Grants**

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## Associations of Polygenic Scores with Phenotypic Delay Discounting and Non-systematic Responding in the Adolescent Brain Cognitive Development Study

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**Abstract**

Delay discounting (DD, i.e., the tendency to select immediate, smaller rewards over larger, future rewards) is a heritable transdiagnostic factor for substance misuse. Less is known about the genetic contributors to DD in children, as well as non-systematic responding, an understudied, but potentially clinically useful feature reflecting executive dysfunction. We investigated whether polygenic scores (PGS) for DD and correlates (i.e., educational attainment, externalizing behaviors, impulsivity) were associated with phenotypic DD and non-systematic responding using the Adolescent Brain Cognitive Development Study. Participants were 9496 children (9–10 years) who had valid genetic data and completed an adjusting delay discounting task, which requires youth to select between a smaller amount of money available now or a larger amount available after a delay and yields summary parameters that reflect youths' DD rate. Non-systematic responding was operationalized as the number of times indifference points increased by greater than 20% across delays. PGS for DD and correlates were generated based on the largest available genome-wide association studies. Greater PGS for DD ($b = 0.25$, $p < 0.0001$), externalizing symptoms ($b = 0.10$, $p = 0.036$), and lower educational attainment ($b = -0.24$, $p < 0.0001$) were associated with greater phenotypic DD. Greater non-systematic responding was similarly associated with greater PRS for DD, externalizing behaviors, and lower educational attainment; however, higher PGS for motor impulsivity was uniquely associated with non-systematic responding. Findings highlight that the genetic architecture of DD and correlates identified in adults are associated with DD and non-systematic responding in children, and suggest that the genetic architecture of non-systematic responding may be partially distinct from DD performance.

**Keywords** Delay discounting, Polygenic risk scores, Behavior genetics

**Grants**

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## Examining the Role of Externalizing Polygenic Risk Scores and Reciprocal Relationship Between Externalizing Behaviors and Substance Use Intentions in Racially/Ethnically Diverse Adolescents

**Angel D. Trevino[1], Belal Jamil[1], Kit K. Elam[2], Rick A. Cruz[1], Kathryn Lemery-Chalfant[1], Kevin J. Grimm[1], Jose M. Causadias[3], Eleanor K. Seaton[4], Jinni Su[1]**

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**Abstract**

This study examines the effect of externalizing polygenic scores (Ext-PGS) on the reciprocal relationship between externalizing behaviors and substance use intentions during early adolescence.

Data from the Adolescent Brain Cognitive Development (ABCD) study's first four waves (ages 9/10 to 12/13) were analyzed using a Random Intercept Cross-Lagged Panel Model separately across non-Hispanic White ($n = 5906$) and Hispanic ($n = 2115$) youth. The model assessed the between-person and within-person bidirectional effects between externalizing behaviors and substance use intentions, with Ext-PGS as a predictor.

Significant associations were found between Ext-PGS and externalizing behaviors and substance use intentions across White and Hispanic youth. At the between-person level, externalizing behaviors and substance use intentions were associated among White youth but not among Hispanic youth. Within-person associations highlighted positive autoregressive effects for externalizing behaviors and substance use intentions across waves, suggesting deviations from an individual's expected level of externalizing behaviors or substance use intentions at a given wave will likely be followed by deviations in the same direction at the following wave. Cross-lagged effects varied by ethnicity: among Hispanic youth, deviations in externalizing behaviors from an individual's expected level at wave 3 predicted greater subsequent substance use intentions; among White youth, deviations in substance use intentions from their expected level at wave 3 predicted later externalizing behaviors.

Our results underscore the intricate relationship between genetic predispositions and the development of externalizing behaviors and substance use intentions, and contribute to our understanding of the developmental pathways to substance use across diverse populations.

**Keywords** Externalizing polygenic scores, Externalizing behaviors, Substance use intentions, Random intercept cross-lagged panel model

**Grants**

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consent is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

## Genetic Epidemiology of Music Engagement from Early to Late Adolescence Using Twin-Parent Modeling

**Hermine H. M. Maes[1,2,3], Daniel E. Gustavson[4], Naomi P. Friedman[4], Robin P. Corley[4], Sally J. Wadsworth[4], Chandra A. Reynolds[4]**

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**Abstract**
Music engagement is an influential factor in many adolescents' lives. However, there are few genetic studies of music engagement in adolescence, a time when music engagement may be changing dramatically. The current study examined data from N = 1863 individual twins/siblings from the Colorado Adoption/Twin Study of Lifespan behavioral development and cognitive aging (CATSLife) sample who completed self-reported measures of music engagement at mean ages 12 and 16 (including measures of interest in musical instruments, talent, lessons, and the number of instruments played) with similar measures obtained from their parents. Results indicated individual differences in music engagement change considerably across this period. The heritability of music engagement was trending upwards between ages 12 and 16 ($a^2$ = 0.20 and 0.28, respectively), as did unique environmental influences ($e^2$ = 0.37 and 0.48), while shared environmental contributions were decreasing ($c^2$ = 0.44 and 0.24). Only a fraction of the additive genetic contribution was accounted for by assortative mating (marital correlations r = 0.15). The largest source of shared environment was twin-specific effects while vertical cultural transmission was not significant. These findings highlight the fact that music engagement is a dynamic phenotype that is increasingly explained by genetic influences between early and late adolescence. Studies examining associations between music engagement and health should carefully consider timing of assessment and the (increasing) role that genetic influences may play in these associations.

**Keywords** Music engagement, Twin-parent modeling, Biological and adoptive siblings, Adolescence

**Grants**
R01 AG046938, U01 DA051037.

## Epigenetic Aging in Siblings Differentially Exposed to Prenatal Maternal Smoking

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**Abstract**
Exposure to maternal smoking during pregnancy (MSDP) has been shown to impact offspring DNA methylation (DNAm) longitudinally into adolescence. We have previously found that global DNAm, along with DNAm in several specific genes (*CNTNAP2, MYO1G, CYP1A1*), was associated with MSDP in 8–15-year-old siblings exposed to more smoking during pregnancy compared to their unexposed (or less exposed) sibling. Additionally, previous studies have reported that prenatal exposure to maternal smoke was associated with increased epigenetic aging acceleration. Extending this work, we tested whether findings of MSDP and epigenetic age acceleration replicate in the sibling-comparison design, which rules out familial confounds that could inflate associations found in the literature (n = 321 families; $M_{age}$ = 13.02 for sibling 1; 10.20 for sibling 2). We quantified smoking exposure severity for each sibling reflecting time and quantity of MSDP. MSDP scores were assessed as sibling average MSDP (i.e., between-family; average severity across siblings, indicating familial risk/confounding related to MSDP) and each individual child's MSDP score relative to the sibling average (i.e., within-family centered MSDP, indicating child-specific effects attributable to exposure to MSDP). We assessed epigenetic aging using the Pediatric-Buccal-Epigenetic (PedBE) clock. Preliminary multilevel models showed a within-family association such that MSDP was associated with increased epigenetic aging in the sibling exposed to more MSDP ($\beta = 0.31$, $p < 0.001$). We plan to expand on these findings by comparing the results of different clocks and controlling for environmental covariates such as second-hand smoke exposure and neighborhood quality.

**Keywords** Prenatal, Epigenetics, Smoking

**Grants**
This work is supported by NIH Grants: DA023134 (Knopik), DA17671 (Knopik).

## Behavioral Inhibition and Parental Supervision: Testing Transactional Pathways Across Early Childhood and Influence of Gene–Environment Correlation

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**Abstract**
Poor monitoring/supervision is a key parenting factor that is related to low behavioral inhibition in child, a risk factor for later substance use and psychopathological symptoms. However, little is known about whether parent's poor supervision can be evoked by inhibition-related genetic factors through child phenotypic behavioral inhibition (i.e.,

evocative gene–environment correlation; evocative rGE). We leveraged data from 561 children (57.2% male, 55.3% non-Latino White) and birth and adoptive parents of the Early Growth and Development Study, a prospective adopted-at-birth design. It was proposed that (i) birth mother behavioral inhibition would predict child behavioral inhibition at age 4.5, 6, and 7; and (ii) lower behavioral inhibition in child would evoke higher parental supervision at the next timepoint, and higher parental supervision increased child behavioral inhibition at the subsequent timepoint. Hypotheses were tested in random-intercept cross-lagged panel models using MPlus 8. Results revealed that random intercepts of primary/second caregiver supervision and child behavioral inhibition were associated [primary caregiver: $\beta = 0.32$, $se = 0.08$, $p < 0.001$; second caregiver: $\beta = 0.18$, $se = 0.07$, $p = 0.018$]. Birth mother behavioral inhibition predicted child behavioral inhibition at age 6 [$\beta = 0.31$, $se = 0.16$, $p = 0.048$], and within-person behavioral inhibition at age 6 was positively associated with within-person second caregiver (mostly adoptive father) supervision concurrently [$\beta = 0.34$, $se = 0.17$, $p = 0.040$]. No evidence for evocative rGE was detected. Findings suggested that early childhood behavioral inhibition and primary caregiver's supervision might not co-develop, but rather correlated at the stable trait level. Age 6 might be a sensitive period for second caregiver to respond to high behavioral inhibition with high supervision.

**Keywords** Behavioral inhibition, Parenting, Early childhood, Adoption design, Gene–environment correlation

**Grants**

R01 HD042608; R01 DA020585; R01 MH092118; R01 DK0 90264; R01 DA035062; R56 HD042608; UH3OD023389; R01D A045108.

## Considering the Risks, Potential Benefits, and Ethical Responsibilities of Social and Behavioral Genomics

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**Abstract**

Social and behavioral genomic research is a field shaped by a fraught and controversial history. This presentation will present the results of a 3-year interdisciplinary effort to examine the risks and potential benefits of social and behavioral genomics (SBG) and to develop recommendations to enhance ethical responsibility in the field. In addition, accompanying the 3-year interdisciplinary academic working group (AWG) was the formation of an 18-month community sounding board (CSB). The CSB was formed to gather public input on the risks and potential benefits of SBG. This presentation describes the process and outcomes of both the 3-year AWG and the 18-month CSB; it offers attendees practical recommendations for enhancing the responsible conduct and communication of SBG.

**Keywords** Social and behavioral genomics, ELSI, Public engagement, Interdisciplinary

**Grants**

NHGRI K01 HG013352-01.

## Simulations of Measurement Non-invariance Effects on PRS Associations with Psychopathology Factors Across Race, Ethnicity, and Sex at Birth

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**Abstract**

Genomics researchers have made concerted efforts to increase inclusion of underrepresented minority individuals (URMs), while finding decreased predictability of Polygenic Risk Scores (PRSs) derived from European ancestry individuals in non-European ancestry individuals. Somewhat analogously, despite efforts to increase inclusion of URMs in studies of psychopathology, we know little about similarities and differences in the structure and measurement of psychopathology. Certain aspects of the measurement of psychopathology may be similar across self-reported race/ethnicity and sex, whereas others differ. Unfortunately, studies testing for measurement invariance tend to be small and non-representative, with limited coverage of psychopathology domains. Herein, I present simulations based on sample characteristics from the *All of Us* Research Program to examine the effects of measurement non-invariance on associations of psychopathology dimensions with a PRS explaining 1%, 4%, or 9% of their variance. I systematically varied (a) diagnostic thresholds, (b) factor loadings, (c) factor variances, (d) factor correlations, and (e) factor means, and examined bias and precision of standardized regression coefficient estimates (ß's) for the PRS associations, how often these estimates contained the true value of ß, and how often the null hypothesis that ß = 0 was rejected. Preliminary findings suggest substantial underestimates of PRS associations in the group with the larger (unmodeled) psychopathology factor variance which increased as the variances diverged, with steeper bias for larger associations. Smaller effects on associations were observed for thresholds and factor loadings. Similar to modeling genomic ancestry differences, modeling differences in the measurement of psychopathology are important for decreasing mental health disparities.

**Keywords** Simulations, Measurement non-invariance, PRS associations, Psychopathology, Mental health disparities

## A Twin Study of Social Media Addiction in the ABCD

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**Abstract**

While the proliferation of ubiquitous internet access and handheld digital devices has fundamentally changed our communication as a species over the past few decades, behaviors associated with digital device usage have received limited investigation through a biological lens. As these technologies become more pervasive in our day-to-day activities, some individuals find themselves in a pattern of compulsive

and persistent usage of technology akin to addiction. While genetic factors are known to contribute to the presentation of addictive behaviors, compulsive engagement with novel technologies is an emergent topic of inquiry.

Using the Adolescent Brain Cognitive Development Study (ABCD) data on novel technologies, we sought to investigate patterns of digital device usage, primarily with self-reported questionnaire data adapted from the Bergen Facebook Addiction Scale (BFAS; Andreassen et al. 2012). BFAS scores were analyzed and incorporated into a classical twin study model on a subsample of the ABCD twins who had endorsed having at least one social media account (N = 1112; 215 MZ pairs, 341 DZ pairs). Models supported moderate evidence of an additive genetic contribution towards addictive social media behavior in young adolescents with social media access. Additionally, alternative models incorporating polygenic risk for general addiction and accounting for parental parameters for their children's device usage were considered.

**Keywords** Twin study, Addiction, Adolescence, Novel technologies

**Reference**

Andreassen CS, Torsheim T, Brunborg GS, Pallesen S (2012) Development of a Facebook addiction scale. Psychol Rep 110(2):501–517

## Epigenetic Markers of Inflammatory Burden Associated with High Frequency Cannabis Use in a High-Risk Sample of Adolescents

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**Abstract**

While the active constituents of cannabis [e.g., tetrahydrocannabinol (THC) and cannabidiol (CBD)] are thought to have anti-inflammatory properties, human studies that directly examine associations between cannabis use and inflammation are few and findings are mixed. C reactive protein (CRP) and interleukin 6 (IL-6) are phasic proteins produced during the immune response and are often measured in blood plasma as markers of acute inflammation. Thus, blood-based markers may not be as representative of chronic inflammation. Epigenetic signatures derived from DNA methylation, however, may provide a more representative measure of chronic inflammation due to the epigenetic marker's longitudinal stability. We examined whether polyepigenetic markers of CRP and IL-6 were associated with frequent cannabis use in a sample of N = 162 adolescents (aged 13–18) hospitalized for suicidal thoughts and behaviors. Peripheral blood DNA methylation was used to calculate polyepigenetic scores using weights from prior studies and regressed on cannabis use (Stevenson et al. 2021; Wielscher et al. 2022). Models controlled for sex, age, genetic principal components, cell type, tobacco smoking, and BMI. Results indicated that youth who used cannabis several times a week or more had higher IL-6 scores ($b = 0.03$, $SE = 0.01$, $p < 0.001$). This effect was not observed for lower levels of cannabis use (e.g., any past 30-day use, lifetime use, etc.) Thus, despite some evidence suggesting potential for use of THC/CBD as a treatment for inflammation, the pattern of results suggests that youth with high levels of cannabis use exhibit epigenetic marks of higher inflammatory burden.

**Keywords** Epigenetics, Inflammation, Cannabis

**Grants**

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## Higher Cigarette Taxes During Adolescence More Effective at Curbing Smoking and Associated Chronic Illnesses in Genetically Vulnerable

**Robel Alemu[1,2], Lauren L. Schmitz[3]**

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**Abstract**

Tobacco use is the leading cause of preventable death worldwide and claims the lives of over half a million people each year in the United States. Higher cigarette taxes, in tandem with other tobacco control policies, have contributed to a substantial drop in adult smoking prevalence since 1965, averting approximately 8 million deaths. In recent years, however, large increases in cigarette taxes have not yielded commensurate reductions in cigarette consumption. The precise mechanisms behind the recent stability in smoking prevalence and the prevailing heterogeneities in demand responses to cigarette taxes are not adequately understood. This study coupled historical data on state-level cigarette excise taxes with polygenic indices (PGIs) for smoking behavior to assess whether genetically driven changes in lifetime smoking varied by the magnitude and timing of cigarette tax exposures. Using data from the U.S. Health and Retirement Study (HRS), we find that higher cigarette taxes in adolescence were more likely to reduce the probability of lifetime smoking in individuals with higher PGIs for smoking behavior and addiction than cigarette taxes in adulthood, whereas smoking intensity appears to be more sensitive to current taxes. The genetic gradient in the efficacy of cigarette taxation is consistent with the predictions of the rational addiction model. These findings suggest that studies that do not account for the heightened price sensitivity of individuals with greater genetic liability for smoking may have underestimated the true effects of cigarette taxes on smoking behavior and subsequent health outcomes.

**Keywords** Smoking, Cigarette excise taxes, Polygenic index (PGI), Gene–environment interaction, Health and retirement study (HRS)

**Grants**

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## Epigenome-Wide Association Study of APOE ε4 Allele Status in the Louisville Twins

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**Abstract**
The APOE ε4 allele is the most well-known genetic risk factor for Alzheimer's disease (AD), with an upper-bound estimate of 25% of the population having at least one copy. Prior epigenome-wide association studies (EWAS) suggest that the APOE gene is hypomethylated in AD cases compared to cognitively normal controls. APOE ε4 status also may be correlated with methylation genes in other DNA regions that may be associated with AD. We present results from an EWAS of APOE ε4 status (0 = absence, 1 = presence) using 837,274 CpG sites in 285 twins from the Louisville Twin Study. The discovery sample included 207 individual twins while complete DZ twin pairs ($N = 39$) were retained for our replication sample to test within-pair associations between APOE ε4 status and DNA methylation. Although no sites were significantly correlated with APOE ε4 status ($p < 5 \times 10^{-8}$), cg06638180 (chromosome 12) and cg17726207 (chromosome 5) emerged as the top two sites. Site cg06638180 contained the KSR2 gene, which is correlated with obesity, insulin resistance, and impaired cellular fuel oxidation. An exploratory gene ontology analysis showed that the MED16 gene, a mediator complex protein coding gene located on chromosome 19, was enriched in sites associated ($p < 0.001$) with APOE ε4 status. The top CpG sites did not replicate in the complete-pair DZ sample, suggesting null association between APOE ε4 status and each probe site within families. Current findings may inform future research attempting to clarify how APOE ε4 status interacts with methylation of genes potentially related to AD.

**Keywords** APOE ε4, Epigenetic, EWAS, DNA methylation, Alzheimer's disease

**Grants**
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## The Genetic and Environmental Contributions to the Associations Between Extracurricular Activities and Substance Initiation in African American Adolescent Twins

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**Abstract**
Participation in extracurricular activities is consistently associated with a reduced odds of substance use initiation (SUI) in adolescents. To date, most genetic epidemiology research of extracurricular activities and SUI has focused on religious involvement. It is unclear whether the genetic and environmental factors contributing to variation in a range of extracurricular activities are also associated with substance use initiation.

Data from a sample of 330 participants (N = 160 pairs) ages 13–20 from the African American Adolescent Twin Study were used to assess the associations between participation in five extracurricular activities in the past 6 months (school activities, athletics, community activities, holding a job, and religious activities) and lifetime initiation of conventional cigarettes, alcohol, and cannabis. Bivariate models were fitted to determine the extent to which genetic and environmental influences were also shared between extracurricular activities and initiation of each substance.

Approximately 24–36% of participants engaged in lifetime initiation across each substance and 35–92% of participants engaged in each of the extracurricular activities. In general, there were negative associations between engagement in extracurricular activities and SUI. However, positive associations were detected between holding a job and SUI across all substances. Shared genetic, common environment, and unique environmental factors contributed to the phenotypic correlations. These results emphasize the idea that extracurricular activities demonstrate varied patterns of association with substance use initiation. Further, the results encourage additional review across multiple activities to develop future substance use prevention strategies.

**Keywords** Drug use, Drug dependence, Drug abuse, High risk, Resilience, Prevention

**Grants**
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## Evidence of Horizontal and Vertical Pleiotropy Between Substance Use and Indices of Aging

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**Abstract**
Substance use and use disorders have been associated with accelerated epigenetic, neuroanatomical, and chronological aging. While it has been theorized that prolonged substance use confers liability to accelerated aging via a combination of biological and correlated environmental/lifestyle factors, no study to date has examined the role of horizontal and vertical pleiotropy between these two trait categories. To address this gap, we examined the degree of association between measures of substance use and aging, and the extent to which they are being driven by horizontal vs. vertical pleiotropy. We first evaluated the genetic correlations between 6 substance-related traits (covering cannabis, alcohol, nicotine, and opioid usage) with 10 aging-related traits using LD score regression, finding that each of our substance use traits (with the exception of cannabis use ever) was genetically correlated with the majority of our aging measures. For all

trait pairs reaching a Bonferroni-adjusted significance level ($p \leq 0.008$), we subsequently tested for vertical pleiotropy using Causal Analysis Using Summary Effect estimates (CAUSE), and evaluated the relative contributions of horizontal and vertical pleiotropy using Mendelian Randomization Pleiotropy RESidual Sum and Outlier (MR-PRESSO). Although results are still largely forthcoming, initial findings indicate that problematic alcohol use (PAU) is causally related to measures of aging; in particular, PAU was found to have a causal effect on frailty, telomere length, and our composite aging factor after correcting for the influence of vertical pleiotropy. Collectively, these preliminary findings underscore the intricate relationship between substance use, aging, and their shared biological/environmental factors.

**Keywords** Substance use, Aging, Mendelian randomization, Pleiotropy, Genetic architecture

## Genetic and Non-genetic Parental Influences on Depressive Symptoms Using Norwegian Mother, Father, and Child Cohort Study

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**Abstract**

Behavioral traits in children are shaped by genetic and environmental influences. Genetic studies focused mostly on direct genetic effects (i.e. nature), although lately, indirect genetic effects (i.e. nurture) started to receive more attention. Here we are testing a novel approach, SEM PGS, that is jointly modelling transmitted and non-transmitted haplotype-specific parental polygenic scores (hPGSs) and their effects on offspring's phenotypes while simultaneously estimating assortative mating and vertical transmission.

Data from Norwegian Mother, Father, and Child Cohort Study were used by selecting full trios with only one child per family. LDpred2 software and summary statistics from the biggest genome-wide association study on depressive symptoms (N = 161,460) were used to calculate the weights needed for hPGSs. Depressive symptoms were measured using Hopkins Symptoms Checklist and Child Behaviour CheckList. Main analyses are being conducted in R software (OpenMX package), using the structural equation modelling implemented in SEM PGS.

Overall, 16,086 mothers, 15,152 fathers, and 4454 children were genotyped with the information on depressive symptoms. In total, 846,168 SNPs were used to calculate weights for hPGSs. Results of main analyses are still to be determined.

To our knowledge, this is the first study estimating both direct and indirect genetic effects involved in the development of depressive symptoms, using SEM PGS. Utilization of genetically informed models, such as SEM PGS, on extensive dataset, such as MoBa, brings us one step closer to understanding how nature and nurture shape the development of behavioral traits.

**Keywords** SEM PGS, Genetic nurture, Neuroticism, MDD

**Grants**

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## Gene–Environment Interaction Study of Adverse Childhood Experiences and DNA Methylation Age Acceleration

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**Abstract**

Adverse childhood experiences (ACEs) encompass a range of early negative life experiences that affect physiological functioning in childhood and adulthood. People exposed to greater numbers of ACEs also have been found to have higher DNA methylation age acceleration (DNAm-AA) values (i.e., biological ages older than predicted from chronological age) in middle adulthood (Kim et al. 2023). Genetic variance contributes to DNAm-AA measures, so we tested whether higher ACEs augmented genetic variance underlying PhenoAge acceleration (PhenoAA) and GrimAge acceleration (GrimAA). Using 101 pairs of twins ($N_{MZ} = 64$; $N_{DZ} = 37$) from the Louisville Twin Study, we fit gene-by-environment interaction models in which we tested whether the number of ACEs (out of 10 possible items) moderated genetic and environmental variance of PhenoAA and GrimAA. Twins reported experiencing 1.51 ($SD = 1.81$) ACEs, on average. Genetic variance accounted for 55.21% of the variance in PhenoAA and 77.62% of the variance in GrimAA. Preliminary results suggest that total ACEs did not significantly account for phenotypic variance in PhenoAA but did for GrimAA ($R^2 = 1.78\%$). ACEs did not moderate the genetic and environmental variance underlying either variable. Future research should consider whether ACEs moderate genetic variance of PhenoAA and GrimAA in twins from low socioeconomic backgrounds, as SES correlates negatively with total ACEs, PhenoAA, and GrimAA.

**Keywords** Adverse childhood experiences, DNA methylation, Age acceleration, Epigenetic clocks

**Reference**

Kim K, Yaffe K, Rehkopf DH et al (2023) Association of adverse childhood experiences with accelerated epigenetic aging in midlife. JAMA Netw Open 6(6):e2317987. https://doi.org/10.1001/jamanetworkopen.2023.17987

**Grants**

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## Lifetime Number of Sexual Partners: A Measure of Risk-Taking Behavior or Reproductive Fitness?

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**Abstract**
This study examines the genetic architecture of risky sexual behavior (RSB), focusing on the number of lifetime sexual partners as an index of RSB. It explores the relationship between genetic predispositions for RSB, risk-taking/risk tolerance, and reproductive fitness using Genomic Structural Equation Modeling (Genomic SEM) and Mendelian randomization (MR). Findings indicate a stronger genetic correlation between the number of sex partners and risk-related phenotypes than with reproductive fitness. Direction of causality was unable to be determined from MR as the effects were significant in both directions for the risk-related phenotypes and only marginally significant for a subset of the fitness traits. Despite the complexity and interrelatedness of these measures, the study contributes to understanding the genetic underpinnings of RSB, highlighting its association with risk-taking behaviors.

**Keywords** Number of lifetime partners, Externalizing, Reproductive fitness

## Multivariate Parental Effects Across the Internalizing–Externalizing Spectrum in Childhood: Results from the Norwegian Mother, Father, and Child Cohort Study

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**Abstract**
Numerous studies have characterized common genetic effects across the spectrum of internalizing–externalizing behaviors in childhood. In general, the underlying genetic contributions have been found to be adequately represented with a simple structure encompassing only a few underlying genetic factors. However, resolving indirect genetic effects of parents form direct genetic effects of children has not received the same attention. These may represent completely different underlying mechanisms.

In this study, we aim to characterize the importance and structure of genetic differences between children and parents across the spectrum of internalizing–externalizing behaviors in children. We address this by modelling multivariate direct genetic effects and indirect maternal genetic effects across seven symptom scores of internalizing and externalizing behaviors. The study relies on symptom measures at age 8 from 42,423 participants in the Norwegian, Mother, Father and Child Cohort Study (MoBa). Genetic effects were identified by combining expected relatedness coefficients derived from pedigrees with empirical coefficients derived from genotype data.

The results were consistent with a maternal contribution across all seven symptoms measures. Moreover, maternal genetic effects could to a large extent be accounted for by a single underlying dimension shared across all symptom measures. We further found that the magnitude of direct genetic effects was smaller, and their dimensionality was greater, when we simultaneously modelled maternal genetic effects.

**Keywords** Internalizing, Externalizing, Parental genetic effects, Direct–indirect effects

## Equity in Genomic Research

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**Abstract**
Since the publication of the first human genome sequence two decades ago, millions of genomes have been sequenced or genotyped. However, the vast majority were European ancestry samples. Despite repeated calls for change, representation of diverse populations has not improved over the last years. Whilst lack of data has received much attention, Euro-centric bias affects genomic research in many complex ways. We used different approaches to identify areas that need improvement in order to ensure genomic medicine is equitable.

Many statistical models assume homogeneity across samples and do not adequately account for cross-population diversity or admixture. This often leads to exclusion samples from minority groups. Through literature review we found that the current landscape of statistical methods aimed at enhancing equity in genomics predominantly centres around genetic ancestry (74%), with a dearth in methods related to other critical social axes of discrimination.

We also assessed bias and its implications in genomic medicine patient data from the 100,000 Genomes Project. Ethnicities in the cancer sub-cohort were largely representative of those in England. However, for both cancer and rare disease patients we found that significantly more genetic variants were prioritised as potentially pathogenic in most ancestry groups compared with the European ancestry group, potentially implying higher rates of variants of uncertain significance.

Finally, we developed a roadmap to enhancing inclusion and ensuring equal health benefits of genomics advances. Our work highlights the importance of sincere concerted global efforts towards genomic equity to achieve the benefits of genomic medicine to all.
**Keyword** Plenary

## Depression: Dissecting Heterogeneity

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**Abstract**
Depression is a highly heterogenous disorder. Although considered as a unitary diagnostic concept by clinicians, depression is extremely variable in terms of its age-at-onset, causal risks and antecedents,

symptom constellation, comorbidities, treatment response and long-term outcomes. These sources of heterogeneity represent major challenges for clinicians who need to select the most effective treatment and estimate outcomes for individuals who differ so greatly. Also, although for clinical purposes, we conceptualise depression as a yes/no diagnostic category, there is much to suggest that it behaves dimensionally.

In this talk, I discuss how genetics has helped us understand more about depression heterogeneity and could aid us shape future conceptualisations of psychopathology. I will review research findings on the genetic architecture of depression, as well as consider how genetics has informed us about the importance of development, antecedents, prognosis and gene–environment interplay. I will then talk about some new genetics research that highlight the neurodevelopmental origins of very early-onset depression.

**Keyword** Plenary

## How Can Molecular Genetic Family-Based Studies Help Behavioural Geneticists?

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**Abstract**

GWAS have discovered thousands of associations between behaviours and other social phenotypes and genetic variation across the genome. Most GWAS have been conducted in samples of unrelated individuals, in which relatedness has been treated as, at best, a nuisance to be controlled. However, we now have substantial evidence that genetic associations for many phenotypes, particularly those that are social and behavioural, are likely due to indirect genetic effects. These indirect effects can confound studies of unrelated individuals aiming to identify the individual-level causal effects of genetic variation. Studies of genotyped related individuals, such as parent–offspring trios or siblings, can overcome this confounding. Behavioural geneticists, if equipped with sufficient molecular genetic data from families, have a unique opportunity to discover new, more credible insights into the causes and consequences of individual, familial and societal risk factors for psychological and psychiatric outcomes. Thus, future population-based studies should prioritise sampling related individuals.

**Keywords** GWAS, Parent–offspring trios, Siblings, Indirect genetic effects

## Executive Function Mediates the Genetic Associations Between ADHD and Comorbid Psychiatric Conditions

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[1]Psychology, University of Ibadan, Ibadan, Nigeria. [2]Institute for Behavioral Genetics, University of Colorado, Boulder, USA. [3]Psychology and Neuroscience, University of Colorado, Boulder, USA

**Abstract**

Previous studies have shown that ADHD has a high comorbidity with other psychiatric disorders, both phenotypically and genetically, and that it is correlated with impairments in cognitive traits such as executive functioning. However, how variations in different cognitive traits influence the relationship between ADHD and comorbid psychiatric conditions remains unclear.

We applied GenomicSEM to investigate the shared genetics of ADHD and five psychiatric disorders: cannabis use disorder (CUD), major depressive disorder (MDD), bipolar disorder (BD), schizophrenia (SCZ), and alcohol use disorder (AUD), alongside the genetic correlation between ADHD and eight cognitive traits including executive function (EF). The data are summary statistics derived from GWAS of individuals of European ancestry. Additionally, we employed mediation analyses to see how the different cognitive traits affect the relationships between ADHD and the comorbid psychiatric disorders.

ADHD negatively correlated with cognitive traits. In simple mediation models, EF had significant indirect effects on ADHD-paired psychiatric disorders (CUD: 0.0301, $p < 0.05$; MDD: 0.014, $p < 0.05$; BIP: 0.0593, $p < 0.001$; SCZ: 0.0737, $p < 0.001$; ALCH: 0.0351, $p < 0.01$). However, all significant indirect effects became nonsignificant in multiple mediation models.

EF significantly influenced the genetic associations between ADHD and comorbid psychiatric disorders, underscoring its importance in understanding these relationships. However, controlling for other cognitive traits in multiple mediation models showed no significant mediating effects. EF's complex interactions with other cognitive traits may contribute to its attenuated effect when controlling for other cognitive traits.

**Keywords** Executive function, ADHD, Comorbidity

## Purported Genetic Basis of Group Differences in Intelligence is a Conspiracy Theory

**Eric Turkheimer**[1]

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**Abstract**

So-called conspiracy theories have always existed and have proliferated in the era of social media. But what, exactly, is a conspiracy theory? To the people who believe in them, "conspiracy theory" is just an epithet wielded to discredit ideas that the establishment dislikes. I suggest that designating an idea as a conspiracy theory has a specific epistemological content. Conspiracy theories are driven by the impossibility of proving an absolute negative. Suspicions that observed behavioral differences between the races have continued unabated since Darwin published Descent of Man. No more genetic evidence for the hypothesis exists now than it did then, yet the belief that there must be something genetic down there somewhere persists unabated. The most recent review in support of the hypothesis (Warne 2021) puts it explicitly: "Between-Group Mean Differences in Intelligence in the United States Are > 0% Genetically Caused". How could such a hypothesis be disproved? Under this evidentiary standard, any scrap of evidence that might be an indicator of genetic causes is sufficient to keep the conspiracy theory alive. I use this analysis to reach some novel conclusions about the never-ending debate regarding group differences in intelligence.

**Keywords** Group differences, Intelligence, Conspiracy theory

**Reference**

Warne RT (2021) Between-group mean differences in intelligence in the United States are > 0% genetically caused: five converging lines of evidence. Am J Psychol 134(4):480–501

## Importance of Caucuses Within Professional Organizations: Creating the Behavior Genetics Association Student Caucus

**Amy M. Loviska[1], Valerie Knopik[1]**

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**Abstract**
Caucuses provide spaces for the support, advocacy, and professional development of members who share similar special interests or identities. Caucuses can increase communication and facilitate discussion of issues and experiences within a comfortable space prior to sharing with a larger organizing body. For students, Caucuses can create opportunities for enhanced involvement with professional organizations by offering leadership and volunteer positions. Within international organizations, caucus membership can promote cultural awareness and cross-cultural collaboration as well as increase access to academic and professional opportunities within other countries.

Given the benefit of caucus membership, the Behavior Genetics Association (BGA) is interested in creating a caucus for early career professionals. The purpose of the student caucus will be to support the goals of BGA while fostering community between early career members through the organization of social, academic, and professional activities. Once established, the student caucus will be run and managed by early career BGA members with mentorship from the past BGA president. To facilitate the genesis of the BGA student caucus, a committee comprised of BGA affiliated early career professionals will be formed. Committee responsibilities will include (a) information gathering through surveys and/or focus groups, (b) generation of caucus bylaws, (c) management of the first BGA student caucus elections, and (d) creation and hosting of first BGA student caucus programming.

**Keywords** Early career, Professional organizations, Leadership

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